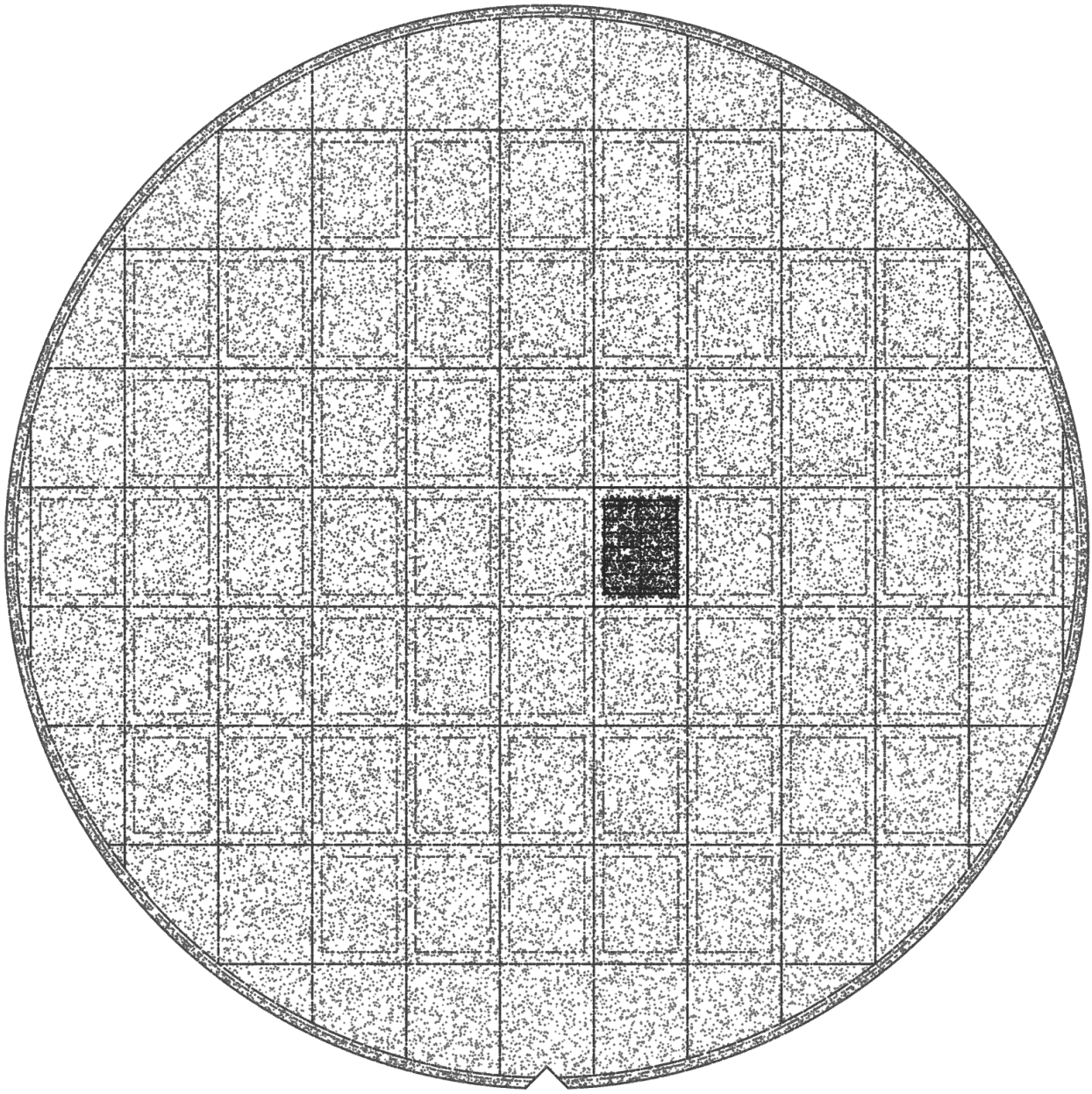


Chip Supply Chain

How AI chips are made

Every stage of the supply chain, and who controls it.

Yashvardhan Sharma

Edition of 28 October 2026 · chipsupplychain.org

Supply Chain

This edition covers each of the fourteen stages of making an AI chip in detail, one chapter per stage, in the order a chip is made, from design software to the data center. Two more chapters at the end look across all the stages, at what a chip costs and who controls the chain. Each chapter ends with questions to answer from memory. Online, every chapter also has a 3D model, and most have an animation.

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These two chapters are not stages. They look across all fourteen, at what a chip costs and at the contest over who controls each stage.

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The chain

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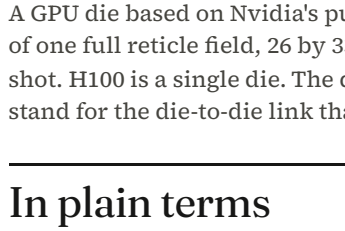
15 The Economics of AI Chips
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Stage 1 of 14 · Extreme concentration · Hard to substitute

Chip Design, EDA and IP

Synopsys, Cadence and Siemens EDA make the software used to design every advanced chip. In 2025 they took over 85 percent of its $18 billion market, licensed blocks included.

Specimen 01 · Chip Design, EDA and IP



A GPU die based on Nvidia's published H100 block diagram. It is drawn at the size of one full reticle field, 26 by 33 mm, the largest area a scanner can print in one shot. H100 is a single die. The drawing adds a copper strip on one short edge to stand for the die-to-die link that two-die designs such as Blackwell carry.

In plain terms

Chip design turns an idea for a new chip into a complete blueprint. A chip holds billions of transistors, tiny electrical switches. The blueprint fixes where each one sits and how it is wired to the others. The chip runs on a clock, and a signal has to cross those wires within one tick, less than a billionth of a second. Software from three companies, Synopsys, Cadence and Siemens, lays out the blueprint, checks it and signs it off. A foundry, a factory that makes chips for other companies, accepts only designs that have passed that software's checks. The foundry prints each layer of the design onto silicon through a stencil called a mask. Changing the blueprint once the masks are made takes new masks and a few months[1].

In short

At advanced nodes, 7 nm and below, Synopsys alone has more than 90 percent of the market for static timing analysis, the check that every signal arrives in time[2]. SemiAnalysis wrote in May 2026 that no startup can replicate a complete design flow by writing it from scratch[2].

Since October 2022 a US license has been needed to export design software specially made for chips with gate-all-around transistors, a transistor design the rule calls key to processes at 3 nm and below[3]. As of 2025, the think tank CSIS found Chinese design tools still far from replacing the full range of advanced capabilities US firms offer, especially for 5 nm and 3 nm chip design[4]. But Chinese design-software companies were winning share

1 The Dark Side Of The Semiconductor Design Renaissance – Fixed Costs Soaring Due... SemiAnalysis, 24 July 2022. newsletter.semianalysis.com

2 EDA Market Primer. SemiAnalysis, 21 May 2026. newsletter.semianalysis.com

3 Implementation of Certain 2021 Wassenaar Arrangement Decisions on Four Section... Federal Register, 15 August 2022. federalregister.gov

4 The Double-Edged Sword of Semiconductor Export Controls: Electronic Design... Center for Strategic and International Studies, 6 October 2025, p. 33. csis-website-prod.s3.amazonaws.com

at older manufacturing processes, the research firm SemiAnalysis reported[2].

Design is the only stage of the supply chain that happens entirely in software. The US government has restricted it with letters to the companies that sell that software. In late May 2025 the government required a license to sell chip design software to anyone in China[5]. Cadence cut off its Chinese customers until the government lifted the requirement on 2 July[5].

Chokepoint card Chip Design, EDA and IP

Concentration

Extreme

Three or fewer firms could supply this stage today, or one firm holds more than half of it.

Substitutability

Hard

Only three complete sets of chip design software exist[6]. A fourth would have to be written from nothing and approved by a foundry[2].

Price or market size

$18 billion

world market for design software, simulation software and licensed circuit blocks in 2025[2]. SemiAnalysis put a single respin, a design revision that needs some new masks, at leading-edge nodes at $50 million to $100 million in May 2026[2]

Who leads	US	**Synopsys**	Over 90 percent of the timing-signoff market at advanced nodes as of 2026[2]
	US	**Cadence**	55 to 60 percent of the hardware emulator market at advanced nodes as of 2026[2]
	US	**Siemens EDA**	Over 85 percent of physical verification, the final layout check, at advanced nodes as of 2026[2]
	GB	**Arm**	About 50 percent of processor compute at the top cloud firms as of May 2026[7]
	US	**Broadcom**	$16.7 billion of AI semiconductor revenue in the third quarter of fiscal 2026, custom accelerators and networking combined[8]
Where it is made	US	**United States**	Synopsys, Cadence, Broadcom, Marvell and Nvidia
	GB	**United Kingdom**	Arm processor and interconnect designs licensed from Cambridge, and since 2026 its own server chip
	TW	**Taiwan**	Alchip, Global Unichip and MediaTek: design services that carry a chip to the factory
	IL	**Israel**	Annapurna Labs, the AWS design house behind Trainium and Graviton
	IN	**India**	Nearly 20 percent of the world's chip design engineers, on the Indian government's count
Why substitution is slow	Changing one tool, such as the synthesis tool that turns a design's code into logic cells, means re-running the later steps of the design[2]. The foundry that makes the chip also specifies which signoff tools its customers must use[2].		

2 **EDA Market Primer.** SemiAnalysis, 21 May 2026. newsletter.semianalysis.com

5 **Cadence Design Systems, current report on Form 8-K dated 2 July 2025.** U.S. Securities and Exchange Commission, 3 July 2025. sec.gov

6 **The Double-Edged Sword of Semiconductor Export Controls: Electronic Design...** Center for Strategic and International Studies, 6 October 2025, p. 13. csis-website-prod.s3.amazonaws.com

7 **Arm delivers record-breaking quarter and full-year results.** Arm, 6 May 2026. newsroom.arm.com

8 **Broadcom, current report on Form 8-K dated 2 September 2026.** U.S. Securities and Exchange Commission, 2 September 2026. sec.gov

Where China stands

China designs its own AI accelerators, but Huawei's most powerful chip in 2026, the Ascend 950, delivers about half the computing performance of Nvidia's H100 from 2022, on Epoch AI's estimate[9]. CSIS found that as of 2025 Chinese design tools were still far from replacing the full range of advanced capabilities US firms offer, especially for 5 nm and 3 nm chips[4].

Where the US stands

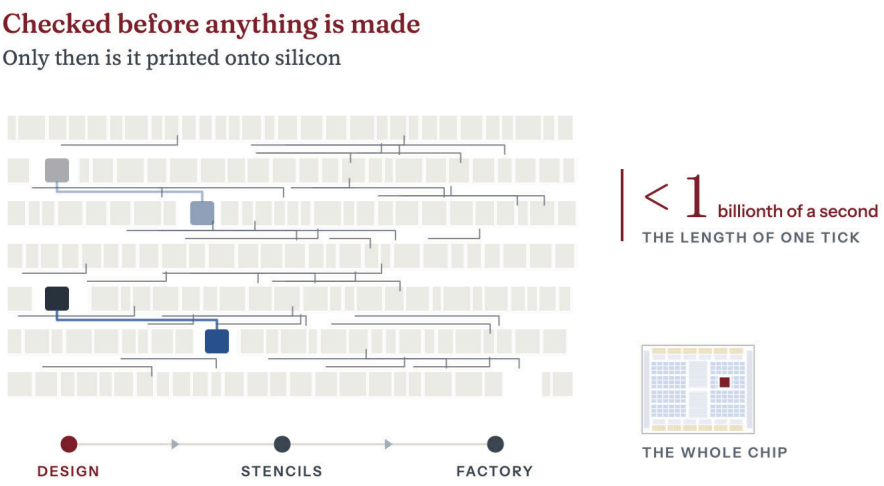
Synopsys and Cadence, the two largest design-software vendors by 2025 revenue, are American[2].

01

How it works

Figure 01.1 · Explainer, 23 seconds

How a chip is designed



1. **A blueprint.** Making a chip starts with design, which turns an idea into a complete blueprint. The blueprint fixes where each of the chip's billions of switches sits and how each one is wired to the others.
2. **One tick.** The chip works in ticks, each less than a billionth of a second. On every tick a signal leaves a register, a small store that holds one number, and it has to cross its wire to the next register before the next tick.
3. **Too far.** If the next register sits far away, the wire is long. The signal is still on its way when the next tick comes, so it arrives too late.
4. **Placed closer.** Design software from three companies, Synopsys, Cadence and Siemens, places the parts and wires them together. Here it moves the register closer, and the signal arrives in time.
5. **Checked.** The software checks that every signal arrives in time and that every shape can be printed, and a chip factory accepts only designs that pass the printing checks. The factory then prints the blueprint onto silicon through stencils, and changing it after that costs new stencils and a few months.

Note: Simplified. Not to scale. The floorplan follows a GPU die. One tick, less than a billionth of a second, is slowed to about two seconds.

2 **EDA Market Primer.** SemiAnalysis, 21 May 2026. newsletter.semianalysis.com

4 **The Double-Edged Sword of Semiconductor Export Controls: Electronic Design...** Center for Strategic and International Studies, 6 October 2025, p. 33. csis-website-prod.s3.amazonaws.com

9 **Will Huawei catch up to Nvidia by 2030?** Epoch AI, 24 September 2026. epoch.ai

Design runs in five stages, all in software, and each stage's output is the next one's input.

- **Architecture.** Engineers fix the shape of the chip: the width of the block that multiplies numbers, the amount of fast on-chip memory beside it and the number of high-bandwidth memory stacks along the chip's edges. The output is a plan.
- **RTL.** Engineers write the plan as register-transfer-level code, RTL, in the hardware language SystemVerilog. A register is a small store on the chip that holds one number. On each tick of the chip's clock, every register takes its next value at the same moment. The code says what every register holds on every tick. The output is a text description of the whole chip.
- **Verification.** A large team checks that the code does what the plan says. They run it in simulation and prove parts of it correct with mathematics. They also load it into emulators, large machines full of reprogrammable chips that behave like the design, so the chip's own software can boot before any silicon exists.
- **Physical design.** Software swaps each piece of the code for a logic cell, a small ready-made circuit the foundry has already proved it can make. It then places the cells on the chip, wires them together and proves that every signal arrives in time.
- **Tape-out.** Engineers hand the finished layout to the foundry, a step called tape-out. After tape-out, software turns the layout into pattern files for the mask shop[10]. First it pre-distorts each shape so that the shape prints as drawn, a step called optical proximity correction. Then it cuts the layout into the files a mask writer reads, a step called mask data preparation.

All that checking exists because each revision after tape-out, called a respin, needs at least some new masks[1]. In 2022 SemiAnalysis put a full mask set at more than $1 million at 28 nm and more than $10 million at 7 nm[1]. The nanometer labels name process generations, called nodes, each finer than the last, and no longer measure anything. In 2022 SemiAnalysis also expected 3 nm sets to push into the $40 million range[1]. Its May 2026 primer still puts a leading-edge mask set at $40 million, and a single leading-edge respin at $50 million to $100 million[2].

The same 2022 SemiAnalysis article reported that startups had taken chips on TSMC 7 nm, then a leading-edge process, from design to tape-out for $50 million to $75 million, everything included[1].

1 The Dark Side Of The Semiconductor Design Renaissance – Fixed Costs Soaring Due... SemiAnalysis, 24 July 2022. newsletter.semianalysis.com

2 EDA Market Primer. SemiAnalysis, 21 May 2026. newsletter.semianalysis.com

10 Unlocking post-tapeout flow scalability and performance with cloud computing. Siemens Digital Industries Software, 8 January 2025. blogs.sw.siemens.com

Figure 01.2

In 2022 SemiAnalysis forecast a 3 nm mask set at $40 million

What one photomask set costs, by node, 2022. The 28 nm and 7 nm figures are minimums, and 3 nm was a forecast



Source: SemiAnalysis, The Dark Side of the Semiconductor Design Renaissance, July 2022. SemiAnalysis's May 2026 EDA Market Primer puts a leading-edge mask set at $40 million [1]

02

Variants and trade-offs

Merchant GPU versus custom ASIC

Nvidia designs its Blackwell GPU once and sells it to everyone, so the buyer gets mature software and a resale market. A custom accelerator, an ASIC, is built for one company's own workload. It handles a narrower range of tasks and is tuned to that workload. Google's tensor processing units (TPUs) are custom accelerators of this kind[11]. Ironwood, the seventh generation, which Google announced in April 2025, carries 192 GB of memory per chip, six times its predecessor's, and delivers twice the performance per watt[11]. In April 2026 Google announced its eighth generation as two chips, the TPU 8t for training and the TPU 8i for inference, with up to twice Ironwood's performance per watt[12].

The EDA flow

Chip design software is known as electronic design automation, or EDA. Three complete EDA flows exist, each a set of tools that covers every step from code to signoff[6]. SemiAnalysis wrote in May 2026 that no startup can replicate one by writing it from scratch[2]. Changing the synthesis tool, which swaps the code for logic cells, means re-running the steps that follow: placing and wiring the cells, signoff and physical verification[2]. Foundries also specify which signoff tools their customers must use for tape-out[2].

For chips at advanced nodes, 7 nm and below, one vendor holds most of the market inside some steps of the flow. Synopsys alone has more than 90 percent of the market for static timing analysis, the check that every signal arrives in time at the chip's rated speed[2]. In May 2026 SemiAnalysis put Siemens EDA's Calibre at more than 85 percent of physical verification, the final layout check[2]. It put Cadence's Palladium at 55 to 60 percent of hardware emulation[2].

1 The Dark Side Of The Semiconductor Design Renaissance – Fixed Costs Soaring Due... SemiAnalysis, 24 July 2022. newsletter.semianalysis.com

2 EDA Market Primer. SemiAnalysis, 21 May 2026. newsletter.semianalysis.com

6 The Double-Edged Sword of Semiconductor Export Controls: Electronic Design... Center for Strategic and International Studies, 6 October 2025, p. 13. csis-website-prod.s3.amazonaws.com

11 Ironwood: The first Google TPU for the age of inference. Google, 9 April 2025. blog.google

12 Our eighth generation TPUs: two chips for the agentic era. Google, 22 April 2026. blog.google

Figure 01.3

In 2026 each segment's leading tool held at least 55 percent

Share held by the leading vendor in each of four design-software segments at advanced nodes, 7 nm and below. Lower end of each reported range



Source: SemiAnalysis, EDA Market Primer, May 2026, table of tool-level shares at advanced nodes [2]

Buy the IP or build it

Design houses license ready-made circuit blocks, known as intellectual property or IP, including the processor cores, the interface controllers and the blocks that drive high-bandwidth memory. The block that drives high-bandwidth memory is hard to design, because its signals have to arrive in time across microscopic solder bumps into a stacked memory die.

Arm licenses processor designs. In fiscal 2026 it reported $4.92 billion of revenue, $2.61 billion of it royalties[7]. It said Arm-based designs held about 50 percent of the computing power of central processors (CPUs) at the top cloud firms[7].

In 2026 Arm also launched the Arm AGI CPU, the first finished chip that Arm sells[7]. By May 2026, servers built on it could be ordered from ASRock, Lenovo, Quanta and Supermicro[7].

RISC-V is an open instruction set, the basic list of commands a processor understands, and it needs no license. Small RISC-V processors handle control tasks on the chip, and Nvidia uses them in every GPU[13].

Chiplets

The reticle limit caps the size of a single die, one piece of silicon[14]. That limit is the largest area a scanner, the machine that prints the chip's pattern, can expose in one shot. Many large accelerators are built from chiplets, several smaller dies wired together inside one package[15]. UCIe is an open industry standard for the links between them[15]. Version 3.0 of UCIe adds two faster data rates, 48 and 64 billion transfers per second, written GT/s for gigatransfers per second[15]. In a December 2024 presentation, Broadcom described its 3.5D XDSiP platform, which stacks dies face to face and sets the stacks side by side on an interposer[14]. It said the platform packs more than 6,000 mm² of silicon and 12 high-bandwidth memory stacks into one package[14].

2 **EDA Market Primer.** SemiAnalysis, 21 May 2026. newsletter.semianalysis.com

7 **Arm delivers record-breaking quarter and full-year results.** Arm, 6 May 2026. newsroom.arm.com

13 **Behind The Scenes of SHD Group's 2026 RISC-V Market Forecast.** RISC-V International, 19 June 2026. riscv.org

14 **3.5D XDSiP Platform Technology.** Broadcom, 4 December 2024. docs.broadcom.com

15 **UCIe specifications.** UCIe Consortium, accessed 9 September 2026. uciexpress.org

03

Who makes it

Design software, simulation software and licensed blocks made up an $18 billion market in 2025, on SemiAnalysis's count[2]. Synopsys, Cadence and Siemens EDA took more than 85 percent of it, counting Ansys as part of Synopsys[2]. No other vendor had more than 5 percent of any of the main categories of design software[2]. Siemens EDA is a US-based subsidiary of the German firm Siemens[16].

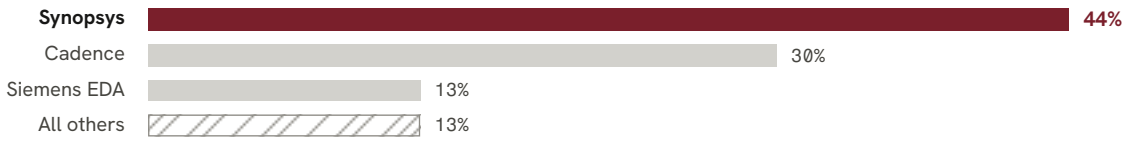
Synopsys is the largest. Its own results put revenue at $7.054 billion for the fiscal year to 31 October 2025, with Ansys, bought that July, contributing $756.6 million[17]. Cadence, whose year ends in December, took $5.297 billion in calendar 2025[18].

The two fiscal years do not line up, so any total for this market depends on which twelve months are counted. It also depends on how much of Ansys the total includes, because Ansys sells to other industries too. SemiAnalysis puts 31 percent of Ansys's business in semiconductors and high technology, 22 percent in aerospace and 18 percent in automotive[2]. It counts Synopsys at $8 billion for calendar 2025 with Ansys included[2]. On the same count Cadence is second, at $5.30 billion, and Siemens EDA third, at an estimated $2.2 billion to $2.5 billion[2].

Figure 01.4

Synopsys took 44 percent of the 2025 market, Cadence 30 percent

Share of revenue from design software, simulation software and licensed blocks, 2025. Synopsys includes part of Ansys's revenue



Source: SemiAnalysis, EDA Market Primer, May 2026, table of calendar 2025 vendor shares [2]

Chip firms such as Broadcom build custom accelerators to a customer's design and sell the finished chips. Broadcom booked $16.7 billion of AI semiconductor revenue in the third quarter of fiscal 2026, up 221 percent year over year[8]. That figure covers both its custom AI accelerators and its AI networking chips[8].

Broadcom's custom-chip customers include OpenAI and Meta. In October 2025 OpenAI and Broadcom said they would deploy OpenAI-designed accelerators drawing 10 gigawatts of power[19]. Broadcom would turn the design into silicon and supply the Ethernet networking around it[19]. In April 2026 Broadcom and Meta extended their

2 EDA Market Primer. SemiAnalysis, 21 May 2026. newsletter.semianalysis.com

8 Broadcom, current report on Form 8-K dated 2 September 2026. U.S. Securities and Exchange Commission, 2 September 2026. sec.gov

16 The Double-Edged Sword of Semiconductor Export Controls: Electronic Design... Center for Strategic and International Studies, 6 October 2025, p. 7. csis-website-prod.s3.amazonaws.com

17 Synopsys, current report on Form 8-K dated 10 December 2025. U.S. Securities and Exchange Commission, 10 December 2025. sec.gov

18 Cadence Design Systems, current report on Form 8-K dated 17 February 2026. U.S. Securities and Exchange Commission, 17 February 2026. sec.gov

19 OpenAI and Broadcom announce strategic collaboration to deploy 10 gigawatts of... Broadcom, 13 October 2025. investors.broadcom.com

partnership to multiple gigawatts of MTIA, Meta's custom chip, including what they called the industry's first 2 nm AI accelerator[20].

Marvell, another US firm, also builds custom AI chips to each customer's specifications[21]. Taiwanese design-service houses carry a customer's circuit list, the logic cells and the wires between them, all the way to a finished mask set on TSMC's newest nodes. One of them, Alchip, reported $992 million of revenue in 2025[22].

India has nearly 20 percent of the world's chip design engineers, the Indian government said in July 2025[23].

04

The chokepoint

Before the 2025 letters, a narrower control already applied. Since 14 October 2022, design software specially made for chips with gate-all-around transistors has needed a US export license for national-security reasons[3]. The rule calls that transistor design key to processes at 3 nm and below[3]. That license requirement covers exports to China[24]. The control carried out a December 2021 decision of the Wassenaar Arrangement, a group of countries that coordinate export controls[3].

In late May 2025 the US Bureau of Industry and Security, the agency that issues export licenses, told the design-software vendors that sales to China now needed a license[5]. Cadence got its letter on 23 May[5]. The letter required a license to export design software and technology classified 3D991 and 3E991 to any party in China, or to a Chinese military end user anywhere[5]. Synopsys got its letter on 29 May and a second letter lifting the requirement on 2 July[25]. Unlike the 2022 control, the 2025 requirement came from the United States alone and took effect immediately.

Synopsys's China revenue fell from 16 percent of its total in fiscal 2024 to 12 percent in fiscal 2025[2]. SemiAnalysis links the decline to tighter export restrictions and to local design-software companies winning share at mature nodes[2]. SemiAnalysis also quoted Synopsys's management saying that when Synopsys cannot sell to a company, that company looks for alternatives, typically local design-software and licensed-block companies[2].

In October 2025 CSIS found that only three companies, Cadence, Synopsys and Siemens, make design software that supports the entire chip design flow of more than 40 steps[6]. In a January 2021 report built on 2019 market data, the Center for Security

2 EDA Market Primer. SemiAnalysis, 21 May 2026. newsletter.semianalysis.com

3 Implementation of Certain 2021 Wassenaar Arrangement Decisions on Four Section... Federal Register, 15 August 2022. federalregister.gov

5 Cadence Design Systems, current report on Form 8-K dated 2 July 2025. U.S. Securities and Exchange Commission, 3 July 2025. sec.gov

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24 Supplement No. 1 to Part 738, Title 15 -- Commerce Country Chart. Electronic Code of Federal Regulations, 30 September 2025. ecfr.gov

25 Synopsys, current report on Form 8-K dated 2 July 2025. U.S. Securities and Exchange Commission, 3 July 2025. sec.gov

and Emerging Technology (CSET) at Georgetown University found that one Chinese firm, Empyrean, could reportedly run a complete design flow only for some analog and mixed-signal chips[26]. Mixed-signal chips combine analog and digital circuits. For all other chips, Chinese designers then relied on US tools[26]. The report found that other Chinese tools covered only narrow parts of the design flow or supplemented US tools, and largely did not support processes at 14 nm and below[26].

Empyrean says it has covered more of the design flow since then. Its 2025 annual report, filed in April 2026, says it now has complete design flows for analog, memory, radio-frequency and flat-panel display circuits[27]. It says it offers 80 percent of the main tools used in digital design[28]. The same report lists foundry certifications for its circuit simulator at 8, 5 and 4 nm and for its physical verification tool, the final layout check, at 28 nm[29]. But CSIS found that as of 2025 Chinese design tools were still far from replacing the full range of advanced capabilities US firms offer, especially for 5 nm and 3 nm chip design[4].

China can design accelerators, but it cannot finish one at the leading edge without foreign tools, or build what it designs without foreign equipment. SMIC, China's leading foundry, makes chips for Huawei[9]. In June 2024 CSET found that SMIC's limited 7 nm capacity forced Huawei to choose between making its most advanced phone chips and its AI chips[30].

Epoch AI estimates that Huawei trails Nvidia both in what each chip can do and in how many chips it makes[9]. On Epoch's estimates, Huawei's flagship chip for 2026, the Ascend 950, delivers about half the computing performance of Nvidia's H100, which began shipping in 2022[9]. Epoch puts Nvidia's B300 at 6.75 times the 950's theoretical peak performance[9]. Epoch's median estimate is that Huawei will make about 1.5 million Ascend chips in 2026, against 5.9 million for Nvidia[9]. On those estimates, Huawei's 2026 output would carry less than 4 percent of the computing power Nvidia ships that year[9].

Epoch puts Huawei's chips three to four years behind Nvidia's until at least 2030, for three reasons[9]. SMIC cannot pack transistors as densely as TSMC, which makes Nvidia's. Chinese memory makers are only starting to produce high-bandwidth memory. And Huawei's software for programming its chips is far less mature than Nvidia's[9].

Five questions to answer from memory

1. What does chip design produce? see How it works
2. Which three companies sell the software used to design leading-edge chips? see Who makes it

4 The Double-Edged Sword of Semiconductor Export Controls: Electronic Design... Center for Strategic and International Studies, 6 October 2025, p. 33. csis-website-prod.s3.amazonaws.com

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3. Why is it hard for a new company to compete with Synopsys, Cadence and Siemens EDA? see The chokepoint
4. How close are Chinese design tools to replacing US tools for the newest chips? see The chokepoint
5. How far behind Nvidia is Huawei's best AI chip? see The chokepoint

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Five things to remember

- Chip design turns an idea into a blueprint, and a factory builds only designs checked in approved software.
- Synopsys, Cadence and Siemens EDA sell the software used to design every leading-edge AI chip.
- To compete with them, a new vendor would have to write a complete set of tools and win a foundry's approval.
- China designs its own AI chips, but in October 2025 CSIS found its design software still far from replacing US tools, especially for 5 nm and 3 nm chips.
- Huawei's best AI chip trails Nvidia's by three to four years, on Epoch AI's estimate.

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The chain

01 Chip Design, EDA and IP
02 Silicon and Wafers
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04 Lithography
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Silicon and Wafers

Firms based in Japan, Taiwan, Germany and South Korea make most 300 mm wafers used for advanced chips. China makes most polysilicon, almost all of it for solar cells.

Specimen 02 · Silicon and Wafers

A silicon ingot, one crystal pulled whole from a melt, with two wafers sawn from it. It is 2.1 m long, 300 mm across and about 265 kg, taller than a door and about as heavy as three grown men. It hangs from a neck about 10 mm wide. Drawn to scale.

In plain terms

Every chip is built on a wafer, a thin round slice of silicon about 30 centimeters across that carries hundreds of chips at once. Making one starts with quartz rock, which is silicon bound to oxygen[1]. The silicon is refined until metal impurities are no more than a few parts per billion[2]. It is then melted, and a small seed crystal is lowered to the surface and drawn slowly back up. The melt freezes onto the seed as one long cylinder, a single crystal with its atoms lined up in one unbroken pattern from end to end. Saws cut the cylinder into discs, and each disc is polished to a mirror finish and made extremely flat across its whole face[2]. In 2025 the top five wafer makers served around 75 percent of the world market, and none of them is American or Chinese[3].

In short

Japanese companies held 53 percent of the world's silicon wafer sales in 2023, by a Fuji Keizai survey that Japan's trade ministry reproduced[4]. Firms based in Taiwan, South Korea and Europe held most of the rest[4]. China's largest maker of 300 mm wafers, Eswin, says it ranked sixth in the world with about 7.7 percent of global 300 mm shipments in the first half of 2026[5]. Chips also need electronic-grade polysilicon, the raw material for wafers[6]. The German wafer maker Siltronic named five major suppliers of it for 2025, and none is Chinese[6].

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2 SUMCO, production processes page. SUMCO, accessed 9 September 2026. sumcosi.com
3 Investor Presentation: Foundation of Digital Life. Siltronic, April 2026, p. 6. siltronic.com
4 METI, Semiconductor and digital industry strategy: future direction... Ministry of Economy, Trade and Industry (Japan), 23 December 2025, p. 8. meti.go.jp
5 西安奕斯伟材料科技股份有限公司2026年半年度报告. CNINFO, 25 August 2026, p. 18. static.cninfo.com.cn
6 Investor Presentation: Foundation of Digital Life. Siltronic, April 2026, p. 5. siltronic.com

GlobalWafers of Taiwan opened a $3.5 billion plant in Texas in May 2025, the first fully integrated plant for advanced silicon wafers built in the United States in over twenty years[7]. US Presidential Proclamation 11052 sets a minimum import price of $100 per kilogram for polysilicon ingots and wafers from 4 December 2026[8].

The world market for silicon wafers was $11.4 billion in 2025, on 12,973 million square inches shipped[9]. That is a small fraction of Nvidia's data center revenue, which reached $193.7 billion in the fiscal year to 25 January 2026[10].

Chokepoint card — Silicon and Wafers

Concentration

High

Four to six firms could supply this stage today, and the top two take most of the revenue.

Substitutability

Moderate

Several firms already make 300 mm wafers that fabs have approved, but customers have to test and approve a new wafer plant before they buy from it[11].

Price or market size

$11.4 billion

world wafer market in 2025, on 12,973 million square inches shipped[9]

Who leads	JP	**Shin-Etsu Handotai**	Its parent's electronics materials segment, wafers included, had ¥1,015.7 billion of sales in the year to March 2026[12]
	JP	**SUMCO**	¥409,670 million of net sales in 2025[13]
	TW	**GlobalWafers**	NT$60.6 billion revenue in 2025[14]
	KR	**SK Siltron**	One of the top five wafer makers by revenue in the first three quarters of 2025[3]
	DE	**Siltronic**	€1,346.7 million revenue in 2025[15]
Where it is made	JP	**Japan**	Shin-Etsu and SUMCO crystal growth and polishing
	TW	**Taiwan**	GlobalWafers headquarters and plants
	DE	**Germany**	Siltronic plants in Burghausen and Freiberg, and Wacker semiconductor-grade polysilicon
	KR	**South Korea**	SK Siltron
	US	**United States**	Hemlock polysilicon in Michigan, and GlobalWafers plants in Sherman, Texas and St Peters, Missouri
Why substitution is possible	Losing one of the top five wafer makers leaves four, and a new wafer plant can still be built[3]. Customer approval of a new plant is called qualification[11]. Siltronic completed key customer qualifications at its new Singapore fab in July 2025[11].		

3 Investor Presentation: Foundation of Digital Life. Siltronic, April 2026, p. 6. siltronic.com

7 GlobalWafers America Officially Opens for Business. GlobalWafers, 15 May 2025. sas-globalwafers.com

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9 SEMI Reports 2025 Annual Worldwide Silicon Wafer Shipments and Revenue Results. PR Newswire, 10 February 2026. prnewswire.com

10 NVIDIA Announces Financial Results for Fourth Quarter and Fiscal 2026. Nvidia, 25 February 2026. nvidianews.nvidia.com

11 Investor Presentation: Foundation of Digital Life. Siltronic, April 2026, p. 12. siltronic.com

12 Consolidated Financial Results for the Fiscal Year Ended March 31, 2026. Shin-Etsu Chemical, 27 April 2026, p. 6. shinetsu.co.jp

13 Notice of Convocation of the 27th Ordinary General Meeting of Shareholders. SUMCO, 6 March 2026, p. 24. sumcosi.com

14 GlobalWafers Reports Full Year 2025 Results. GlobalWafers, 3 March 2026. sas-globalwafers.com

15 Siltronic AG: Robust business performance in 2025 demonstrates resilience... Siltronic, 27 August 2026. siltronic.com

Where China stands

China made about 93 percent of the world's polysilicon in 2023, and an estimated 98 percent of China's output was solar-grade [16]. Eswin, mainland China's largest maker of 300 mm wafers, says it had about 7.7 percent of world 300 mm shipments in the first half of 2026 [5].

Where the US stands

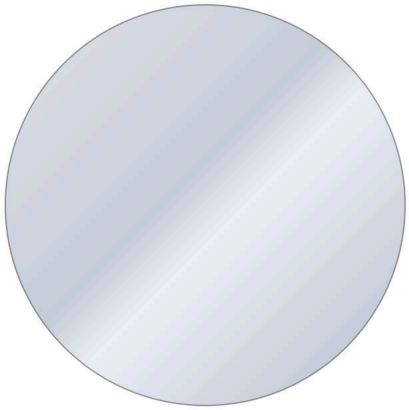
None of the five largest wafer makers in 2025 was American [3]. GlobalWafers of Taiwan won CHIPS awards of up to $406 million in December 2024 for wafer plants in Texas and Missouri [17].

01

How it works

Figure 02.1 · Explainer, 20 seconds

From sand to wafer



POLISHED TO A MIRROR FINISH

A few parts per billion
METAL IMPURITIES

1,420 °C
MELTING POINT, HOTTER THAN LAVA

30 cm
ACROSS ONE FINISHED WAFER

1. **Refine.** Silicon starts as quartz rock. It is refined into rods so pure that metal impurities are no more than a few parts per billion.
2. **Grow.** The rods are melted. A seed crystal touches the melt and is drawn slowly up, turning, and the silicon freezes onto it as one long crystal.
3. **Slice.** A wire saw slices the crystal, called an ingot, into thin discs.
4. **Polish.** Each disc is ground, etched and polished to a mirror finish, flat right out to the edge.

Note: Simplified. The ingot and the machines are not drawn to scale.

An arc furnace melts the quartz rock with carbon, and the carbon takes the oxygen away as gas, leaving rough silicon metal. Refiners turn the metal into a gas, trichlorosilane, and distill it, because the impurities boil at other temperatures and separate out.

In the next step, the Siemens process, the clean gas flows over silicon filaments heated inside a bell jar. On the hot surface the gas breaks apart and its silicon settles onto the filaments. Over days the filaments thicken into gray rods of polysilicon. Everything downstream is made from those rods. Polysilicon comes in two ultra-high-purity

3 Investor Presentation: Foundation of Digital Life. Siltronic, April 2026, p. 6. siltronic.com
5 西安奕斯伟材料科技股份有限公司2026年半年度报告. CNINFO, 25 August 2026, p. 18. static.cninfo.com.cn
16 2023 Minerals Yearbook: China (advance release). U.S. Geological Survey, February 2026, p. 7. pubs.usgs.gov
17 Biden-Harris Administration Announces CHIPS Incentives Awards with GlobalWafers... NIST, 17 December 2024. nist.gov

grades, one for semiconductors and one for solar cells, and chips use the semiconductor grade[1].

A grower melts the polysilicon at about 1,420 degrees Celsius, hotter than lava, and pulls a single crystal out of it by the Czochralski method[2]. The grower lowers a small seed crystal on the end of a rod to the surface of the melt and draws it back up, slowly turning. Silicon freezes onto the seed in the seed's own atomic pattern, and a cylinder, the ingot, grows beneath it. The pull starts with a thin neck so that flaws from the seed's first contact with the melt stay in the neck and never reach the ingot. Boron or phosphorus is stirred into the melt, a step called doping. The added element sets how readily the wafer conducts and whether negative or positive charges carry its current.

A wire saw, a web of fine wire running through abrasive, slices the ingot into discs. The maker then laps each disc, grinding both faces flat between rotating plates. It etches off the damaged surface with chemicals, then polishes the disc to a mirror with colloidal silica, a slurry of silica particles far too small to see. Almost every step throws material away.

Epitaxial wafers get one more layer of near-perfect silicon, grown on top at about 1,200 degrees Celsius[2]. That is still hot enough to glow orange.

02

Variants and trade-offs

Polished versus epitaxial

A polished wafer is a doped single crystal polished to a mirror finish[2]. An epitaxial wafer is a polished wafer with a thin layer of single-crystal silicon grown on top[2]. The transistors are built in that layer, so flaws in it can make them leak current. SUMCO says an epitaxial wafer suits customers who need a near-perfect crystal or layers that differ in electrical resistance[2]. SEMI, the industry association, credits the 2025 rise in shipments to AI-driven demand for advanced epitaxial wafers for logic chips and polished wafers for high-bandwidth memory[9].

Silicon on insulator

In a silicon-on-insulator (SOI) wafer, a buried layer of oxide, which is glass, separates a thin working layer of silicon from the bulk beneath. Soitec says nearly all SOI wafers sold worldwide are made with its Smart Cut process[18]. Soitec's SOI product lines include RF-SOI for radio chips and fully depleted SOI (FD-SOI)[18]. Its markets run from mobile communications to cloud AI[18]. But for logic chips, SEMI credited the AI-driven growth in 2025 to a different kind of wafer, the advanced epitaxial wafer[9].

1 Mineral Commodity Summaries 2026: Silicon. U.S. Geological Survey, 5 February 2026. pubs.usgs.gov

2 SUMCO, production processes page. SUMCO, accessed 9 September 2026. sumcosi.com

9 SEMI Reports 2025 Annual Worldwide Silicon Wafer Shipments and Revenue Results. PR Newswire, 10 February 2026. prnewswire.com

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03

Who makes it

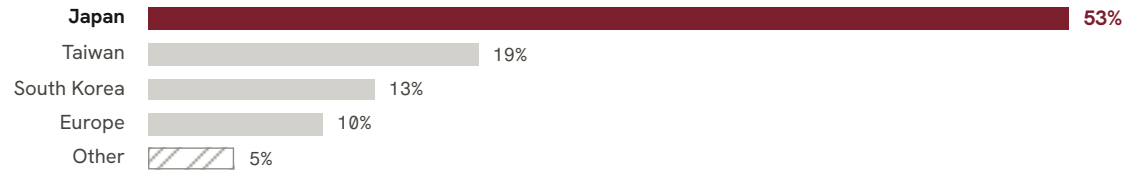
The Center for Security and Emerging Technology (CSET) at Georgetown University named five key wafer firms in a January 2021 brief[19]. They were Shin-Etsu and SUMCO of Japan, GlobalWafers of Taiwan, Siltronic of Germany and SK Siltron of South Korea[19]. CSET also listed nine Chinese-owned producers, among them Zhonghuan, Simgui, ZingSEMI and Okmetic, a Finnish firm bought by a Chinese state investment fund[19]. It said Chinese wafer producers made little revenue compared with the five key firms[20].

China's 300 mm wafer industry has grown since 2018. As of 2018, five Chinese firms were making or planning 300 mm wafers[20]. In its October 2025 listing prospectus, Eswin, a Chinese wafer maker, counts seven Chinese firms, itself included, making 300 mm wafers at scale[21]. Their combined capacity was 2.94 million wafers a month at the end of 2024[21].

Figure 02.2

Companies based in Japan held 53 percent of 2023 wafer sales

Silicon wafer sales by supplier headquarters, 2023. The source shows no separate share for China or the United States



Source: METI, December 2025, from Fuji Keizai data (2023). Japan's and Europe's shares are printed, and the others are read off METI's chart [4]

Japanese companies held 53 percent of the world's silicon wafer sales in 2023[4]. The shares come from a survey of 2023 sales by Fuji Keizai, a Japanese research firm[4]. Japan's Ministry of Economy, Trade and Industry reproduced the figures in a December 2025 strategy deck[4]. The ministry's chart of the survey prints only Japan's 53 percent, so the other shares are measured from its bars[4]. An earlier page of the ministry's deck prints Europe's share in the same Fuji Keizai survey as 10 percent[22]. The ministry's figures give no separate share to Chinese or US companies[4].

Siltronic's April 2026 investor presentation says the top five wafer makers serve around 75 percent of the market, based on revenue in the first three quarters of 2025[3]. It names the five as Shin-Etsu, SUMCO, GlobalWafers, SK Siltron and Siltronic itself[3]. Siltronic describes itself as the only Western-based maker among the five[3].

3 **Investor Presentation: Foundation of Digital Life.** Siltronic, April 2026, p. 6. siltronic.com

4 **METI, Semiconductor and digital industry strategy: future direction...** Ministry of Economy, Trade and Industry (Japan), 23 December 2025, p. 8. meti.go.jp

19 **The Semiconductor Supply Chain - Issue Brief.** Center for Security and Emerging Technology, 21 January 2021, p. 55. cset.georgetown.edu

20 **The Semiconductor Supply Chain - Issue Brief.** Center for Security and Emerging Technology, 21 January 2021, p. 57. cset.georgetown.edu

21 **西安奕材首次公开发行股票并在科创板上市招股说明书.** CNINFO, 22 October 2025, p. 138. static.cninfo.com.cn

22 **METI, Semiconductor and digital industry strategy: future direction...** Ministry of Economy, Trade and Industry (Japan), 23 December 2025, p. 4. meti.go.jp

The wafer makers' 2025 results show strong AI demand and weak demand for chips made on older, mature nodes.

- **Shin-Etsu.** Shin-Etsu Chemical's electronics materials segment, which includes its wafer business, sold ¥1,015.7 billion in the year to March 2026, up 9 percent[12]. AI-related demand stayed strong and demand in other sectors began to rise, lifting sales of silicon wafers, photoresist and photomask blanks[12]. The segment also sells rare earth magnets and other materials, and the results release does not break out wafer sales[23]. So the figure cannot be compared with the wafer revenue of the other firms listed.
- **SUMCO.** Sold ¥409,670 million[24]. Its operating profit fell from ¥36,924 million to ¥1,342 million, and it ended the year with a net loss of ¥11,751 million[24]. In the same year it added leading-edge 300 mm capacity and reorganized production of 200 mm and smaller wafers, where demand stayed weak[13].
- **Siltronic.** Lost €77.9 million on €1,346.7 million of sales while investing €369.1 million, with its new Singapore fab as the focus[15].
- **GlobalWafers.** NT$60.6 billion of revenue, down 3.24 percent in local currency[14].

Worldwide, the wafer area shipped rose 5.8 percent in 2025, but revenue fell 1.2 percent[9]. SEMI attributes the softer revenue mostly to traditional chip uses outside AI, where demand and pricing had yet to improve[9].

Polysilicon is the raw material for both wafers and solar cells, and almost all of China's is made for solar cells. In 2023 China made 1.50 million metric tons of polysilicon, about 93 percent of world output[16]. Solar-grade material was an estimated 98 percent of that output and electronic-grade 2 percent[16]. The US Commerce Department found in 2026 that semiconductor-grade polysilicon makes up only 2.4 percent of global production, close to the electronic-grade share of China's output[8]. These figures do not show how much of China's electronic-grade polysilicon ends up in chips.

Figure 02.3

An estimated 98 percent of China's 2023 polysilicon was solar-grade

China's polysilicon output by grade, 2023. China made 1.50 million metric tons that year, about 93 percent of world output



Source: USGS Minerals Yearbook, China, 2023 [16]

Siltronic put the 2025 market for electronic-grade silicon at $1.4 billion, with five major suppliers, none of them Chinese: Wacker, Hemlock, OCI, Tokuyama and Mitsubishi[6].

6 Investor Presentation: Foundation of Digital Life. Siltronic, April 2026, p. 5. siltronic.com
8 Adjusting Imports of Polysilicon and Its Derivatives Into the United States. Federal Register, 11 August 2026. federalregister.gov
9 SEMI Reports 2025 Annual Worldwide Silicon Wafer Shipments and Revenue Results. PR Newswire, 10 February 2026. prnewswire.com
12 Consolidated Financial Results for the Fiscal Year Ended March 31, 2026. Shin-Etsu Chemical, 27 April 2026, p. 6. shinetsu.co.jp
13 Notice of Convocation of the 27th Ordinary General Meeting of Shareholders. SUMCO, 6 March 2026, p. 24. sumcosi.com
14 GlobalWafers Reports Full Year 2025 Results. GlobalWafers, 3 March 2026. sas-globalwafers.com
15 Siltronic AG: Robust business performance in 2025 demonstrates resilience... Siltronic, 27 August 2026. siltronic.com
16 2023 Minerals Yearbook: China (advance release). U.S. Geological Survey, February 2026, p. 7. pubs.usgs.gov
23 Consolidated Financial Results for the Fiscal Year Ended March 31, 2026. Shin-Etsu Chemical, 27 April 2026, p. 20. shinetsu.co.jp
24 Notice of Convocation of the 27th Ordinary General Meeting of Shareholders. SUMCO, 6 March 2026, p. 26. sumcosi.com

GlobalWafers of Taiwan has expanded its wafer making in the United States. It opened a $3.5 billion plant in Sherman, Texas in May 2025, the first fully integrated plant for advanced silicon wafers built in the United States in over twenty years[7]. The same day it announced another $4 billion for phases three and four in Sherman[7]. The US government had awarded it up to $406 million under the CHIPS incentives program for wafer plants in Texas and Missouri[17].

US import rules issued in 2026 on polysilicon products also cover the single-crystal wafers that chips are built on. US Presidential Proclamation 11052, signed 6 August 2026, sets a minimum import price of $100 per kilogram for polysilicon ingots and wafers from 4 December 2026[8]. From the same date it adds an extra 15 percent duty on imported ingots and other listed products made from polysilicon[8]. A September 2026 US Commerce Department rule meant to stop stockpiling before the duties start lists tariff code 3818.00.0040 among the products it covers[25]. That code covers round, doped, single-crystal silicon wafers[26].

The US Commerce Department may approve plans to build US polysilicon, ingot, wafer or solar cell plants whose construction starts by 20 January 2029[8]. Firms with approved plans can import production equipment, and polysilicon, ingots, wafers and solar cells in amounts the department sets to match their new investment, without the new duties[8].

04

The chokepoint

China makes most of the world's polysilicon, but as of 2019 Chinese firms held under 5 percent of the wafer market and under 1 percent of the 300 mm market[19]. As of 2018, Chinese producers made 12 percent of the wafers for Chinese-headquartered 300 mm fabs, against 18 percent for 100 to 200 mm fabs[27].

Eswin, mainland China's largest maker of 300 mm wafers, says it ranked sixth in the world with about 7.7 percent of global 300 mm shipments in the first half of 2026[5]. Its October 2025 Shanghai listing prospectus, drawing on SEMI data, estimates that the five largest makers shipped 79.08 percent of 300 mm wafers in 2024[28]. It puts Eswin at 6.01 percent and all other makers at 14.91 percent[28].

Making the smoothest, high-purity 300 mm wafers takes considerable know-how that is hard to write down[20]. Customers have to qualify a new wafer plant before they buy

5 西安奕斯伟材料科技股份有限公司2026年半年度报告. CNINFO, 25 August 2026, p. 18. static.cninfo.com.cn

7 GlobalWafers America Officially Opens for Business. GlobalWafers, 15 May 2025. sas-globalwafers.com

8 Adjusting Imports of Polysilicon and Its Derivatives Into the United States. Federal Register, 11 August 2026. federalregister.gov

17 Biden-Harris Administration Announces CHIPS Incentives Awards with GlobalWafers... NIST, 17 December 2024. nist.gov

19 The Semiconductor Supply Chain - Issue Brief. Center for Security and Emerging Technology, 21 January 2021, p. 55. cset.georgetown.edu

20 The Semiconductor Supply Chain - Issue Brief. Center for Security and Emerging Technology, 21 January 2021, p. 57. cset.georgetown.edu

25 Measures To Restrict Stockpiling of Polysilicon and Polysilicon Derivatives... Federal Register, 24 September 2026. federalregister.gov

26 Harmonized Tariff Schedule, heading 3818.00.00. U.S. International Trade Commission, accessed 9 September 2026. hts.usitc.gov

27 The Semiconductor Supply Chain - Issue Brief. Center for Security and Emerging Technology, 21 January 2021, p. 58. cset.georgetown.edu

28 西安奕材首次公开发行股票并在科创板上市招股说明书. CNINFO, 22 October 2025, p. 145. static.cninfo.com.cn

from it, and Siltronic completed key customer qualifications at its new Singapore fab in July 2025[11].

Almost all fab capacity for advanced chips runs on 300 mm wafers. In SEMI figures from November 2020 cited by CSET, 99.7 percent of world fab capacity at 45 nm and below used 300 mm wafers[20]. SEMI estimated that 300 mm wafers made up about 78.8 percent of all wafer area shipped worldwide in 2025, according to Eswin's 2025 annual report summary[29].

The last increase in wafer size, to 300 mm, happened in 2002[30]. Wafers are still produced in diameters of up to 300 mm[9]. Bigger wafers cut the cost of deposition, etching and other steps, because those steps have to be run fewer times for the same number of chips[30]. So printing, or photolithography, takes a bigger share of the processing cost[30]. A 2013 University of Texas lecture cited by CSET put photolithography at half the cost of processing a 300 mm wafer, against a quarter for a 150 mm one[30].

Large GPU dies and the newest processes need better wafers. Some GPU dies fill the whole area a lithography tool can print in one exposure. One fatal defect on such a die ruins far more silicon than one on a small chip. Extreme-ultraviolet printing also stays in focus over only a tiny range of heights (see Lithography), so the disc has to be flat at every point. SEMI's wafer group says the move to sub-3nm processes is raising requirements for wafer quality and consistency[9].

Four questions to answer from memory

1. What is a wafer? `see How it works`
2. Which five companies make most of the world's silicon wafers? `see Who makes it`
3. Why can a fab not switch wafer suppliers quickly? `see The chokepoint`
4. Why does making most of the world's polysilicon give China little hold over chip wafers? `see The chokepoint`

Four things to remember

- Every chip starts on a wafer, a polished disc of pure silicon about 30 centimeters across.
- The top five wafer makers, in Japan, Taiwan, Germany and South Korea, served around 75 percent of the market in 2025.
- Customers have to test and approve a new wafer plant before they buy from it.
- China makes most of the world's polysilicon, but none of the five major suppliers of electronic-grade silicon, the grade chips need, is Chinese.

9 SEMI Reports 2025 Annual Worldwide Silicon Wafer Shipments and Revenue Results. PR Newswire, 10 February 2026. `prnewswire.com`

11 Investor Presentation: Foundation of Digital Life. Siltronic, April 2026, p. 12. `siltronic.com`

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30. A The Semiconductor Supply Chain - Issue Brief. Center for Security and Emerging Technology (CSET) · 21 January 2021 · p. 80
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The chain

01 Chip Design, EDA and IP
02 Silicon and Wafers
03 **Chemicals, Gases and Photoresist**
04 Lithography
05 Photomasks and Pellicles
06 Deposition and Etch
07 Metrology and Inspection
08 Foundries and Fabs
09 Memory and HBM
10 Advanced Packaging
11 Substrates and PCBs
12 Test and Assembly
13 Systems and Networking
14 Data Centers and Power

15 The Economics of AI Chips
16 Who Controls the Chain

Stage 3 of 14 · High concentration · Moderately hard to substitute

Chemicals, Gases and Photoresist

Japanese companies held 78 percent of world photoresist sales in 2023. Supply of chip chemicals, gases and metals has been disrupted three times since 2019.

Specimen 03 · Chemicals, Gases and Photoresist

A specialty gas cabinet with its door off, and a bulk liquid cylinder standing behind it. The cabinet is about 1 m wide and 1.9 m tall and holds three gas cylinders. The drawing is to scale. The steel-gray panel on the back wall holds the valves, regulator and gauges that control the gas flow.

In plain terms

Photoresist is a coating that changes wherever light hits it. A machine spins a coat of it, thinner than a soap bubble, across the wafer. Light shines through a stencil onto the coat, and a liquid wash then takes away the parts the light reached. What remains is a pattern, and the next machine cuts that pattern into the wafer beneath, a step called etching. Hundreds of other chemicals, gases and materials lay down layers, strip them off and clean the wafer between steps[1]. The cleaning chemicals are purified until metal and other impurities are down to parts per trillion, the Japanese chemical maker Sumitomo Chemical said in 2021[2]. A chipmaker has to test and approve each new resist in its own factory before using it[3]. Even a resist that passes the first time takes seven to 17 months to approve, the Semiconductor Industry Association wrote in 2023[3]. Japanese companies held 78 percent of world photoresist sales in 2023, by a Fuji Keizai survey that Japan's trade ministry reproduced[4].

In short

In July 2019 Japan began requiring an export license for each contract to sell photoresist, hydrogen fluoride or fluorinated polyimide to South Korea[5]. Japan's hydrogen fluoride exports to South Korea then fell 87.9 percent, but

1 Testimony of the Semiconductor Industry Association (SIA) before the Senate... U.S. Senate Committee on Environment and Public Works, 4 March 2026, p. 6. epw.senate.gov

2 Sumitomo Chemical to Expand Production Capacity for High-Purity Chemicals for... Sumitomo Chemical, 24 August 2021. sumitomo-chem.co.jp

3 PFAS-Containing Photo-Acid Generators Used in Semiconductor Manufacturing. Semiconductor Industry Association, 8 June 2023, p. 22. semiconductors.org

4 METI, Semiconductor and digital industry strategy: future direction... Ministry of Economy, Trade and Industry (Japan), 23 December 2025, p. 8. meti.go.jp

5 The impact of export controls on international trade: evidence from the... Research Institute of Economy, Trade and Industry, 8 May 2023. rieti.go.jp

its photoresist exports did not[5]. Production of all three chemicals rose inside South Korea[5].

China banned exports of the metals gallium, germanium and antimony to the United States in December 2024[6]. In Europe, germanium metal averaged an estimated $4,100 a kilogram in 2025, against $1,392 in 2023, on Argus Media prices reported by the US Geological Survey[7]. In November 2025 China suspended the ban until 27 November 2026[8].

A chip fab draws on hundreds of chemicals, gases and other materials[1]. A shortage of any one of them can stop it. The French gas company Air Liquide alone supplies more than 200 different chemicals and roughly 50,000 cylinders a year[9].

Chokepoint card — Chemicals, Gases and Photoresist

Concentration

High

Four to six firms could supply this stage today, and the top two take most of the revenue.

Substitutability

Moderate

This rating is based on photoresist, the hardest input to replace at this stage[10].

Price or market size

$45.8 billion

market for wafer fabrication materials in 2025[11]. SEMI's count includes silicon wafers and photomasks[12]

Who leads	JP	**JSR**	A vehicle of the state-backed Japan Investment Corporation bought 84.36 percent of JSR in April 2024[13]
	JP	**Tokyo Ohka Kogyo**	Resist maker with ¥139,668 million of sales in the first half of fiscal 2026[14]
	FR	**Air Liquide**	€2,465 million of electronics revenue in 2025, 9.1 percent of group sales[9]
	US	**Entegris**	Bought the polishing-slurry maker CMC Materials in July 2022[15]
	JP	**JX Advanced Metals**	Says it holds about 65 percent of the world market for semiconductor sputtering targets as of fiscal 2026[16]
Where it is made	JP	**Japan**	Photoresist, ultra-pure hydrofluoric acid, sputtering targets, polishing slurries
	US	**United States**	Entegris polishing slurries, Qnity Electronics resist, Air Products gases
	FR	**France**	Air Liquide electronics gases and precursors
	KR	**South Korea**	Dongjin resist, with output of resist and hydrogen fluoride rising after 2019
	CN	**China**	Gallium, germanium and antimony refining

1 Testimony of the Semiconductor Industry Association (SIA) before the Senate... U.S. Senate Committee on Environment and Public Works, 4 March 2026, p. 6. epw.senate.gov

5 The impact of export controls on international trade: evidence from the... Research Institute of Economy, Trade and Industry, 8 May 2023. rieti.go.jp

6 商务部公告2024年第46号 关于加强相关两用物项对美国出口管制的公告. Ministry of Commerce of the People's Republic of China, 3 December 2024. mofcom.gov.cn

7 Mineral Commodity Summaries 2026: Germanium. U.S. Geological Survey, 5 February 2026. pubs.usgs.gov

8 商务部公告2025年第72号 公布调整实施商务部公告2024年第46号的决定. Ministry of Commerce of the People's Republic of China, 9 November 2025. mofcom.gov.cn

9 Air Liquide, electronics page. Air Liquide, accessed 9 September 2026. airliquide.com

10 PFAS-Containing Photo-Acid Generators Used in Semiconductor Manufacturing. Semiconductor Industry Association, 8 June 2023, p. 21. semiconductors.org

11 Global Semiconductor Materials Market Revenue Reaches Record $73.2 Billion in... PR Newswire, 12 May 2026. prnewswire.com

12 Materials Market Data Subscription (MMDS). SEMI, accessed 20 October 2026. semi.org

13 Announcement Regarding Result of Tender Offer for JSR Corporation (Securities... Japan Investment Corporation, 17 April 2024. jiccapital.co.jp

14 Consolidated Financial Results for the Second Quarter (Interim) of the Fiscal... Tokyo Ohka Kogyo, 6 August 2026, p. 4. tok.co.jp

15 Entegris, current report on Form 8-K dated 6 July 2022. U.S. Securities and Exchange Commission, 6 July 2022. sec.gov

16 Quick Guide to JX Advanced Metals. JX Advanced Metals, accessed 9 September 2026. jx-nmm.com

Why substitution is possible

Each formula is tuned to the layers it prints and has to pass qualification in the fab [10]. So a new supplier has to develop its version together with the fab, one layer at a time [3]. When Japan required individual export licenses for resist and two other chemicals sold to South Korea in 2019, South Korea bought more hydrogen fluoride from the United States and Taiwan and more resist from Belgium [5].

Where China stands

In March 2026 the Chinese resist maker Hubei Dinglong said China had yet to reach large-scale production of advanced resist for 248 nm, 193 nm or extreme-ultraviolet printing [17]. Since August 2023 China has required licenses to export gallium and germanium [18].

Where the US stands

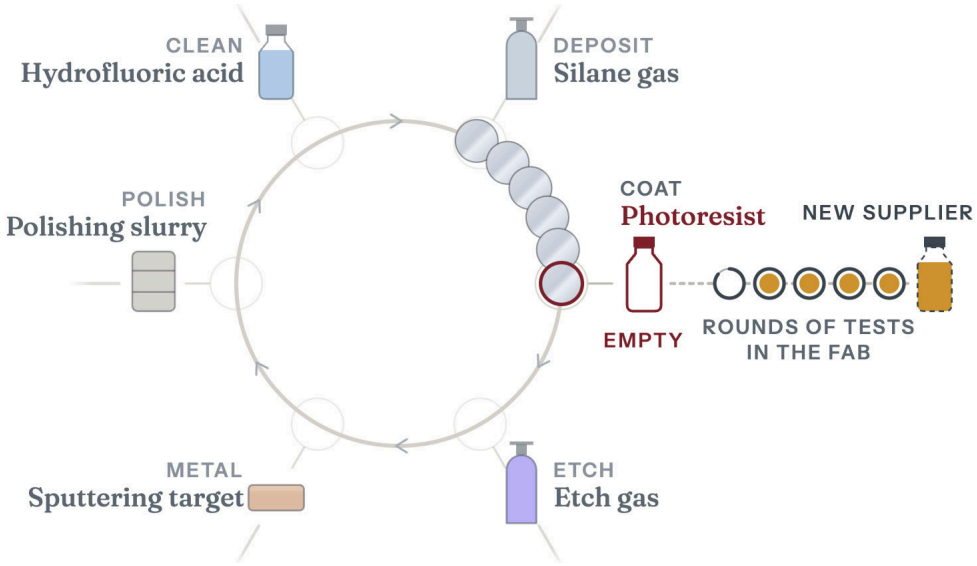
In photoresist, American companies ranked a distant second to Japanese companies by sales in 2023 [4]. In 2021 JSR of Japan agreed to buy Inpria, an Oregon-based US maker of resist for extreme ultraviolet [19].

01

How it works

Figure 03.1 · Explainer, 25 seconds

Why one missing material can stop a fab



1. **One layer.** A chip is built up in layers. For each layer the wafer goes around a loop of steps like this one, and each step uses its own chemical, gas or metal.
2. **Many wafers.** A fab runs many wafers at once, so every material is always in use. Deliveries keep each stock topped up.
3. **One runs out.** If one material stops arriving, the fab works through its stock.
4. **The fab stops.** When the stock runs out, wafers pile up at that step, and the rest of the fab stands idle.

3 PFAS-Containing Photo-Acid Generators Used in Semiconductor Manufacturing. Semiconductor Industry Association, 8 June 2023, p. 22. semiconductors.org

4 METI, Semiconductor and digital industry strategy: future direction... Ministry of Economy, Trade and Industry (Japan), 23 December 2025, p. 8. meti.go.jp

5 The impact of export controls on international trade: evidence from the... Research Institute of Economy, Trade and Industry, 8 May 2023. rieti.go.jp

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17 湖北鼎龙控股股份有限公司2025年年度报告摘要. CNINFO, 27 March 2026, p. 2. static.cninfo.com.cn

18 商务部 海关总署公告2023年第23号 关于对镓、锗相关物项实施出口管制的公告. Ministry of Commerce of the People's Republic of China, 3 July 2023. mofcom.gov.cn

19 JSR Agrees to Acquire EUV Pioneer Inpria Corporation. JSR Corporation, 17 September 2021. jsr.co.jp

5. **A new supplier.** A new supplier's resist goes through round after round of tests in the fab. Even one that passes the first time takes seven to 17 months to approve.

Note: Simplified diagram. Six steps and their materials stand for the many a fab uses, and each real layer uses its own mix.

The resist has to change only where the light lands. It also has to work with very little light, so the scanner, the machine that projects the pattern, can expose each patch of the wafer in milliseconds.

Since the 1990s the standard has been chemically amplified resist. In resists for light with wavelengths of 248 nm and 193 nm, a photon absorbed in the film can free an acid molecule, and little more happens until the wafer is baked. In the bake, each acid molecule sets off hundreds of reactions around itself that turn the film soluble, so a little light does a lot of chemistry[20]. The wash then takes away the exposed film and leaves the rest.

The 248 nm light comes from a krypton fluoride (KrF) laser. The 193 nm light comes from an argon fluoride (ArF) laser. Immersion scanners also pass it through a layer of water, which sharpens the image.

Chemically amplified resist works less well with extreme ultraviolet (EUV) light (see Lithography). An absorbed EUV photon instead knocks electrons loose, and a cascade of lower-energy secondary electrons makes the chemical changes in the resist[21]. A 13.5 nm EUV photon carries about fourteen times the energy of a 193 nm one, so the same dose of light arrives as far fewer photons. With fewer photons, random variation in where each one lands matters more. It shows up as ragged line edges and missing contact holes, the tiny holes that link one layer of a chip to the next.

Resist makers responded with metal-oxide resists, which are built on clusters with a tin-oxide core and absorb more of the EUV light than chemically amplified resists[22]. In 2021 the Japanese resist maker JSR agreed to buy Inpria, an Oregon-based US maker of these resists[19].

The other chemicals and gases put material on the wafer, take it off or keep it clean. Bulk gases such as nitrogen come by pipeline or from plants built at the fab, and specialty gases come in cylinders[9]. Silane gas carries the silicon for new layers into the chamber that holds the wafer. A plasma splits nitrogen trifluoride, and the fluorine it frees scrubs the chamber walls clean between wafers. Hydrofluoric acid cleans the wafer itself. Polishing slurries grind each new layer flat before the next goes on.

Sputtering targets are slabs of pure metal. A plasma knocks atoms off them, and the atoms settle on the wafer as thin metal films, including the barrier and seed layers for copper wiring[23]. Electroplating then fills in the copper wiring itself[24]. Filters and wafer carriers keep contamination out.

9 Air Liquide, electronics page. Air Liquide, accessed 9 September 2026. airliquide.com

19 JSR Agrees to Acquire EUV Pioneer Inpria Corporation. JSR Corporation, 17 September 2021. jsr.co.jp

20 PFAS-Containing Photo-Acid Generators Used in Semiconductor Manufacturing. Semiconductor Industry Association, 8 June 2023, p. 6. semiconductors.org

21 How Scientists Are Accelerating the Development of Next-Gen Microelectronics. Lawrence Berkeley National Laboratory, 21 August 2023. newscenter.lbl.gov

22 Inpria, company website. Inpria, accessed 9 September 2026. inpria.com

23 Applied Materials, PVD technology page. Applied Materials, accessed 9 September 2026. appliedmaterials.com

24 SABRE Product Family. Lam Research, 11 August 2017. lamresearch.com

02

Variants and trade-offs

Resist class	Wavelength	Where it is used	Limit
i-line novolak	365 nm	Packaging, mature nodes	Resolution
KrF chemically amplified	248 nm	Implant layers, thick films	Resolution
ArF immersion chemically amplified	193 nm	Most critical layers below 28 nm	Needs several exposures per layer
EUV chemically amplified	13.5 nm	Leading-edge single-exposure layers	Shot noise from too few photons
EUV metal-oxide	13.5 nm	High-NA tools, tightest pitches	Cost, defects, outgassing

Each resist class is made for one kind of lithography, such as EUV, ArF immersion, ArF, KrF or i-line[25].

03

Who makes it

Japanese companies sell most of the world's photoresist, by two different counts. In a January 2021 brief, Georgetown University's Center for Security and Emerging Technology (CSET) put Japan's share of the semiconductor photoresist market at 90 percent, citing a 2019 estimate[25]. The brief put most of the rest with US and South Korean firms[25]. In a December 2025 strategy presentation, Japan's Ministry of Economy, Trade and Industry (METI) published figures from a separate survey by Fuji Keizai, a Japanese market research firm[4]. That survey counts sales by company nationality and puts Japanese companies at 77 percent in 2021 and 78 percent in 2023, with American and South Korean companies next[4].

Figure 03.2

Japanese companies held 78 percent of photoresist sales in 2023

Photoresist sales share by company nationality, 2023



Source: METI, December 2025, from Fuji Keizai data [4]

CSET's January 2021 brief lists five Japanese resist makers: JSR, Tokyo Ohka Kogyo, Shin-Etsu Chemical, Fujifilm and Sumitomo Chemical[26]. It also lists Dongjin of South Korea, four Chinese firms and three firms it counts as American: DuPont, EMD Performance Materials and Inpria[26]. Qnity Electronics, the former DuPont electronics business, says in its 2025 annual report that it offers lithography materials for every wavelength, including EUV[27].

4 METI, Semiconductor and digital industry strategy: future direction... Ministry of Economy, Trade and Industry (Japan), 23 December 2025, p. 8. meti.go.jp

25 The Semiconductor Supply Chain - Issue Brief. Center for Security and Emerging Technology, 21 January 2021, p. 59. cset.georgetown.edu

26 The Semiconductor Supply Chain - Issue Brief. Center for Security and Emerging Technology, 21 January 2021, p. 56. cset.georgetown.edu

27 Qnity Electronics, annual report on Form 10-K for the fiscal year ended 31... U.S. Securities and Exchange Commission, 26 February 2026. sec.gov

In its 2025 annual report summary, the Chinese firm Hubei Dinglong said China had yet to reach large-scale production of advanced resist for 248 nm, 193 nm or EUV printing[17]. But it said three of its own resists for 248 nm and 193 nm printing had moved into stable volume supply[28].

A Japanese state-backed fund, Japan Investment Corporation, took control of JSR in April 2024. A company the fund controls bought 84.36 percent of JSR's shares at ¥4,350 each in an offer to all shareholders that closed on 16 April 2024[13]. JSR's shares were delisted from the Tokyo Stock Exchange on 25 June 2024[29].

Two Japanese suppliers tie their growth to demand for AI chips. Tokyo Ohka Kogyo sold ¥139,668 million in the first half of fiscal 2026, up 25.1 percent, as strong demand for AI servers in data centers raised demand for chips[14]. Fujifilm planned to raise its polishing-slurry capacity at Kumamoto in Japan by about 30 percent from January 2025, to meet demand driven by AI chips[30]. It also scheduled new production and quality-evaluation equipment for EUV resist at Shizuoka in Japan and Pyeongtaek in South Korea to start in October 2025[31].

World sales of wafer fabrication materials reached $45.8 billion in 2025, and sales of packaging materials reached $27.4 billion[11]. SEMI counts silicon wafers and photomasks among wafer fabrication materials, along with resists, gases, chemicals and polishing materials[12].

In electronic gases, CSET named Merck, Air Products and Air Liquide as the market leaders in January 2021[32]. Air Liquide's electronics business sold €2,465 million in 2025, 9.1 percent of group revenue[9]. SK Specialty has made nitrogen trifluoride since 2001, when it became the first company to produce the gas in South Korea[33]. The Japanese firm Kanto Denka Kogyo refines its own fluorine gas and turns it into specialty gases for deposition, etching and cleaning[34]. Stella Chemifa, also Japanese, makes ultra-high-purity hydrofluoric acid for chipmaking[35]. Another Japanese firm, Morita Chemical Industries, sells an electronic-grade hydrofluoric acid[36].

A few firms held most of the markets for wet chemicals and polishing slurries in 2019. CSET's January 2021 brief, drawing on a 2019 report by the market research firm Mordor Intelligence, put five firms, KMG Chemicals, BASF, Avantor, Honeywell and Kanto Chemical, at more than 60 percent of the wet chemicals market[32]. Citing a 2019

9 Air Liquide, electronics page. Air Liquide, accessed 9 September 2026. airliquide.com

11 Global Semiconductor Materials Market Revenue Reaches Record $73.2 Billion in... PR Newswire, 12 May 2026. prnewswire.com

12 Materials Market Data Subscription (MMDS). SEMI, accessed 20 October 2026. semi.org

13 Announcement Regarding Result of Tender Offer for JSR Corporation (Securities... Japan Investment Corporation, 17 April 2024. jiccapital.co.jp

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17 湖北鼎龙控股股份有限公司2025年年度报告摘要. CNINFO, 27 March 2026, p. 2. static.cninfo.com.cn

28 湖北鼎龙控股股份有限公司2025年年度报告摘要. CNINFO, 27 March 2026, p. 8. static.cninfo.com.cn

29 Announcement of Delisting of Shares of JSR Corporation. JSR Corporation, 24 June 2024. jsr.co.jp

30 Fujifilm to Enhance its Production Capacity of Advanced Semiconductor Material... Fujifilm, 5 December 2024. fujifilm.com

31 Fujifilm Launches EUV Resist and EUV Developer. Fujifilm, 29 October 2024. fujifilm.com

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report by the market research firm Market Reports World, it put DuPont and Cabot Microelectronics at about 56 percent of a $790 million market for polishing slurries[25]. Counting by company nationality, the same Fuji Keizai survey, as published by METI, puts Japanese companies at 51 percent of polishing-slurry sales and 59 percent of high-purity cleaning-liquid sales in 2023[4]. Both shares were 49 percent in 2021[4].

The slurry businesses of both leaders CSET named have since moved to other companies. Cabot Microelectronics renamed itself CMC Materials in October 2020[37]. The American firm Entegris completed its purchase of CMC Materials on 6 July 2022[15]. DuPont separated its electronics business into an independent company, Qnity Electronics, on 1 November 2025[38]. Qnity sells polishing slurries and pads[39].

In sputtering targets for semiconductors, JX Advanced Metals puts its own share of the world market at about 65 percent as of fiscal 2026[16].

04

The chokepoint

Supply at this stage has been disrupted three times since 2019, twice by governments and once by a war.

In July 2019 Japan put photoresist, hydrogen fluoride and fluorinated polyimide under individual export licenses for South Korea, so each export contract needed its own license[5]. South Korea's chip industry depended heavily on Japan for all three chemicals, and Japanese firms supplied more than 90 percent of South Korea's imports of two of them[5]. Japan's hydrogen fluoride exports to South Korea then fell 87.9 percent[5]. Japanese photoresist exports to South Korea did not fall, possibly in part because Japan allowed three-year bulk licenses for some resist sales in December 2019[5].

South Korea bought more hydrogen fluoride from the United States and Taiwan and more photoresist from Belgium[5]. Production of all three chemicals rose inside South Korea[5]. The rise came both from South Korean firms and from Japanese companies' affiliates in South Korea[5].

Researchers at Japan's Research Institute of Economy, Trade and Industry (RIETI) found that Japanese hydrogen fluoride exports to the United States rose[5]. They say some Japanese supply may have reached South Korea by way of the United States[5]. Japan ended the individual licenses on 23 March 2023, when it put exports of the three chemicals to South Korea under its bulk export license system[40].

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Russia's invasion of Ukraine in 2022 put chipmakers' supply of neon at risk. Before the invasion, Ukraine supplied about 70 percent of the world's neon and 90 percent of the semiconductor-grade neon used by US industry[41]. The fighting forced at least two of Ukraine's three major gas producers, Cryoin and Ingas, to shut their operations in Odesa and Mariupol[41]. That halted a significant part of the world's exports of neon, krypton and xenon[41]. Gas mixtures containing neon fill the lasers used in lithography, but EUV scanners use a tin light source instead[42]. Many chipmakers had added to their gas stockpiles after Russia annexed Crimea in 2014[41]. Samsung, SK hynix, Renesas and Rohm reported adequate neon supplies from China[41].

For chipmaking materials, China's export controls cover metals. On 3 July 2023 China announced that exports of gallium and germanium would need licenses from 1 August[18]. On 3 December 2024 it said it would in principle grant no licenses to export gallium, germanium, antimony or superhard materials to the United States[6]. On 9 November 2025 it suspended that ban until 27 November 2026[8]. In February 2026 the US Geological Survey put China's share of world production of primary low-purity gallium at 99 percent[43].

The White House set out a trade deal with China on 1 November 2025[44]. It said that under the deal China would issue general licenses for exports of gallium, germanium and antimony for the benefit of US end users and their suppliers[44].

Most gallium used in the United States goes into chips, and both metals are recovered as byproducts of processing other minerals. Of the gallium used in the United States in 2025, 73 percent went into integrated circuits built on gallium arsenide or gallium nitride[43]. Another 26 percent went into optoelectronic devices such as LEDs and laser diodes[43]. In the United States, germanium goes mainly into fiber optics, infrared optics, semiconductors and solar cells, and radiation detectors[7]. Gallium is recovered mostly as a byproduct of processing bauxite, the ore that aluminum is made from[43]. Germanium is recovered from certain zinc smelter residues and from coal ash[45]. Because both metals are economical to produce only as byproducts, their supply may not rise much when their prices do[45].

In Europe, germanium metal averaged an estimated $4,100 a kilogram in 2025, against $1,392 in 2023, on Argus Media prices reported by the US Geological Survey[7]. US imports of the metal fell an estimated 67 percent in 2025 because of China's ban[7]. In October 2024 the US Geological Survey estimated that a total Chinese ban on gallium and germanium exports could cut US GDP by $3.4 billion[46].

6 商务部公告2024年第46号 关于加强相关两用物项对美国出口管制的公告. Ministry of Commerce of the People's Republic of China, 3 December 2024. mofcom.gov.cn

7 Mineral Commodity Summaries 2026: Germanium. U.S. Geological Survey, 5 February 2026. pubs.usgs.gov

8 商务部公告2025年第72号 公布调整实施商务部公告2024年第46号的决定. Ministry of Commerce of the People's Republic of China, 9 November 2025. mofcom.gov.cn

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41 Russia's Invasion of Ukraine Impacts Gas Markets Critical to Chip Production. Center for Strategic and International Studies, 14 March 2022. csis.org

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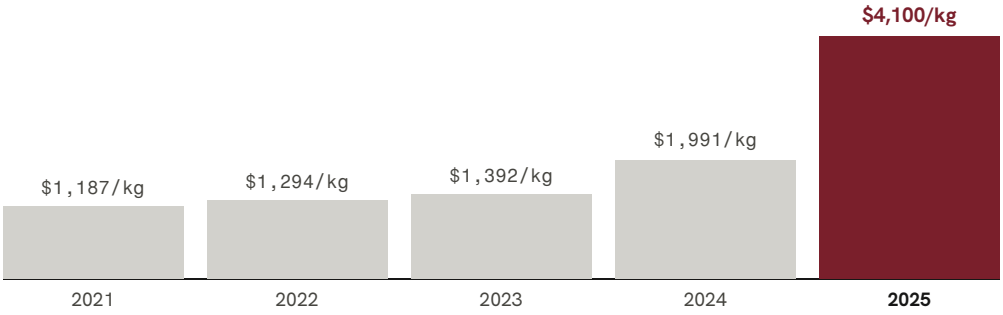
A slower risk is the European Union's proposed restriction on PFAS, the long-lived fluorinated chemicals. These chemicals are used throughout a fab, from some etch gases to seals, filters and the plastic linings of chemical tanks and pipes. Most of the photoacid generators that free the acid in resist contain PFAS too[47].

Each resist made without PFAS would need full qualification in the chipmaker's fab[3]. The Semiconductor Industry Association wrote in 2023 that photoresist needs the most rigorous qualification of any chemical a chipmaker uses, because each resist is tuned to the layers it prints[10]. The resist maker adjusts its formula over rounds of tests in the fab[3]. A fab could not qualify all the new resists at once, mainly because testing several together can muddle the yield results[3]. So some would have to be qualified one after another[3].

Figure 03.3

Germanium in Europe rose to an estimated $4,100 a kilogram in 2025

Germanium metal, annual average price in Europe. China required export licenses from August 2023, banned shipments to the US in December 2024 and suspended the ban from November 2025 to 27 November 2026



Source: USGS Mineral Commodity Summaries, February 2026 (Argus Media European prices, 2025 estimated) [7]

The EU proposal may still let chipmakers keep using PFAS where the risks can be controlled. The national authorities of Denmark, Germany, the Netherlands, Norway and Sweden prepared the proposal and updated it after evaluating more than 5,600 comments[48]. ECHA, the EU chemicals agency, published the updated proposal on 20 August 2025[48]. The updated proposal assessed options that would let electronics and semiconductor makers keep using PFAS where the risks can be controlled[48].

In March 2026 ECHA's two scientific committees backed an EU-wide restriction with targeted derogations, or exemptions[49]. One committee gave its backing in a final opinion and the other in a draft[49]. Their evaluation is due to finish by the end of 2026[49]. The European Commission will decide in consultation with the member states[48].

3 PFAS-Containing Photo-Acid Generators Used in Semiconductor Manufacturing. Semiconductor Industry Association, 8 June 2023, p. 22. semiconductors.org

7 Mineral Commodity Summaries 2026: Germanium. U.S. Geological Survey, 5 February 2026. pubs.usgs.gov

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Four questions to answer from memory

1. What does photoresist do? see How it works
2. Which country's companies sell most of the world's photoresist? see Who makes it
3. Why is it slow to switch to a new photoresist supplier? see The chokepoint
4. Which two metals has China required a license to export since August 2023? see The chokepoint

Four things to remember

— Photoresist holds the pattern printed by light so the next step can etch it into the wafer.

— Japanese companies held 77 percent of world photoresist sales in 2021 and 78 percent in 2023.

— A chipmaker has to approve each new resist in its own fab, and even one that passes the first time takes seven to 17 months.

— China has required licenses to export gallium and germanium since August 2023.

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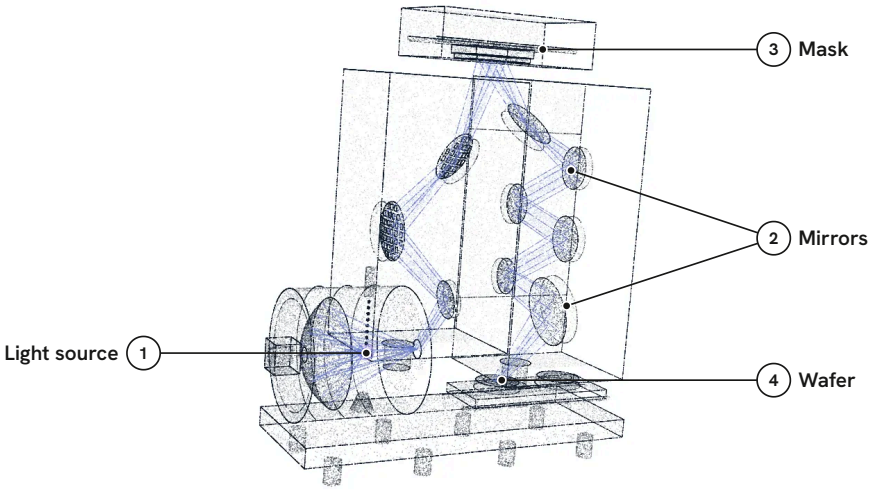
The chain

01 Chip Design, EDA and IP
02 Silicon and Wafers
03 Chemicals, Gases and Photoresist
04 Lithography
05 Photomasks and Pellicles
06 Deposition and Etch
07 Metrology and Inspection
08 Foundries and Fabs
09 Memory and HBM
10 Advanced Packaging
11 Substrates and PCBs
12 Test and Assembly
13 Systems and Networking
14 Data Centers and Power

15 The Economics of AI Chips
16 Who Controls the Chain

Lithography

ASML, a Dutch firm, is the only maker of extreme ultraviolet scanners, the machines that print the finest layers of a chip. It sold 48 of them in 2025.



Specimen 04 · Lithography

An extreme ultraviolet scanner cut open along its light path. The machine is the size of a bus and weighs about 180 metric tons. The blue-violet thread is the 13.5 nm light. The mask and the wafer are drawn about twice their true size.

Parts, as numbered on the drawing

1. Light source: ASML San Diego (US), laser by Trumpf (DE)
2. Mirrors: Zeiss SMT (DE)
3. Mask: see stage 05, Photomasks and Pellicles
4. Wafer: see stage 02, Silicon and Wafers

In plain terms

Lithography prints the pattern of a circuit onto the wafer. A machine holds a stencil of one layer of the circuit up to a light source and projects the image, four times smaller, onto the silicon. The shorter the wavelength of the light, the finer the lines a machine can print. Even so, a machine can print lines narrower than its light's wavelength. Immersion scanners, for example, shine the light through a layer of water between lens and wafer, which lets them print smaller features with the same light[1]. ASML rates one of its immersion scanners, the

1 Lenses & mirrors - Lithography principles. ASML, accessed 9 September 2026. asml.com

NXT:2150i, at features of 38 nanometers or less, printed with argon fluoride laser light at 193 nanometers[2]. Air soaks up extreme ultraviolet light, the shortest in use, so machines that print with it work in a vacuum. Only ASML, a Dutch company, makes them. Export controls are government rules on who a company may sell to. Every advanced chip passes through ASML's machines, so export controls on ASML affect all advanced chipmaking.

In short

Beyond extreme ultraviolet (EUV), ASML holds above 80 percent of the market for deep ultraviolet (DUV) lithography, which uses light of longer wavelength[3]. In July 2026 ASML put its capacity for 2026 at around 65 standard EUV scanners, which caps how many fabs anywhere can add that year[4]. ASML says the production capacity of Zeiss, its sole supplier of optics, limits how many lithography systems it can build[5].

Export controls set up by the United States with allies including the Netherlands block EUV scanners from Chinese chip foundries[6]. In December 2025 Reuters reported a Chinese prototype that had made EUV light but had printed no chip[7]. In July 2026 Reuters reported, citing one unnamed source, that a Shanghai firm with two state shareholders has started producing immersion DUV tools, which shine their light through water[8].

ASML sold 327 lithography systems in 2025[9]. Tools shipped to China made up 33 percent of its net system sales that year, more than any other region[10].

Chokepoint card Lithography

Concentration

Extreme

Three or fewer firms could supply this stage today, or one firm holds more than half of it.

Substitutability

Very hard

ASML is the only maker of EUV scanners, and Zeiss is its sole supplier of their mirrors[5].

Price or market size

About $200 million

for a standard EUV scanner, an estimated $350 million to $400 million for the newer High-NA version and roughly $60 million for a tool in the DUV segment, according to Reuters in July 2026[3]

Who leads	NL	**ASML**	327 lithography systems sold in 2025, including all 48 EUV (extreme ultraviolet) systems[9]
	JP	**Canon**	234 chip lithography systems sold in 2025[11]
	JP	**Nikon**	22 new chip lithography systems sold in the year to March 2026[12]

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Where it is made	NL	**Netherlands**	ASML, the Dutch maker of every EUV scanner
	DE	**Germany**	Zeiss SMT optics, Oberkochen, and Trumpf carbon dioxide drive lasers, Ditzingen
	US	**United States**	ASML light-source research and manufacturing, San Diego
	JP	**Japan**	Nikon and Canon lithography scanners and steppers

Why substitution is slow

A newcomer would first have to make mirrors a meter wide, polished smooth to within tens of picometers, a picometer being a trillionth of a meter [1]. It would also need a light source that hits tin droplets with a laser 50,000 times a second [13].

Where China stands

In July 2026 Reuters reported, citing one unnamed source, that a Shanghai firm with two state shareholders had started producing immersion DUV tools [8]. The Center for Technology & Statecraft found in September 2026 that Chinese firms had begun small-batch production of such scanners, but none had yet been shown fit for producing advanced chips [14]. Its central estimate in 2026 was that China would have a commercially viable immersion scanner for 7 nm chips in the mid-2030s, with the early 2030s plausible, and EUV not before the second half of the 2030s [14]. Meanwhile Chinese fabs had built a stockpile of more than 330 immersion DUV systems by the first quarter of 2026, most of them ASML's NXT:1980i [14].

Where the US stands

CSET found in 2021 that no American firm made lithography scanners [15]. In September 2026 the Center for Technology & Statecraft found that ASML held about 99 percent of the market for immersion DUV scanners and Japan's Nikon the rest [14]. But every immersion DUV scanner holds a light source from Cymer, a US company now part of ASML, or from Japan's Gigaphoton, so the two governments could stop sales by controlling those parts, the Center for Technology & Statecraft found in September 2026 [14]. A US rule published in October 2023 already covers immersion scanners with an alignment error between 1.5 nm and 2.4 nm bound for advanced chipmaking if they hold any American-made parts [16]. The US entity list also requires a license to supply the firms it names [17].

01

How it works

The stencil is called a photomask, and the machine a scanner (see Photomasks and pellicles).

The wafer is coated with photoresist, a film that changes wherever light touches it. The mask holds one layer of the circuit as a pattern of clear and dark areas, drawn at four times its size on the chip. The machine shines light through the mask, shrinks the image four times with a lens, and lands it on the film.

A wash then removes the film where the light hit, or, with some films, where it did not, leaving the pattern standing on the wafer [18]. The film then serves as a stencil for the next step, which either etches the pattern into the layer beneath or fires ions into the open areas to change how the silicon conducts [18]. Then the film is stripped and the next mask goes in. Chips are made up of dozens of layers, ASML says [18]. Photronics, a US mask maker, says masks are typically sold in sets, with each layer's pattern on a different mask [19].

1 Lenses & mirrors - Lithography principles. ASML, accessed 9 September 2026. asml.com

8 China starts production of home-grown immersion DUV chipmaking tools, source... Reuters, 28 July 2026. reuters.com

13 All about light and lasers in lithography. ASML, accessed 9 September 2026. asml.com

14 DUV Immersion Lithography: The Foundation of the US AI Advantage. Center for Technology & Statecraft, September 2026. techstatecraft.org

15 The Semiconductor Supply Chain - Issue Brief. Center for Security and Emerging Technology, 21 January 2021, p. 4. cset.georgetown.edu

16 Export Controls on Semiconductor Manufacturing Items. Federal Register, 25 October 2023. federalregister.gov

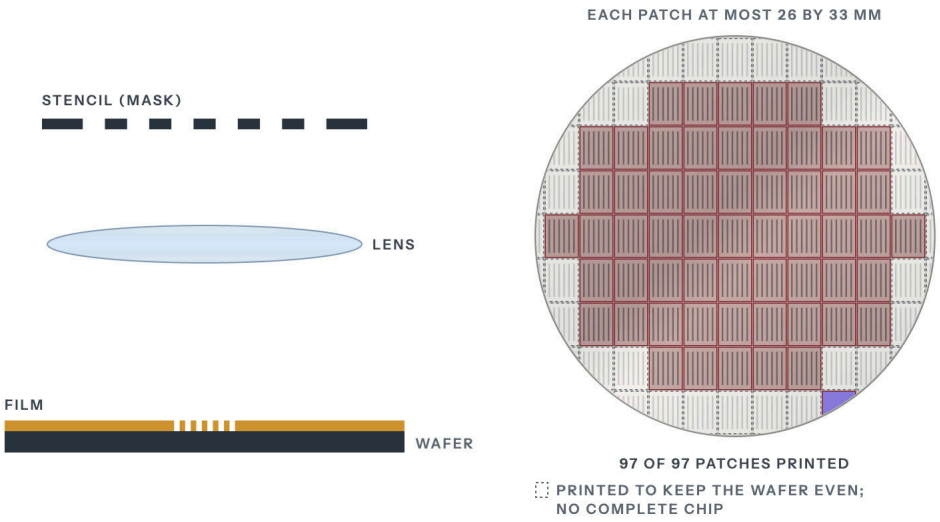
17 Expansion of End-User Controls To Cover Affiliates of Certain Listed Entities. Federal Register, 30 September 2025. federalregister.gov

18 6 crucial steps in semiconductor manufacturing. ASML, 4 October 2023. asml.com

19 Photronics, annual report on Form 10-K for the fiscal year ended 31 October 2025. U.S. Securities and Exchange Commission, 17 December 2025. sec.gov

Figure 04.1 · Explainer, 19 seconds

How lithography prints a chip



1. **Coat.** The wafer is coated with a thin film that changes wherever light touches it.
2. **Stencil.** A stencil, called a mask, holds the pattern for one layer of the chip, drawn four times larger than life.
3. **Shrink.** Light shines through the stencil, and a lens shrinks the image four times and lands it on the film.
4. **Wash.** A wash removes the film wherever the light landed, and the pattern is left on the wafer.
5. **Repeat.** The machine steps across the wafer, printing one patch at a time, each about the size of a postage stamp.

Note: Simplified. The newest machines, which use extreme ultraviolet light, carry the image with mirrors in place of a lens.

The image from one mask covers a patch of up to 26 by 33 mm[20]. The wafer moves under the optics one patch at a time, close to a hundred patches per wafer. ASML's own wafers-per-hour rating for an EUV scanner assumes 96 of them[20]. Before the wafer is exposed, the machine measures alignment marks printed in earlier layers so the new layer lines up with them[20]. For its NXE:3400B EUV scanner, ASML rates the matched-machine overlay, a measure of that alignment, at 2 nm[20]. That model first shipped in 2017[21].

Finer lines need shorter light. Glass and air both absorb the shortest light in use, at 13.5 nm[22]. So an EUV scanner works in a vacuum and uses mirrors instead of lenses, and its masks are mirrors too[22]. Deep ultraviolet (DUV) machines use light with a wavelength more than fourteen times longer[13]. ASML calls its immersion DUV scanners the workhorses of the industry, used in high-volume manufacturing of the most advanced logic and memory chips[23].

13 All about light and lasers in lithography. ASML, accessed 9 September 2026. asml.com
20 TWINSCAN NXE:3400B: EUV lithography systems. ASML, accessed 27 September 2026. asml.com
21 Stronger than expected demand drives ASML Q1 sales. U.S. Securities and Exchange Commission, 19 April 2017. sec.gov
22 ASML EUV lithography systems. ASML, accessed 9 September 2026. asml.com
23 See ASML's DUV lithography systems. ASML, accessed 9 September 2026. asml.com

When one exposure cannot draw a pattern finely enough, chipmakers use multipatterning. In the litho-etch kind, the pattern is split between two or more masks printed in turn[24]. Each added mask brings more process steps, more cost and a risk that the parts of the pattern misalign[25]. Self-aligned double and quadruple patterning (SADP and SAQP) print the lines with one exposure and then multiply them with deposition and etch steps[24]. In SAQP each printed line ends up as four[24]. Because one exposure places all those lines, they cannot misalign with each other[24]. A later exposure still prints blocks that break them where the circuit needs gaps[24]. ASML says one EUV exposure per layer cuts masks and process steps and lowers cost[26].

02

Variants and trade-offs

Deep ultraviolet

Deep ultraviolet (DUV) light comes from gas lasers called excimer lasers: 248 nanometers from krypton fluoride (KrF), 193 from argon fluoride (ArF), shone through air or water[13]. Cymer, an independently run ASML business in San Diego, makes these excimer lasers, which generate the light that scanners use to print patterns on wafers[27]. Gigaphoton, a Komatsu company in Japan, also makes excimer lasers for lithography tools[28].

Of the systems ASML sold in 2025, 131 were ArF immersion, 78 KrF and 16 ArF dry, all deep ultraviolet, and another 54 were older i-line tools[10]. The i-line tools use 365 nm light from a mercury lamp, and ASML says they print features as small as 220 nm[13].

Reuters gives one price for the whole DUV segment, with tools costing roughly $60 million each[3]. ASML's own figures split its DUV sales by tool type. In 2025 the 131 ArF immersion tools it sold made up 42 percent of its system sales, and the 78 KrF tools only 4 percent[10]. In DUV, ASML competes with Nikon and Canon, and it still holds above 80 percent of that market[3].

Chipmakers without EUV build their finest layers on immersion DUV scanners, in several passes per layer[26]. From 1 September 2023, Dutch export licensing covered the most advanced of these scanners, ASML's NXT:2000i and later models[29].

Extreme ultraviolet

The 13.5 nm light comes from molten tin droplets shot from a generator[13]. A laser hits each droplet twice, first with a low-intensity pulse that flattens it and then with a more powerful one that vaporizes it into plasma[13]. The process repeats 50,000 times a second[13]. Mirrors made by Zeiss, a German optics company, carry the light to the

3 Explainer: The $400 million ASML 'printers' key for the AI chip boom. Reuters, 28 July 2026. reuters.com
10 ASML 2025 fourth-quarter and full-year results (investor presentation). ASML, 28 January 2026, p. 9. ourbrand.asml.com
13 All about light and lasers in lithography. ASML, accessed 9 September 2026. asml.com
24 EUV lithography: weighing the options. Imec, 6 June 2019. imec-int.com
25 Applied Materials' Innovative Pattern-Shaping Technology Reduces the Cost... Applied Materials, 28 February 2023. ir.appliedmaterials.com
26 Strategic report: From ASML's Annual Report 2025. ASML, 24 February 2026, p. 16. ourbrand.asml.com
27 Explore ASML San Diego. ASML, accessed 9 September 2026. asml.com
28 Gigaphoton, company profile. Gigaphoton, accessed 9 September 2026. gigaphoton.com
29 Statement regarding export control regulations Dutch government. ASML, 30 June 2023. asml.com

wafer[1]. The largest are a meter across and smooth to within tens of picometers, a picometer being a thousandth of a nanometer[1].

Brighter light means more wafers an hour, and ASML demonstrated the first 1,000-watt EUV source in April 2025[30]. Its TWINSCAN NXE:3800E ships at its full 220 wafers an hour, 37 percent more than the NXE:3600D[31].

High-NA EUV

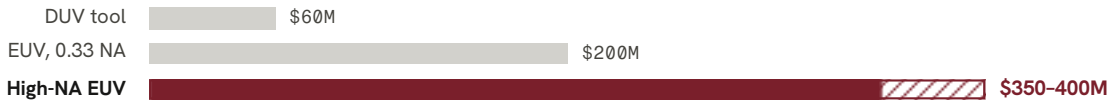
High-NA scanners raise the numerical aperture (NA), a measure of how well the optics collect and focus light, from 0.33 to 0.55[32]. That cuts the smallest printable feature from 13 nm to 8 nm[32]. Their optics shrink the mask image four times in one direction and eight in the other, halving the patch of wafer one exposure covers and doubling the exposures per wafer[32]. The second-generation TWINSCAN EXE:5200B runs at 175 wafers an hour, 60 percent above the EXE:5000[26].

Reuters puts a standard EUV tool at around $200 million and a High-NA machine at an estimated $350 to $400 million[3].

Figure 04.2

High-NA EUV scanners cost an estimated $350 million to $400 million in 2026

What one lithography system costs. The DUV bar is one figure for the whole segment



Source: Reuters explainer on ASML's lithography machines, first published 28 January 2026 and updated 28 July 2026 [3]

ASML sold four High-NA (EXE) systems in 2025, against two in 2024[33]. In its 2025 annual report, filed in February 2026, ASML said it expected High-NA to start supporting high-volume manufacturing in 2027[5].

Intel and TSMC have chosen differently on High-NA:

- **Intel Foundry** says it uses High-NA in volume production for select layers of some Intel 18A products[34]. It installed the first EXE:5200B[35]. By September 2026 it had run more than one million wafers on High-NA tools, a count that includes tool testing and research as well as volume production[34].
- **TSMC** waited, and in September 2026 said it will use High-NA in high-volume manufacturing from 2030[36].

On 8 September 2026 ASML and TSMC announced a move from 6-inch photomasks to 12-inch ones, to remove stitching constraints[36]. Stitching constraints are the design

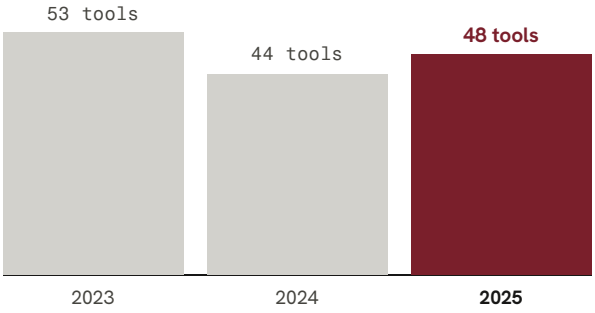
1 Lenses & mirrors - Lithography principles. ASML, accessed 9 September 2026. asml.com
3 Explainer: The $400 million ASML 'printers' key for the AI chip boom. Reuters, 28 July 2026. reuters.com
5 ASML Holding, 2025 annual report, filed on Form 6-K. U.S. Securities and Exchange Commission, 25 February 2026. sec.gov
26 Strategic report: From ASML's Annual Report 2025. ASML, 24 February 2026, p. 16. ourbrand.asml.com
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31 Strategic report: From ASML's Annual Report 2025. ASML, 24 February 2026, p. 17. ourbrand.asml.com
32 5 things you should know about High NA in EUV. ASML, 25 January 2024. asml.com
33 Financial performance: From ASML's Annual Report 2025. ASML, 24 February 2026, p. 7. ourbrand.asml.com
34 Intel Foundry and ASML Accelerate Industry Readiness for High-NA EUV. Intel, 7 September 2026. intel.com
35 High NA EUV reaches new readiness milestone with first high-volume Logic product. ASML, 15 July 2026. asml.com
36 ASML and TSMC Announce Initiative to Pioneer Industry Transition to... TSMC, 8 September 2026. pr.tsmc.com

limits that come from printing a large chip in two halves. ASML and TSMC aim for a pilot line for the new masks by 2031 and for High-NA systems that use them to enter production by 2033[36].

Figure 04.3

ASML sold 48 EUV systems in 2025, up from 44 in 2024

EUV systems ASML sold each year



Source: ASML 2025 annual report for 2024 and 2025, and ASML's Q4 2023 investor presentation for 2023 [9]

03

Who makes it

ASML's net sales were EUR 32.7 billion in 2025, at a 52.8 percent gross margin[9]. In October 2026 it expected 2026 net sales of EUR 44 to 45 billion, at a gross margin of around 56 percent[37]. Its 2025 annual report lists 5,100 suppliers[5]. Three of them and ASML's own San Diego site make the main parts of the EUV machine:

- **Zeiss SMT**, Oberkochen, Germany, makes the illumination system and the six-mirror projection optics[38].
- **Trumpf**, Ditzingen, Germany, makes the carbon dioxide drive laser that hits the tin droplets[39]. It amplifies a few watts to 40 kilowatts[39].
- **ASML San Diego**, which includes the light-source maker Cymer that ASML bought in 2013, designs the source where the laser turns tin into plasma, and builds the droplet generator[27].
- **VDL ETG** builds the frames that suspend and position the mirrors[40].

ASML says the production capacity of Zeiss, its sole supplier of optics, limits how many lithography systems it can build[5].

Lithography is the stage of the chain that European firms lead, with ASML the only maker of EUV scanners[26]. The bulk of ASML's suppliers are in the Netherlands and the

5 ASML Holding, 2025 annual report, filed on Form 6-K. U.S. Securities and Exchange Commission, 25 February 2026. sec.gov
9 Financial performance: From ASML's Annual Report 2025. ASML, 24 February 2026, p. 5. ourbrand.asml.com
26 Strategic report: From ASML's Annual Report 2025. ASML, 24 February 2026, p. 16. ourbrand.asml.com
27 Explore ASML San Diego. ASML, accessed 9 September 2026. asml.com
36 ASML and TSMC Announce Initiative to Pioneer Industry Transition to... TSMC, 8 September 2026. pr.tsmc.com
37 ASML reports €11.7 billion total net sales and €5.0 billion net income in Q3... U.S. Securities and Exchange Commission, 14 October 2026. sec.gov
38 EUV lithography and technology. Carl Zeiss, accessed 9 September 2026. zeiss.com
39 Good things come in ever-smaller packages. TRUMPF, accessed 9 September 2026. trumpf.com
40 VDL ETG builds complex frames for Zeiss. VDL Groep, 22 March 2019. web.archive.org

rest of Europe, the Middle East and Africa, the company says[41].

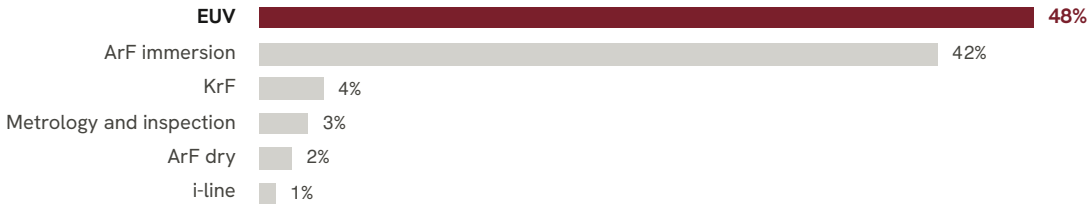
In Japan, Nikon and Canon also make lithography tools. Nikon sold 22 new chip lithography systems in the year to March 2026, in a market it sizes at 570 units for 2025[12]. Its Precision Equipment business made an operating loss in the year to March 2026[42]. Canon sold 234 chip lithography systems in 2025, 188 of them i-line and 46 KrF[11]. Its range runs from KrF scanners to i-line steppers, including some built for advanced packaging[43]. Canon also makes a nanoimprint system, the FPA-1200NZ2C, which stamps the pattern instead of projecting it and can print lines down to 14 nm wide[44]. Canon delivered one in 2024 to the Texas Institute for Electronics, to be used for research and prototypes[44].

In January 2021 CSET, a research center at Georgetown University, found that the United States lacked firms in a few key parts of the chip supply chain, above all lithography tools[15].

Figure 04.4

EUV made up 48 percent of ASML's 2025 system sales

ASML net system sales by technology, 2025



Source: ASML Q4 2025 investor presentation [10]

04

Controls and China's answer

Export controls set up by the United States with allies including the Netherlands block ASML from shipping EUV scanners to Chinese foundries[6]. In September 2025 CSIS judged that EUV remained "firmly out of reach" for China despite massive state investment[6]. For DUV scanners and other tools, the restrictions came in this order:

- **Dutch national licensing** of advanced DUV took effect on 1 September 2023[45].
- **A US rule** published on 25 October 2023 covered immersion scanners that print features of 45 nm or smaller with an alignment error between 1.5 nm and 2.4 nm[16].

6 Breakthroughs or Boasts? Assessing Recent Chinese Lithography Advancements. Center for Strategic and International Studies, 24 September 2025. csis.org

10 ASML 2025 fourth-quarter and full-year results (investor presentation). ASML, 28 January 2026, p. 9. ourbrand.asml.com

11 Canon Inc. FY 2025 Results. Canon, 29 January 2026, p. 22. global.canon

12 Financial Results: The Year Ended March 31, 2026. Nikon, 8 May 2026, p. 8. nikon.com

15 The Semiconductor Supply Chain - Issue Brief. Center for Security and Emerging Technology, 21 January 2021, p. 4. cset.georgetown.edu

16 Export Controls on Semiconductor Manufacturing Items. Federal Register, 25 October 2023. federalregister.gov

41 Sourcing & Supply Chain. ASML, accessed 27 October 2026. asml.com

42 Financial Results: The Year Ended March 31, 2026. Nikon, 8 May 2026, p. 6. nikon.com

43 Semiconductor Lithography Equipment. Canon, accessed 18 October 2026. global.canon

44 Canon delivers FPA -1200NZ2C nanoimprint lithography system for semiconductor... Canon, 26 September 2024. global.canon

45 The Netherlands expands export control measure for advanced semiconductor... Government of the Netherlands, 6 September 2024. government.nl

If such a scanner was bound for advanced chipmaking, the rule put it under US export rules when it held any American content at all[16]. The rule did not apply to scanners first exported from a country that lists them on its own export control list[16].

- **A partial license revocation** by the Dutch government, disclosed by ASML on 1 January 2024, stopped shipments of NXT:2050i and NXT:2100i immersion scanners to a few Chinese customers[46].
- **The Netherlands widened its licensing** to more types of deep ultraviolet lithography equipment from 7 September 2024[45]. On 15 January 2025 it announced that from 1 April 2025 the license requirement would also cover a very limited number of further goods, among them specific measuring and inspection equipment[47].
- **The BIS Affiliates Rule**, issued by the US Bureau of Industry and Security, extended entity-list controls to affiliates owned 50 percent or more by listed entities from 29 September 2025[17]. BIS then suspended it until 9 November 2026[48]. Exporting goods covered by US export rules to a firm on the entity list needs a US license[17].
- **The EU** added certain Dutch controls to the EU Control List in November 2025[5]. ASML says this changed nothing for its immersion tools, because they already needed a Dutch license[5].

China's share of ASML's net system sales fell from 41 percent in 2024 to 33 percent in 2025[10].

On DUV, Reuters reported in July 2026, citing one unnamed source, that Shanghai Aishengna Electronic Technology Group has started producing home-grown immersion DUV tools[8]. The firm was registered in August 2023 with 7 billion yuan of capital and two state shareholders, Shanghai Electric Holding and a Shanghai International Trust subsidiary[8]. It has no website and has taken in teams from SMEE, China's older scanner maker, and the startup Yuliangsheng[8]. SMEE is strongest in i-line, the coarsest kind of lithography tool, and even there holds only four percent of the world market[6].

The Center for Technology & Statecraft, a research group, found in September 2026 that none of China's home-grown immersion scanners had yet been shown fit for producing advanced chips[14]. Its central estimate was a commercially viable Chinese immersion scanner for 7 nm chips in the mid-2030s, and EUV not before the second half of the 2030s[14]. Until then Chinese fabs rely on imported tools, and by the first quarter of 2026

5 ASML Holding, 2025 annual report, filed on Form 6-K. U.S. Securities and Exchange Commission, 25 February 2026. sec.gov

6 Breakthroughs or Boasts? Assessing Recent Chinese Lithography Advancements. Center for Strategic and International Studies, 24 September 2025. csis.org

8 China starts production of home-grown immersion DUV chipmaking tools, source... Reuters, 28 July 2026. reuters.com

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17 Expansion of End-User Controls To Cover Affiliates of Certain Listed Entities. Federal Register, 30 September 2025. federalregister.gov

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48 One Year Suspension of Expansion of End-User Controls for Affiliates of Certain... Federal Register, 12 November 2025. federalregister.gov

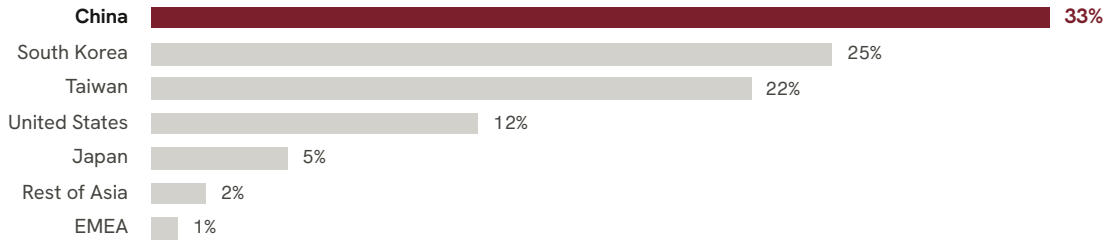
they had built a stockpile of more than 330 immersion DUV systems, most of them ASML's NXT:1980i[14].

There is less evidence of Chinese progress on EUV. Reuters reported in December 2025 that a Shenzhen team of former ASML engineers had built a prototype that made extreme ultraviolet light but had printed no chip[7]. To print a chip, a machine also needs mirrors that carry that light to the wafer, and ASML's sole supplier of those mirrors is Zeiss[5].

Figure 04.5

China took 33 percent of ASML's 2025 system sales, the largest share

ASML net system sales by the region tools shipped to, 2025



Source: ASML Q4 2025 investor presentation [10]

Note: Net system sales leave out service and field option sales, which ASML counts in its total net sales. The Export Controls page measures China's share against total net sales.

05

The chokepoint

If ASML stopped shipping, no fab anywhere could add EUV capacity, because no other company makes EUV scanners[26]. The scanners already in fabs would keep printing only as long as they were serviced. ASML has around 10,000 customer support employees who maintain and upgrade those machines around the clock[49]. Serviced this way, ASML's machines stay in use for a long time. In its 2025 annual report ASML said about 95 percent of the systems it had sold in the previous 30 years were still in use[5]. The same report warned that export controls may limit the service ASML can provide for machines already installed[50].

ASML has a chokepoint of its own. Zeiss SMT is its only supplier of lenses, mirrors and other critical optics[51]. ASML says that if Zeiss could not supply them for a long period, ASML would effectively be unable to run its business[51].

ASML's capacity caps how many EUV scanners fabs anywhere can add each year. In July 2026 ASML put its 2026 capacity at around 65 standard (low-NA) EUV scanners and

5 ASML Holding, 2025 annual report, filed on Form 6-K. U.S. Securities and Exchange Commission, 25 February 2026. sec.gov

7 How China built its 'Manhattan Project' to rival the West in AI chips. Reuters, 17 December 2025. reuters.com

10 ASML 2025 fourth-quarter and full-year results (investor presentation). ASML, 28 January 2026, p. 9. ourbrand.asml.com

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said it plans to add 30 percent for 2027[4].

Leading-edge AI accelerators are made with EUV, which prints their finest layers, the ones that set how densely transistors and wires pack[22]. TSMC expects the number of layers needing High-NA to rise, driven mainly by the more complex transistors that AI chips need[36].

Four questions to answer from memory

1. What does lithography do? `see How it works`
2. Why do chipmakers print the finest layers with extreme ultraviolet (EUV) light? `see How it works`
3. Who makes EUV machines? `see Who makes it`
4. How do export controls keep EUV machines out of China? `see Controls and China's answer`

Four things to remember

- Lithography prints each layer of a chip's circuit onto the wafer.
- Extreme-ultraviolet (EUV) light prints the finest layers in fewer steps than deep ultraviolet, but air and glass absorb it, so EUV machines work in a vacuum and use mirrors.
- ASML of the Netherlands is the only maker of EUV machines, and Zeiss alone supplies their mirrors.
- Export controls block EUV sales to Chinese chipmakers, and a Chinese prototype reported in December 2025 had printed no chip.

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The chain

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08 Foundries and Fabs
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Photomasks and Pellicles

Only three firms in the world make the blank plates for the masks used to print the finest layers of leading-edge chips. Japan's AGC and Hoya are two of them.

Specimen 05 · Photomasks and Pellicles

An EUV photomask, or reticle, with its pellicle dust cover. The glass plate is drawn to scale. It is a 152 mm square, 6.35 mm thick, about the size of a small floor tile. The dark blue band around the top edge is the mirror coating. The real coating is about forty pairs of layers, each pair 7 nm thick, far thinner than drawn.

In plain terms

A photomask is the stencil that lithography prints from, at least one for each layer of a chip[1]. The same mask prints its layer on wafer after wafer, so one mask shapes millions of chips. The most complicated layers are printed with extreme ultraviolet (EUV) light, and the rest can often be printed with deep ultraviolet light[2]. For deep ultraviolet the mask is a quartz plate that the light shines through. No glass lets EUV light through, so the mask for EUV layers is a mirror. The mirror is built up from about forty pairs of thin layers, and the light bounces off it. Japan's AGC and Hoya are two of the three firms that make these mirrors. A dark pattern drawn on top of the mirror soaks up the light wherever no line should print. Any speck of dust on the mask repeats on every wafer it prints. So a thin membrane on a frame, called a pellicle, sits just above the mask and catches dust where it is too far out of focus to print.

In short

On CSET's count from TechInsights data, Austria's IMS Nanofabrication held 46.6 percent of the 2025 world market for electron-beam mask writers, which draw the pattern onto a mask[3]. Hoya says its share of the mask blank market is exceptionally high[4].

SemiAnalysis put a full set of masks for a leading-edge chip at $40 million in May 2026[5]. In July 2022 it wrote that more companies will lack the volume

1 **Photronics, annual report on Form 10-K for the fiscal year ended 31 October 2025.** U.S. Securities and Exchange Commission, 17 December 2025. sec.gov

2 **See ASML's DUV lithography systems.** ASML, accessed 9 September 2026. asml.com

3 **Supply Chain Explorer: Advanced Chips.** Emerging Technology Observatory (ETO), accessed 9 September 2026. chipexplorer.eto.tech

4 **HOYA, Information Technology business, review of operations, Integrated Report...** HOYA, accessed 9 September 2026. hoya.com

5 **EDA Market Primer.** SemiAnalysis, 21 May 2026. newsletter.semianalysis.com

to spread the fixed cost of mask sets[6]. On 8 September 2026 ASML and TSMC announced a plan to move from 6-inch to 12-inch masks[7]. They expect the change to raise fab productivity and lower chipmaking costs[7].

Each EUV layer of an AI accelerator is printed from a mask built on a blank plate. Only three companies worldwide supply those plates, and Japan's AGC is one of them[8]. Hoya, also Japanese, is another[4]. AGC says it is the only company in the world that takes an EUV blank all the way from the glass to the finished coating[9].

Chokepoint card Photomasks and Pellicles

Concentration

Extreme

Three or fewer firms could supply this stage today, or one firm holds more than half of it.

Substitutability

Hard

AGC said in December 2024 that only three firms supply the blank plates for EUV masks[8].

Price or market size

$40 million

for a leading-edge mask set, SemiAnalysis's May 2026 figure[5]. In July 2022 it put a 7 nm set above $10 million[6]

Who leads	JP	**AGC**	Ranks itself No.2 worldwide in EUV mask blanks on its own fiscal 2026 estimates[10]
	JP	**Hoya**	Said in its 2026 report that its share of mask blanks is exceptionally high[4]
	JP	**Lasertec**	Launched the world's first tool that inspects patterned EUV masks at the scanner's own wavelength in 2019[11]
	AT	**IMS Nanofabrication**	CSET puts its 2025 share of the market for all electron-beam mask writers at 46.6 percent, using TechInsights data[3]
	US	**Photronics**	$849.3 million of revenue in fiscal 2025[12]
Where it is made	JP	**Japan**	Mask blanks (AGC, Hoya), DUV blanks (Shin-Etsu), inspection (Lasertec), writers (NuFlare), pellicles (Mitsui Chemicals)
	AT	**Austria**	IMS Nanofabrication multi-beam mask writers
	DE	**Germany**	Zeiss mask checking in EUV light (AIMS EUV) and mask repair
	US	**United States**	Photronics merchant mask shops, KLA inspection
	TW	**Taiwan**	TSMC, which held 10 percent of IMS at the end of 2024
Why substitution is slow			AGC says it spent more than a decade meeting the EUV blank's requirements, such as near-zero thermal expansion, and began mass production only in 2017[8]. Germany's Zeiss calls its AIMS the only system that tests how a mask will print under conditions equivalent to the scanner's[13].

3 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO), accessed 9 September 2026. chipexplorer.eto.tech

4 HOYA, Information Technology business, review of operations, Integrated Report... HOYA, accessed 9 September 2026. hoya.com

5 EDA Market Primer. SemiAnalysis, 21 May 2026. newsletter.semianalysis.com

6 The Dark Side Of The Semiconductor Design Renaissance – Fixed Costs Soaring Due... SemiAnalysis, 24 July 2022. newsletter.semianalysis.com

7 ASML and TSMC Announce Initiative to Pioneer Industry Transition to... TSMC, 8 September 2026. pr.tsmc.com

8 The Japanese glass behind next-generation chips. AGC, 13 December 2024. agc.com

9 AGC to Boost Production Capacity of EUVL Photomask Blanks. AGC, 27 April 2023. agc.com

10 AGC Data Book. AGC Inc., 29 May 2026, p. 8. agc.com

11 Lasertec Releases ACTIS A200HiT Series Actinic EUV Patterned Mask Inspection... Lasertec, 31 October 2025. lasertec.co.jp

12 Photronics Reports Full Year and Fourth Quarter Fiscal 2025 Results. GlobeNewswire (release by Photronics), 10 December 2025. globenewswire.com

13 Mask Qualification Solutions by ZEISS SMT. Carl Zeiss, accessed 9 September 2026. zeiss.com

Where China stands

As of 2026 SMIC runs two mask shops of its own in Shanghai [14]. No Chinese firm appears in CSET's 2025 market shares for electron-beam mask writers, the machines that draw the pattern onto a mask[3].

Where the US stands

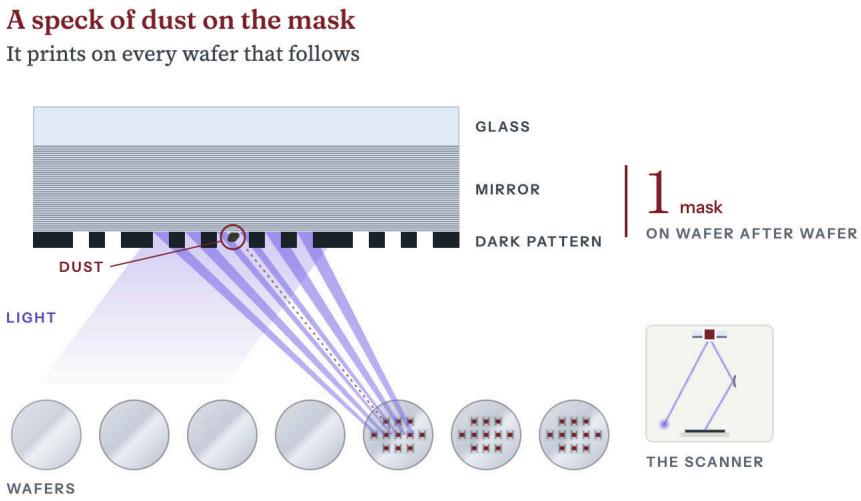
In December 2025 Photronics called itself a leading worldwide maker of photomasks [12]. As of 2026 KLA sells tools that inspect EUV masks and optical masks, the quartz masks used with deep ultraviolet light [15].

01

How it works

Figure 05.1 · Explainer, 23 seconds

How a photomask prints its pattern



1. **The scanner.** A photomask is the stencil that lithography prints from, at least one for each layer of a chip. Inside the scanner, light bounces off the mask and carries its pattern down to the wafer.
2. **A mirror.** No glass lets extreme ultraviolet (EUV) light through. So the mask for EUV layers is a mirror, built up from about forty pairs of very thin layers laid on glass, and the light bounces off it.
3. **The pattern.** A dark pattern drawn on the mirror soaks up the light wherever no line should print. The light that bounces back carries the pattern of one layer.
4. **Wafer after wafer.** The same mask prints its layer on wafer after wafer, so one plate shapes millions of chips.
5. **One speck.** A speck of dust on the mask prints on every wafer, until someone finds it. So a thin membrane on a frame, called a pellicle, covers the mask's patterned side and catches dust where it is too far out of focus to print.

Note: Simplified. The mask is drawn face down, as it sits in the scanner. The mirror layers and the dark pattern are drawn far thicker than they are, and the light's path through the scanner's mirrors is cut to one bounce.

A mask starts as a blank, a polished plate with no pattern on it yet. For deep ultraviolet the blank is a window of quartz carrying a light-blocking film such as molybdenum

3 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO), accessed 9 September 2026. chipexplorer.eto.tech

12 Photronics Reports Full Year and Fourth Quarter Fiscal 2025 Results. GlobeNewswire (release by Photronics), 10 December 2025. globenewswire.com

14 Mask Service. SMIC, accessed 9 September 2026. smics.com

15 Reticle Qualification. KLA Corporation, accessed 20 October 2026. kla.com

silicide[16]. Japan's Shin-Etsu is one maker of those blanks[16]. For EUV the blank is a mirror of about 40 molybdenum and silicon layer pairs, 7 nm to a pair, laid on glass[17]. A dark tantalum film on top, called the absorber, will carry the pattern[18].

The blank is coated with a sensitive film called resist. A mask writer then draws the design into the resist with electrons. A single-beam writer flashes one shape at a time, while NuFlare's multi-beam writer for the most advanced masks controls 260,000 beams at once[19].

A wash removes the drawn resist. An etch then removes the blocking film or absorber wherever the resist is gone, and the rest of the resist is stripped. The plate now passes light, or reflects it, in the shape of one layer of the chip. On standard EUV scanners the pattern is drawn four times larger than it will print[20]. An inspection tool compares the plate to the design, and a repair tool fixes the stray defects it finds.

EUV masks bring problems of their own:

- **Mask 3D effects.** Light strikes the mirror at a slant, and the absorber has height, so its edges cast small shadows that distort the printed shape. Software has to correct for the shadows in advance.
- **Write precision.** Every edge of the absorber has to sit within a few nanometers of where the design says.
- **Particles.** Anything on the mirror prints on every wafer until someone finds it.

High-NA scanners, the newest EUV machines, print finer lines with larger mirrors. To keep masks at their usual size, the scanners' mirrors shrink the pattern four times in one direction and eight times in the other[20]. So each exposure covers half the area of a standard exposure, and a wafer takes twice as many exposures[20].

On 8 September 2026 ASML and TSMC announced a plan to move from the current 6-inch masks to 12-inch masks[7]. They expect the change to raise fab productivity, lower chipmaking costs and remove stitching constraints, the design limits that come from printing a large chip in two halves[7]. They aim to set up a 12-inch mask pilot line by 2031, so that High-NA scanners using 12-inch masks can enter advanced chip production by 2033[7].

02

Variants and trade-offs

Blanks

AGC says it spent more than a decade meeting the blank's requirements, which include near-zero thermal expansion, perfectly smooth and flat surfaces and no particle contamination[8]. It says it began mass production only in 2017[8].

7 ASML and TSMC Announce Initiative to Pioneer Industry Transition to... TSMC, 8 September 2026. pr.tsmc.com

8 The Japanese glass behind next-generation chips. AGC, 13 December 2024. agc.com

16 Photomask blanks. Shin-Etsu Chemical, 3 April 2019. shinetsu.co.jp

17 The refined EUV mask model. ArXiv, 19 December 2019, p. 2. arxiv.org

18 The refined EUV mask model. ArXiv, 19 December 2019, p. 1. arxiv.org

19 EB Mask Writer. NuFlare Technology, accessed 9 September 2026. nuflare.co.jp

20 5 things you should know about High NA in EUV. ASML, 25 January 2024. asml.com

AGC publishes no market share figure for EUV blanks. In its data book, AGC ranks itself No.2 worldwide in EUV blanks, based on its own fiscal 2026 estimates[10]. Hoya says its share of the mask blank market is exceptionally high, and it calls itself the primary supplier for leading-edge EUV nodes[4]. It also says customers may gradually start buying EUV blanks from more than one supplier[4].

AGC planned to add EUV blank capacity by 2025, and it says its sales reached their target a year early. In April 2023 AGC said its EUV blank capacity would grow by about 30 percent in 2025[9]. The growth would come from a line in Fukushima[9]. METI, Japan's industry ministry, chose the project for a program that supports new plants and equipment in Japan to strengthen supply chains[9]. AGC also said it was aiming for more than ¥40 billion of sales in the business by 2025[9]. Its EUV blank sales grew 50 percent in 2023, AGC said in December 2024[8]. In February 2025 AGC reported that the sales reached ¥40 billion in 2024, which it called a year ahead of its initial target[21].

AGC plans to add capacity again after a dip in sales. Its EUV blank sales declined in 2025 and were recovering in 2026, AGC said in June 2026[22]. In August 2026 AGC set out capacity expansions of 20 to 50 percent across EUV blanks, polishing slurry and lens materials for lithography tools[23]. It gave no separate figure for the blanks[23]. Mass production on the added capacity is to begin in 2028[23].

Writers

An electron-beam writer draws the pattern into the resist on the blank. Multi-beam writers fire many beams at once, and Japan's NuFlare sells them for the 3 nm node[19].

In September 2023 Intel, the majority owner of Austria's IMS Nanofabrication, called it the established industry leader in the multi-beam mask writing tools that advanced EUV requires[24]. Counting all electron-beam mask writers, CSET, a research center at Georgetown University, put IMS's share of a $994 million world market at 46.6 percent in 2025, using TechInsights data[3]. NuFlare had 39.2 percent and JEOL 14.2 percent, in the same CSET analysis of TechInsights data[3].

Intel sold a combined 32 percent of IMS in 2023 and still held the other 68 percent on 27 December 2025[25]. In June 2023 it agreed to sell about 20 percent of IMS to Bain Capital[24]. Intel agreed in September 2023 to sell roughly 10 percent of IMS to TSMC at a valuation of about $4.3 billion[24]. TSMC's subsidiary TSMC Development held that 10 percent at the end of 2024, at a fair value of $432.8 million in TSMC's accounts[26].

3 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO), accessed 9 September 2026. chipexplorer.eto.tech

4 HOYA, Information Technology business, review of operations, Integrated Report... HOYA, accessed 9 September 2026. hoya.com

8 The Japanese glass behind next-generation chips. AGC, 13 December 2024. agc.com

9 AGC to Boost Production Capacity of EUVL Photomask Blanks. AGC, 27 April 2023. agc.com

10 AGC Data Book. AGC Inc., 29 May 2026, p. 8. agc.com

19 EB Mask Writer. NuFlare Technology, accessed 9 September 2026. nuflare.co.jp

21 Financial Results for FY2024. AGC Inc., 7 February 2025, p. 45. agc.com

22 Semiconductor-related Business Briefing: Event Summary. AGC Inc., 2 June 2026, p. 17. agc.com

23 AGC Company Overview. AGC Inc., August 2026, p. 52. agc.com

24 Intel to Sell Minority Stake in IMS Nanofabrication Business to TSMC. Intel, 12 September 2023. intc.com

25 Intel, annual report on Form 10-K for the fiscal year ended 27 December 2025. U.S. Securities and Exchange Commission, 23 January 2026. sec.gov

26 TSMC, report on Form 6-K dated 31 December 2024. U.S. Securities and Exchange Commission, 27 February 2025. sec.gov

Inspection and repair

A mask defect cuts the number of good chips on every wafer that mask prints, so masks are checked as blanks, after patterning and again at the fab. Actinic inspection uses 13.5 nm light, the wavelength of EUV lithography[11]. In 2019 Lasertec launched the ACTIS A150, which it calls the world's first actinic inspection system for patterned EUV masks[11]. In October 2025 it launched the ACTIS A200HiT series, which it says triples the A150's inspection speed[11]. Lasertec reported net sales of ¥230,485 million in the year to June 2026[27].

As of 2026 KLA's Teron systems for mask shops detect pattern and particle defects on both EUV masks and optical masks, the quartz masks used with deep ultraviolet light[15].

Zeiss makes AIMS EUV, which checks a mask in EUV light under the scanner's own imaging conditions to judge how a defect will affect printing[13]. Zeiss calls AIMS the only system that tests how a mask will print under conditions equivalent to the scanner's[13]. Zeiss says the tool can be extended to ASML's High-NA scanner, the EXE:5000[13]. Zeiss also sells electron-beam repair tools that work on EUV masks[28].

Pellicles

A pellicle is a thin membrane on a frame above the mask. Particles land on it, too far out of focus to print, and the pattern below stays clean.

Every photon a pellicle absorbs is light that never reaches the wafer[29]. The light crosses the pellicle twice, on the way in and on the way back off the mirror, so absorption counts double. In 2022 ASML described its most advanced EUV pellicle as a membrane 13 nanometers thick that can withstand up to 500 degrees Celsius in a high vacuum[29]. ASML released its first EUV pellicles in 2016 and calls the pellicle one of the breakthroughs that brought EUV lithography to high-volume manufacturing[29].

Carbon nanotube membranes are a newer kind of pellicle, aimed at next-generation High-NA, high-output EUV scanners[30]. In 2020 imec, a Belgian chip research center, mounted them on masks and exposed them in an ASML NXE:3300 scanner[31]. The membranes passed up to 97 percent of the light in a single crossing, and imec found their effect on imaging low and correctable[31].

Mitsui Chemicals began commercial production of silicon-based EUV pellicles under an ASML license in 2021[30]. In 2024 it said it would build a carbon nanotube membrane plant at Iwakuni-Ohtake, tentatively due to finish in December 2025[30]. In December 2025 Mitsui Chemicals said the plant was scheduled for completion in its fiscal 2025, in line with its initial plan[32]. The same presentation lists the plant under March 2026 in

11 Lasertec Releases ACTIS A200HiT Series Actinic EUV Patterned Mask Inspection... Lasertec, 31 October 2025. lasertec.co.jp

13 Mask Qualification Solutions by ZEISS SMT. Carl Zeiss, accessed 9 September 2026. zeiss.com

15 Reticle Qualification. KLA Corporation, accessed 20 October 2026. kla.com

27 Lasertec, business report. Lasertec, September 2026. lasertec.co.jp

28 Mask Repair Solutions by ZEISS SMT. Carl Zeiss, accessed 9 September 2026. zeiss.com

29 Indistinguishable from magic: the EUV pellicle. ASML, 14 September 2022. asml.com

30 Mitsui Chemicals Sets up Production Facilities for CNT Pellicles to Be Used in... Mitsui Chemicals, 28 May 2024. jp.mitsuichemicals.com

31 Imec demonstrates CNT pellicle utilization on EUV scanner. Imec, 6 October 2020. imec-int.com

32 Business Strategy Presentation (L&HC, Mobility, ICT) in December 2025. Mitsui Chemicals, 17 December 2025, p. 31. jp.mitsuichemicals.com

its table of key investments[33]. The membranes are designed to pass 92 percent or more of the light and to withstand exposure power above 1 kW[30].

03

Cost and design economics

A mask set costs more at each newer node, by SemiAnalysis's figures. In July 2022 SemiAnalysis put mask sets above $1 million at 28 nm and above $10 million at 7 nm[6]. It expected 3 nm mask sets to reach the $40 million range[6]. Its May 2026 primer still puts a leading-edge mask set at $40 million[5]. The same primer puts a single respin, remaking a chip after a design flaw, at $50 million to $100 million at leading-edge nodes[5].

Mask costs restrict which firms can use a node. SemiAnalysis wrote that at 5 nm and 3 nm, only firms with large wafer volume see the cost per transistor fall[6]. It wrote that more companies will lack the volume to spread the fixed cost of mask sets[6]. It expected failed chip designs to show up first among AI chip startups, which work on much tighter budgets[6].

Figure 05.2

In 2022 SemiAnalysis forecast a 3 nm mask set at $40 million

What one photomask set costs, by node, 2022. The 28 nm and 7 nm figures are minimums, and 3 nm was a forecast



Source: SemiAnalysis, The Dark Side of the Semiconductor Design Renaissance, July 2022. SemiAnalysis's May 2026 EDA Market Primer puts a leading-edge mask set at $40 million [6]

04

Who makes it

Mask making splits between chipmakers that make masks for their own use and independent merchant firms that sell to them and others[1]. Photronics says that chipmakers' own shops once supplied most of the market, and that production has more recently tended to move toward the independents[1]. TSMC makes masks too, and lists mask making among the services it offers besides making wafers[34].

1 Photronics, annual report on Form 10-K for the fiscal year ended 31 October 2025. U.S. Securities and Exchange Commission, 17 December 2025. sec.gov

5 EDA Market Primer. SemiAnalysis, 21 May 2026. newsletter.semianalysis.com

6 The Dark Side Of The Semiconductor Design Renaissance – Fixed Costs Soaring Due... SemiAnalysis, 24 July 2022. newsletter.semianalysis.com

30 Mitsui Chemicals Sets up Production Facilities for CNT Pellicles to Be Used in... Mitsui Chemicals, 28 May 2024. jp.mitsuichemicals.com

33 Business Strategy Presentation (L&HC, Mobility, ICT) in December 2025. Mitsui Chemicals, 17 December 2025, p. 28. jp.mitsuichemicals.com

34 TSMC, annual report on Form 20-F for the fiscal year ended 31 December 2025. U.S. Securities and Exchange Commission, 16 April 2026. sec.gov

Photronics calls itself a leading worldwide maker of masks for chips and flat panel displays[12]. It had $849.3 million of revenue in fiscal 2025, $615.1 million of it from chip masks[12]. The competitors it names in chip and display masks include Dai Nippon Printing, Hoya, Tekscend Photomask and two Chinese firms, Shenzhen Newway Photomask and Shenzhen Qingyi Photomask[1]. It also competes with chipmakers' own mask shops[1].

Chinese and foreign firms make masks in China, but no Chinese firm appears in CSET's 2025 market shares for electron-beam mask writers, the machines that draw the pattern onto a mask[3]. SMIC, a Chinese foundry, runs two in-house mask shops in Shanghai that make masks for deep ultraviolet and older lithography machines[14]. Photronics began operating two mask plants in China, in Hefei and Xiamen, in 2019[1]. Its chip mask business in Xiamen is a joint venture in which Dai Nippon Printing holds 49.99 percent[1].

05

The chokepoint

Every EUV mask depends on a handful of firms at each step, from the blank plate to the tool that tests how it will print. Only three firms supply the blank plates for EUV masks[8]. IMS Nanofabrication held 46.6 percent of the 2025 market for electron-beam mask writers and NuFlare 39.2 percent, by CSET's count[3]. In 2019 Lasertec launched what it calls the first tool that inspects patterned EUV masks in EUV light[11]. Zeiss calls its AIMS the only system that tests how a mask will print under conditions equivalent to the scanner's[13].

Pellicles already work in high-volume EUV manufacturing[29]. But ASML says no material is completely transparent to EUV light, so any pellicle in the light's path costs customers productivity[29]. In 2024 Mitsui Chemicals said it would set up production of carbon nanotube membranes, designed to pass 92 percent or more of the light, for next-generation High-NA, high-output EUV scanners[30].

Four questions to answer from memory

1. What is a photomask? see How it works
2. How many firms make the blank plates for EUV masks? see Blanks
3. Why is it hard for a new firm to start making EUV mask blanks? see The chokepoint
4. What kind of masks does China's SMIC make in its own shops? see Who makes it

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1 Photronics, annual report on Form 10-K for the fiscal year ended 31 October 2025. U.S. Securities and Exchange Commission, 17 December 2025. sec.gov

3 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO), accessed 9 September 2026. chipexplorer.eto.tech

8 The Japanese glass behind next-generation chips. AGC, 13 December 2024. agc.com

11 Lasertec Releases ACTIS A200HiT Series Actinic EUV Patterned Mask Inspection... Lasertec, 31 October 2025. lasertec.co.jp

12 Photronics Reports Full Year and Fourth Quarter Fiscal 2025 Results. GlobeNewswire (release by Photronics), 10 December 2025. globenewswire.com

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29 Indistinguishable from magic: the EUV pellicle. ASML, 14 September 2022. asml.com

30 Mitsui Chemicals Sets up Production Facilities for CNT Pellicles to Be Used in... Mitsui Chemicals, 28 May 2024. jp.mitsuichemicals.com

Four things to remember

- A photomask is the stencil for one layer of a chip, and any flaw on it repeats on every wafer.
- Three firms make EUV mask blanks, and Lasertec launched the first tool that checks finished masks in EUV light.
- AGC spent more than a decade learning to make EUV blanks before mass production began in 2017.
- China makes masks for deep ultraviolet machines and older ones, in SMIC's own mask shops and at independent firms such as Shenzhen Qingyi and Shenzhen Newway.

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The chain

01 Chip Design, EDA and IP
02 Silicon and Wafers
03 Chemicals, Gases and Photoresist
04 Lithography
05 Photomasks and Pellicles
06 Deposition and Etch
07 Metrology and Inspection
08 Foundries and Fabs
09 Memory and HBM
10 Advanced Packaging
11 Substrates and PCBs
12 Test and Assembly
13 Systems and Networking
14 Data Centers and Power

15 The Economics of AI Chips
16 Who Controls the Chain

Stage 6 of 14 · High concentration · Moderately hard to substitute

Deposition and Etch

Deposition and etch tools build a chip layer by layer. China-based toolmakers' share of deposition tools rose from 2 to 7 percent between 2019 and 2024, by CSET's count.

Specimen 06 · Deposition and Etch

A plasma etch chamber cut in half. The vessel is 500 mm across inside and holds one 300 mm wafer clamped to a holder, called a chuck, at the bottom. The violet haze above the wafer is the plasma. The gap it fills is drawn about twice its real height of 30 to 40 mm. The rest is to scale.

In plain terms

Deposition and etch build the chip itself, one layer at a time. Deposition lays down a thin film, sometimes only a few atoms thick. Lithography prints a pattern on it. Etch then eats away everything the pattern does not protect, and the next film goes down on top. Making one chip takes 1,000 to 1,500 steps, the Semiconductor Industry Association estimated in 2025[1]. Some of the holes the etch cuts in memory chips are more than 50 times deeper than they are wide, like a finger-wide shaft about a meter deep[2]. The film has to coat them right to the bottom. The etch has to eat one material and leave the one beside it untouched. Applied Materials and Lam Research in the United States, Tokyo Electron in Japan and ASM International in the Netherlands make most of these machines. Chinese toolmakers have gained share in both kinds of tool since 2019.

In short

American companies held 60.3 percent of the world market for deposition tools in 2025, Japanese companies 14.0 percent and China-based companies 9.8 percent, by CSET's count from TechInsights data[3]. In etch and cleaning tools that year, American companies held 52.6 percent, Japanese companies 32.6 percent and China-based companies 10.1 percent[4].

1 Testimony of David Isaacs, Semiconductor Industry Association, hearing on... U.S. Senate Committee on Finance, 14 May 2025, p. 2. finance.senate.gov

2 Foreign-Produced Direct Product Rule Additions, and Refinements to Controls for... U.S. Government Publishing Office, 5 December 2024, p. 17. govinfo.gov

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4 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO), accessed 9 September 2026. chipexplorer.eto.tech

Atomic layer deposition tools coat a hole's walls and floor as evenly as its rim[5]. China-based companies held only 2.6 percent of the world market for them in 2025, by CSET's count from TechInsights data[6]. The December 2024 US export rule targets tools for atomic layer deposition of the metals molybdenum and ruthenium, and for coating holes more than 200 times deeper than they are wide[7].

Applied Materials and Lam Research are the two largest deposition and etch suppliers. Applied Materials' revenue across all its businesses was $28.368 billion in fiscal 2025[8]. Lam Research's was $23.233 billion in fiscal 2026, the year to June 2026[9].

Chokepoint card Deposition and Etch

Concentration

High

Four to six firms could supply this stage today, and the top two take most of the revenue.

Substitutability

Moderate

In 2025 Lam Research held 41.1 percent of dry etch tools, Tokyo Electron 23.4 percent and Applied Materials 17.4 percent, by CSET's count from TechInsights data[10].

Price or market size

$26.2 billion

deposition tool market in 2025, by CSET's count from TechInsights data[3]. Dry etch tools were a $21.9 billion market in 2025[10]

Who leads	US	**Applied Materials**	82.4 percent of physical vapor deposition tools in 2025, by CSET's count from TechInsights data[11]
	US	**Lam Research**	51.2 percent of conductor etch tools in 2025[12]
	JP	**Tokyo Electron**	55.9 percent of dielectric etch tools in 2025[13]
	NL	**ASM International**	54.5 percent of atomic layer deposition tools in 2025[6]
	CN	**Naura**	14.7 percent of physical vapor deposition tools in 2025[11]
Where it is made	US	**United States**	Applied Materials, Lam Research
	JP	**Japan**	Tokyo Electron, Kokusai Electric
	NL	**Netherlands**	ASM International, atomic layer deposition
	KR	**South Korea**	Large equipment buyer
	CN	**China**	Naura, AMEC and Piotech, in the largest equipment market ($49.3 billion in 2025)
Why substitution is possible	Buyers have alternatives[5]. At least five firms sell atomic layer deposition tools, which lay down a film a fraction of a layer of atoms at a time[6].		

3 **Supply Chain Explorer: Advanced Chips.** Emerging Technology Observatory (ETO), accessed 9 September 2026. chipexplorer.eto.tech

5 **Export Controls on Semiconductor Manufacturing Items.** U.S. Government Publishing Office, 25 October 2023, p. 16. govinfo.gov

6 **Supply Chain Explorer: Advanced Chips.** Emerging Technology Observatory (ETO), accessed 9 September 2026. chipexplorer.eto.tech

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Where China stands

Chinese toolmakers held 12.4 percent of dry etch in 2025, according to CSET's analysis of TechInsights data [10]. They held 2.6 percent of atomic layer deposition that year, by the same count [6].

Where the US stands

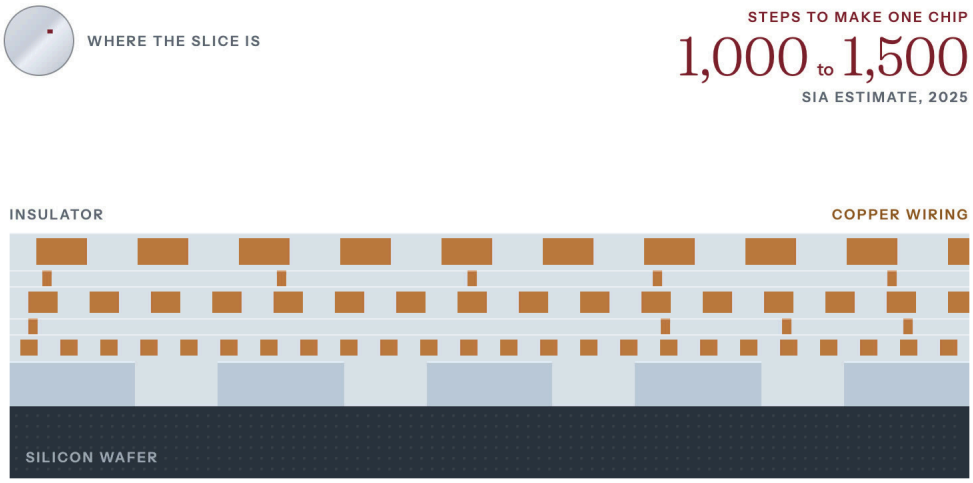
A December 2024 US rule lists some dry etch, atomic layer deposition and deep-hole deposition tools as controlled exports [14]. Its foreign direct product rule reaches some tools built outside the United States [15].

01

How it works

Figure 06.1 · Explainer, 24 seconds

One round of deposition and etch



1. **The wafer.** A slice through one spot on a wafer, magnified until its layers show.
2. **Deposit.** A thin film is laid over the whole wafer.
3. **Coat.** A light-sensitive coating, the resist, is spun on top.
4. **Expose.** Light shines through a stencil and changes the resist wherever it lands.
5. **Develop.** A wash removes the resist the light reached, leaving the stencil's pattern.
6. **Etch.** A plasma cuts straight down through the film wherever the resist is gone.
7. **Strip.** The leftover resist is removed. The pattern now sits in the film.
8. **Repeat.** Insulator fills the gaps and the next layer goes on. Making one chip takes 1,000 to 1,500 steps, the Semiconductor Industry Association estimated in 2025.

Note: Simplified cross-section. The layers are drawn much thicker than they are.

Most etch steps, and many deposition steps, happen in a vacuum chamber that holds one wafer at a time. A pump empties the chamber of air, and a measured flow of gas comes in. A radio-frequency field, an electric field that reverses millions of times a second, turns the gas into a plasma. The field splits the gas into charged ions and into neutral fragments called radicals, which are more reactive than whole molecules [16].

6 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO), accessed 9 September 2026. chipexplorer.eto.tech

10 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO), accessed 9 September 2026. chipexplorer.eto.tech

14 Foreign-Produced Direct Product Rule Additions, and Refinements to Controls for... U.S. Government Publishing Office, 5 December 2024, p. 32. govinfo.gov

15 Commerce Strengthens Export Controls to Restrict China's Capability to Produce... U.S. Bureau of Industry and Security, 2 December 2024. bis.gov

16 Etch Essentials: The Building Blocks of AI Era Microchips (Semi 101). Lam Research, 12 June 2024. newsroom.lamresearch.com

In most deposition steps, the gas carries the material the film will be made of. The molecules, or the radicals a plasma splits from them, land on the wafer and stick. The film builds up a few atoms at a time.

To coat the floor of a deep hole as thickly as its rim, the machine can work in cycles, a method called atomic layer deposition. First one gas settles in a single layer on every surface, and the pump removes the rest. Then a second gas reacts with that layer and nothing else, and the pump empties the chamber again. Each cycle adds a fraction of a layer of atoms wherever the gas reached, so the film grows as evenly on a hole's walls and floor as on its rim[5].

Not every step runs one wafer at a time. Batch furnaces such as Tokyo Electron's TELINDY PLUS heat as many as 125 wafers at once, usually without plasma[17]. They grow oxide films and run low-pressure chemical vapor deposition and some atomic layer deposition[17].

For etch, the gas attacks one material and leaves the one beside it. Wherever the photoresist covers the wafer, it shields the film beneath. Wherever it does not, the radicals react with the surface and turn it into a gas, which the pump carries away. A second field pulls the ions straight down at the wafer, so the etch cuts vertically and a narrow hole keeps its width as it deepens[16]. When the etch reaches the material underneath, which the gas was chosen not to touch, it slows almost to a stop, and the tool ends the step.

02

Variants and trade-offs

Deposition families

The market sizes and leading shares below are 2025 figures from Georgetown University's Center for Security and Emerging Technology (CSET), based on data from the research firm TechInsights. CSET puts the deposition tool market at $26.2 billion[3]. It puts the dry etch market at $21.9 billion[10].

- **Physical vapor deposition** (PVD), which knocks atoms off a solid target and lands them on the wafer, was a $4.9 billion market in 2025[11]. Applied Materials held 82.4 percent of it and China's Naura 14.7 percent[11].
- **Chemical vapor deposition** (CVD), which reacts gases at the wafer surface to grow a film, was an $11.2 billion market in 2025, led by Applied Materials at 47 percent[18].

3 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO), accessed 9 September 2026. chipexplorer.eto.tech

5 Export Controls on Semiconductor Manufacturing Items. U.S. Government Publishing Office, 25 October 2023, p. 16. govinfo.gov

10 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO), accessed 9 September 2026. chipexplorer.eto.tech

11 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO), accessed 9 September 2026. chipexplorer.eto.tech

16 Etch Essentials: The Building Blocks of AI Era Microchips (Semi 101). Lam Research, 12 June 2024. newsroom.lamresearch.com

17 TELINDY™ Series. Tokyo Electron, accessed 9 September 2026. tel.com

18 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO), accessed 9 September 2026. chipexplorer.eto.tech

The plasma-enhanced version (PECVD) runs cooler and made up most of it, $7.5 billion[19].

- **Atomic layer deposition** (ALD) lays down less than one atomic layer at a time. Each step stops once every surface is covered, so it coats vertical walls as evenly as horizontal floors. The US Bureau of Industry and Security (BIS), which writes the export rules, says that even coating makes it important for building 3D structures, such as 3D DRAM, 3D NAND, and the FinFET and gate-all-around transistors in logic chips[5]. ASM International leads it with 54.5 percent of a $4.1 billion market in 2025[6].
- **Epitaxy** grows layers that line up with the crystal beneath them, using high-temperature CVD tools[20]. **Electroplating** deposits the copper used for wiring across the wafer[21].

Etch families

- **Conductor etch**, which cuts gates, metals and silicon, was a $12.2 billion market in 2025, led by Lam at 51.2 percent, by CSET's count from TechInsights data[12].
- **Dielectric etch**, which cuts the insulating oxides and nitrides between the conductors, was an $8.7 billion market in 2025, led by Tokyo Electron at 55.9 percent, by CSET's count from TechInsights data[13].
- **High-aspect-ratio etch** cuts the deep, narrow channels that let 3D NAND stack layers of memory cells[22]. A channel's aspect ratio is its depth divided by its width. Lam says slight departures from a channel's intended shape can hurt yield[22]. Lam runs this etch well below 0 degrees Celsius[22]. It says its Cryo 3.0 technology can etch channels up to 10 microns deep whose width varies from top to bottom by less than 0.1 percent of their depth[22].

AI hardware needs sideways cuts, deeper holes and a second stack of metal wiring on the back of the wafer.

- **Gate-all-around** transistors need highly selective isotropic etch, which cuts sideways[23]. BIS controls isotropic dry etch tools that remove silicon-germanium at least 100 times faster than silicon, a selectivity of 100:1 or more[24].
- **HBM** (high-bandwidth memory) needs holes etched straight through the silicon, called through-silicon vias (TSVs), so chips can be stacked and wired[25]. BIS controls the tools that etch these holes at aspect ratios (depth to width) of 10:1 or more and at

5 Export Controls on Semiconductor Manufacturing Items. U.S. Government Publishing Office, 25 October 2023, p. 16. govinfo.gov
6 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO), accessed 9 September 2026. chipexplorer.eto.tech
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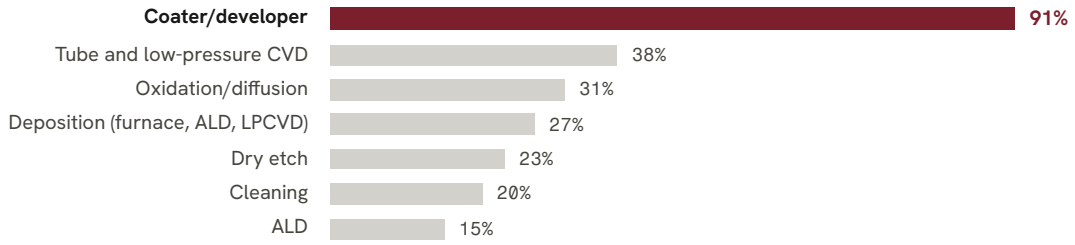
rates above 7 microns a minute[14]. The control covers only tools that also keep the holes' depth even to within 2 percent across the wafer[14].

- **Backside power delivery** moves the wires that carry power to the back of the wafer[26]. In July 2024 Applied Materials said it expected backside power to add $1 billion to the market for its wiring tools, for every 100,000 wafers a month of new fab capacity[27].

Figure 06.2

Tokyo Electron held 91 percent of coater/developers in 2025, on Gartner's figures

Tokyo Electron world market share by tool type, 2025. Some types overlap, because deposition includes furnace, ALD and low-pressure CVD tools



Source: Gartner, as reported in Tokyo Electron's FY2026 Q4 results presentation [28]

03

Who makes it

The industry association SEMI put world sales of chipmaking equipment at $135.1 billion in 2025, up 15 percent from $117.1 billion[29]. Sales of wafer processing equipment, the category that includes deposition and etch tools, rose 12 percent[29].

No one firm dominates deposition and etch the way ASML dominates lithography. In dry etch, CSET's count for 2025 from TechInsights data puts Lam Research at 41.1 percent, Tokyo Electron at 23.4 percent and Applied Materials at 17.4 percent[10]. Tokyo Electron's own results presentation, citing Gartner, gives it 23 percent of dry etch and 27 percent of deposition systems in 2025, a category that includes furnace, ALD and low-pressure CVD tools[28]. Within that category, Tokyo Electron's share of ALD was 15 percent[28]. The same presentation, again citing Gartner, gives it 91 percent of coater/developers in 2025[28]. These machines coat wafers with photoresist and develop the pattern after lithography.

ASM International, the atomic layer deposition specialist, reached a record EUR 3.2 billion of revenue in 2025 as customers built 2 nm gate-all-around capacity[30]. Molybdenum ALD, a new use of its tools that deposits the metal molybdenum, entered volume production that year[30].

10 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO), accessed 9 September 2026. chipexplorer.eto.tech

14 Foreign-Produced Direct Product Rule Additions, and Refinements to Controls for... U.S. Government Publishing Office, 5 December 2024, p. 32. govinfo.gov

26 PowerVia Test Shows Industry-Leading Performance. Intel, 5 June 2023. download.intel.com

27 Applied Materials Unveils Chip Wiring Innovations for More Energy-Efficient... GlobeNewswire (release by Applied Materials), 8 July 2024. globenewswire.com

28 FY2026 (April 2025-March 2026) Financial Announcement, transcript. Tokyo Electron, 30 April 2026, p. 28. tel.com

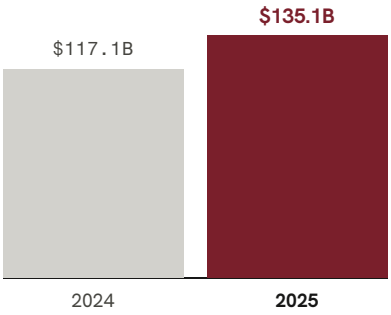
29 SEMI Reports Global Semiconductor Equipment Billings Reached $135 Billion in... SEMI, 7 April 2026. semi.org

30 ASM reports Q4 and full-year 2025 results. ASM International, 3 March 2026, p. 2. asm.com

Figure 06.3

Chipmaking equipment sales rose to $135.1 billion in 2025, from $117.1 billion

What the world spent on chipmaking equipment



Source: SEMI, Worldwide Semiconductor Equipment Market Statistics [29]

04

China and the controls

China is the tool vendors' largest market, buying $49.3 billion of equipment in 2025, half a percent less than in 2024[29]. Taiwan's spending rose 90 percent to $31.5 billion[29]. South Korea's rose 26 percent to $25.8 billion[29]. China still takes a large share of every major vendor's sales.

- **Tokyo Electron**: 34.1 percent of sales in the year to March 2026, down from 41.7 percent[31]. It was 30.4 percent in the quarter to June 2026[32].
- **Lam Research**: about 34 percent of revenue in the year to June 2026, the same as a year earlier[33].
- **Applied Materials**: 28 percent in the third fiscal quarter of 2026, down from 35 percent a year earlier[34].
- **ASM International**: more than 30 percent of 2025 revenue[30].

Between 2019 and 2024, by CSET's analysis of TechInsights data, China-based suppliers went from 2 to 7 percent of deposition and from under 3 to almost 9 percent of etch and clean[35]. In ALD they gained no share and held less than 1 percent in 2024[35].

CSET's newer analysis, of data that TechInsights published in July 2026, puts China-based suppliers at 9.8 percent of deposition in 2025[3]. They held 10.1 percent of etch and

3 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO), accessed 9 September 2026. chipexplorer.eto.tech

29 SEMI Reports Global Semiconductor Equipment Billings Reached $135 Billion in... SEMI, 7 April 2026. semi.org

30 ASM reports Q4 and full-year 2025 results. ASM International, 3 March 2026, p. 2. asm.com

31 FY2026 (April 2025-March 2026) Financial Announcement, transcript. Tokyo Electron, 30 April 2026, p. 27. tel.com

32 Q1 FY2027 (April-June 2026) Financial Announcement, transcript. Tokyo Electron, 30 July 2026, p. 6. tel.com

33 Lam Research, annual report on Form 10-K for the fiscal year ended 28 June 2026. U.S. Securities and Exchange Commission, 7 August 2026. sec.gov

34 Applied Materials Announces Third Quarter 2026 Results. GlobeNewswire (release by Applied Materials), 13 August 2026. globenewswire.com

35 Inside Beijing's Chipmaking Offensive. Center for Security and Emerging Technology, 14 July 2025. cset.georgetown.edu

clean that year[4]. In dry etch alone their share was 12.4 percent[10]. In ALD they held 2.6 percent in 2025, most of it from Piotech[6].

The Chinese toolmaker AMEC alone held 12.4 percent of dielectric etch in 2025, by CSET's count from TechInsights data[13]. Its revenue was RMB 12.385 billion that year, up about 36.62 percent[36]. It spent RMB 3.744 billion on research and development, 30.2 percent of sales[36].

The United States tightened its export controls at least five times between 2022 and 2025. The rules describe each deposition or etch tool by the process it is designed to run[24].

- **October 2022.** BIS issued a rule on 7 October that required a license to export advanced chipmaking equipment to China[37]. The rule listed that equipment under a new entry, ECCN 3B090[37]. It also required US persons to get a license to ship, transfer or service items used at advanced fabs in China[38]. It defined these as fabs that make logic chips with non-planar transistors or at 16/14 nm or below, NAND memory with 128 or more layers, or DRAM at 18 nm half-pitch or below[38]. The rule extended US license rules to more foreign-made items bound for 28 Chinese entities on the Entity List[39]. That list names the foreign firms that exporters need a license to supply.
- **October 2023.** A rule issued in October 2023 and effective 17 November rewrote the US control list's entries for chipmaking equipment[40]. It added tools for four processes[5]. The first is dry etch, both sideways (isotropic) and vertical (anisotropic), under ECCN 3B001.c.1[23]. The second is spatial ALD (3B001.d.9), whose design keeps stray particles down to the level advanced chips need[5]. The third is low-fluorine tungsten ALD and CVD (3B001.d.10), which fills deep, narrow gaps in advanced chips with tungsten while keeping resistance low[5]. The fourth is carbon hard mask PECVD (3B001.d.5), which lays down the etch-resistant carbon films that smaller features need[5].
- **December 2024.** BIS controlled 24 more types of equipment and three types of software, and added 140 entities to the Entity List[15]. The new equipment entries

4 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO), accessed 9 September 2026. chipexplorer.eto.tech

5 Export Controls on Semiconductor Manufacturing Items. U.S. Government Publishing Office, 25 October 2023, p. 16. govinfo.gov

6 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO), accessed 9 September 2026. chipexplorer.eto.tech

10 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO), accessed 9 September 2026. chipexplorer.eto.tech

13 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO), accessed 9 September 2026. chipexplorer.eto.tech

15 Commerce Strengthens Export Controls to Restrict China's Capability to Produce... U.S. Bureau of Industry and Security, 2 December 2024. bis.gov

23 Export Controls on Semiconductor Manufacturing Items. U.S. Government Publishing Office, 25 October 2023, p. 15. govinfo.gov

24 Export Controls on Semiconductor Manufacturing Items. U.S. Government Publishing Office, 25 October 2023, p. 28. govinfo.gov

36 AMEC holds its 2025 annual results briefing (中微公司成功举办2025年度业绩说明会). Advanced Micro-Fabrication Equipment (AMEC), 1 April 2026. amec-inc.com

37 Implementation of Additional Export Controls: Certain Advanced Computing and... U.S. Government Publishing Office, 13 October 2022, p. 7. govinfo.gov

38 Implementation of Additional Export Controls: Certain Advanced Computing and... U.S. Government Publishing Office, 13 October 2022, p. 14. govinfo.gov

39 Implementation of Additional Export Controls: Certain Advanced Computing and... U.S. Government Publishing Office, 13 October 2022, p. 1. govinfo.gov

40 Export Controls on Semiconductor Manufacturing Items. U.S. Government Publishing Office, 25 October 2023, p. 1. govinfo.gov

include TSV etch, and deposition into features with aspect ratios above 200:1[41]. BIS also wrote a foreign direct product rule that reaches some tools built outside the United States[15]. The rule largely exempts foreign-made tools exported by firms located in Japan, the Netherlands and the 31 other countries it lists, whose governments can control these tools themselves[42]. Japan, one of those countries, had added advanced chipmaking equipment to its own control list on 23 July 2023[43].

- **September 2025.** BIS removed the Chinese units of Samsung and SK hynix, and Intel Semiconductor (Dalian), from the Validated End-User program, effective 31 December 2025[44]. From that date, US tools bound for them need a license[44]. BIS estimated that the change would add 1,000 license applications a year[44].
- **September 2025.** The Affiliates Rule extended Entity List restrictions to any company at least 50 percent owned by a listed entity[45]. BIS then suspended it until 9 November 2026[46].

BIS enforces the rules on toolmakers too. In February 2026 BIS said Applied Materials had shipped ion implanters, tools that fire atoms into silicon, to a company in China on the Entity List in 2021 and 2022 without the license the Entity List requires[47]. The tools, worth about $126 million, went to China through Applied's Korean unit[47]. BIS set the penalty at about $252 million, twice their value and the most the law allows[47]. In fiscal 2026 Applied Materials took a $253 million charge to settle an export controls compliance matter with BIS[34].

Among deposition and etch tool segments, China-based suppliers have their largest share in dry stripping, which clears leftover photoresist off the wafer[48]. Dry stripping was a $314.6 million market in 2025, by CSET's count from TechInsights data[48]. Japan's Kokusai held 35.7 percent of it, Mattson Technology 22.6 percent and Naura 8.3 percent[48]. Mattson is based in the United States, but Beijing E-Town bought it in 2016[35]. CSET counts Mattson's share as China's[35].

15 Commerce Strengthens Export Controls to Restrict China's Capability to Produce... U.S. Bureau of Industry and Security, 2 December 2024. bis.gov

34 Applied Materials Announces Third Quarter 2026 Results. GlobeNewswire (release by Applied Materials), 13 August 2026. globenewswire.com

35 Inside Beijing's Chipmaking Offensive. Center for Security and Emerging Technology, 14 July 2025. cset.georgetown.edu

41 Foreign-Produced Direct Product Rule Additions, and Refinements to Controls for... U.S. Government Publishing Office, 5 December 2024, p. 16. govinfo.gov

42 Foreign-Produced Direct Product Rule Additions, and Refinements to Controls for... U.S. Government Publishing Office, 5 December 2024, p. 28. govinfo.gov

43 Foreign-Produced Direct Product Rule Additions, and Refinements to Controls for... U.S. Government Publishing Office, 5 December 2024, p. 21. govinfo.gov

44 Revocation of Validated End-User Authorizations in the People's Republic of... U.S. Government Publishing Office, 2 September 2025. govinfo.gov

45 Expansion of End-User Controls To Cover Affiliates of Certain Listed Entities. U.S. Government Publishing Office, 30 September 2025, p. 1. govinfo.gov

46 One Year Suspension of Expansion of End-User Controls for Affiliates of Certain... U.S. Government Publishing Office, 12 November 2025. govinfo.gov

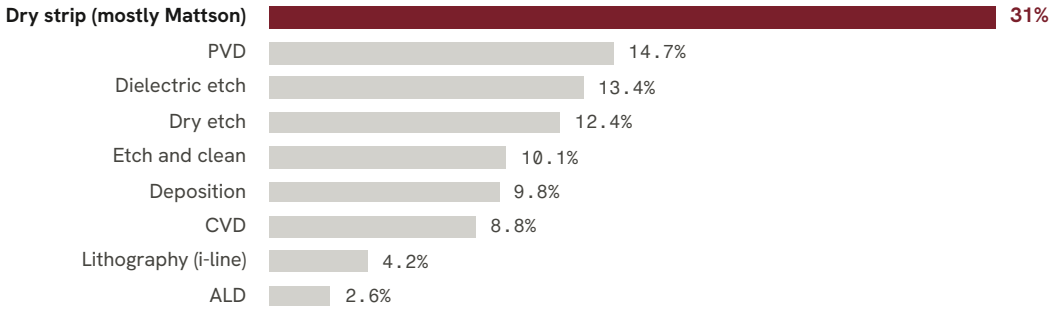
47 Applied Materials to Pay $252 Million Penalty to BIS for Illegally Exporting... U.S. Bureau of Industry and Security, 11 February 2026. bis.gov

48 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO), accessed 9 September 2026. chipexplorer.eto.tech

Figure 06.4

China-based suppliers held 31 percent of dry strip in 2025, mostly Mattson's

China-based suppliers' share of global tool segments, 2025



Source: CSET analysis of TechInsights data (published July 2026), ETO Chip Explorer [3]

05

The chokepoint

Deposition and etch tools are a chokepoint, but a weaker one than lithography, because chipmakers can buy each main tool from more than one maker. In dry etch, Lam Research held 41.1 percent in 2025, Tokyo Electron 23.4 percent and Applied Materials 17.4 percent, by CSET's count[10]. Atomic layer deposition has at least five makers[6].

Four questions to answer from memory

1. What do deposition and etch do? see How it works
2. Which four companies make most deposition and etch tools? see Who makes it
3. What share of the deposition and dry etch tool markets did Chinese toolmakers hold in 2025? see China and the controls
4. What share of atomic layer deposition tools did Chinese toolmakers hold in 2025? see China and the controls

Four things to remember

- Deposition lays down thin films and etch cuts them into patterns. Making one chip takes 1,000 to 1,500 steps, by a 2025 industry estimate.
- Four American, Japanese and Dutch firms make most of these tools, and none dominates.
- Chinese makers held 12.4 percent of dry etch and 9.8 percent of deposition in 2025, on CSET's count.
- Chinese makers held 2.6 percent of atomic layer deposition in 2025, most of it from Piotech.

3 **Supply Chain Explorer: Advanced Chips.** Emerging Technology Observatory (ETO), accessed 9 September 2026. chipexplorer.eto.tech

6 **Supply Chain Explorer: Advanced Chips.** Emerging Technology Observatory (ETO), accessed 9 September 2026. chipexplorer.eto.tech

10 **Supply Chain Explorer: Advanced Chips.** Emerging Technology Observatory (ETO), accessed 9 September 2026. chipexplorer.eto.tech

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The chain

01 Chip Design, EDA and IP
02 Silicon and Wafers
03 Chemicals, Gases and Photoresist
04 Lithography
05 Photomasks and Pellicles
06 Deposition and Etch
07 **Metrology and Inspection**
08 Foundries and Fabs
09 Memory and HBM
10 Advanced Packaging
11 Substrates and PCBs
12 Test and Assembly
13 Systems and Networking
14 Data Centers and Power

15 The Economics of AI Chips
16 Who Controls the Chain

Stage 7 of 14 · High concentration · Moderately hard to substitute

Metrology and Inspection

KLA, an American firm, held 56.8 percent of the market for tools that measure and inspect wafers in 2025, by CSET's count.

Specimen 07 · Metrology and Inspection

An electron-beam inspection column, cut in half down its axis, standing over a 300 mm wafer. The column is about a meter tall, drawn to scale. The green line is the electron beam, running from the emitter at the top through the lenses to the spot where it lands on the wafer.

In plain terms

Metrology (measurement) and inspection are a chip factory's quality control. Between the steps that build a chip, another machine looks the wafer over. It checks whether the lines are the right width, whether this layer landed squarely on the last one, and whether a speck of dirt has killed a circuit. The main chip in an AI accelerator, the one that does the calculating, is one of the largest cut from a wafer. A bigger chip is a bigger target for a speck. In a textbook example with the same density of specks, 49.2 percent of chips 1 centimeter on a side come out flawless, but only 11.3 percent of chips 2 centimeters on a side[1]. In 2025 one American firm, KLA, held 56.8 percent of the market for these tools, by CSET's count from TechInsights data[2]. So a chip factory, or fab, cut off from KLA would find fewer of the faults that make its chips fail.

In short

By CSET's count from TechInsights data, defect inspection was the largest kind of measuring and inspection tool in 2025, a $6.7 billion market[2]. KLA held 85.9 percent of it[2]. Chinese makers held 0.7 percent of the whole market that year by the same count[2]. Yet China took 26 percent of KLA's revenue in the June 2026 quarter[3].

Since October 2022 the US has required a license for any tool subject to US export rules bound for Chinese fabs that make advanced logic or memory

1 Behrooz Parhami, Dependable Computing: A Multilevel Approach, part 2. University of California, Santa Barbara, 11 October 2020, p. 8. web.ece.ucsb.edu

2 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO), accessed 9 September 2026. chipexplorer.eto.tech

3 Letter to Shareholders Q4 Fiscal 2026. KLA Corporation, 28 July 2026, p. 5. d1io3yog0oux5.cloudfront.net

chips[4]. A December 2024 rule added specific tools to the export control list, among them inspection tools that can find defects of 21 nm or smaller[5].

Process control tools, the machines that measure and inspect wafers and masks, were a $15.7 billion market in 2025, by CSET's count from TechInsights data[2]. CSET is a research center at Georgetown University. KLA puts the whole wafer equipment market, including advanced packaging, at about $120 billion that year[6].

KLA held more than half of this market in 2025, several times the share of any rival. CSET's count puts KLA at 56.8 percent and Applied Materials, the next largest, at 11.2 percent[2]. At its March 2026 investor day KLA cited Gartner's figures for 2024, which put its share at 56.5 percent, 6.5 times its nearest competitor's[7]. In April 2026 KLA said its 2025 share was about 7 times the nearest competitor's[8].

Chokepoint card Metrology and Inspection

Concentration

High

Four to six firms could supply this stage today, and the top two take most of the revenue.

Substitutability

Moderate

Every kind of measuring and inspecting tool had a second maker in 2025[2].

Price or market size

$15.7 billion

process control tool market in 2025[2]. KLA puts the 2025 wafer equipment market at about $120 billion[6]

Who leads	US	**KLA**	56.8 percent of process control tools in 2025 by CSET's count[2]
	US	**Applied Materials**	11.2 percent of process control tools in 2025[2]
	JP	**Lasertec**	6.4 percent of process control tools in 2025[2]
	JP	**Hitachi High-Tech**	6.2 percent of process control tools in 2025[2]
	NL	**ASML**	5.9 percent of process control tools in 2025[2]
	IL	**Nova**	4.3 percent of process control tools in 2025[2]
	US	**Onto Innovation**	2.6 percent of process control tools in 2025[2]
Where it is made	US	**United States**	KLA (Milpitas), Onto Innovation (Wilmington, MA), Applied Materials
	NL	**Netherlands**	ASML YieldStar and HMI e-beam, Veldhoven and San Jose
	JP	**Japan**	Hitachi High-Tech (CD-SEM), Lasertec (mask inspection), Rigaku (X-ray)
	IL	**Israel**	Nova (Rehovot) and Camtek (Migdal Haemek)
	CN	**China**	Skyverse (Shenzhen), RSIC and other domestic makers, with 0.7 percent of the market in 2025 by CSET's count
Why substitution is possible	Eight firms besides KLA each held more than 1 percent of the market in 2025[2]. But KLA alone held 85.9 percent of defect inspection, the largest kind, that year[2].		

Where China stands

Chinese firms held 0.7 percent of the world market for tools that measure and inspect wafers in 2025, by CSET's count[2]. Skyverse, a Shenzhen maker of inspection and metrology tools, reported 2,053,298,171.74 yuan of revenue in 2025, up 48.75 percent on 2024[9].

Where the US stands

In 2025 the American firm KLA was the world's largest maker of metrology and inspection tools[2]. Citing Gartner, KLA puts its 2024 share at 56.5 percent, 6.5 times its nearest rival's share[7].

2 **Supply Chain Explorer: Advanced Chips.** Emerging Technology Observatory (ETO), accessed 9 September 2026. chipexplorer.eto.tech

4 **Implementation of Additional Export Controls: Certain Advanced Computing and...** U.S. Government Publishing Office, 13 October 2022, p. 5. govinfo.gov

5 **Foreign-Produced Direct Product Rule Additions, and Refinements to Controls for...** U.S. Government Publishing Office, 5 December 2024, p. 18. govinfo.gov

6 **Letter to Shareholders Q4 Fiscal 2026.** KLA Corporation, 28 July 2026, p. 7. d1io3yog0oux5.cloudfront.net

7 **Compounding Sustainable Outperformance.** KLA Corporation, 12 March 2026, p. 35. d1io3yog0oux5.cloudfront.net

8 **Letter to Shareholders Q3 Fiscal 2026.** KLA Corporation, 29 April 2026, p. 3. d1io3yog0oux5.cloudfront.net

9 **深圳中科飞测科技股份有限公司2025年年度报告摘要.** CNINFO, 24 April 2026, p. 12. static.cninfo.com.cn

01

How it works

An inspection tool sweeps a beam of light across the wafer, photographs it, and compares the image of each chip with the image of its neighbors. The chips on a wafer are meant to be identical, so the tool flags anything that shows in one and not the others as a possible defect. It records where each one is. For faults too small for light to show, a second tool scans a beam of electrons across a small patch and builds a picture from the electrons the beam knocks loose from the surface.

A metrology tool measures known shapes. It shines light on a test pattern printed beside the circuits and works out from the reflected light how wide the lines are and how thick the film is. It also reads marks printed in this layer and the last to measure how far the two are out of line, which chipmakers call overlay.

The fab pays for all this to protect yield, the share of chips on a wafer that work. A wafer loses value two ways: whole regions ruined by mishandling, misalignment or over-etching, and single circuits killed by stray material or an airborne particle[10].

A bigger chip is a bigger target for the second kind, so the cost of each good chip grows faster than its area[1]. Chipmakers call each chip on a wafer a die. In the standard formula, die yield is (1 + D0A/a) to the power minus a. Here D0 is defects per square centimeter, A is die area, and a is a clustering constant of 3 to 4 that says how much defects bunch together. At a defect density of 0.8 per square centimeter, with a set to 3, a 1 square centimeter die yields 49.2 percent and a 2 by 2 centimeter die 11.3 percent[1]. Because the bigger die yields less and fits fewer times on a wafer, it costs 20.1 times as much as the small one[1].

One percentage point of yield can be worth millions of dollars a year. In another textbook example, a fab makes 5,000 wafers a week, each holding 200 dies, and earns $15 from each good die[11]. Each percentage point of yield is then worth $7.8 million a year[11]. Engineers raise yield by finding the process step that is losing dies and fixing it, work called yield learning.

Chipmakers can sell more dies than the formula suggests, because its percentages count only dies with no killer defect. A chip built from many identical blocks can carry spares and switch off a faulty block[12]. Nvidia's full H100 design has 144 of its main computing blocks, called streaming multiprocessors, and the SXM5 version it ships has 132 turned on[13]. But the formula shows how fast flawless dies get scarce as chips grow.

1 Behrooz Parhami, Dependable Computing: A Multilevel Approach, part 2. University of California, Santa Barbara, 11 October 2020, p. 8. web.ece.ucsb.edu

10 Behrooz Parhami, Dependable Computing: A Multilevel Approach, part 2. University of California, Santa Barbara, 11 October 2020, p. 10. web.ece.ucsb.edu

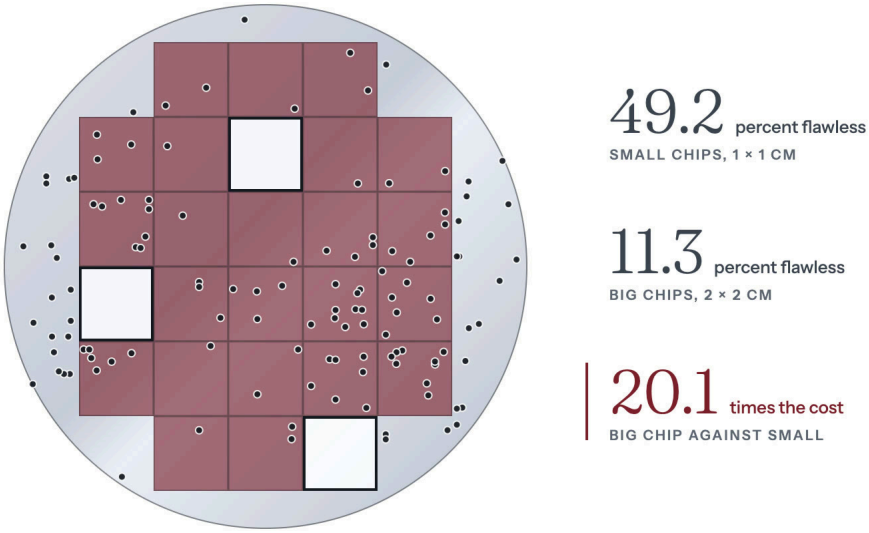
11 Behrooz Parhami, Dependable Computing: A Multilevel Approach, part 2. University of California, Santa Barbara, 11 October 2020, p. 7. web.ece.ucsb.edu

12 Behrooz Parhami, Dependable Computing: A Multilevel Approach, part 2. University of California, Santa Barbara, 11 October 2020, p. 23. web.ece.ucsb.edu

13 NVIDIA Hopper Architecture In-Depth. Nvidia, 22 March 2022. developer.nvidia.com

Figure 07.1 · Explainer, 16 seconds

Why big chips lose more to defects



1. **Specks land.** Specks of dust and tiny flaws land at random on every wafer. A textbook example assumes 0.8 per square centimeter.
2. **Small chips.** Of chips 1 centimeter on a side, 49.2 percent miss every speck and come out flawless.
3. **Big chips.** Chips 2 centimeters on a side are bigger targets for the same specks. Only 11.3 percent come out flawless.
4. **The cost.** The big chip yields less and fits fewer times on a wafer, so it costs 20.1 times as much as the small one.

Note: Simulated wafer laid out like the textbook example: 120 chips 1 centimeter on a side, or 26 chips 2 centimeters on a side. Specks land at random in loose clusters, 0.8 per square centimeter. The share of flawless chips follows the formula above, with a clustering constant of 3, as closely as whole chips allow.

02

Variants and trade-offs

All the market sizes and shares below are CSET's, for 2025, from TechInsights data[2].

- **Defect inspection** scans a whole wafer with light, or small areas of it with electrons, and compares each die against its neighbors. It is by far the largest family, at $6.7 billion, and KLA held 85.9 percent of it. ASML's HMI eScan 1100 runs 25 electron beams at once, up to 15 times the throughput of single-beam inspection tools, and finds patterning defects down to 7 nm[14].
- **Film and shape metrology** measures how thick each film is and how flat the wafer is, without cutting it open. In this $2.6 billion family KLA held 45.6 percent, Nova 24.6 and Onto Innovation 10.9. Onto paid about $720 million in August 2026 for 27 percent of the Japanese X-ray tool maker Rigaku[15].
- **Mask inspection and repair** has its own tools, because a defect on a mask, the stencil the circuit pattern is printed from, repeats on every die. This $2.2 billion family and

2 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO), accessed 9 September 2026. chipexplorer.eto.tech

14 The HMI eScan 1100. ASML, accessed 9 September 2026. asml.com

15 Onto Innovation Completes Strategic Investment in Rigaku Holdings Corporation. Onto Innovation, 10 August 2026. investors.ontoinnovation.com

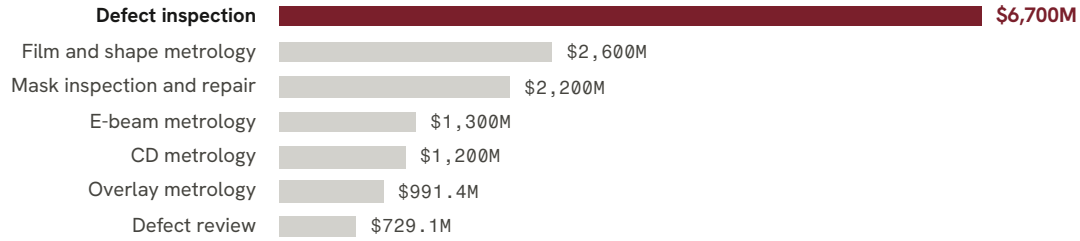
overlay metrology are the two where KLA has a close rival. KLA held 45.9 percent, Lasertec 42 and Zeiss 10.1. Lasertec also makes actinic inspection tools, which check EUV masks with the same 13.5 nm light that prints them[16]. Lasertec calls its ACTIS A150 the world's first EUV patterned mask inspection system[16]. KLA's own actinic mask inspection tool was still at an early demo stage at its March 2026 investor day[17].

- **Overlay metrology** measures how well this layer lines up with the one below. Its $991.4 million market was split almost evenly between KLA at 51.5 percent and ASML at 47.9.
- **Critical dimension metrology**, also called line-width measurement, measures how wide a printed feature came out. It is the one family Japan leads. Hitachi held 71.9 percent of this $1.2 billion market and Applied Materials 27.5.
- **Electron-beam metrology** resolves detail too fine for light. Applied Materials led this $1.3 billion family at 45.1 percent, with ASML at 35.6 and KLA at 18.8.

Figure 07.2

Defect inspection was the largest tool family in 2025, at $6,700 million

Process control market by tool family, 2025



Source: ETO Supply Chain Explorer, CSET [2]

03

Who makes it

KLA says its share grew 3.6 percentage points from 2021 to 2025[8]. It says it gained share again in 2025 across mask, optical wafer and e-beam inspection[8]. Its revenue for fiscal 2026, the year to June 2026, was $13.579 billion, up from $12.156 billion in fiscal 2025[18].

2 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO), accessed 9 September 2026. chipexplorer.eto.tech

8 Letter to Shareholders Q3 Fiscal 2026. KLA Corporation, 29 April 2026, p. 3. d1io3yog0oux5.cloudfront.net

16 ACTIS A150. Lasertec, 12 September 2019. lasertec.co.jp

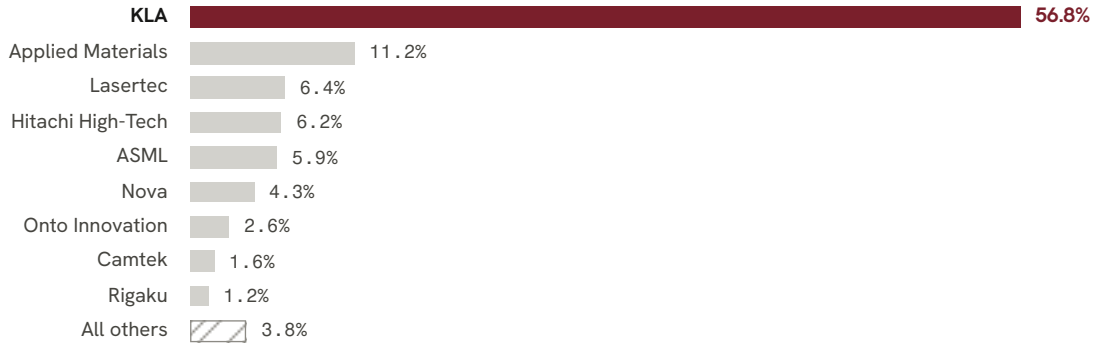
17 Compounding Sustainable Outperformance. KLA Corporation, 12 March 2026, p. 91. d1io3yog0oux5.cloudfront.net

18 KLA, revenue reported to the SEC (XBRL company data). U.S. Securities and Exchange Commission, accessed 9 September 2026. data.sec.gov

Figure 07.3

KLA held 56.8 percent of process control tool revenue in 2025

Process control tool revenue share, 2025



Source: ETO Supply Chain Explorer, CSET [2]

KLA's rivals are far smaller, and none holds more than 11.2 percent, by CSET's count from TechInsights data for 2025[2]. Eight of them each held more than 1 percent of the market that year[2].

- **Applied Materials** is the largest of them. It sells e-beam defect review tools, CD-SEMs (electron microscopes that measure line width) and e-beam metrology tools next to the deposition and etch tools whose work they check.
- **Lasertec** holds 6.4 percent, almost all of it mask inspection[2]. **Hitachi High-Tech** holds 6.2 percent, most of it from CD-SEMs[2]. **ASML** holds 5.9 percent through YieldStar overlay and HMI e-beam metrology[2].
- **Nova** holds 4.3 percent, through film metrology[2]. It took $880.6 million of revenue in 2025[19].
- **Onto Innovation** holds 2.6 percent[2]. It took $1,005.3 million of revenue in 2025[20].
- **Camtek** holds 1.6 percent[2]. It took $496.1 million of revenue in 2025[21].

KLA's patterning line, which includes several metrology products and mask inspection, grew 61 percent year on year in the June 2026 quarter[3].

KLA says process control's share of spending on chipmaking equipment is still rising[22]. For memory chips, KLA credits the rise to higher performance specifications, fewer spare circuits and more logic circuitry in devices built for high-performance computing[23].

KLA says its systems use AI to analyze data and suppress noise, and that this helps chipmakers reach high yield sooner[23]. In April 2026 KLA said it had added AI to every product line over the previous five years[23].

2 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO), accessed 9 September 2026. chipexplorer.eto.tech

3 Letter to Shareholders Q4 Fiscal 2026. KLA Corporation, 28 July 2026, p. 5. d1io3yog0oux5.cloudfront.net

19 Nova, revenue reported to the SEC (XBRL company data). U.S. Securities and Exchange Commission, accessed 9 September 2026. data.sec.gov

20 Onto Innovation Reports 2025 Fourth Quarter and Full Year Results. Onto Innovation, 19 February 2026. investors.ontoinnovation.com

21 Camtek, report on Form 6-K dated 18 February 2026. U.S. Securities and Exchange Commission, 18 February 2026. sec.gov

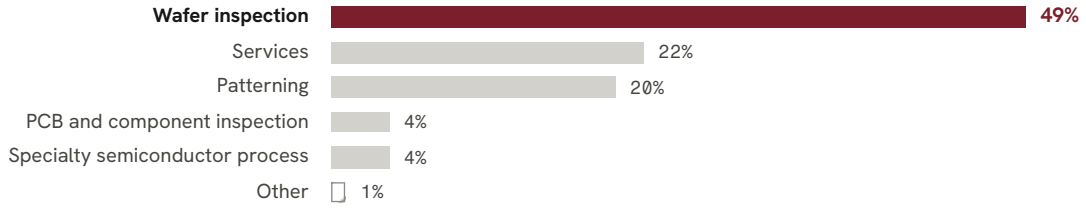
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23 Letter to Shareholders Q3 Fiscal 2026. KLA Corporation, 29 April 2026, p. 4. d1io3yog0oux5.cloudfront.net

Figure 07.4

Wafer inspection was 49 percent of KLA's June 2026 quarter revenue

KLA revenue by product line, June 2026 quarter



Source: KLA letter to shareholders, Q4 FY2026 [3]

04

Packaging, AI and China

KLA says the AI build-out is making advanced packaging, and the process control it needs, more important[24]. KLA says it became the largest supplier of process control tools for advanced wafer-level packaging in 2025, adding 14 percentage points of share[8]. It expects about $1.1 billion of packaging process control revenue in 2026, up more than 70 percent[24]. In February 2026 Onto said it had signed a volume purchase agreement estimated at over $240 million with a high-bandwidth memory (HBM) maker[20]. The deal covers 2D inspection and 3D bump metrology, which measures the tiny metal bumps that join stacked chips, through 2027[20].

Chinese makers are still small in process control, but one of them is growing fast. CSET's count, built on TechInsights data, puts all Chinese firms at 0.7 percent of the market in 2025[2]. Skyverse (Shenzhen Zhongke Feice), a maker of wafer inspection and metrology tools, reported 2,053,298,171.74 yuan of revenue in 2025[9]. That was up 48.75 percent on 2024[9]. In its 2025 annual report Skyverse says Chinese makers still hold a low share of China's market for these tools and that KLA holds the largest share of it[25]. In the June 2026 quarter China took 26 percent of KLA's revenue and Taiwan 31 percent[3].

Chinese firms held 2 percent or less of each of five kinds of tool in 2025. By CSET's 2025 count from TechInsights data, they held 0.6 percent of defect inspection tools[26]. They held 0.4 percent of line-width measurement tools in 2025[27]. In film and shape metrology they held 2 percent in 2025, all of it from Raintree Scientific Instruments, or

2 **Supply Chain Explorer: Advanced Chips.** Emerging Technology Observatory (ETO), accessed 9 September 2026. chipexplorer.eto.tech

3 **Letter to Shareholders Q4 Fiscal 2026.** KLA Corporation, 28 July 2026, p. 5. d1io3yog0oux5.cloudfront.net

8 **Letter to Shareholders Q3 Fiscal 2026.** KLA Corporation, 29 April 2026, p. 3. d1io3yog0oux5.cloudfront.net

9 **深圳中科飞测科技股份有限公司2025年年度报告摘要.** CNINFO, 24 April 2026, p. 12. static.cninfo.com.cn

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25 **深圳中科飞测科技股份有限公司2025年年度报告摘要.** CNINFO, 24 April 2026, p. 11. static.cninfo.com.cn

26 **Supply Chain Explorer: Advanced Chips.** Emerging Technology Observatory (ETO), accessed 9 September 2026. chipexplorer.eto.tech

27 **Supply Chain Explorer: Advanced Chips.** Emerging Technology Observatory (ETO), accessed 9 September 2026. chipexplorer.eto.tech

RSIC[28]. They held no share of mask inspection and repair tools in 2025[29]. Nor did they hold any share that year of the tools that inspect wafers during advanced packaging[30].

In October 2022 the US began requiring a license for any item subject to US export rules, process control tools included, bound for certain Chinese fabs[4]. The rule covered fabs that made logic chips at 16/14 nm or below, NAND memory with 128 or more layers, or DRAM at 18 nm half-pitch or below[31].

A December 2024 rule added specific process control tools to the export control list. It controls tools for defect inspection and measurement of patterned 300 mm wafers if they can find defects of 21 nm or smaller[32]. The control covers only tools that do this using light with a wavelength below 400 nm, an electron beam with a resolution of 1.65 nm or better, a cold field emission source, or two or more electron-beam sources[32]. The same rule controls overlay metrology accurate to 0.5 nm or better in two cases[32]. One is a tool built into the track, the machine that coats and develops the light-sensitive resist[33]. The other is a tool that can change its measurement wavelength and take a measurement in less than 25 ms[33]. It also controls stand-alone wafer shape metrology that feeds corrections to immersion or EUV scanners[32]. A separate new entry, 3B994, covers optical film and line-width metrology for 300 mm wafers with software for measuring non-planar transistors[33].

BIS, the US Commerce Department bureau that wrote the rule, placed all of these tools in two new entries, 3B993 and 3B994, for items it says also have legitimate uses in older chipmaking[34]. For that reason BIS decided that, unlike tools built only for advanced chips, they do not warrant a license requirement covering a whole country or a presumption that BIS will refuse a license[34]. A new foreign direct product rule extends US jurisdiction to specified foreign-made chipmaking tools bound for Macau or for a Country Group D:5 destination such as China[35].

05

The chokepoint

A fab that can buy scanners and etchers but not sensitive inspection tools finds fewer of the defects that lower its yield, so its yield rises more slowly. Most of those tools come from one firm, KLA, which held more than half the process control market in 2025[2]. KLA led four kinds of process control tool that year, by CSET's count[2]. It held

2 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO), accessed 9 September 2026. chipexplorer.eto.tech

4 Implementation of Additional Export Controls: Certain Advanced Computing and... U.S. Government Publishing Office, 13 October 2022, p. 5. govinfo.gov

28 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO), accessed 9 September 2026. chipexplorer.eto.tech

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32 Foreign-Produced Direct Product Rule Additions, and Refinements to Controls for... U.S. Government Publishing Office, 5 December 2024, p. 37. govinfo.gov

33 Foreign-Produced Direct Product Rule Additions, and Refinements to Controls for... U.S. Government Publishing Office, 5 December 2024, p. 38. govinfo.gov

34 Foreign-Produced Direct Product Rule Additions, and Refinements to Controls for... U.S. Government Publishing Office, 5 December 2024, p. 17. govinfo.gov

35 Commerce Strengthens Export Controls to Restrict China's Capability to Produce... U.S. Bureau of Industry and Security, 2 December 2024. bis.gov

85.9 percent of defect inspection, the largest kind[2]. Still, every kind of process control tool had a second maker that year, by CSET's count from TechInsights data[2]. KLA says its share of the whole market grew from 2021 to 2025[8].

Four questions to answer from memory

1. What do metrology and inspection tools do? see How it works
2. Why do AI chips depend so much on finding defects? see How it works
3. Which company leads the market for tools that measure and inspect wafers? see Who makes it
4. What share of the market for wafer measuring and inspection tools did Chinese makers hold in 2025? see Packaging, AI and China

Four things to remember

- Metrology and inspection tools check each wafer between steps and help a fab find out why its chips fail.
- AI chips are among the largest made, so finding defects matters most for them.
- CSET's count puts KLA of the United States at 56.8 percent of this tool market in 2025, and Applied Materials, the next largest, at 11.2 percent.
- Chinese makers held 0.7 percent of this tool market in 2025 on CSET's count, though one Chinese maker, Skyverse, grew its revenue 48.75 percent that year.

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The chain

01 Chip Design, EDA and IP
02 Silicon and Wafers
03 Chemicals, Gases and Photoresist
04 Lithography
05 Photomasks and Pellicles
06 Deposition and Etch
07 Metrology and Inspection
08 Foundries and Fabs
09 Memory and HBM
10 Advanced Packaging
11 Substrates and PCBs
12 Test and Assembly
13 Systems and Networking
14 Data Centers and Power

15 The Economics of AI Chips
16 Who Controls the Chain

Stage 8 of 14 · Extreme concentration · Very hard to substitute

Foundries and Fabs

The newest chips need the know-how to build the newest transistors and a fab that costs tens of billions of dollars. Only TSMC, Intel and Samsung have both.

Specimen 08 · Foundries and Fabs

A gate-all-around transistor, the switch in a 2 nm-class chip, drawn at true proportions. The three silicon sheets are each 6 nm thick and 30 nm wide. Each sheet is a few dozen atoms from top to bottom, roughly ten thousand times thinner than a human hair. The gold cage is the gate metal wrapping all four sides of each sheet.

In plain terms

A chip factory, or fab, builds the transistors, the billions of tiny switches in every chip, and then wires them together. Each switch works like a valve. Current runs along a narrow strip of silicon called the channel, and a small voltage on a gate above the strip opens or shuts the flow. For decades chipmakers made the strips shorter, so the transistors switched faster and more of them fit on a chip. But once the strip gets short enough, current leaks through even when the gate is off, and the chip burns power doing nothing. The newest kind of transistor, called gate-all-around, cuts this leak by wrapping the gate around all four sides of the strip. A chipmaker needs two things to make the newest chips, and having one does not bring the other. The first is the know-how to build the newest switches, a long sequence of steps that have to run in exact order. In 2026 only TSMC, Intel and Samsung had it. The second is factories that cost tens of billions of dollars each, and in 2022 most of the world's capacity for the most advanced logic chips was in Taiwan. China's SMIC runs factories, but without EUV machines or the know-how for the newest switches, its most advanced chips are 7 nm-class. IBM developed 2 nm technology, but it yields no chips until Rapidus finishes a factory to run it, with mass production planned for 2027. No chip company sells enough of one product to fill such a factory on its own, so chip companies share a handful of them, run by foundries that make chips for other companies. TSMC runs the biggest, and all six of its largest plants are in Taiwan.

In short

Since September 2024 the United States has required a license to share the know-how for building gate-all-around transistors, except with countries

that impose matching controls[1]. China has announced $43 billion to develop its own extreme ultraviolet (EUV) scanners, but in September 2025 the Center for Strategic and International Studies judged that EUV remains firmly out of reach[2].

SEMI projected in 2026 that China would keep the world's largest installed fab capacity in 2026 and 2027[3]. In July 2026 TSMC raised its planned investment in Arizona to $265 billion across 12 facilities[4]. SIA and BCG projected in May 2024 that the US share of the world's logic capacity below 10 nm would rise from almost none in 2022 to 28 percent by 2032[5].

Nvidia's Blackwell GPU holds 208 billion switches on two dies, each die the largest a lithography machine can print in one shot[6]. Every switch has to turn on fully, let almost no current through when off, and keep doing it for years. Blackwell is made by TSMC, on a custom process called 4NP[6]. TSMC calls a plant a GIGAFAB once it can run more than 100,000 wafers a month, each a 300 mm disc, twelve inches across[7]. It operates six of them, and every one is in Taiwan[8].

Chokepoint card Foundries and Fabs

Concentration

Extreme

Three or fewer firms could supply this stage today, or one firm holds more than half of it.

Substitutability

Very hard

Intel Foundry earned only $293 million of its $5.765 billion revenue in the second quarter of 2026 from outside customers[9].

Price or market size

$35 billion

to $43 billion over ten years to own a new 2 nm-class fab completed in 2026, BCG's 2023 estimate[10]. A 300 mm wafer on TSMC's 4NP process cost $17,000, Epoch AI estimated in December 2025[11].

Who leads			
	TW	**TSMC**	N2, its first nanosheet node, in volume production since 4Q25[12]
	US	**Intel**	18A in high-volume production in the United States, as of 2026[13]
	KR	**Samsung Foundry**	3 nm gate-all-around process in initial production since June 2022[14]
	CN	**SMIC**	$9.327 billion revenue in 2025, with its lines 93.5 percent full[15]
	JP	**Rapidus**	267.6 billion yen raised in February 2026 to reach 2 nm mass production by 2027[16]

1 Commerce Control List Additions and Revisions; Implementation of Controls on... Federal Register, 6 September 2024. federalregister.gov

2 Breakthroughs or Boasts? Assessing Recent Chinese Lithography Advancements. Center for Strategic and International Studies, 24 September 2025. csis.org

3 World Fab Forecast. SEMI, accessed 9 September 2026. semi.org

4 TSMC Announces Additional $100 Billion Investment In Arizona. City of Phoenix, 16 July 2026. phoenix.gov

5 Emerging Resilience in the Semiconductor Supply Chain. Boston Consulting Group and Semiconductor Industry..., May 2024, p. 14. semiconductors.org

6 NVIDIA Blackwell Architecture. Nvidia, 18 December 2025. nvidia.com

7 GIGAFAB® Facilities. TSMC, accessed 9 September 2026. tsmc.com

8 Fab Capacity. TSMC, accessed 9 September 2026. tsmc.com

9 Intel, quarterly report on Form 10-Q for the quarter ended 27 June 2026. U.S. Securities and Exchange Commission, 24 July 2026. sec.gov

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11 NVIDIA's B200 costs around $6,400 to produce. Epoch AI, 10 December 2025. epoch.ai

12 2nm Technology. TSMC, accessed 9 September 2026. tsmc.com

13 Intel 18A: See Our Biggest Process Innovation. Intel, accessed 18 October 2026. intel.com

14 Samsung Begins Chip Production Using 3nm Process Technology With GAA... Samsung Global Newsroom, 30 June 2022. news.samsung.com

15 SMIC Announces 2025 Annual Results. SMIC, 26 March 2026. smics.com

16 Rapidus Secures 267.6 Billion Yen in Funding from Japan Government and Private... Rapidus, 27 February 2026. rapidus.inc

Where it is made	TW	**Taiwan**	TSMC's six gigafabs, with N2 at Fab 20 Hsinchu and Fab 22 Kaohsiung, and UMC
	KR	**South Korea**	Samsung Foundry, Hwaseong and Pyeongtaek
	US	**United States**	Intel 18A at Fab 52 in Arizona, TSMC Arizona, Samsung Taylor and GlobalFoundries
	JP	**Japan**	Rapidus Chitose, Hokkaido, and TSMC Kumamoto
	CN	**China**	SMIC Shanghai and Beijing 300 mm lines, and Hua Hong

Why substitution is slow

Since September 2024 the United States has required a license to share the know-how for building gate-all-around transistors, except with countries that impose matching controls [1]. In March 2026 the Semiconductor Industry Association (SIA) put the cost to build and equip a leading-edge fab at as much as $20 to $25 billion [17]. SemiAnalysis judged in December 2025 that finding the highly skilled engineers a foundry needs remained a major challenge for TSMC in Arizona [18].

Where China stands

China's SMIC made its most advanced chips on N+3, a 7 nm-class process built without extreme ultraviolet scanners, SemiAnalysis found in June 2026 [19]. SMIC cannot buy those scanners, so it prints its tightest layers in several passes on older deep-ultraviolet machines [19]. China will keep the world's largest installed fab capacity in 2026 and 2027, SEMI projected in its third-quarter 2026 forecast [3].

Where the US stands

Intel is the only chipmaker that both develops and manufactures leading-edge logic chips in the United States, by its own account in August 2025 [20]. The United States had almost no logic capacity below 10 nm in 2022, and SIA and BCG projected in May 2024 that its share would reach 28 percent by 2032 [5]. In July 2026 TSMC raised its planned investment in Arizona to $265 billion across 12 facilities [4]. The US government agreed in August 2025 to buy a passive 9.9 percent stake in Intel, with no board seat [20].

1 Commerce Control List Additions and Revisions; Implementation of Controls on... Federal Register, 6 September 2024. federalregister.gov

3 World Fab Forecast. SEMI, accessed 9 September 2026. semi.org

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17 Testimony of the Semiconductor Industry Association (SIA) before the Senate... U.S. Senate Committee on Environment and Public Works, 4 March 2026, p. 3. epw.senate.gov

18 TSMC Overseas Fabs – A Success? SemiAnalysis, 1 December 2025. newsletter.semianalysis.com

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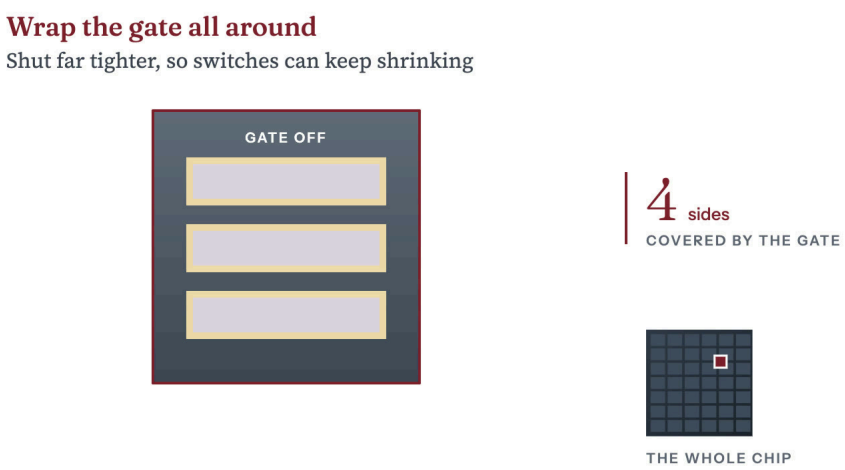
20 Intel and Trump Administration Reach Historic Agreement to Accelerate American... Intel, 22 August 2025. intc.com

01

How it works

Figure 08.1 · Explainer, 23 seconds

Why the gate wraps around the channel



1. **Inside a chip.** An AI chip does all its work with transistors, tiny switches built into the silicon. One Nvidia Blackwell GPU holds 208 billion of them. Making the switches is the front end of the factory, and wiring them together comes after.
2. **On.** Each one is a switch. Seen end on, current flows through a strip of silicon, the channel, and the gate sits on top.
3. **Off, leaking.** Switching off, the gate squeezes the flow shut, but only near itself. In a tiny transistor, current still leaks through the part it cannot reach.
4. **Fin.** Stand the channel on edge as a fin, and the gate covers three sides. Much less gets through.
5. **All around.** Split it into thin sheets and wrap the gate around every side, and it shuts far tighter. That lets the switches keep shrinking, so each new chip can pack in more of them.

Note: Simplified. The channel is seen end on, cut across the gate, with current drawn as points of light. Not to scale.

Current leaks in two main ways, straight through the gate insulator and along the channel from end to end while the gate is off. The second leak grows as the channel gets shorter. Three fixes deal with the two leaks:

- **Better insulator.** A thicker hafnium-based film, paired with a metal gate, lets fewer electrons slip straight through yet controls the channel as tightly as before. Intel replaced the silicon dioxide insulator with it at 45 nm in 2007[21]. It said the new film cut leakage through the insulator more than tenfold[21].
- **More gate.** A planar transistor lays the gate on one side of a flat channel, and as the channel shortens the gate loses control of it. A FinFET stands the channel on edge as a fin, so the gate covers three sides, and gate-all-around wraps it around all four.
- **Thinner channel.** The fab thins the silicon until no current can flow where the gate cannot reach it.

21 Intel's Transistor Technology Breakthrough Represents Biggest Change to... Intel, 27 January 2007. intel.com

02

How a transistor is built

Making the switches is called the front end of line. It builds them in a fixed order, starting with the channel. In a FinFET, the older design, the fab etches the silicon into thin fins that stand up from the wafer, and the gate goes over each one.

In a gate-all-around transistor, the fab instead grows alternating layers of silicon and silicon-germanium and cuts the stack into narrow bars[22]. It lays a placeholder, called a dummy gate, across each bar where the real gate will go[23]. On either side of the placeholder it cuts deep trenches where the source and drain, the two ends of the channel, will go[24]. Through those trenches it etches the silicon-germanium layers back a little and fills the gaps with an insulator, the inner spacer, which keeps the gate apart from the source and drain[22]. Then it grows the source and drain as crystal inside the trenches, a step called epitaxy[24].

Only then does the fab remove the placeholder and etch away the silicon-germanium, leaving thin silicon sheets that hang between the source and drain like shelves[22]. The inner spacer shields the source and drain from that etch[23]. The fab coats every sheet with an insulating film a few atoms thick, so the gate can control the channel without touching it. Gate metal then fills the gaps above, below and beside each sheet, covering all four sides[25]. Last come the contacts, small metal plugs down to the source, the drain and the gate, ready for the wiring layers above.

The steps have to run in exact order, because later layers cover earlier ones. Once gate metal surrounds a sheet, the insulating film can no longer go between the sheet and the metal. Each step is a separate deposition, etch or implant. An implant step, called ion implantation, fires atoms of boron or phosphorus into chosen patches of silicon so they conduct the way the design needs.

03

The newest designs

Gate-all-around nanosheets

A nanosheet transistor lays the channel flat again, stacks two to four thin sheets and wraps gate metal around each one. Each fin in a FinFET adds a fixed amount of current, so a designer who needs more current has to add a whole fin. Nanosheets can be drawn at different widths, so current can be set in finer steps.

TSMC's N2, its first nanosheet node, entered volume production in the fourth quarter of 2025[12]. TSMC rates the follow-on N2P at 18 percent more speed at the same power and 36 percent less power at the same speed than its earlier N3E[26]. It gives N2P 1.2 times the logic density of N3E and 1.15 times its chip density, but its technology page

12 2nm Technology. TSMC, accessed 9 September 2026. tsmc.com

22 Entering the nanosheet transistor era. Imec, 11 February 2022. imec-int.com

23 Development of SiGe Indentation Process Control for Gate-All-Around FET... ArXiv, 12 January 2022. arxiv.org

24 Applied Materials Unveils Next-Gen Chipmaking Products to Supercharge AI... Applied Materials, 7 October 2025. ir.appliedmaterials.com

25 Applied Materials Technologies Enable 2D Scaling with EUV and 3D... Applied Materials, 21 April 2022. ir.appliedmaterials.com

26 HPC Platform – Advanced Technologies. TSMC, accessed 9 September 2026. tsmc.com

does not define either measure[26]. These numbers are TSMC's own claims until someone measures the silicon. Each foundry picks its own baseline, and none publishes a density figure that can be compared with a rival's.

Intel and Samsung also make gate-all-around transistors. Intel's version, called RibbonFET, is built into its 18A process, which Intel says is in high-volume production in the United States[13]. Samsung got there first, starting initial production of its 3 nm gate-all-around process in June 2022[14]. But SemiAnalysis reported in February 2023 that, because of yield problems, high-volume chips on Samsung's 3 nm gate-all-around technology were expected only in 2024[27].

SemiAnalysis reported in 2024 that the SRAM cell, which holds one bit in the fast memory on a logic die, had barely shrunk since the 5 nm node[28]. TSMC said in 2025 that its 2 nm SRAM uses a 0.021 square-micrometer cell and gets 1.1 times the density of its previous node by tuning design and process together[29]. See Memory and HBM.

Backside power delivery

In a conventional chip, power and signal share one stack of copper wiring above the transistors, where wide power rails crowd out the signal wires. With backside power, the fab grinds the wafer thin and builds a second network for power underneath it.

Intel's version, PowerVia, is part of 18A, and Intel calls it an industry first[13]. It cut routed area, the space a block of circuits takes once its wiring is laid out, by 11 percent and dynamic voltage droop tenfold[30]. Droop is the sag in supply voltage when a block of logic switches on at once. TSMC's Super Power Rail connects the backside network directly to each transistor's source and drain[28]. Against N2P, TSMC rates A16, which uses Super Power Rail, at 8 to 10 percent more speed, 15 to 20 percent less power and up to 1.10 times the chip density[26].

TSMC plans to start A14 production in 2028, with more than 20 percent higher logic density than N2[31]. It said in April 2026 that two more processes, A13 and A12, are scheduled to enter production in 2029[32].

13 Intel 18A: See Our Biggest Process Innovation. Intel, accessed 18 October 2026. intel.com

14 Samsung Begins Chip Production Using 3nm Process Technology With GAA... Samsung Global Newsroom, 30 June 2022. news.samsung.com

26 HPC Platform – Advanced Technologies. TSMC, accessed 9 September 2026. tsmc.com

27 The Future of the Transistor. SemiAnalysis, 21 February 2023. newsletter.semianalysis.com

28 Clash of the Foundries: Gate All Around + Backside Power at 2nm. SemiAnalysis, 1 October 2024. newsletter.semianalysis.com

29 A 38.1Mb/mm2 SRAM in a 2nm-CMOS-Nanosheet Technology for High-Density and... TSMC, 2025. research.tsmc.com

30 Intel Foundry Details Process Milestones and Future Innovation at VLSI Symposium. Intel, 16 June 2026. intc.com

31 TSMC Unveils Next-Generation A14 Process at North America Technology Symposium. TSMC, 24 April 2025. pr.tsmc.com

32 TSMC Debuts A13 Technology at 2026 North America Technology Symposium. TSMC, 23 April 2026. pr.tsmc.com

Figure 08.2

TSMC plans four logic nodes beyond N2 for 2028 and 2029

TSMC's logic roadmap beyond N2, from its own announcements

- 2028
  N2U, 3-4% faster or 8-10% lower power than N2P
- 2028
  A14, the base node for A13 and A12
- 2029
  A13, a direct shrink of A14 for 6% area savings
- 2029
  A12, the A14 platform with Super Power Rail backside power

Source: TSMC, 2026 North America Technology Symposium [32]

CFET and what comes after

A basic logic circuit needs two transistors that switch on opposite signals, and in production chips the two sit side by side. Complementary field-effect transistors, or CFETs, stack the two on top of each other[33]. The research institute imec sees them as an attractive design for logic nodes beyond 1 nm[33].

04

The back end of line

Wiring the transistors together is called the back end of line, and it has become as hard to make as the transistor. Copper, the usual wiring metal, seeps into the insulator around it, so every wire needs a barrier, a liner and a cap[34]. As the most critical wiring levels drop below a 20 nm pitch, those extra layers take a large share of the wire's width[34]. Pitch is the distance from the center of one wire to the center of the next. Two metals are moving into those narrow levels.

- **Ruthenium** shows promise for use without a barrier, which would keep the wire's resistance low[34]. imec has demonstrated ruthenium lines at a 16 nm pitch[35].
- **Molybdenum** in a thin film has up to 30 percent lower resistivity than tungsten, the metal it is replacing in contacts[36]. It needs no barrier either, so chipmakers are putting it into some contacts and local wiring layers[36].

32 TSMC Debuts A13 Technology at 2026 North America Technology Symposium. TSMC, 23 April 2026. pr.tsmc.com
33 CFET (complementary FET). Imec, 16 June 2022. imec-int.com
34 Semi-damascene metallization. Imec, 31 October 2024. imec-int.com
35 16nm Ru lines using semi-damascene integration approach. Imec, 3 June 2025. imec-int.com
36 Molybdenum's Role in Ultra-Fast Computing: The Metal Behind the Speed. Entegris, 18 May 2026. blog.entegris.com

05

The foundry and the fab

A foundry makes chips designed by other companies, running all their designs on the same tools. TSMC and UMC, both based in Taiwan, make chips only for other companies. TSMC, founded in 1987, pioneered this pure-play model[37].

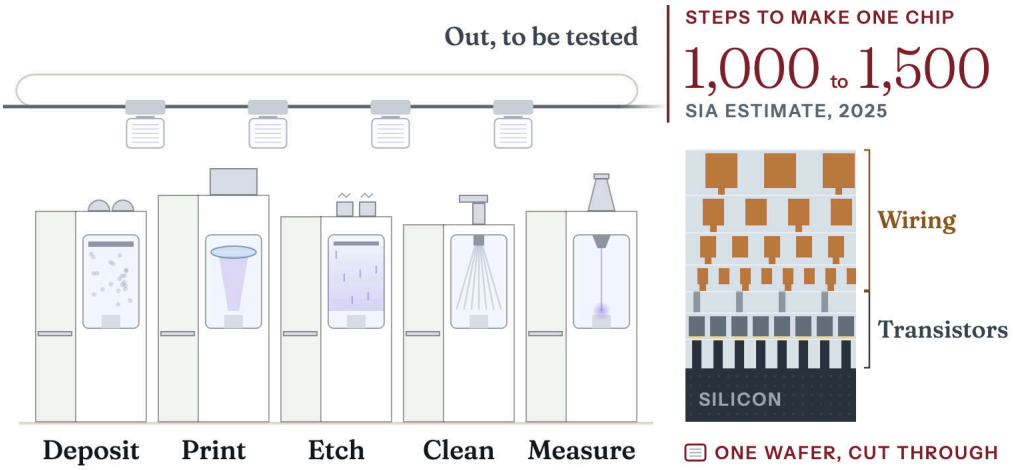
Inside a fab, a chip is built up in layers, and a similar loop of steps repeats for each one. Wafers travel in batches, called lots, inside a sealed pod called a FOUP, short for front-opening unified pod. Entegris, a US maker of these pods, offers a standard version that holds 25 wafers[38]. An overhead track carries the pod from tool to tool.

At each tool a robot lifts one wafer out and the tool does its one job, such as depositing a film, printing a pattern, etching it, cleaning it or measuring it. Then the robot puts the wafer back, and the pod moves on to the next tool, often across the building. When a layer is finished, the loop starts again with the next one.

Turning silicon into a finished, packaged chip takes 1,000 to 1,500 steps, depending on how complex the chip is, the Semiconductor Industry Association said in 2025[39]. In 2021 the association put the time to make a finished wafer at about 12 weeks on average, and up to 14 to 20 weeks for advanced processes[40]. The pod keeps the wafer clean for all of that time, because the wafer never meets the room's air.

Figure 08.3 · Explainer, 20 seconds

How a fab builds a chip, layer by layer



1. **The fab.** Sealed pods carry wafers from tool to tool on a track near the ceiling.
2. **Deposit.** The first tool lays a thin film on the wafer.
3. **Print.** The next prints a pattern on it with light.

37 TSMC, company profile page. TSMC, accessed 9 September 2026. tsmc.com

38 A300 FOUPs: Front opening unified pod platform with super microenvironment... Entegris, 2021. entegris.com

39 Testimony of David Isaacs, Semiconductor Industry Association, hearing on... U.S. Senate Committee on Finance, 14 May 2025, p. 2. finance.senate.gov

40 Chipmakers Are Ramping Up Production to Address Semiconductor Shortage. Here's... Semiconductor Industry Association, 26 February 2021. semiconductors.org

4. **Etch.** An etch tool cuts the pattern into the film.
5. **Clean, measure.** The wafer is cleaned, then measured to check the layer came out right.
6. **Next layer.** Then the loop starts again for the next layer. The transistors come first, then the wiring above them.
7. **Finished.** Making one chip takes 1,000 to 1,500 steps, the Semiconductor Industry Association estimated in 2025.

Note: Simplified. A real fab has many tools of each kind, spread across the building, and a chip has far more layers than the seven shown. The layers are drawn far thicker than they are.

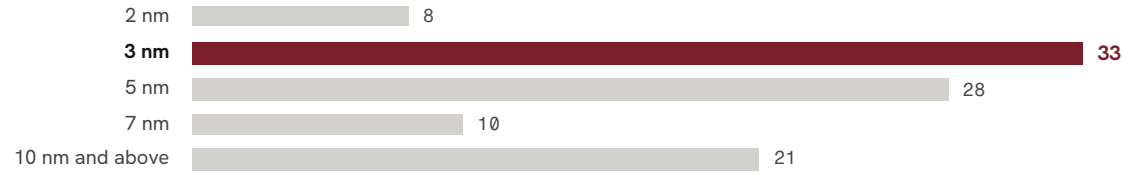
Each tool is set by a recipe, the list of gases, temperatures and times for one step. A foundry offers every customer the same list of recipes, so every design in the building can run on the same shared tools. The tools run day and night, because an idle tool earns nothing. The finished wafer is tested (see Test and assembly), then cut and packaged.

Chips at 7 nm and below, meaning the newest processes, were 79 percent of TSMC's wafer revenue in the third quarter of 2026[41]. That share split into 33 percent at 3 nm, 28 at 5 nm, 10 at 7 nm and 8 at 2 nm[41]. TSMC's revenue for the quarter was $46.72 billion, at a 66.4 percent gross margin[41].

Figure 08.4

3 nm led TSMC's wafer revenue in third-quarter 2026, at 33 percent

TSMC wafer revenue by node, 3Q26 % of wafer revenue



Source: TSMC 3Q26 results, filed with the SEC [41]

Intel and Samsung design and sell their own chips and rent out capacity too. Renting out capacity is harder for them, because customers dislike handing their designs to a competitor and each company's own products take its best capacity. Intel Foundry reported $5.8 billion of segment revenue in the second quarter of 2026 and an operating loss of $2.089 billion[42]. Only $293 million of that revenue came from outside customers[9].

06

Capacity and cost

A new leading-edge fab costs tens of billions of dollars. In 2023 the Boston Consulting Group (BCG) estimated that a new fab for chips at 2 nm and below, completed in 2026, would cost $35 to $43 billion to build, equip and run over ten years[10]. Building and

9 Intel, quarterly report on Form 10-Q for the quarter ended 27 June 2026. U.S. Securities and Exchange Commission, 24 July 2026. sec.gov

10 Navigating the Costly Economics of Chip Making. Boston Consulting Group, 28 September 2023. bcg.com

41 TSMC Reports Third Quarter EPS of NT$29.43. U.S. Securities and Exchange Commission, 15 October 2026. sec.gov

42 Intel Reports Second-Quarter 2026 Financial Results. Intel, 23 July 2026. intc.com

equipping alone takes as much as $20 to $25 billion of capital for a leading-edge fab, the Semiconductor Industry Association (SIA) told the US Senate in March 2026[17].

Most of TSMC's capacity is in its six GIGAFABs. Together, the plants TSMC and its subsidiaries run had capacity for more than 17 million wafers a year in 2025, counted as if each were the standard 12-inch size[8]. The six largest plants had a combined capacity of more than 13 million[7]. The other plants are five smaller fabs in Taiwan, two in China, two in the United States and one in Japan[8]. In July 2026 TSMC announced another $100 billion for four more fabs at 2 nm or below, taking its planned investment in Arizona to $265 billion across 12 facilities[4].

In 2025 the biggest AI chip designers used most of the world's supply of two inputs but only a small share of its advanced chipmaking. Epoch AI estimated that the four largest used around 90 percent of the world's CoWoS packaging capacity and HBM memory supply[43]. CoWoS is TSMC's packaging for AI chips, and HBM is the stacked memory that sits beside them. The same four used only 12 percent of advanced logic production[43]. See Advanced packaging.

AI accelerators have usually trailed the leading edge. A die as large as theirs is more likely to hold a defect[44]. So the share of dies that work, called the yield, gets high enough only once a process has run long enough to cut its defect rate.

Nvidia's Blackwell is made on an older TSMC process than AMD's MI455X. Epoch AI estimated in December 2025 that a 300 mm wafer on Blackwell's 4NP process costs $17,000[11]. AMD's MI455X, launched on 23 July 2026, is made on TSMC's 2 nm and 3 nm processes[45]. It spreads its compute across eight separate dies[46].

07

Who makes it

TSMC, Intel and Samsung build gate-all-around nodes with extreme ultraviolet (EUV) lithography. China's SMIC builds 7 nm-class nodes without it[19]. SemiAnalysis reported in September 2026 that Intel's Panther Lake, Samsung's Exynos 2600 and Apple's A20 Pro, which is made on TSMC's N2, all use gate-all-around transistors[47].

In August 2025 Intel called itself the only semiconductor company that both develops and manufactures leading-edge logic chips in the United States[20]. That month the US government agreed to buy 433.3 million newly issued Intel shares at $20.47 each, or $8.9 billion in all, for a 9.9 percent stake[20]. The stake is passive, with no board seat[20]. The deal also gave the government a five-year warrant to buy a further 5 percent of

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46 AMD Instinct MI455X GPU. AMD, 20 July 2026. amd.com

47 Intel Panther Lake Teardown. SemiAnalysis, 26 September 2026. newsletter.semianalysis.com

Intel at $20 a share[20]. It can use the warrant only if Intel's ownership of its foundry business falls below 51 percent[20].

In 2026 Intel began high-volume manufacturing of some of its Panther Lake processors using ASML's High-NA extreme ultraviolet scanners[42]. But it is building fewer fabs than it planned. In July 2025 Intel said it would no longer go ahead with planned projects in Germany and Poland and would slow construction in Ohio[48].

In December 2024 the US Commerce Department awarded Samsung up to $4.745 billion of CHIPS Act funding, to support more than $37 billion of planned investment in Central Texas[49]. That plan covers two new leading-edge logic fabs and a research fab in Taylor, and an expansion of Samsung's existing plant in Austin[49].

China's SMIC has the factories without the know-how for the newest transistors. Its revenue rose 16.2 percent to $9.327 billion in 2025, at a 21 percent gross margin[15]. Its lines ran 93.5 percent full, on monthly capacity above a million wafers counted in the smaller 8-inch size[15].

A node's name, such as 7 nm, does not measure any feature on the chip. SMIC's N+3, its third-generation 7 nm process, sets its narrowest wires 32.5 nm apart, tighter than the 36 nm in Intel's Panther Lake chips on 18A[19]. But the 18A process itself allows wires 32 nm apart[19]. SemiAnalysis called the comparison an incomplete, cherry-picked metric[19]. N+3 packs 113.4 million transistors per square millimeter, against 107.7 for TSMC's N6[19].

Two research groups disagree about where Huawei's Ascend AI chips are made. Georgetown University's Center for Security and Emerging Technology (CSET) reported in June 2024 that the Ascend 910B was believed to be made on SMIC's N+2 process[50]. But the research firm SemiAnalysis judged in April 2025 that most Ascend 910B and 910C chips were made on TSMC's 7 nm process[51]. It said Huawei got around the sanctions that barred it from TSMC by buying about $500 million of 7 nm wafers through another company, Sophgo[51].

IBM shows the opposite split. Its 2 nm technology yields no chips until Rapidus, a state-backed newcomer in Japan, finishes a factory to run it. The two firms agreed in December 2022 to develop that technology further for production at Rapidus's plant in Japan[52].

In July 2025 Rapidus said it had started making 2 nm prototypes at the plant, in Chitose, Hokkaido, and would begin mass production in 2027[53]. In February 2026 it raised 267.6 billion yen from the Japanese government and 32 companies to reach 2 nm

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19 Is SMIC N+3's Metal Pitch Smaller than Intel 18A's? SemiAnalysis, 14 June 2026. newsletter.semianalysis.com

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53 Rapidus Achieves Significant Milestone at its State-of-the-Art Foundry with... Rapidus, 18 July 2025. rapidus.inc

mass production by 2027[16]. By June 2026 Rapidus had raised 424.95 billion yen in equity[54].

08

The chokepoint

A firm that wants to make the newest chips has to develop its own process, because TSMC, Intel and Samsung do not sell their leading-edge process flows. A process flow sets the steps, their order and the tools for each. In September 2024 the United States began requiring a license to share the know-how for building gate-all-around transistors, except with countries that impose matching controls[1].

China has money for chipmaking but cannot buy EUV scanners. It launched its third National Integrated Circuit Fund in May 2024 with 344 billion yuan, about $47 billion[55]. China's 7 nm chips have been made on deep-ultraviolet machines, a generation behind EUV[2]. China had stockpiled those machines from ASML, the Dutch scanner maker, before export restrictions took effect[2]. SMIC gets fine features out of them by building its tightest layers in several patterning steps[19].

In July 2026 Reuters reported that a Shanghai state firm had begun making home-grown immersion deep-ultraviolet scanners[56]. Reuters's own source said the firm's machine still needed testing and was far from matching ASML's models[56].

Domestic EUV is further off. China has announced $43 billion to develop it[2]. In September 2025 the Center for Strategic and International Studies (CSIS) judged that EUV remains firmly out of reach despite massive state investment[2]. See Lithography.

The factories are concentrated too. In 2022 South Korea and Taiwan held almost all of the world's logic capacity below 10 nm, and the United States had almost none[57]. Taiwan alone held 69 percent[5]. In May 2024 SIA and BCG projected that the US share of that capacity would reach 28 percent by 2032[5].

China leads the world in capacity for logic chips made at 28 nm or older, but little of its capacity is advanced[58]. Most of the world's wafers come off such older lines, which make power devices, analog chips and microcontrollers. Across all nodes, old and new, the industry association SEMI forecast in June 2024 that China would have installed capacity of 10.1 million wafers a month in 2025, the most of any region[59]. It forecast 5.8 million for Taiwan, out of a world total of 33.7 million, all counted in the 8-inch size[59].

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2 Breakthroughs or Boasts? Assessing Recent Chinese Lithography Advancements. Center for Strategic and International Studies, 24 September 2025. csis.org

5 Emerging Resilience in the Semiconductor Supply Chain. Boston Consulting Group and Semiconductor Industry..., May 2024, p. 14. semiconductors.org

16 Rapidus Secures 267.6 Billion Yen in Funding from Japan Government and Private... Rapidus, 27 February 2026. rapidus.inc

19 Is SMIC N+3's Metal Pitch Smaller than Intel 18A's? SemiAnalysis, 14 June 2026. newsletter.semianalysis.com

54 Rapidus Completes 150 Billion Yen Funding Round from Japan Government. Rapidus, 5 June 2026. rapidus.inc

55 China launches $47bn chip fund to counter U.S. restrictions. Nikkei Asia, 27 May 2024. asia.nikkei.com

56 China starts production of home-grown immersion DUV chipmaking tools, source... Reuters, 28 July 2026. reuters.com

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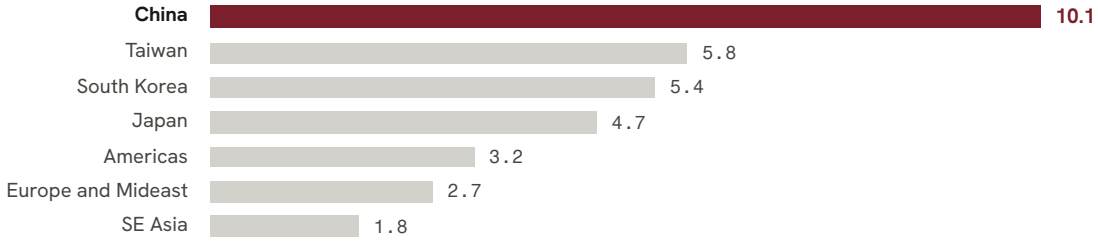
59 Global Semiconductor Fab Capacity Projected to Expand 6% in 2024 and 7% in... SEMI, 18 June 2024. semi.org

In the third-quarter 2026 update of its World Fab Forecast, SEMI projected that China would keep the largest installed capacity in 2026 and 2027[3].

Figure 08.5

China led SEMI's 2025 capacity forecast at 10.1 million 8-inch-equivalent wafers monthly

SEMI's June 2024 forecast of installed wafer fab capacity by region in 2025, all nodes `million 8-inch-equivalent wafers per month`



Source: SEMI World Fab Forecast, June 2024 [59]

Chip production in Taiwan faces three risks:

- **Seismic.** Taiwan sits on an active plate boundary, and after an earthquake in January 2025 TSMC had to scrap some wafers in process[60].
- **Power.** TSMC says power outages, dips or surges caused by problems at its electricity supplier have at times interrupted its operations[61].
- **Political.** In December 2025 SemiAnalysis put Taiwan's share of the world's advanced-node capacity at over 90 percent, though it did not say which node sizes count as advanced[18]. So a blockade or conflict would cut most of that output off from the world market at once. See Who Controls the Chain.

Building capacity outside Taiwan is slow, and TSMC has yet to build a gigafab anywhere else[8]. SemiAnalysis wrote in December 2025 that a foundry needs highly skilled engineers to get the most working chips from each wafer[18]. It judged that finding enough workers remained a major challenge for TSMC in Arizona[18].

Five questions to answer from memory

1. What two things does a chipmaker need to make the newest chips? `see How it works`
2. Why do the newest transistors wrap the gate around the channel? `see How it works`
3. What is a foundry? `see The foundry and the fab`
4. Where are TSMC's largest plants? `see Capacity and cost`
5. How does China's SMIC make advanced chips without an EUV machine? `see The chokepoint`

3 World Fab Forecast. SEMI, accessed 9 September 2026. `semi.org`
8 Fab Capacity. TSMC, accessed 9 September 2026. `tsmc.com`
18 TSMC Overseas Fabs – A Success? SemiAnalysis, 1 December 2025. `newsletter.semianalysis.com`
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60 TSMC January 2025 Revenue Report and the Statement on the Impact of Earthquake. TSMC, 10 February 2025. `pr.tsmc.com`
61 TSMC, annual report on Form 20-F for the fiscal year ended 31 December 2025. U.S. Securities and Exchange Commission, 16 April 2026. `sec.gov`

Five things to remember

- The newest chips need both the know-how to build the newest transistors and factories that cost tens of billions of dollars, and in 2026 only TSMC, Intel and Samsung had both.
- The newest transistors wrap the gate around all four sides of the channel to cut the current that leaks when the switch is off.
- A foundry makes chips for other companies, because no chip company sells enough of one product to fill a fab alone.
- All six of TSMC's largest plants are in Taiwan, which held 69 percent of the world's logic capacity below 10 nm in 2022.
- China's SMIC makes 7 nm-class chips without EUV by building its tightest layers in several patterning steps.

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The chain

01 Chip Design, EDA and IP
02 Silicon and Wafers
03 Chemicals, Gases and Photoresist
04 Lithography
05 Photomasks and Pellicles
06 Deposition and Etch
07 Metrology and Inspection
08 Foundries and Fabs
09 Memory and HBM
10 Advanced Packaging
11 Substrates and PCBs
12 Test and Assembly
13 Systems and Networking
14 Data Centers and Power

15 The Economics of AI Chips
16 Who Controls the Chain

Stage 9 of 14 · Extreme concentration · Hard to substitute

Memory and HBM

SK hynix, Samsung and Micron make nearly all high bandwidth memory (HBM), the stacked memory beside AI chips. SK hynix took 63.2 percent of HBM revenue in 2025.

Specimen 09 · Memory and HBM

A twelve-high HBM stack pulled apart: a logic die at the base, shown in copper, and twelve memory dies above it. The finished part is 11 by 11 mm and 0.72 mm tall. Each memory die is ground to about 30 micrometers, thinner than a human hair. The footprint is to scale, but the heights are exaggerated about fifteen times.

In plain terms

High bandwidth memory, or HBM, is where an AI chip keeps the numbers it is working on, and it is built to hand them over fast. A memory maker etches tiny holes into each memory wafer, fills them with copper and grinds the wafer until the copper shows through its back. It tests each of the wafer's chips and stacks up to sixteen good ones into a cube, with every layer wired to the ones below. The cube then sits beside the processor, the chip that does the calculating. Stacking is the hard part because each chip is ground thinner than a human hair and the finished cube has to stay within a height limit of less than a millimeter. One chip that fails after stacking can scrap the whole cube. Only SK hynix, Samsung and Micron make HBM in volume. One US export rule covers HBM from all three, including the two based in South Korea.

In short

US-based Micron went from 5.8 percent of HBM revenue in 2024 to 23.1 percent in the first quarter of 2026, according to IDC figures in SK hynix's US prospectus[1]. Over the same period, Samsung's share fell from 37.8 percent to 20.5 percent[1].

Since December 2024, a US license has been required, unless an exception applies, to export to China any HBM moving more than 2 GB per second per square millimeter of its area[2]. The Institute for Progress reported in October 2025 that China's CXMT planned to produce HBM3 in 2026[3]. Even so, working from TrendForce's estimate of world HBM shipments, the institute projected that producers outside China would make 70 times as

1 SK hynix, prospectus on Form 424B4 dated 10 July 2026. U.S. Securities and Exchange Commission, 10 July 2026. sec.gov
2 Foreign-Produced Direct Product Rule Additions, and Refinements to Controls for... Federal Register, 5 December 2024. federalregister.gov
3 Should the US Sell Blackwell Chips to China? Institute for Progress, 25 October 2025. ifp.org

much HBM as China in 2026[3]. In June 2026 SemiAnalysis judged that CXMT was still struggling to stabilize supply of eight-high HBM3[4].

An AI accelerator moves numbers from memory to its arithmetic units and back. The US export control office wrote in December 2024 that a processor cannot reach its full capabilities unless its memory capacity and bandwidth rise with its speed[2].

Chokepoint card Memory and HBM

Concentration

Extreme

Three or fewer firms could supply this stage today, or one firm holds more than half of it.

Substitutability

Hard

In June 2026 SemiAnalysis judged that CXMT, China's strongest DRAM maker, was still struggling to stabilize supply of eight-high HBM3[4].

Price or market size

About $35 billion

HBM market in 2025, Micron's December 2025 estimate[5]

Who leads	KR	**SK hynix**	63.2 percent of HBM revenue in 2025[1]
	KR	**Samsung**	19.3 percent of HBM revenue in 2025[1]
	US	**Micron**	17.4 percent of HBM revenue in 2025[1]
Where it is made	KR	**South Korea**	SK hynix in Cheongju and Icheon, Samsung in Pyeongtaek and Hwaseong
	US	**United States**	Micron headquarters and R&D, and an SK hynix packaging plant under construction in Indiana
	TW	**Taiwan**	Micron DRAM and HBM production
	JP	**Japan**	Micron Hiroshima, 1-gamma DRAM
	SG	**Singapore**	Micron HBM advanced packaging plant, adding capacity from 2027
Why substitution is slow			SK hynix says only a few firms have the expertise and resources to design, make and package memory chips[1]. It says HBM also needs advanced processes such as packaging with through-silicon vias (TSVs), the copper-filled holes that link the layers, and must pass customers' rigorous testing and approval[1].

Where China stands

The Institute for Progress estimated in October 2025 that China would make virtually no HBM that year[3]. Huawei said in September 2025 that its Ascend 950 chips would use two HBM designs of its own, without saying who would manufacture the memory[6].

Where the US stands

Micron is the only American HBM maker[1]. As of October 2026, License Exception HBM lets US and allied exporters ship HBM below 3.3 GB per second per square millimeter without a license, including to China when the chip designer buys it directly and it goes straight to the packaging site[7].

01

How it works

Each layer in the cube is a DRAM chip, the ordinary kind of computer memory. Each layer is ground thin and bonded to the one below through tiny metal bumps called microbumps. The whole stack sits on a base die, a chip that manages it. The copper-

1 SK hynix, prospectus on Form 424B4 dated 10 July 2026. U.S. Securities and Exchange Commission, 10 July 2026. sec.gov

2 Foreign-Produced Direct Product Rule Additions, and Refinements to Controls for... Federal Register, 5 December 2024. federalregister.gov

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4 China's CXMT Is Set to Challenge DRAM Incumbents. SemiAnalysis, 23 June 2026. newsletter.semianalysis.com

5 Fiscal Q1 2026 Earnings Call Prepared Remarks. Micron Technology, 17 December 2025. s25.q4cdn.com

6 Groundbreaking SuperPoD Interconnect: Leading a New Paradigm for AI... Huawei, 18 September 2025. huawei.com

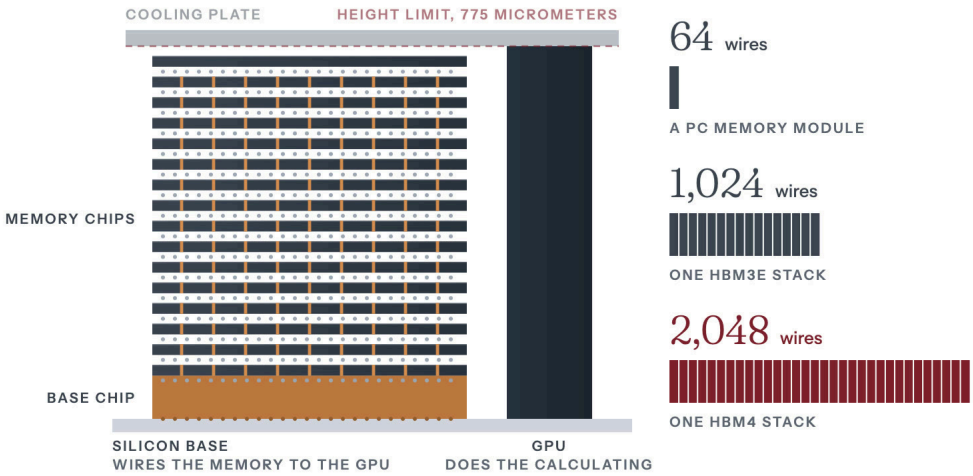
7 15 CFR 740.25 -- License Exception High Bandwidth Memory (HBM). Electronic Code of Federal Regulations, 5 December 2024. ecfr.gov

filled holes are called through-silicon vias, or TSVs. They let a signal reach any layer without a wire running around the outside.

An HBM3E cube has a 1,024-bit interface, meaning it exchanges data with the processor over 1,024 wires at once[8]. That is sixteen times wider than a standard DDR5 memory module, the memory stick in a PC[8]. HBM4 doubles that to 2,048 wires[9]. Ordinary memory cannot be made that wide, because the pins would not fit on a circuit board. The stack can, since its wires run down through the TSVs and across a silicon interposer to the processor beside it, with no board in the way[8]. The interposer is a slice of silicon carrying fine wiring, set under both the memory and the processor (see Advanced packaging).

Figure 09.1 · Explainer, 21 seconds

Inside a memory stack



1. **Base chip.** A memory stack sits beside the GPU on a shared silicon base. It starts with a base chip, which controls the memory chips above it.
2. **Thin.** Each memory chip is ground to about 30 micrometers, thinner than a human hair.
3. **Stack.** Twelve chips are joined by dots of solder, and copper-filled holes run straight down through all but the top one.
4. **Height.** The stack has to fit under the cooling plate. Twelve chips reach 0.72 mm, the old limit. HBM4 raised the limit to 775 micrometers and fits up to sixteen chips.
5. **Width.** An HBM3E stack talks to the processor over 1,024 wires at once, sixteen times as many as a PC memory module. HBM4 doubles that to 2,048.

Note: Simplified cross-section. Heights follow the HBM height limits of 720 and 775 micrometers and are drawn to one vertical scale. Widths are not to scale.

A chip needs the width most when it answers a prompt. Training reuses each stored number across a huge multiplication, so a chip can fetch little and compute a lot. But to generate one word of an answer, the chip reads the whole model from memory, plus the whole record of the conversation so far. Then it does a little arithmetic and waits for the next read. That record grows with the length of the conversation and the

8 High-bandwidth memory (HBM). Micron Technology, accessed 9 September 2026. micron.com

9 JEDEC® and Industry Leaders Collaborate to Release JESD270-4 HBM4 Standard... JEDEC Solid State Technology Association, 16 April 2025. jedec.org

number of users served at once. So memory size and speed usually set how many people one chip can serve.

02

Variants and trade-offs

Generation	Interface	Bandwidth per stack	Capacity per stack
HBM3E	1,024-bit, 16 channels (32 pseudo-channels)	over 1.2 TB/s	24 GB (8 chips), 36 GB (12 chips)
HBM4	2,048-bit, 32 channels	over 2.0 TB/s, up to 3.3 TB/s	up to 64 GB (16 chips, 32 Gb dies)

HBM3E

Micron's HBM3E comes in two sizes, 24 GB with eight chips stacked and 36 GB with twelve[10].

HBM4

All three HBM makers run their HBM4 faster than the 8 Gb/s per pin in the industry standard. The standards body JEDEC published JESD270-4 in April 2025, setting the 2,048-bit interface and 2 TB/s per stack at 8 Gb/s per pin[9]. SK hynix says it was first to complete HBM4 development, in September 2025, at over 10 Gb/s per pin[11]. Samsung says it began mass production and shipped the first commercial HBM4 in February 2026, at 11.7 Gb/s per pin, with a 4 nm logic base die[12]. It gives that HBM4 a maximum of 3.3 TB/s per stack[12]. Micron's parts run above 11 Gb/s for more than 2.8 TB/s per stack[13]. Micron says it began volume production of its 36 GB twelve-high HBM4 in fiscal 2026 and expects to ramp volume of HBM4E, the next generation, in calendar 2027[14].

Through HBM3E, SK hynix made its own base dies, the chips at the bottom of each stack[15]. For HBM4 it chose TSMC's advanced logic process, so the base die can hold more functions[15]. As a result, TSMC, which does not make HBM, makes the base die for SK hynix's HBM4.

9 JEDEC® and Industry Leaders Collaborate to Release JESD270-4 HBM4 Standard... JEDEC Solid State Technology Association, 16 April 2025. jedec.org

10 Micron, HBM3E product page. Micron Technology, accessed 9 September 2026. micron.com

11 SK hynix Completes World's First HBM4 Development and Readies Mass Production. SK hynix, 12 September 2025. news.skhynix.com

12 Samsung Ships Industry-First Commercial HBM4 With Ultimate Performance for AI... Samsung Global Newsroom, 12 February 2026. news.samsung.com

13 Micron, HBM4 product page. Micron Technology, accessed 9 September 2026. micron.com

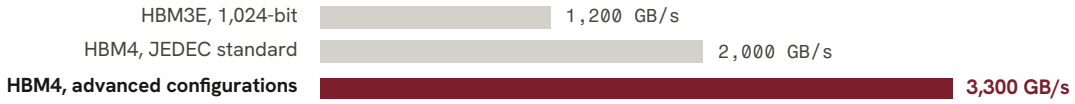
14 Micron Technology, annual report on Form 10-K for the fiscal year ended 3... U.S. Securities and Exchange Commission, 9 October 2026. sec.gov

15 SK hynix Partners with TSMC to Strengthen HBM Technological Leadership. SK hynix, 19 April 2024. news.skhynix.com

Figure 09.2

As of April 2026, HBM4 reaches up to 3,300 GB/s per stack, against 2,000 in the JEDEC standard

Bandwidth per HBM stack



Source: Siemens EDA, HBM3E and HBM4 IC design guide, April 2026 [16]

03

How a stack is built

- **Etch and thin.** Etch vertical holes, the through-silicon vias, into a memory wafer, fill them with copper, then grind the wafer from behind until the copper comes through.
- **Stack and join.** Makers join the chips through metal bumps by methods such as TC-NCF and MR-MUF[17]. In MR-MUF, short for mass reflow molded underfill, the chips are first connected through microbumps and stacked, and a liquid underfill is then injected to fill every gap at once and cured[17]. SK hynix uses its own version, Advanced MR-MUF, which stacks the chips, injects a liquid protective material between them and hardens it[11]. SK hynix says the process controls warping and dissipates heat well[11]. Samsung said in February 2024 that its twelve-high HBM3E uses TC-NCF, thermocompression with a thin non-conductive film between the chips[18]. Samsung expected TC-NCF to help limit warping as chips get thinner[18].
- **Stay under the height limit.** The package specification for HBM4 raised the height allowed for a finished cube from 720 micrometers to 775, still under a millimeter[17].

Hybrid bonding is the next step, but it is not ready. It would join the chips copper to copper and insulator to insulator, with no solder and no filler[17]. A tech note on SK hynix's newsroom gives about 20 micrometers as the typical spacing between microbumps, against less than 1 micrometer for hybrid bonds[17]. It says full-scale adoption in HBM is most likely at HBM4E or HBM5, where stacks are projected to pass twenty layers[17]. Taller stacks trap more heat, and removing it becomes a major challenge at 16 to 20 layers[17]. The same tech note says hybrid bonding manages that heat more efficiently[17].

04

Who makes it

Three firms took more than 90 percent of the revenue from ordinary computer memory, or DRAM, in the first quarter of 2026, according to IDC figures in SK hynix's

11 SK hynix Completes World's First HBM4 Development and Readies Mass Production. SK hynix, 12 September 2025. news.skhynix.com

16 HBM3e and HBM4: IC design guide for next-generation high bandwidth memory. Siemens Digital Industries Software, 24 April 2026. blogs.sw.siemens.com

17 [Tech Note] Hybrid Bonding: Evolving into a Foundational Technology for... SK hynix, 25 August 2026. news.skhynix.com

18 Samsung Develops Industry-First 36GB HBM3E 12H DRAM. Samsung Global Newsroom, 27 February 2024. news.samsung.com

US prospectus[1]. HBM comes from the same three, SK hynix, Samsung and Micron[1]. No other company has both leading-edge DRAM and a stacking line of its own. SK hynix says only a few firms have the expertise and resources to design, make and package memory chips[1]. It says HBM also needs advanced production processes, such as TSV packaging, and must pass the rigorous testing and approval processes its customers set[1].

SK hynix says it was the first company to mass-produce several generations of HBM, including HBM3 and HBM3E[1].

SK hynix and Micron are both adding HBM capacity. SK hynix broke ground in August 2026 on an advanced packaging plant in West Lafayette, Indiana, an investment of over $4 billion[19]. Micron is modernizing and expanding its DRAM and HBM production capacity in Taiwan[14]. It has also broken ground on an HBM advanced packaging plant in Singapore to expand its packaging capacity from early 2027[14].

SK hynix led HBM revenue in 2024, in 2025 and in the first quarter of 2026, and in that quarter Micron passed Samsung for second place. In 2024 SK hynix took 56.4 percent, Samsung 37.8 and Micron 5.8[1]. In 2025 the three took 63.2, 19.3 and 17.4 percent[1]. In the first quarter of 2026 they stood at 56.4, 20.5 and 23.1 percent, which put SK hynix back at its 2024 share[1]. The split is public because SK hynix registered shares in the United States, and its US prospectus credits the research firm IDC[1].

Figure 09.3

SK hynix took 63.2 percent of global HBM revenue in 2025

Global HBM revenue by supplier, 2025

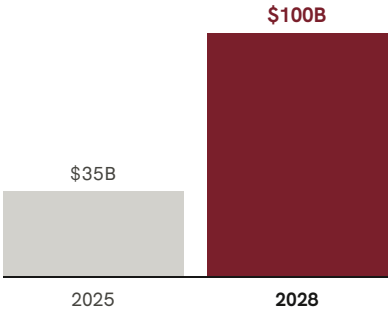


Source: SK hynix Form 424B4 prospectus, 10 July 2026, on IDC data [1]

Figure 09.4

Micron sees HBM reaching $100 billion in 2028, from an estimated $35 billion

HBM market, Micron's estimate for 2025 and forecast for 2028



Source: Micron prepared remarks, fiscal first quarter 2026, 17 December 2025 [5]

1 SK hynix, prospectus on Form 424B4 dated 10 July 2026. U.S. Securities and Exchange Commission, 10 July 2026. sec.gov

5 Fiscal Q1 2026 Earnings Call Prepared Remarks. Micron Technology, 17 December 2025. s25.q4cdn.com

14 Micron Technology, annual report on Form 10-K for the fiscal year ended 3... U.S. Securities and Exchange Commission, 9 October 2026. sec.gov

19 SK hynix breaks ground on $4 billion advanced packaging production facility in... Purdue University, 27 August 2026. purdue.edu

As HBM grows, memory makers have less factory capacity left for traditional DRAM. In December 2025 Micron put the HBM market at about $35 billion in 2025 and forecast that it would reach around $100 billion in 2028[5]. Micron says HBM needs more wafers than traditional DRAM, the kind used in PCs and phones, to make the same number of bits[14]. SK hynix says the industry's shift of production capacity to HBM has left traditional DRAM undersupplied[1]. The shortage, it says, helped drive a significant price recovery for that DRAM from the third quarter of 2025[1].

05

The chokepoint

The leading accelerators made outside China use HBM from SK hynix, Samsung or Micron. Nvidia's GB300 NVL72 rack carries 20 TB of HBM3E at up to 576 TB/s[20]. Each Nvidia Rubin GPU carries up to 288 GB of HBM4 at up to 22 TB/s[21]. AMD's MI355X carries 288 GB of HBM3E at 8 TB/s[22].

A US export rule of December 2024 covers every HBM stack that was in production when it was published. The Bureau of Industry and Security (BIS), the US export control office, published the rule on 5 December 2024[2]. It added an export classification, ECCN 3A090.c, for "High bandwidth memory (HBM) having a 'memory bandwidth density' greater than 2 gigabytes per second per square millimeter"[2]. Its technical note defines that density as the memory bandwidth in gigabytes per second divided by the area of the package or stack in square millimeters[2]. The rule also covers stacks made outside the United States, through a foreign direct product rule, which applies US controls to goods made abroad with American technology[2]. BIS stated in the rule that all HBM stacks then in production exceeded that threshold[2].

BIS reviews license applications to ship controlled HBM to Macau or a Country Group D:5 destination such as China according to where the buyer is headquartered[23]. It presumes it will deny a license when the buyer or its ultimate parent is headquartered in one of those places, and presumes approval otherwise[23].

Two provisions narrow the control. License Exception HBM, at 15 CFR 740.25, applies to exporters headquartered in the United States or a Country Group A:5 ally[7]. It lets them ship HBM below 3.3 GB per second per square millimeter without a license, including to Macau and Country Group D:5 destinations such as China[7]. HBM sent to those destinations must be bought directly by the designer of the chip it will be packaged into[7]. Every shipment under the exception must go straight to the packaging site[7]. The rule also excludes co-packaged chips that combine HBM and logic and whose main function is processing, which other chip controls may cover instead[2].

1 SK hynix, prospectus on Form 424B4 dated 10 July 2026. U.S. Securities and Exchange Commission, 10 July 2026. sec.gov

2 Foreign-Produced Direct Product Rule Additions, and Refinements to Controls for... Federal Register, 5 December 2024. federalregister.gov

5 Fiscal Q1 2026 Earnings Call Prepared Remarks. Micron Technology, 17 December 2025. s25.q4cdn.com

7 15 CFR 740.25 -- License Exception High Bandwidth Memory (HBM). Electronic Code of Federal Regulations, 5 December 2024. ecfr.gov

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21 Inside NVIDIA Rubin GPU Architecture: Powering the Era of Agentic AI. Nvidia, 21 July 2026. developer.nvidia.com

22 AMD Instinct™ MI355X GPUs. AMD, accessed 9 September 2026. amd.com

23 15 CFR 742.6, Regional stability. Electronic Code of Federal Regulations, 23 July 2026. ecfr.gov

The think tank CSIS said in December 2024 that the rules still allowed HBM2, a generation introduced in 2016, to go to China, with limits on who could buy it and what for[24]. It also said Nvidia's H20, whose HBM3 is packaged with the processor, was then still exportable under the performance limits for AI chips[24].

China's strongest DRAM maker, CXMT, was still struggling to make HBM in steady supply by June 2026. In December 2024 CSIS called CXMT one of only two Chinese firms with a credible path to producing advanced HBM, though it did not name the other[24]. The Institute for Progress reported in October 2025 that CXMT was planning to produce HBM3 in 2026[3]. It also estimated that China would make virtually no HBM in 2025[3]. In June 2026 SemiAnalysis judged that CXMT was still struggling to stabilize supply of eight-high HBM3, with even greater challenges at twelve-high[4].

Huawei plans to use HBM of its own design in its Ascend 950 AI chips, but when it announced the designs in September 2025 it did not say who would make the memory[6]. The two designs are called HiBL 1.0 and HiZQ 2.0[6]. Huawei said the 950DT, due in the fourth quarter of 2026, would carry 144 GB of HiZQ 2.0 at 4 TB/s[6].

Four questions to answer from memory

1. What is HBM? `see How it works`
2. Why does an AI chip need such fast memory? `see How it works`
3. Who makes HBM? `see Who makes it`
4. How much HBM did China make in 2025? `see The chokepoint`

Four things to remember

- HBM is memory stacked into a cube beside the processor and wired to pass data fast.
- An AI chip reads its whole model from memory for each word, so memory speed limits it.
- SK hynix, Samsung and Micron make nearly all HBM.
- Since December 2024 US controls have covered the HBM stacks then in production, with two exceptions. The Institute for Progress estimated that China made virtually no HBM in 2025.

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The chain

01 Chip Design, EDA and IP
02 Silicon and Wafers
03 Chemicals, Gases and Photoresist
04 Lithography
05 Photomasks and Pellicles
06 Deposition and Etch
07 Metrology and Inspection
08 Foundries and Fabs
09 Memory and HBM
10 Advanced Packaging
11 Substrates and PCBs
12 Test and Assembly
13 Systems and Networking
14 Data Centers and Power

15 The Economics of AI Chips
16 Who Controls the Chain

Stage 10 of 14 · Extreme concentration · Hard to substitute

Advanced Packaging

Epoch AI estimates that in 2025 Nvidia, Google, AMD and Amazon used around 90 percent of the capacity for CoWoS, TSMC's packaging for AI chips.

Specimen 10 · Advanced Packaging

A CoWoS-L package, pulled apart. Two logic dies and eight HBM stacks sit on the interposer, shown in silver with its copper wiring. In CoWoS-L the interposer is fine redistribution wiring with small embedded silicon bridges. It sits on a resin substrate with a grid of solder balls underneath. The chips together cover about 2,700 mm^2, a little over three reticle fields. Footprints are to scale. Thicknesses and the gaps between layers are exaggerated.

In plain terms

Packaging joins finished chips into one component and mounts them on a small circuit board, called a substrate, wired far more coarsely than the chips themselves. A lithography machine, which prints a chip's circuits, can cover only a rectangle about the size of a postage stamp in one shot. Many leading accelerators, Nvidia's Blackwell among them, need more logic than fits in that rectangle. So they are built from several chips, called dies, set side by side on an interposer, a layer of much finer wiring that links them and sits on the substrate. Stacks of memory chips sit on the same interposer, a few millimeters away. The interposer is either one piece of silicon or layers of fine copper wiring that often carry small silicon bridges where the links are densest. Silicon expands less than the resin substrate beneath it when heated, so the package can warp and its joints crack. Preventing that in high volume is hard. Nvidia, Google, AMD and Amazon all use TSMC's interposer packaging, called CoWoS, made in Taiwan.

In short

Epoch AI estimates that in 2025 Nvidia used 60.3 percent of all capacity for CoWoS, TSMC's interposer packaging[1]. On the July 2026 earnings call, TSMC's chief executive, C.C. Wei, said its packaging capacity was so tight that it limited customers' growth[2]. Asked about Intel's EMIB-T packaging, he said he hoped it would take some of the load off TSMC[2].

In a TrendForce ranking by 2024 revenue, which CSET cites, four of the top ten outsourced firms that package and test chips are headquartered in

1 Advanced packaging and HBM, not logic dies, were the bottlenecks on AI chip... Epoch AI, 12 March 2026. epoch.ai

2 Q2 2026 Taiwan Semiconductor Manufacturing Co Ltd Earnings Call (Chinese... LSEG StreetEvents, 16 July 2026, p. 10. investor.tsmc.com

China[3]. But in 2025 Chinese firms had 5.3 percent of the market for advanced packaging fabrication tools, against 65.8 percent for US firms, by CSET's analysis of TechInsights data[4].

A lithography scanner prints a rectangle 26 mm by 33 mm, or 858 mm², in one exposure[5]. That area is called the reticle field. Every leading AI accelerator, counting its memory stacks, covers more area than that. Packaging is the stage that lets an accelerator be larger than one reticle field, and the packaging method limits how much larger it can be.

Chokepoint card Advanced Packaging

Concentration

Extreme

Three or fewer firms could supply this stage today, or one firm holds more than half of it.

Substitutability

Hard

Rated on CoWoS capacity, which TSMC said in July 2026 was so tight that it limited its customers' growth[2].

Price or market size

About $1,100

for the CoWoS-L packaging of one Nvidia B200, plus about $1,000 for the dies and memory lost in failed packages, against about $900 for its two logic dies, by Epoch AI's December 2025 estimate[6]

Who leads	TW	**TSMC**	CoWoS packaging used by Nvidia, Google, AMD and Amazon in 2025[1]
	US	**Intel**	Foveros 3D packaging in volume at Rio Rancho, New Mexico, since January 2024[7]
	TW	**ASE Technology**	NT$389.2 billion assembly, test and materials revenue, 2025[8]
	US	**Amkor**	$6.71 billion net sales, 2025[9]
	CN	**JCET**	RMB 38.87 billion revenue, 2025[10]
	SG	**ASMPT**	Thermocompression bonders, with $532.1 million of advanced packaging revenue in 2025[11]
Where it is made	TW	**Taiwan**	TSMC advanced packaging fabs and a cluster of outsourced assembly and test firms
	US	**United States**	Intel Foveros packaging in volume at Rio Rancho, New Mexico
	CN	**China**	Four of the top ten outsourced assembly and test firms by 2024 revenue are Chinese, in a TrendForce ranking cited by CSET
	NL	**Netherlands**	Besi hybrid bonders
	JP	**Japan**	Disco grinders and dicers

1 Advanced packaging and HBM, not logic dies, were the bottlenecks on AI chip... Epoch AI, 12 March 2026. epoch.ai

2 Q2 2026 Taiwan Semiconductor Manufacturing Co Ltd Earnings Call (Chinese... LSEG StreetEvents, 16 July 2026, p. 10. investor.tsmc.com

3 Inside Beijing's Chipmaking Offensive. Center for Security and Emerging Technology, 14 July 2025. cset.georgetown.edu

4 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO), accessed 20 October 2026. chipexplorer.eto.tech

5 TWINSCAN NXE:3400C. ASML, accessed 9 September 2026. asml.com

6 NVIDIA's B200 costs around $6,400 to produce. Epoch AI, 10 December 2025. epoch.ai

7 Intel Opens Fab 9 in New Mexico. Intel, 24 January 2024. intel.com

8 ASE Technology Holding Co., Ltd. Reports Its Unaudited Consolidated Financial... PR Newswire, 5 February 2026. prnewswire.com

9 Amkor Technology, current report on Form 8-K dated 9 February 2026. U.S. Securities and Exchange Commission, 9 February 2026. sec.gov

10 JCET Releases 2025 Annual Report, Posts Record-High Full-Year Revenue and... JCET Group, 9 April 2026. jcetglobal.com

11 ASMPT Announces 2025 Annual Results AI-Driven Structural Growth Underpins Group... ASMPT, 4 March 2026. asmpt.com

Why substitution is slow

CoWoS packaging has been outsourced to assembly firms such as ASE and Amkor before [12]. Intel's EMIB, which links chips through silicon bridges embedded in the substrate, is another option, and Amazon's Trainium and Google's TPU were both adopting it as of March 2026 [12]. But a newcomer would have to match TSMC, which said in April 2026 that it was producing CoWoS at 5.5 times the area a scanner prints in one shot [13].

Where China stands

Chinese firms had 5.3 percent of the market for fabrication tools built for advanced packaging in 2025, by CSET's analysis of TechInsights data [4]. Between 2019 and 2024 they gained 5 percentage points of share in test tools but lost share in bonding and other conventional assembly and packaging tools [3].

Where the US stands

Intel opened Fab 9 in Rio Rancho, New Mexico, in January 2024 to make its Foveros 3D packages in volume [7]. Amkor broke ground in October 2025 on a $7 billion packaging and test campus in Peoria, Arizona, and expects production to start in early 2028 [14].

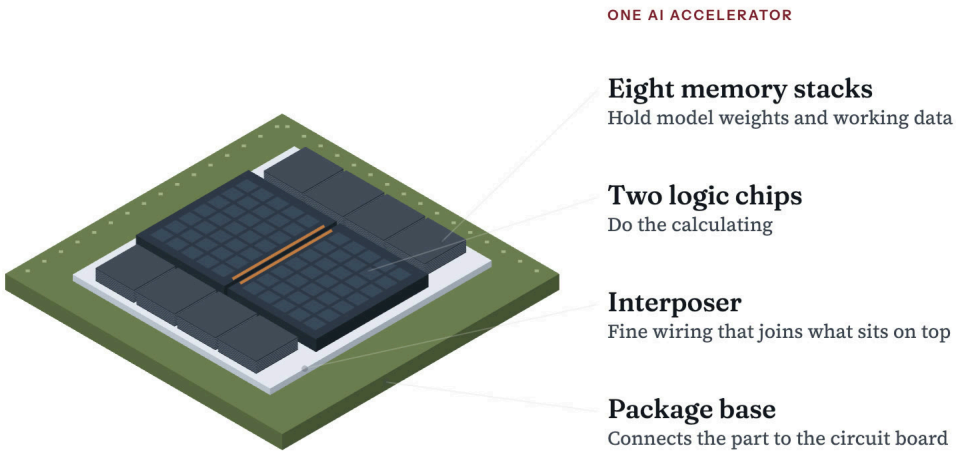
01

How it works

A second reason for the interposer is high bandwidth memory (HBM). An HBM4 stack connects through 2,048 data signals, packed too tightly for a standard resin substrate, so the memory sits beside the logic on the interposer [15]. Even Nvidia's H100, whose single logic die is 814 mm² and fits inside one reticle field, carries five HBM stacks [16]. Epoch AI counts TSMC's CoWoS packaging among the three inputs the H100 needs, along with its logic die and its HBM [1].

Figure 10.1 · Explainer, 19 seconds

The parts of an AI accelerator



1 Advanced packaging and HBM, not logic dies, were the bottlenecks on AI chip... Epoch AI, 12 March 2026. epoch.ai

3 Inside Beijing's Chipmaking Offensive. Center for Security and Emerging Technology, 14 July 2025. cset.georgetown.edu

4 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO), accessed 20 October 2026. chipexplorer.eto.tech

7 Intel Opens Fab 9 in New Mexico. Intel, 24 January 2024. intel.com

12 The Great AI Silicon Shortage. SemiAnalysis, 12 March 2026. newsletter.semianalysis.com

13 TSMC Debuts A13 Technology at 2026 North America Technology Symposium. TSMC, 23 April 2026. pr.tsmc.com

14 Amkor Technology Breaks Ground on New Semiconductor Advanced Packaging and Test... Amkor Technology, 6 October 2025. ir.amkor.com

15 New JEDEC® SPHBM4 Standard Enables HBM4-Class Bandwidth on Organic Substrates. JEDEC Solid State Technology Association, 13 July 2026. jedec.org

16 NVIDIA Hopper Architecture In-Depth. Nvidia, 22 March 2022. developer.nvidia.com

1. **Package base.** An AI chip is built from several parts. First comes the package base, a layered board that connects everything to the circuit board below.
2. **Interposer.** On it sits an interposer, a thin layer of very fine wiring that joins everything placed on top.
3. **Logic chips.** Two logic chips go in the middle. They do the calculating, and each is as large as a chipmaking machine can print.
4. **Memory.** Eight stacks of memory go beside them, a few millimeters away. They hold the AI model's weights and the record of each conversation so far.
5. **One package.** Together they make one AI accelerator. Most are put together by TSMC, in Taiwan.

Note: Simplified view of a two-die, eight-stack accelerator. Footprints follow the sizes given in this chapter and chapter 9. Heights are exaggerated so each layer shows.

CoWoS stands for chip on wafer on substrate, the order in which the parts are joined. For CoWoS-S, the version whose interposer is one piece of silicon, TSMC makes a silicon wafer that carries wiring and vertical copper vias but no transistors. Vias are metal-filled holes that join one layer of wiring to the next.

TSMC bonds the logic dies and memory stacks face down onto that wafer with microbumps. These are dots of solder, and each dot is one electrical joint. In Intel's Foveros 2.5D packaging, which can also set dies side by side on a silicon interposer, the microbumps sit 36 micrometers apart, less than half a hair's width[17]. The spacing differs by design, and SK hynix gives about 20 micrometers as typical for microbumps[18]. Then TSMC cuts the interposer out of the wafer and attaches it to the resin substrate, which later goes onto a circuit board.

One cracked joint can scrap a package holding eight tested HBM stacks and two large, expensive logic dies. Epoch AI estimated in December 2025 that CoWoS-L packaging, the version built with small silicon bridges, added about $1,100 to the cost of building one Nvidia B200[6]. That is more than its estimate of about $900 for the B200's two logic dies[6]. Epoch AI adds about $1,000 more to each B200 for the packages that fail testing, whose logic dies and memory stacks are lost[6].

02

Variants and trade-offs

Family	Carrier	Size limit	Used for
CoWoS-S	Full silicon interposer	Up to 3.3× reticle	H100-class parts
CoWoS-R	Resin interposer with fine redistribution wiring	Above 3.3× reticle	Cost-sensitive, fewer HBM stacks
CoWoS-L	Redistribution wiring with local silicon bridges	5.5× reticle in production, 9.5× in development	Blackwell

2.5D: silicon, RDL and bridges

In 2.5D packaging the chips sit side by side and connect through an interposer or silicon bridges beneath them. TSMC has run CoWoS in volume since 2012[19]. CoWoS-S

6 NVIDIA's B200 costs around $6,400 to produce. Epoch AI, 10 December 2025. epoch.ai

17 Foveros Direct 3D Technology Brief. Intel, 6 February 2026. intel.com

18 [Tech Note] Hybrid Bonding: Evolving into a Foundational Technology for... SK hynix, 25 August 2026. news.skhynix.com

19 TSMC, CoWoS technology page. TSMC, accessed 9 September 2026. 3dfabric.tsmc.com

has the finest wiring, but its interposer is one unbroken piece of silicon whose yield (the share of interposers that work) falls as the area grows. CoWoS-L replaces that piece of silicon with redistribution layers (RDL) of fine copper wiring and embeds small silicon bridges where the wiring is densest[20].

Intel's EMIB embeds similar silicon bridges in the resin substrate[17]. SemiAnalysis reported in March 2026 that EMIB was gaining traction, with Amazon's Trainium and Google's TPU both adopting it[12].

Blackwell was the first high-volume design on CoWoS-L, and packaging problems at TSMC and in Nvidia's design limited its supply[21]. The bridges that carry the 10 TB/s link between the two logic dies must be placed very precisely[21]. The dies, bridges, interposer and substrate also expand at different rates, and that mismatch caused warpage[21].

3D: SoIC and Foveros

In 3D packaging one die sits on top of another, and the joints between them are moving from solder microbumps to hybrid bonding. Intel's Foveros stacks dies face to face and joins them with solder microbumps[17]. Its newer Foveros Direct drops the solder and bonds the copper pads directly, a method called hybrid bonding[17]. Hybrid bonding joins the insulating layer around the pads as well as the copper, with no bumps in between[18]. That lets the connections sit far closer together than microbumps allow[17]. TSMC's SoIC also uses hybrid bonding, and AMD's 3D V-Cache processors use SoIC to attach an extra cache die to the processor die[22]. Stacking also changes how power reaches the top die, which has to draw it up through vertical wires in the die below, called through-silicon vias (TSVs)[22].

TSMC put SoIC into volume production in 2022, stacking SRAM memory chips on logic wafers[23]. Its 3 nm SoIC entered volume production in 2025[20]. An A14-on-A14 version, stacking dies made on TSMC's 1.4 nm process, is due in 2029[13]. It will carry 1.8 times the die-to-die connection density of SoIC that stacks 2 nm dies on 2 nm dies[13].

SoIC bonding also brings optics into the package. TSMC's COUPE uses SoIC to join a silicon photonics chip, which sends data as light, to its electrical control chip[20]. In April 2026 TSMC planned to begin producing COUPE on substrate later that year[13]. It also said that putting COUPE inside the package gives twice the power efficiency of a pluggable version on the circuit board[13].

Two ways off the silicon interposer

Panel-level packaging builds packages on a large rectangular panel instead of a round wafer, so rectangular packages fit it with less waste at the edges. See Test and assembly. JEDEC, the semiconductor standards body, published a standard called SPHBM4 on 13

12 The Great AI Silicon Shortage. SemiAnalysis, 12 March 2026. newsletter.semianalysis.com

13 TSMC Debuts A13 Technology at 2026 North America Technology Symposium. TSMC, 23 April 2026. pr.tsmc.com

17 Foveros Direct 3D Technology Brief. Intel, 6 February 2026. intel.com

18 [Tech Note] Hybrid Bonding: Evolving into a Foundational Technology for... SK hynix, 25 August 2026. news.skhynix.com

20 TSMC 2025 Annual Report, chapter 5. TSMC, 16 April 2026, p. 3. investor.tsmc.com

21 Nvidia's Blackwell Reworked - Shipment Delays & GB200A Reworked Platforms. SemiAnalysis, 4 August 2024. semianalysis.com

22 Hybrid Bonding Process Flow - Advanced Packaging Part 5. SemiAnalysis, 9 February 2024. newsletter.semianalysis.com

23 TSMC 2022 Annual Report, page 97. TSMC, accessed 18 October 2026. investor.tsmc.com

July 2026. It puts HBM4 memory on a base chip carrying 512 data signals instead of 2,048, each running four times as fast to keep the same bandwidth[15]. With fewer signals, the connections can be spaced widely enough to mount the stack on a standard resin substrate instead of the silicon interposer[15].

03

Capacity

TSMC does not publish its CoWoS capacity[1]. Industry estimates from SemiWiki and TrendForce, compiled by Epoch AI, put it at about 65,000 to 75,000 wafers a month at the end of 2025, up from 35,000 to 40,000 a year earlier[1].

In 2024 TSMC was more than doubling CoWoS capacity and still could not keep up with demand. On the July 2024 earnings call, TSMC's chief executive, C.C. Wei, said TSMC was more than doubling CoWoS capacity in 2024 and would more than double it again in 2025[24]. He said he could not yet balance supply and demand[24]. At its 2024 symposium TSMC had said CoWoS capacity would grow at a 60 percent compound annual rate over the next few years[24]. Asked about that rate on the call, Wei did not repeat it[24].

In October 2025 Wei would say only that TSMC was still working to increase CoWoS capacity for 2026, and he gave no figure[25]. He said front-end capacity (making the chips) and back-end capacity (packaging them) were both, in his words, very tight[25].

By Epoch AI's estimate, four chip designers used most of the world's CoWoS capacity in 2025 but only a small share of its advanced chipmaking. Epoch AI compared each chip designer's use of components with world supply[1]. It puts Nvidia at 60.3 percent of all CoWoS capacity in 2025, with a 90 percent confidence interval of 56.5 to 64.3[1]. Google took 13.5 percent, AMD 8.4 and Amazon 7.4[1]. Together those four used around 90 percent of global CoWoS capacity and HBM supply, with HBM measured by value[1]. But they used only 12 percent of advanced logic die production[1].

Figure 10.2

Nvidia used an estimated 60.3 percent of CoWoS capacity in 2025

Estimated share of CoWoS capacity used, 2025



Source: Epoch AI, advanced packaging and HBM were the bottlenecks on AI chip production in 2025 [1]

1 Advanced packaging and HBM, not logic dies, were the bottlenecks on AI chip... Epoch AI, 12 March 2026. epoch.ai

15 New JEDEC® SPHBM4 Standard Enables HBM4-Class Bandwidth on Organic Substrates. JEDEC Solid State Technology Association, 13 July 2026. jedec.org

24 Q2 2024 Taiwan Semiconductor Manufacturing Co Ltd Earnings Call (Chinese... Refinitiv StreetEvents, 18 July 2024, p. 6. investor.tsmc.com

25 Q3 2025 Taiwan Semiconductor Manufacturing Co Ltd Earnings Call (Chinese... Refinitiv StreetEvents, 16 October 2025, p. 7. investor.tsmc.com

In March 2026 the research firm SemiAnalysis judged that front-end capacity had become the dominant bottleneck and that CoWoS constraints were easing[12]. It said TSMC was planning CoWoS capacity around how many wafers N3, its 3 nm process, could produce, because added packaging capacity is of no use without front-end wafers to fill it[12]. It also noted that CoWoS packaging can be, and has been, outsourced to assembly firms such as ASE and Amkor[12].

TSMC gave a different account four months later. On the July 2026 earnings call, Wei said TSMC's packaging capacity was so tight that it limited customers' growth[2]. Asked about Intel's EMIB-T, he said he welcomed the extra flexibility and hoped it would take some of the load off TSMC[2]. Later in the call he said back-end capacity was still in shortage[26].

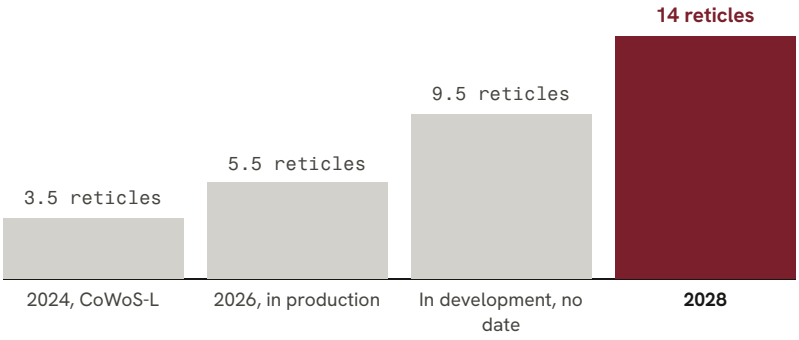
04

Who makes it

Figure 10.3

TSMC plans 14-reticle CoWoS interposers for 2028, up from 3.5 in 2024

CoWoS interposer size on TSMC's roadmap



Source: TSMC 2025 annual report and 2026 technology symposium [13]

Note: The 2024 column is CoWoS-L. CoWoS-S, whose interposer is one piece of silicon, reaches up to 3.3 reticles.

Three kinds of firm package chips. TSMC, which makes chips for others, and Intel, which makes its own, do it in their own back-end fabs. Outsourced assembly and test firms do the rest. Intel opened Fab 9 in Rio Rancho, New Mexico, in January 2024 to make its Foveros 3D packages in volume[7]. It called the plant the only US factory producing the most advanced packaging at scale[7].

ASE leads the outsourced assembly and test firms, and Amkor is the largest one headquartered in the United States. ASE Technology Holding had revenue of NT$645.4 billion in 2025, about 40 percent of it from its electronics manufacturing arm, a business separate from assembly and test[8]. Its assembly, test and materials business

2 Q2 2026 Taiwan Semiconductor Manufacturing Co Ltd Earnings Call (Chinese... LSEG StreetEvents, 16 July 2026, p. 10. investor.tsmc.com

7 Intel Opens Fab 9 in New Mexico. Intel, 24 January 2024. intel.com

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26 Q2 2026 Taiwan Semiconductor Manufacturing Co Ltd Earnings Call (Chinese... LSEG StreetEvents, 16 July 2026, p. 17. investor.tsmc.com

brought in NT$389.2 billion, up from NT$325.9 billion in 2024[8]. Amkor had net sales of $6.71 billion in 2025[9].

Each of the key packaging tools comes from a small number of makers.

- **Thermocompression bonding.** Pressing dies together under heat and force. ASMPT's advanced packaging revenue reached $532.1 million in 2025, with thermocompression bonder revenue up about 146 percent[11]. ASMPT targets 35 to 40 percent of a market it puts at $1.6 billion in 2028[11].
- **Hybrid bonding.** Copper pads and the insulator around them bond directly, with no solder. Applied Materials co-developed its Kinex hybrid bonding system with Besi, which Applied calls the industry leader in hybrid bonding[27].
- **Grinding and dicing.** Disco held 87 percent of the $1.9 billion world market for dicing tools in 2025, by CSET's count from TechInsights data[28]. The category covers the grinders that thin wafers and the saws that cut them into chips[28]. See Test and assembly.

TSMC and Amkor both plan advanced packaging plants in Arizona. TSMC's Arizona commitment reached $265 billion in July 2026 and covers two advanced packaging plants alongside 10 fabs[29]. Amkor started building a $7 billion campus in Peoria, Arizona, in October 2025 and expects production from early 2028[14]. The US Commerce Department awarded Amkor up to $407 million in CHIPS Act funding for the site in December 2024[30]. In October 2025 Wei said a large outsourced packaging partner, which he did not name, was already breaking ground in Arizona[31]. He said its plant was on an earlier schedule than TSMC's own two packaging fabs there[31].

05

The chokepoint

Four of the top ten outsourced assembly and test firms by 2024 revenue are headquartered in China, by a TrendForce ranking that CSET cites[3]. JCET, a leading Chinese assembly and test firm, had record revenue of RMB 38.87 billion in 2025, up 8.1 percent[10]. JCET says its XDFOI packaging platform has entered high-volume manufacturing with redistribution-layer interposer, silicon interposer and silicon bridge options[32]. Those are the same three kinds of carrier that TSMC sells as CoWoS-R, CoWoS-S and CoWoS-L.

3 Inside Beijing's Chipmaking Offensive. Center for Security and Emerging Technology, 14 July 2025. cset.georgetown.edu

8 ASE Technology Holding Co., Ltd. Reports Its Unaudited Consolidated Financial... PR Newswire, 5 February 2026. prnewswire.com

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14 Amkor Technology Breaks Ground on New Semiconductor Advanced Packaging and Test... Amkor Technology, 6 October 2025. ir.amkor.com

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31 Q3 2025 Taiwan Semiconductor Manufacturing Co Ltd Earnings Call (Chinese... Refinitiv StreetEvents, 16 October 2025, p. 13. investor.tsmc.com

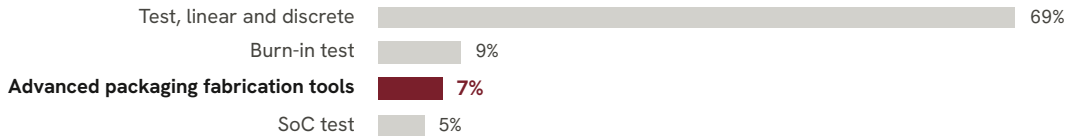
32 JCET Accelerates Strategic Shift Toward High-End Advanced Packaging, 2025... JCET Group, 21 April 2026. jcetglobal.com

Chinese firms had a small share of the market for advanced packaging tools in 2024 and 2025, and they lost ground in bonding. In 2024 Chinese firms had 7 percent of the market for advanced packaging fabrication tools[3]. Between 2019 and 2024, they gained 5 percentage points of share in test tools and 4 points in fabrication tools built for advanced packaging, by CSET's analysis of TechInsights data[3]. But they lost share in bonding and other conventional assembly and packaging tools, the category where CSET counts hybrid and thermocompression bonders[3]. Within the advanced packaging fabrication tools, Chinese firms gained 10 points in etch and clean tools and 3 points in deposition tools[3]. They lost 2 points each in lithography and polishing (CMP) tools[3]. CSET's Chip Explorer, a separate data tool also built on TechInsights data, puts Chinese firms at 5.3 percent of the $6.2 billion world market for advanced packaging fabrication tools in 2025[4]. It puts US firms at 65.8 percent and Japanese firms at 15.3 percent[4].

Figure 10.4

China-based suppliers held 7 percent of the market for advanced packaging fabrication tools in 2024

China-based suppliers' share of advanced packaging fabrication tools and three test tool segments, 2024



Source: CSET, Inside Beijing's Chipmaking Offensive [3]

TSMC said in April 2026 that it was producing CoWoS at 5.5 times the size of a reticle field and planning larger versions[13]. Each generation on TSMC's roadmap needs a larger interposer.

- **2028.** A 14-reticle interposer, about the area of a compact disc, carrying about ten large compute dies and 20 HBM stacks[13].
- **2029.** More than 14 reticles, plus a 40-reticle System-on-Wafer[13]. TSMC describes System-on-Wafer as joining different data center chips at the scale of a whole wafer[33].

Four questions to answer from memory

1. Why is an AI accelerator built from several chips in one package? see How it works
2. Which company do Nvidia, Google, AMD and Amazon rely on to package their AI accelerators? see Who makes it
3. Why has packaging limited how many AI chips can be made? see Capacity
4. Where are advanced chip packages made in volume in the United States? see Who makes it

3 **Inside Beijing's Chipmaking Offensive.** Center for Security and Emerging Technology, 14 July 2025. cset.georgetown.edu
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33 **TSMC 2025 Annual Report, chapter 5.** TSMC, 16 April 2026, p. 4. investor.tsmc.com

Four things to remember

- Many leading accelerators, Nvidia's Blackwell among them, need more logic than one chip can hold, so packaging joins several chips and their memory.
- Nvidia, Google, AMD and Amazon all rely on TSMC, in Taiwan, to package their accelerators.
- Packaging was one of the main limits on AI chip output in 2025. In July 2026 TSMC said its packaging capacity was still so tight that it limited customers' growth.
- No US site yet packages Nvidia-class accelerators with HBM on CoWoS, though Intel makes its own Foveros 3D packages in volume in New Mexico.

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31. A **Q3 2025 Taiwan Semiconductor Manufacturing Co Ltd Earnings Call (Chinese, English) — edited transcript.** Refinitiv StreetEvents (transcript of a TSMC earnings call), via TSMC · 16 October 2025 · p. 13
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33. A **TSMC 2025 Annual Report, chapter 5.** TSMC · 16 April 2026 · p. 4
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The chain

01 Chip Design, EDA and IP
02 Silicon and Wafers
03 Chemicals, Gases and Photoresist
04 Lithography
05 Photomasks and Pellicles
06 Deposition and Etch
07 Metrology and Inspection
08 Foundries and Fabs
09 Memory and HBM
10 Advanced Packaging
11 **Substrates and PCBs**
12 Test and Assembly
13 Systems and Networking
14 Data Centers and Power

15 The Economics of AI Chips
16 Who Controls the Chain

Stage 11 of 14 · Extreme concentration · Hard to substitute

Substrates and PCBs

Ajinomoto, a Japanese food company, told investors in June 2026 that its build-up film, the insulation inside high-end chip packages, has held over 95 percent of its market since its launch.

Specimen 11 · Substrates and PCBs

A build-up substrate, the board a chip sits on, with a quarter cut away. It is about 100 by 72 mm and 1.5 mm thick, with eighteen copper layers inside. The olive-green bands are the build-up film between the copper layers. The footprint is to scale, but the thickness is exaggerated about twenty times.

In plain terms

A package substrate is the adapter between a chip and the circuit board under it. The chip's connections are finer than a human hair, but the board is wired in millimeters. The substrate is built in layers, like plywood, around a stiff core of glass cloth and resin. Thin sheets of epoxy film are pressed onto both faces of the core. A laser burns holes through each sheet, and copper plated into the holes carries signals from one layer to the next. Every added layer is another set of holes that can fail. A handful of firms build these substrates, and two of the materials inside them come mainly from Japanese makers. Ajinomoto dominates the film, and Nitto Boseki dominates the low-expansion glass cloth in the core.

In short

Ajinomoto's third site for making its build-up film, in Gifu Prefecture, is scheduled to start operating in 2032 to meet demand after 2030[1]. Nitto Boseki says market demand seems to have risen about two to three times since it set its plan for fiscal 2024 to 2027[2]. In January 2026 Nikkei Asia reported that Apple and Qualcomm were among the customers worried about Japanese glass cloth supply[3]. In August 2026 Ibiden said it had prospects of a third and fourth glass cloth supplier and expected its material procurement risks to decrease[4].

The silicon interposer wires chips together inside a package. Beyond the interposer, the wiring runs through a build-up substrate made of epoxy

1 Ajinomoto, Growth Strategy for the Electronic Materials Business, 30 June 2026. Ajinomoto, 30 June 2026, p. 20. ajinomoto.co.jp
2 Nittobo Group Integrated Report 2025. Nitto Boseki (Nittobo), 16 September 2025, p. 4. nittobo.co.jp
3 Apple and Qualcomm fret over strained supplies of Japan's glass cloth. Nikkei Asia, 14 January 2026. asia.nikkei.com
4 Ibiden, questions and answers from the financial presentation for the first... Ibiden, 5 August 2026. ibiden.com

and glass, then a printed circuit board that can have more than 30 layers, then a server rack[5].

Chokepoint card

Substrates and PCBs

Concentration

Extreme

Three or fewer firms could supply this stage today, or one firm holds more than half of it.

Substitutability

Hard

Rated on the insulating film inside the substrate, because chipmakers and device makers would have to approve any rival film before substrate makers could use it[6].

Price or market size

$7.5 billion

market for FC-BGA substrates in 2025, up 18 percent, in a Prismark estimate cited by AT&S[7]

Who leads	JP	**Ajinomoto**	Told investors in June 2026 that its build-up film has held over 95 percent of its market since its launch[8]
	JP	**Nitto Boseki**	Claimed an overwhelming advantage in low thermal expansion glass in its 2025 integrated report[2]
	JP	**Ibiden**	Said in May 2026 that it typically takes close to 100 percent of each new generation of interposer-type substrates at launch[9]
	TW	**Unimicron**	Build-up substrates were 52 percent of sales in Q2 2026[10]
	KR	**Samsung Electro-Mechanics**	Package solution sales KRW 771.6bn in Q2 2026, up 37 percent[11]
Where it is made	JP	**Japan**	Build-up film, low-expansion glass cloth, Ibiden and Shinko substrate plants in Gifu and Nagano
	TW	**Taiwan**	Unimicron and Nan Ya PCB, substrate makers that sell to outside customers
	KR	**South Korea**	Samsung Electro-Mechanics and a planned glass-core joint venture with Sumitomo Chemical's Dongwoo Fine-Chem
	CN	**China**	Most of TTM's circuit board competitors are based in China and Taiwan
	AT	**Austria**	AT&S
Why substitution is slow			Ajinomoto makes the thin insulating film between the substrate's wiring layers[12]. Chipmakers and device makers approve substrate materials such as laminate, so a rival film would need their approval too[6].

Where China stands

TTM, an American circuit board maker, said in its annual report for 2025 that its circuit board competitors are mostly based in China and Taiwan[5]. Nitto Boseki, a Japanese glass fiber maker, said in its 2025 integrated report that Chinese and Taiwanese companies are aggressively catching up with it in the special glass sector[2].

Where the US stands

The White House's 2021 supply chain review listed no American firm among the key substrate makers[13]. In December 2024 the US Commerce Department finalized an award of up to $75 million to Absolics, an affiliate of Korea's SKC, for a glass substrate plant in Covington, Georgia[14].

2 Nittobo Group Integrated Report 2025. Nitto Boseki (Nittobo), 16 September 2025, p. 4. nittobo.co.jp

5 TTM Technologies, annual report on Form 10-K for the fiscal year ended 29... U.S. Securities and Exchange Commission, 17 February 2026. sec.gov

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7 AT&S Annual Report 2025/26. AT&S, 10 June 2026, p. 33. ats.net

8 Ajinomoto, Growth Strategy for the Electronic Materials Business, 30 June 2026. Ajinomoto, 30 June 2026, p. 3. ajinomoto.co.jp

9 Ibiden, questions and answers from the financial presentation for the year... Ibiden, 12 May 2026. ibiden.com

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12 Ajinomoto, Growth Strategy for the Electronic Materials Business, 30 June 2026. Ajinomoto, 30 June 2026, p. 9. ajinomoto.co.jp

13 Building Resilient Supply Chains, Revitalizing American Manufacturing, and... The White House (Biden administration archive), June 2021, p. 43. bidenwhitehouse.archives.gov

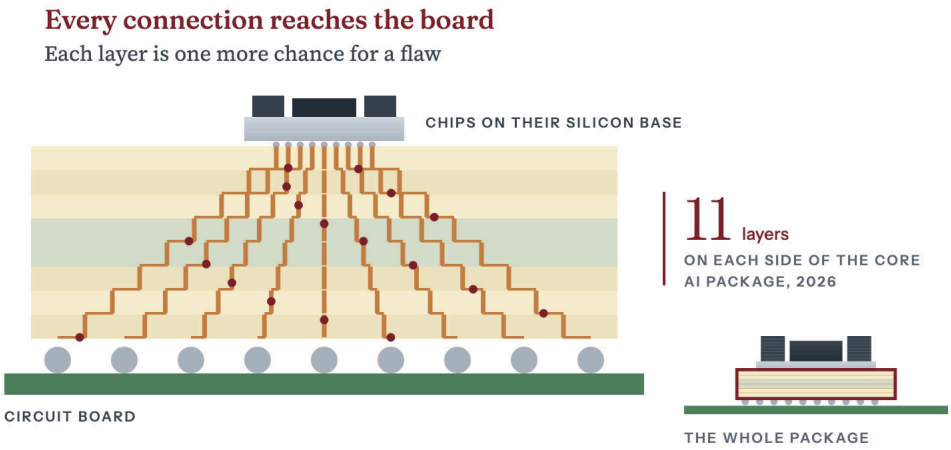
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01

How it works

Figure 11.1 · Explainer, 22 seconds

How a substrate fans out



1. **The package.** An AI chip is more than the chip itself. It sits in a package: the chips on a silicon base, and under them the substrate, which connects everything to the circuit board.
2. **Tight to wide.** The connections on top are packed tight. The circuit board below is wired in millimeters. The substrate has to join the two.
3. **Layer by layer.** The substrate does it in layers, built outward from a stiff core. Each layer carries the copper a little further out.
4. **Connected.** Stacked together, the layers fan the tight grid out to the board's wide one. Ajinomoto counts 11 layers of its film on each side of the core in a 2026 AI package substrate, against three on each side for a PC in 1999. Every layer is another chance for a flaw, which is what makes these substrates hard to build.

Note: Simplified. Three layers of film are drawn on each side of the core, and every thickness is exaggerated.

The substrate maker drills the core and plates it with copper before the build-up starts. In each cycle the maker then presses on a film of epoxy, burns holes through it with a laser to connect the layers, and adds copper wiring on top[15]. Each film is about 10 micrometers thick, and a human hair is about 80 micrometers across[16].

Substrates for AI chips are larger and carry more layers of film than older ones. Ajinomoto counts 11 layers of its film on each side of the core in a 2026 AI package substrate about 100 mm square[17]. A high-performance computing substrate about 70 mm square had nine on each side around 2023[17]. PC substrates had three layers of the film on each side when PCs first adopted it in 1999[17].

Silicon and epoxy expand at different rates when heated, so the larger the package, the more it warps. Nitto Boseki says its low-expansion T-glass cloth is highly rated as an

15 Ajinomoto, Growth Strategy for the Electronic Materials Business, 30 June 2026. Ajinomoto, 30 June 2026, p. 18. ajinomoto.co.jp
16 The Ajinomoto Group's Unexpected Role in Semiconductor Manufacturing: The... Ajinomoto, 5 March 2026. ajinomoto.com
17 Ajinomoto, Growth Strategy for the Electronic Materials Business, 30 June 2026. Ajinomoto, 30 June 2026, p. 11. ajinomoto.co.jp

effective fix for that warping[18]. It says use of the cloth in substrates has increased significantly, without giving a figure[18].

02

Variants and trade-offs

BT substrates

Substrates built on bismaleimide-triazine laminate, or BT, are the other main family. Mitsubishi Gas Chemical, which makes BT materials, says they go mostly into thin packages for smartphone processors, radio modules and memory[19]. The company says BT also goes into flip-chip substrates for computer CPUs and GPUs[19]. In these the chip is mounted face down, and BT laminate also serves as the core of a build-up substrate, between layers of build-up film[20]. In an October 2024 investor presentation, the company estimated that its BT resins held about 40 percent of what it calls the substrate materials market, the top share[21].

ABF build-up substrates

One company, Ajinomoto, supplies nearly all the insulating film inside package substrates for logic chips. Ajinomoto Build-up Film, or ABF, is the insulating film between the layers of package substrates for logic chips, used in PCs, network equipment, servers and AI equipment[12]. Ajinomoto first developed it in 1999 and says major chipmakers have used it as the de facto standard ever since[22]. On Ajinomoto's own site, its electronic materials division said practically all high-performance computers and servers rely on the film[16]. In June 2026 Ajinomoto told investors that the film has held over 95 percent of its market since its launch[8].

8 Ajinomoto, Growth Strategy for the Electronic Materials Business, 30 June 2026. Ajinomoto, 30 June 2026, p. 3. ajinomoto.co.jp

12 Ajinomoto, Growth Strategy for the Electronic Materials Business, 30 June 2026. Ajinomoto, 30 June 2026, p. 9. ajinomoto.co.jp

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18 Nittobo Group Integrated Report 2025. Nitto Boseki (Nittobo), 16 September 2025, p. 17. nittobo.co.jp

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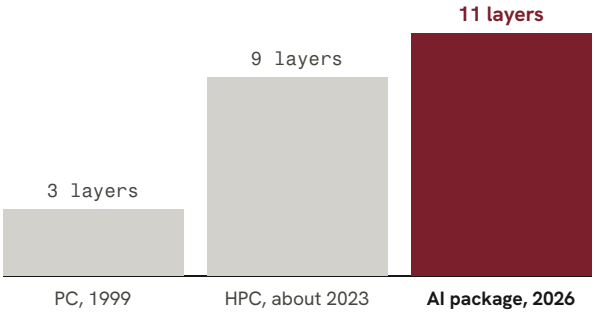
21 Electronics Materials Business (Mitsubishi Gas Chemical business briefing). Mitsubishi Gas Chemical, 2 October 2024, p. 11. mgc.co.jp

22 Ajinomoto, Business Briefing: ABF-Based Growth Strategy in ICT, 12 June 2023. Ajinomoto, 12 June 2023, p. 4. ajinomoto.co.jp

Figure 11.2

Substrate film layers rose from 3 in 1999 to 11 in 2026

Layers of build-up film on each side of a package substrate's core



Source: Ajinomoto, Growth Strategy for the Electronic Materials Business, June 2026 [17]

Glass core substrates

A glass core expands less with heat than a resin core and stays flatter, so chipmakers and substrate makers are both trying it[23].

- **Absolics**, a subsidiary of Korea's SKC, has a final US Commerce Department award of up to $75 million for a 120,000-square-foot glass substrate plant in Covington, Georgia[24]. The department finalized the award in December 2024[14]. As of October 2026, the award's project page at NIST, a Commerce Department agency, still said first deliveries to customers were expected in 2025 and production capacity in 2027, without saying whether those deliveries had happened[24].
- **Intel** said in September 2023 that it planned glass substrates for the latter part of the 2020s[25].
- **Samsung Electro-Mechanics** signed a joint venture agreement with Sumitomo Chemical's subsidiary Dongwoo Fine-Chem in July 2026 to make glass core substrates[26]. The partners plan to have a supply system in place by the second half of fiscal 2027[26].

03

The materials underneath

Ajinomoto's electronic materials revenue is rising, and it is adding film capacity, but its third production site is not scheduled to open until 2032. Ajinomoto's Healthcare and Others segment, which includes electronic materials such as the film, lifted business profit 45.1 percent to JPY 66.2 billion in the year to March 2026[27]. Ajinomoto attributes the rise mainly to higher electronic materials revenue and higher profit in Bio-Pharma Services & Ingredients[27]. It has set an investment budget of JPY 25 billion

14 Biden-Harris Administration Announces CHIPS Incentives Awards with Absolics and... NIST, 5 December 2024. nist.gov

17 Ajinomoto, Growth Strategy for the Electronic Materials Business, 30 June 2026. Ajinomoto, 30 June 2026, p. 11. ajinomoto.co.jp

23 Samsung Electro-Mechanics Signs MOU with Sumitomo Chemical Group to Establish a... Samsung Electro-Mechanics, 5 November 2025. samsungsem.com

24 Absolics (Georgia). NIST, 5 July 2024. nist.gov

25 Intel Unveils Industry-Leading Glass Substrates to Meet Demand for More... Intel, 18 September 2023. download.intel.com

26 Sumitomo Chemical Subsidiary to Establish of a Joint Venture with Samsung... Sumitomo Chemical, 2 July 2026. sumitomo-chem.co.jp

27 Consolidated Financial Results for the Fiscal Year Ended March 31, 2026 (Under... Ajinomoto, 7 May 2026, p. 8. ajinomoto.co.jp

between 2023 and 2030 to expand film production[28]. It says it has no concerns about its current supply chain overall and will steadily expand its supply capacity over the medium to long term[8]. In May 2026 it announced the acquisition of land in Gifu Prefecture for a third production site, scheduled to start operating in 2032 to meet demand after 2030[1].

The glass cloth in the core also has a dominant supplier. Nitto Boseki's T-glass is stiff and expands little when heated, and the company calls it essential for making high-density package substrates[29]. Laminate makers such as Mitsubishi Gas Chemical coat the glass cloth with resin and press it with copper foil into the core laminate that substrate makers build on[30].

Nitto Boseki says market demand seems to have risen about two to three times since it set its plan for fiscal 2024 to 2027[2]. That plan included about JPY 80 billion of capital spending[31]. The company claims an overwhelming advantage in quality and capacity for low-expansion glass[2]. In August 2025 it decided to expand glass fiber cloth production at its Fukushima Enterprise Center, with production scheduled to begin in the fourth quarter of its fiscal 2026[32]. In May 2026 it forecast capital investment of JPY 45,000 million for the year to March 2027, up from JPY 21,702 million in the year to March 2026[33].

In January 2026 Nikkei Asia reported that Apple and Qualcomm were among the customers worried about Japanese glass cloth supply[3]. In August 2026 Ibiden, a Japanese substrate maker, said it had used only two glass cloth suppliers, in Japan and Taiwan, but now had prospects of a third and fourth for mass production[4]. Because a Japanese supplier, which Ibiden did not name, also plans to add capacity, Ibiden said it expected its material procurement risks to decrease[4].

Nitto Boseki says Chinese and Taiwanese companies are aggressively catching up with it in the special glass sector[2]. In July 2026 Nikkei Asia reported that Chinese chip materials makers were competing with Japanese rivals in the market for chip materials, which the industry association SEMI put at $73.2 billion in 2025[34].

04

Who makes it

Most of the firms that build substrates for others are Asian. The White House's 2021 hundred-day supply chain review listed build-up substrate among the materials that commenters called vulnerable[35]. It named key substrate makers including Ibiden,

1 Ajinomoto, Growth Strategy for the Electronic Materials Business, 30 June 2026. Ajinomoto, 30 June 2026, p. 20. ajinomoto.co.jp

2 Nittobo Group Integrated Report 2025. Nitto Boseki (Nittobo), 16 September 2025, p. 4. nittobo.co.jp

3 Apple and Qualcomm fret over strained supplies of Japan's glass cloth. Nikkei Asia, 14 January 2026. asia.nikkei.com

4 Ibiden, questions and answers from the financial presentation for the first... Ibiden, 5 August 2026. ibiden.com

8 Ajinomoto, Growth Strategy for the Electronic Materials Business, 30 June 2026. Ajinomoto, 30 June 2026, p. 3. ajinomoto.co.jp

28 Ajinomoto, Growth Strategy for the Electronic Materials Business, 30 June 2026. Ajinomoto, 30 June 2026, p. 21. ajinomoto.co.jp

29 Nittobo Group Integrated Report 2025. Nitto Boseki (Nittobo), 16 September 2025, p. 3. nittobo.co.jp

30 Electronics Materials Business (Mitsubishi Gas Chemical business briefing). Mitsubishi Gas Chemical, 2 October 2024, p. 28. mgc.co.jp

31 Nittobo Group Integrated Report 2025. Nitto Boseki (Nittobo), 16 September 2025, p. 5. nittobo.co.jp

32 Nittobo Group Integrated Report 2025. Nitto Boseki (Nittobo), 16 September 2025, p. 12. nittobo.co.jp

33 Consolidated Financial Results for the Fiscal Year Ended March 31, 2026. Nitto Boseki (Nittobo), 12 May 2026, p. 5. ssl4.eir-parts.net

34 China chip material makers battle Japan rivals for $73bn market. Nikkei Asia, 1 July 2026. asia.nikkei.com

35 Building Resilient Supply Chains, Revitalizing American Manufacturing, and... The White House (Biden administration archive), June 2021, p. 45. bidenwhitehouse.archives.gov

Nanya, Shinko, Samsung, Unimicron, Shennan Circuits, Zhuhai Yueya and AKM, none of them American[13].

A May 2024 report by the Semiconductor Industry Association and Boston Consulting Group found that every major package substrate vendor was headquartered in Japan, Taiwan or mainland China[36]. It counted two in Japan, two in Taiwan, one in mainland China and none in the United States[36]. But TTM Technologies, an American circuit board maker, says it offers package substrates too[5].

- **Ibiden** will invest about JPY 500 billion in high-performance package substrates from fiscal 2026 to fiscal 2028, starting with about JPY 220 billion, mainly at its Gama plant in Gifu Prefecture[37]. In August 2026 it said the Gama plant was scheduled to start operating in stages from the end of fiscal 2027 and to add to its results from fiscal 2028[4]. In May 2026 management said Ibiden typically takes close to 100 percent of each new generation of what it calls interposer-type substrates at launch[9]. It added that this share may decline by 20 to 30 percent after about three to six months[9]. On silicon-bridge substrates, it said wiring the underside is hard enough that Ibiden expects to keep its technical lead at least through fiscal 2030[9].
- **Unimicron** reported NT$42.9 billion of sales in the second quarter of 2026[10]. By technology, 52 percent came from build-up substrates, and by application, 61 percent came from AI data center work[10].
- **Samsung Electro-Mechanics** grew package solution sales 37 percent year on year to KRW 771.6 billion in the second quarter of 2026, on substrates for AI accelerators, server processors and cars[11].
- **AT&S**, an Austrian firm, calls itself a globally leading maker of high-end package substrates[38]. It makes them in China and Malaysia, and since its 2025/26 financial year also in Austria[39]. Citing the research firm Prismark, its annual report for 2025/26 valued the market for flip-chip ball grid array (FC-BGA) substrates at $7.5 billion in 2025, up 18 percent from 2024[7].

In February 2025 a company owned mainly by funds of JIC Capital launched a tender offer to take Shinko, a Japanese substrate maker, private[40]. JIC Capital is a wholly owned subsidiary of Japan Investment Corporation, an investment company set up in September 2018 under Japan's Industrial Competitiveness Enhancement Act to pursue policy goals such as stronger industrial competitiveness[41]. Fujitsu, which owned 50.02

4 Ibiden, questions and answers from the financial presentation for the first... Ibiden, 5 August 2026. ibiden.com

5 TTM Technologies, annual report on Form 10-K for the fiscal year ended 29... U.S. Securities and Exchange Commission, 17 February 2026. sec.gov

7 AT&S Annual Report 2025/26. AT&S, 10 June 2026, p. 33. ats.net

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36 Emerging Resilience in the Semiconductor Supply Chain. Boston Consulting Group and Semiconductor Industry..., May 2024, p. 19. semiconductors.org

37 Ibiden, Notice Regarding Capital Investment Plan for High-Performance IC... Ibiden, 3 February 2026. ibiden.com

38 AT&S Annual Report 2025/26. AT&S, 10 June 2026, p. 67. ats.net

39 AT&S Annual Report 2025/26. AT&S, 10 June 2026, p. 53. ats.net

40 Announcement Regarding Commencement of the Tender Offer for the Shares of... Japan Investment Corporation, 17 February 2025, p. 1. jiccapital.co.jp

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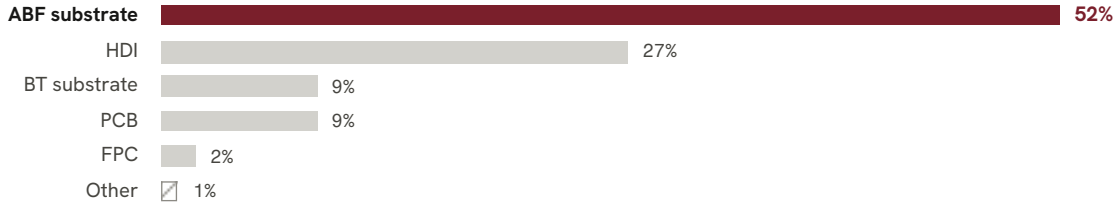
percent of Shinko, agreed to sell that stake back to Shinko, and the offer covered the shares Fujitsu did not own[42]. The tender offer succeeded in March 2025[43].

In September 2026 Shinko agreed to buy a plant in Mobara, Chiba Prefecture, from Japan Display, a display maker[44]. Shinko said it would make the site a new core plant for its flip-chip package substrates and would also expand capacity at its plants in Nagano[44]. Production at the Mobara plant is scheduled to start in fiscal 2028[44].

Figure 11.3

ABF substrates were 52 percent of Unimicron's second-quarter 2026 sales

Unimicron sales by technology, second quarter 2026



Source: Unimicron 2026 Q2 earnings conference [10]

05

The boards under the GPU

TTM Technologies says it regularly builds printed circuit boards of more than 30 layers and can go past 70[5]. Those layers carry power to the GPUs and route fast signals between them, so how much signal the laminate absorbs is a main design constraint. Nitto Boseki also makes NE-glass, a low-dielectric glass cloth that absorbs less signal, used in switches and routers in data centers[45].

Data center computing was 24 percent of TTM's fiscal 2025 revenue of about $2.9 billion, against 14 percent two years earlier[5].

Board makers are adding plants outside China and Taiwan, where TTM says most of its circuit board competitors are based[5]. In its annual report for 2025, TTM said it was building a plant in Syracuse, New York, to make ultra-high-density circuit boards for national security needs[5]. It expected volume production there in the second half of 2026[5]. Nikkei Asia reported in September 2025 that Thailand aimed to become the biggest circuit board hub outside China as board makers moved to diversify production[46].

5 TTM Technologies, annual report on Form 10-K for the fiscal year ended 29... U.S. Securities and Exchange Commission, 17 February 2026. sec.gov

10 Unimicron 2026 Q2 Earnings Conference. Unimicron Technology, July 2026. unimicron.com

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45 Nittobo Group Integrated Report 2025. Nitto Boseki (Nittobo), 16 September 2025, p. 7. nittobo.co.jp

46 Thailand carves out a less-glamorous AI niche: printed circuit boards. Nikkei Asia, 3 September 2025. asia.nikkei.com

06

The chokepoint

The few firms that build substrates are spending to add capacity, but it takes years. In August 2026 Ibiden said that, with supply short across the industry, the average price of package substrates had risen more than it expected[4]. The materials inside the substrate are harder to replace than the firms that build it. Mitsubishi Gas Chemical, a laminate maker, says chipmakers and electronic device makers further down the chain approve the materials it sells to substrate makers[6]. So a rival to Ajinomoto's film or to Nitto Boseki's low-expansion glass cloth would first need approval from those chipmakers and device makers.

Packages are also getting bigger. TSMC told investors in July 2026 that its CoWoS packaging roadmap goes beyond 14 times the area a lithography machine can print in one shot[47]. A bigger package warps more and needs more layers, so it has more holes that can break the circuit. That lowers yield, the share of substrates that come out working.

Four questions to answer from memory

1. What does a package substrate do? see How it works
2. Which company makes nearly all the insulating film inside chip package substrates? see The materials underneath
3. Why is Ajinomoto's film hard to replace? see The chokepoint
4. Where are the major package substrate makers based? see Who makes it

Four things to remember

- A package substrate connects a chip's fine contacts to a circuit board wired in millimeters.
- One Japanese supplier leads the market for the substrate's insulating film, and another leads the market for its glass cloth.
- Replacing the film would mean re-approval by every substrate maker and chip customer.
- The White House's 2021 supply chain review named no American firm among the key substrate makers. Absolics is building a glass substrate plant in Georgia, and a Commerce Department agency expects its production capacity to come online in 2027.

4 Ibiden, questions and answers from the financial presentation for the first... Ibiden, 5 August 2026. ibiden.com

6 Electronics Materials Business (Mitsubishi Gas Chemical business briefing). Mitsubishi Gas Chemical, 2 October 2024, p. 12. mgc.co.jp

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The chain

01 Chip Design, EDA and IP
02 Silicon and Wafers
03 Chemicals, Gases and Photoresist
04 Lithography
05 Photomasks and Pellicles
06 Deposition and Etch
07 Metrology and Inspection
08 Foundries and Fabs
09 Memory and HBM
10 Advanced Packaging
11 Substrates and PCBs
12 Test and Assembly
13 Systems and Networking
14 Data Centers and Power

15 The Economics of AI Chips
16 Who Controls the Chain

Stage 12 of 14 · High concentration · Moderately hard to substitute

Test and Assembly

Testing finds bad chips before they go into an expensive package. Japan's Advantest took about 65 percent of 2025 spending on chip testers, by its own estimate.

Specimen 12 · Test and Assembly

A vertical probe card over a wafer. It is called vertical because its needles point straight down at the wafer. The card is round, about 305 mm across, a little wider than the 300 mm wafer beneath it. Thousands of needles under it land on four dies at once, and those four dies are picked out in blue-violet. The gap between card and wafer is exaggerated. The real gap is a millimeter or two.

In plain terms

Before a chip can be sold, a machine called a tester has to check that it works. While the chips are still on the wafer, thousands of needles press onto small metal pads on each one. The tester sends signals through the needles and reads what comes back, and any chip that fails goes no further. A modern AI accelerator is a dozen expensive pieces bonded into one package, and if one piece is bad, all of them are thrown away. So the industry now tests chips earlier, at higher temperatures and for longer, to make weak ones fail during testing before they reach a finished product. Two firms, Advantest in Japan and Teradyne in the United States, sell most of the testers.

In short

In a TrendForce ranking cited by CSET, four of the ten largest chip assembly and test contractors by 2024 revenue are headquartered in China[1]. Georgetown's CSET, analyzing TechInsights data, finds that between 2019 and 2024 Chinese toolmakers gained 5 percentage points of world market share in test tools[1]. Across assembly, test and packaging tools, they still lag significantly behind foreign suppliers in market share[1]. Advantest says that switching to another tester maker means rebuilding a customer's whole test setup, at significant extra cost[2].

Turning silicon into a finished, packaged chip takes 1,000 to 1,500 steps, depending on how complex the chip is, the Semiconductor Industry Association estimated in 2025[3]. Of all those steps, only test puts electrical signals through the chip, Advantest's investor guide of July 2026 says[4]. The

1 **Inside Beijing's Chipmaking Offensive.** Center for Security and Emerging Technology, 14 July 2025. cset.georgetown.edu

2 **Investors Guide July 29, 2026.** Advantest, 30 July 2026, p. 12. advantest.com

3 **Testimony of David Isaacs, Semiconductor Industry Association, hearing on…** U.S. Senate Committee on Finance, 14 May 2025, p. 2. finance.senate.gov

4 **Investors Guide July 29, 2026.** Advantest, 30 July 2026, p. 10. advantest.com

earlier stages of chipmaking make dies, the individual chips on a wafer. The back end of the industry, assembly and test, checks which of them work and puts the good ones into packages.

Chokepoint card Test and Assembly

Concentration

High

Four to six firms could supply this stage today, and the top two take most of the revenue.

Substitutability

Moderate

Rated on test tools, where Advantest held 62.7 percent of the market in 2025 and Teradyne 21.8 percent, by CSET's analysis of TechInsights data[5].

Price or market size

About $9.0 billion

market for chip testers in 2025, by Advantest's estimate[6]

Who leads	JP	**Advantest**	About 65 percent of the tester market in 2025, by its own estimate[6]
	US	**Teradyne**	$2.52 billion of semiconductor test revenue in 2025, service included[7]
	TW	**ASE**	Ran 8,348 testers in Q2 2026, up from 6,797 a year earlier[8]
	US	**Amkor**	$6.71 billion of 2025 revenue[9]
	JP	**Disco**	87 percent of the $1.9 billion market for dicing saws and grinders in 2025, on CSET's analysis of TechInsights data[10]
Where it is made	TW	**Taiwan**	Taiwanese firms such as ASE and Powertech lead outsourced assembly and test
	JP	**Japan**	Advantest testers, Accretech and Tokyo Electron probers, Disco and Accretech dicing and grinding
	US	**United States**	Test equipment from Teradyne, FormFactor, Cohu and Aehr, and Amkor's planned Arizona packaging plant
	CN	**China**	Four of the top ten assembly and test firms by 2024 revenue, in a TrendForce ranking cited by CSET
	MY	**Malaysia**	Malaysia said in July 2026 that it ships 13 percent of the world's packaged chips
Why substitution is possible	Customers such as ASE already buy testers from both firms[11]. The barrier is cost, because switching means rebuilding a customer's whole test setup, Advantest says[2].		

2 **Investors Guide July 29, 2026.** Advantest, 30 July 2026, p. 12. advantest.com

5 **Supply Chain Explorer: Advanced Chips.** Emerging Technology Observatory (ETO), accessed 20 October 2026. chipexplorer.eto.tech

6 **Advantest, presentation notes for the FY2025 (year ended 31 March 2026) results...** Advantest, 27 April 2026, p. 13. advantest.com

7 **Teradyne, annual report on Form 10-K for the fiscal year ended 31 December 2025.** U.S. Securities and Exchange Commission, 19 February 2026. sec.gov

8 **ASE Technology Holding, report on Form 6-K dated 30 July 2026.** U.S. Securities and Exchange Commission, 30 July 2026. sec.gov

9 **Amkor Technology, current report on Form 8-K dated 9 February 2026.** U.S. Securities and Exchange Commission, 9 February 2026. sec.gov

10 **Supply Chain Explorer: Advanced Chips.** Emerging Technology Observatory (ETO), accessed 9 September 2026. chipexplorer.eto.tech

11 **ASE Technology Holding, annual report on Form 20-F for the fiscal year ended 31...** U.S. Securities and Exchange Commission, 1 April 2026. sec.gov

Where China stands

Georgetown's CSET, analyzing TechInsights data, finds that between 2019 and 2024 Chinese toolmakers gained 5 percentage points of world market share in test tools and 4 points in fabrication tools built for advanced packaging [1]. But they lost share in bonding and other conventional assembly and packaging tools [1].

Where the US stands

American firms held 23.5 percent of the market for test tools in 2025 and Japanese firms 63.0 percent, by CSET's analysis of TechInsights data [5]. Amkor raised its planned investment in an Arizona packaging and test campus to $7 billion in October 2025 [12]. In June 2026 it announced a long-term partnership with TSMC, which will buy packaging and test services from it in Arizona [13].

01

How it works

Figure 12.1 · Explainer, 23 seconds

How a chip is tested



1. **The wafer.** Earlier stages make the chips. Before any of them can be sold, a machine tests each one, starting while the chips are still on the wafer.
2. **Needles.** Thousands of needles on a probe card press onto one chip. They touch small metal pads on its surface.
3. **Questions and answers.** Through the needles, the tester sends signals into the chip and reads what comes back. It checks every answer.
4. **Chip by chip.** The wafer moves under the needles, chip by chip. The tester marks any chip that answers wrong, and that chip goes no further.
5. **Into the package.** Only chips that pass go on to be packaged. A modern accelerator bonds together a dozen expensive pieces, and one bad piece throws away all of them, so each piece is tested before it goes in.

Note: Simplified. Nine needles stand for thousands. A real prober, the machine that moves the wafer under the card, holds the card still and raises the wafer to meet the needles. The failed chips are placed for illustration and do not reflect a real yield. The package follows chapter 10: two logic chips and eight memory stacks on an interposer and a package base.

1 Inside Beijing's Chipmaking Offensive. Center for Security and Emerging Technology, 14 July 2025. cset.georgetown.edu

5 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO), accessed 20 October 2026. chipexplorer.eto.tech

12 Amkor Technology Breaks Ground on New Semiconductor Advanced Packaging and Test... Amkor Technology, 6 October 2025. ir.amkor.com

13 TSMC and Amkor Technology Announce Long Term Partnership to Accelerate Advanced... Amkor Technology, 16 June 2026. ir.amkor.com

Each round of test catches failures the round before it cannot see.

- **Wafer sort.** Thousands of needles land on each die while it is still on the wafer. A tester sends in test patterns and checks the answers, and dies that fail go no further.
- **Burn-in.** The part runs hot and at raised voltage for minutes to days, on the wafer or in the package, so that early-life failures happen in the factory, before the part ships[14].
- **Final test.** The tester checks the packaged part again, because packaging adds failures of its own.
- **System-level test.** The finished part sits in a socket and runs something close to a real workload, because some failures appear only when everything runs at once.

As the parts in a package grow more valuable, finding a bad die before packaging matters more[15]. Aehr, an American maker of burn-in equipment, sells wafer-level burn-in systems that look for early-life and hidden failures before dies are cut from the wafer[15]. Catching those failures early lowers the risk of packaging high-bandwidth memory and an expensive substrate with a processor that may later fail[15]. Aehr says its lead AI customer is moving all burn-in of that customer's AI accelerators from the system level, after packaging, to the wafer[16]. The FOX-XP systems that customer ordered from Aehr in August 2026 each burn in nine 300 mm wafers at a time[15].

02

Variants and trade-offs

Wafer sort and known-good die

The hard part of wafer sort is the probe card, the circuit board that holds the needles. For a large logic die every needle must sit in the same plane, survive repeated touchdowns without drifting, and do it hot. Each card is built for one chip design, so every new design needs a new card and a new qualification. FormFactor, an American probe card maker, says its cards work at high enough frequencies to compete well in testing high-bandwidth memory, a stack of eight, twelve or sixteen memory dies[17].

Final test, burn-in and system-level test

The bigger the package, the more one bad die can cost. TSMC told investors in July 2026 that its CoWoS roadmap includes packages larger than 14 times the area a lithography machine can print in one shot[18]. A die that fails after assembly wastes the memory and substrate joined to it[15].

14 Aehr Test Systems, annual report on Form 10-K for the fiscal year ended 29 May... U.S. Securities and Exchange Commission, 27 July 2026. sec.gov

15 Aehr Receives $22 Million Follow-On Order for AI Processor Wafer-Level Burn-In... Aehr Test Systems, 12 August 2026. aehr.com

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17 FormFactor, annual report on Form 10-K for the fiscal year ended 27 December... U.S. Securities and Exchange Commission, 20 February 2026. sec.gov

18 Q2 2026 Taiwan Semiconductor Manufacturing Co Ltd Earnings Call (Chinese... LSEG StreetEvents, 16 July 2026, p. 16. investor.tsmc.com

Figure 12.2

Advantest put its tester market share at about 65 percent in 2025

Automated test equipment: share of the ~$9.0 billion tester market, 2025



Source: Advantest FY2025 results briefing, April 2026 [6]

03

Who makes it

Advantest puts its own share at about 65 percent of an estimated $9.0 billion tester market in 2025[6]. Its share of system-on-chip (SoC) testers, the machines that test processors and other non-memory chips, rose about ten points in a year to 66 percent[6]. It held about 60 percent of memory test[6].

Teradyne, Advantest's main rival, took $2.52 billion of semiconductor test revenue in 2025, service included[7]. Of that, $1.89 billion came from SoC testers and $505 million from memory testers[7]. Teradyne's total revenue grew 13 percent to $3.19 billion in 2025, with fourth-quarter revenue up 44 percent on AI demand[19].

Georgetown University's Center for Security and Emerging Technology (CSET) makes a second estimate, from data by the research firm TechInsights[5]. It gives Advantest 62.7 percent of a $9.1 billion market for test tools in 2025 and Teradyne 21.8 percent[5]. Japanese firms held 63.0 percent of that market and American firms 23.5 percent[5].

Figure 12.3

SoC testers took an estimated $6.9 billion in 2025, memory $2.1 billion

Where the tester money went in calendar 2025



Source: Advantest FY2025 results briefing, April 2026 [6]

The rest of the test and assembly equipment market splits into niches, each with its own suppliers.

- **Probe cards.** FormFactor took $785.0 million of revenue in fiscal 2025, most of it from probe cards[17]. Technoprobe's revenue grew 15.7 percent to EUR 628.4 million in 2025, about 38 percent of it from AI[20]. It aims to double its manufacturing capacity by the end of 2027[20].

5 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO), accessed 20 October 2026. chipexplorer.eto.tech

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20 Technoprobe, press release: the Board of Directors approves the draft statutory... Technoprobe, 18 March 2026, p. 1. technoprobe.com

- **Probers**, the machines that move each wafer under the probe card and keep it at a set temperature. Tokyo Seimitsu, which sells under the Accretech name, says its own research gives it the largest global share in probers[21]. Tokyo Electron makes probers too[22]. CSET's analysis of TechInsights data counts handlers and probers as one market, worth $2.9 billion in 2025, and gives Tokyo Electron 19.9 percent of it and Accretech 17.1 percent[23].
- **Handlers and burn-in**, the machines that feed packaged parts into a tester and hold them at a set temperature, and the systems that run burn-in. Cohu took $453.0 million of revenue in fiscal 2025[24]. Aehr's revenue was $50.0 million in fiscal 2026, but it booked a record $60.7 million of orders in the fourth quarter alone[16]. It expects revenue of $130 million to $150 million in fiscal 2027[16].
- **Dicing and grinding.** Dicing saws cut wafers into dies, and grinders thin them. CSET's analysis of TechInsights data found that in 2025 Disco held 87 percent and Accretech 8.1 percent of the $1.9 billion market for these machines[10]. Panel-level packaging builds packages on large panels in place of 300 mm wafers, to make more of them at once[25]. In December 2025 Disco announced a saw that cuts such panels, as large as 400 by 400 mm, into individual packages[25].
- **Bonding.** Kulicke and Soffa took $654.1 million of revenue in fiscal 2025[26]. Besi's second-quarter 2026 revenue rose 68.7 percent to EUR 249.9 million on hybrid bonding and data center demand[27].

Thermocompression bonders use heat and force to press memory dies into a stack and to attach chiplets to substrates. ASMPT says it remained the leader in that market in 2025 and aims for a 35 to 40 percent share[28]. In its 2025 results it said it expects the market to reach $1.6 billion by 2028[28]. In July 2026 ASMPT announced new bulk orders for more than 50 of its chip-to-substrate bonders for AI chips, from outsourced assembly and test firms[29]. Hanmi Semiconductor in South Korea also sells thermocompression bonders for high-bandwidth memory[30]. See Memory and HBM.

04

The back end's geography

Taiwanese firms lead outsourced assembly and test, the OSAT business, and Chinese firms are large challengers. Taiwan's ASE billed $20,573.4 million in 2025[11]. Packaging made up about 48 percent of that, testing about 11 percent and its electronics

10 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO), accessed 9 September 2026. chipexplorer.eto.tech

11 ASE Technology Holding, annual report on Form 20-F for the fiscal year ended 31... U.S. Securities and Exchange Commission, 1 April 2026. sec.gov

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22 Precio™ Series. Tokyo Electron, accessed 9 September 2026. tel.com

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25 DISCO develops the DFD6080 fully automatic dicing saw for work sizes up to 400... DISCO Corporation, 15 December 2025. disco.co.jp

26 Kulicke and Soffa, results for the fourth quarter and fiscal 2025 (Form 8-K... U.S. Securities and Exchange Commission, 19 November 2025. sec.gov

27 BE Semiconductor Industries N.V. Announces Q2-26 and H1-26 Results. BE Semiconductor Industries (Besi), 23 July 2026. besi.com

28 ASMPT Announces 2025 Annual Results AI-Driven Structural Growth Underpins Group... ASMPT, 4 March 2026. asmpt.com

29 ASMPT Secures Significant Wins for More than 50 Chip-to-Substrate... ASMPT, 29 July 2026. asmpt.com

30 Hanmi Semiconductor, products page. Hanmi Semiconductor, accessed 9 September 2026. hanmisemi.com

manufacturing arm about 40 percent[31]. In a TrendForce ranking that CSET cites, four of the top ten OSATs by 2024 revenue are headquartered in China[1].

Malaysia is a large center for chip assembly and test. Its investment agency said in July 2026 that the country ships 13 percent of the world's packaged chips[32]. It said a consortium of five Malaysian firms aims to help the country capture 7 percent of world advanced packaging shipments by 2035[32]. A 2024 study by the Semiconductor Industry Association and Boston Consulting Group used a different measure, capacity[33]. It gave Malaysia 7 percent of global assembly, test and packaging capacity in 2022, the largest share in Southeast Asia[33]. The study measured capacity by factory floor space at contractors and chipmakers' own plants[34].

Taiwan's assembly and test contractors are expanding. ASE ran 6,797 testers in the second quarter of 2025 and 8,348 a year later[8]. It spent $2.7 billion on machinery and $1.4 billion on buildings, facilities and automation in the first half of 2026[35]. In July 2026 it said revenue from leading-edge advanced packaging was running ahead of its $3.5 billion forecast for 2026[35]. Powertech, another Taiwanese assembly and test firm, aims to run the first panel-level packaging line for AI chips in 2027[36].

Amkor, an American assembly and test contractor, took $6.71 billion of revenue in 2025[9]. Its revenue in the second quarter of 2026 alone was $1.90 billion, up 26 percent year on year[37]. By October 2025 it had raised the planned investment in its Arizona campus to $7 billion[12].

In October 2024 TSMC and Amkor signed a memorandum of understanding for TSMC to buy packaging and test services from Amkor's planned plant in Peoria, Arizona[38]. TSMC would use those services for its customers, especially those using its Phoenix fabs[38]. The two announced a long-term partnership in June 2026[13].

1 **Inside Beijing's Chipmaking Offensive.** Center for Security and Emerging Technology, 14 July 2025. cset.georgetown.edu

8 **ASE Technology Holding, report on Form 6-K dated 30 July 2026.** U.S. Securities and Exchange Commission, 30 July 2026. sec.gov

9 **Amkor Technology, current report on Form 8-K dated 9 February 2026.** U.S. Securities and Exchange Commission, 9 February 2026. sec.gov

12 **Amkor Technology Breaks Ground on New Semiconductor Advanced Packaging and Test...** Amkor Technology, 6 October 2025. ir.amkor.com

13 **TSMC and Amkor Technology Announce Long Term Partnership to Accelerate Advanced...** Amkor Technology, 16 June 2026. ir.amkor.com

31 **ASE Technology Holding, report on Form 6-K dated 5 February 2026.** U.S. Securities and Exchange Commission, 5 February 2026. sec.gov

32 **Advanced Packaging: How Malaysia is Packaging the Future of AI.** Malaysian Investment Development Authority, 1 July 2026. mida.gov.my

33 **Emerging Resilience in the Semiconductor Supply Chain.** Boston Consulting Group and Semiconductor Industry..., May 2024, p. 21. semiconductors.org

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35 **ASE Technology Holding, report on Form 6-K dated 30 July 2026.** U.S. Securities and Exchange Commission, 30 July 2026. sec.gov

36 **Powertech eyes world's first panel-level packaging for AI chips in 2027.** Nikkei Asia, 27 August 2026. asia.nikkei.com

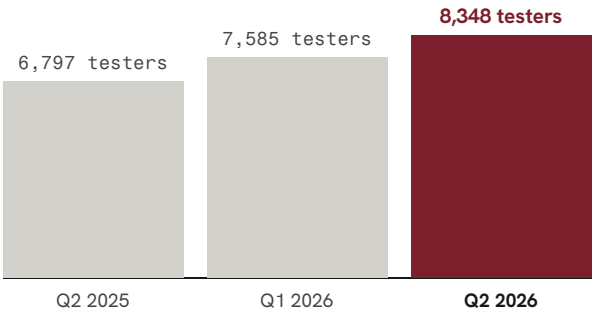
37 **Amkor Technology, current report on Form 8-K dated 27 July 2026.** U.S. Securities and Exchange Commission, 27 July 2026. sec.gov

38 **Amkor and TSMC to Expand Partnership and Collaborate on Advanced Packaging in...** TSMC, 4 October 2024. pr.tsmc.com

Figure 12.4

ASE ran 8,348 testers in mid-2026, up from 6,797 a year earlier

Testers installed at ASE, a Taiwanese assembly and test contractor



Source: ASE Technology Holding, second quarter 2026 earnings release (SEC Form 6-K) [8]

05

The chokepoint

Customers can switch tester makers, but at a cost. Advantest says that switching to another tester maker means rebuilding the whole test setup used for development, evaluation and mass production, at significant extra cost to the customer[2]. Still, ASE says it buys testers with similar functions from different makers, including Teradyne and Advantest[11].

Chinese toolmakers are gaining share in test and advanced packaging tools, but CSET finds they still lag significantly behind foreign suppliers in market share[1]. Between 2019 and 2024 they gained 5 percentage points of world market share in test tools and 4 points in fabrication tools built for advanced packaging, by CSET's analysis of TechInsights data[1]. But they lost share in bonding and other conventional assembly and packaging tools[1].

Assembly and test contractors such as ASE and Amkor earn a small part of what the chip industry sells. ASE's packaging business took $9,829.2 million from outside customers in 2025, and its testing business $2,292.0 million[11]. Amkor's revenue was $6.71 billion[9]. Global semiconductor sales that year were $791.7 billion[39].

Three questions to answer from memory

1. Why do chipmakers test chips early and often? see How it works
2. Who makes most chip testers? see Who makes it

1 Inside Beijing's Chipmaking Offensive. Center for Security and Emerging Technology, 14 July 2025. cset.georgetown.edu

2 Investors Guide July 29, 2026. Advantest, 30 July 2026, p. 12. advantest.com

8 ASE Technology Holding, report on Form 6-K dated 30 July 2026. U.S. Securities and Exchange Commission, 30 July 2026. sec.gov

9 Amkor Technology, current report on Form 8-K dated 9 February 2026. U.S. Securities and Exchange Commission, 9 February 2026. sec.gov

11 ASE Technology Holding, annual report on Form 20-F for the fiscal year ended 31... U.S. Securities and Exchange Commission, 1 April 2026. sec.gov

39 Global Annual Semiconductor Sales Increase 25.6% to $791.7 Billion in 2025. Semiconductor Industry Association, 6 February 2026. semiconductors.org

3. How many of the ten largest chip assembly and test contractors are based in China?
   see The back end's geography

Three things to remember

— Chips are tested early and often because one bad chip can ruin a package of expensive ones.

— Advantest and Teradyne account for most spending on chip testers.

— In a TrendForce ranking cited by CSET, four of the top ten outsourced assembly and test firms by 2024 revenue are based in China.

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The chain

01 Chip Design, EDA and IP
02 Silicon and Wafers
03 Chemicals, Gases and Photoresist
04 Lithography
05 Photomasks and Pellicles
06 Deposition and Etch
07 Metrology and Inspection
08 Foundries and Fabs
09 Memory and HBM
10 Advanced Packaging
11 Substrates and PCBs
12 Test and Assembly
13 **Systems and Networking**
14 Data Centers and Power

15 The Economics of AI Chips
16 Who Controls the Chain

Stage 13 of 14 · High concentration · Moderately hard to substitute

Systems and Networking

Nvidia's and AMD's racks wire 72 AI chips together to run as one computer. Many firms can assemble the racks, but lasers for the optical links were scarce in 2026.

Specimen 13 · Systems and Networking

An eight-GPU baseboard, the board that carries eight accelerator modules and the switch chips joining them. It is about 510 by 400 mm and drawn to scale, with the module frames standing about 30 mm above the board. The two nickel-gray squares in the middle are the NVLink switch chips.

In plain terms

One AI chip is not enough to train a model. So seventy-two of them are wired together in a rack, a cabinet the size of a wardrobe, and run as one computer. Many such racks are then linked to train a large model. Each chip sits on its own board and draws more than a thousand amps, several times the current a whole house is wired to carry. The chips have to swap results with each other all the time. A copper wire can carry those signals only a few meters before they fade, so the longer links send them as light down glass fiber. Many companies can build the rack, but in 2026 the lasers that send the light were scarce.

In short

In March 2026 Nvidia agreed to invest $2 billion in the laser maker Lumentum, alongside a multibillion-dollar purchase commitment and rights to future capacity for advanced laser components[1]. Indium phosphide is used in high-speed lasers for optical communications, and China produced 70 percent of the world's indium in 2025[2].

On 4 February 2025 China added indium phosphide substrates, the wafers the lasers are grown on, to its export control list[3]. AXT, a US-listed maker of those wafers, does all its manufacturing in China[3]. In its filing for the quarter to June 2026, AXT said it could not estimate when it would resume shipping the wafers to the United States[3].

Before a chip package can compute, it has to be built into a machine. Building that machine is a separate industry from making the package.

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Chokepoint card

Systems and Networking

Concentration

High

Four to six firms could supply this stage today, and the top two take most of the revenue.

Substitutability

Moderate

Rated on NVLink and its NVSwitch chips, which Nvidia controls and which join the accelerators in a rack[4]. Separately, lasers for the optical links were scarce in 2026[5]. Nvidia secured future capacity rights for them from Lumentum in March 2026[1].

Price or market size

$3.1 million

for one GB200 NVL72 rack bought by a typical hyperscaler, or about $3.9 million with networking, storage and other items, by SemiAnalysis's August 2025 estimate[6]

Who leads	US	**Nvidia**	NVLink, InfiniBand and Spectrum-X Ethernet, which it credited for data center growth in the quarter to April 2026[7]
	US	**Broadcom**	$16.7 billion of AI semiconductor revenue in the quarter to August 2026, up 221 percent[8]
	TW	**Foxconn**	World's largest AI server provider, its chairman said in March 2026[9]
	CN	**InnoLight**	High-speed optical modules made in Suzhou and, since 2019, Thailand[10]
	US	**Coherent**	Lasers and optical networking products, with a $2 billion Nvidia investment agreed in March 2026[11]
Where it is made	US	**United States**	Switch and accelerator silicon, system design, indium phosphide lasers
	TW	**Taiwan**	Contract design, rack assembly and high-layer-count circuit boards
	JP	**Japan**	Low-loss board laminate
	CN	**China**	Optical module assembly and test, and most of the world's indium
	TH	**Thailand**	Chinese module makers' offshore transceiver plants
Why substitution is possible			UALink is an open specification for the same job as NVLink[12]. AMD's Helios rack, launched in July 2026, runs it over Ethernet[13]. Since May 2025 Nvidia's NVLink Fusion program has let other firms' custom chips use NVLink, which keeps those firms tied to Nvidia's interconnect[14].

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Where China stands

Chinese firms such as InnoLight assemble high-speed optical modules in volume, at sites including Suzhou and, since 2019, Thailand [10]. In February 2025 China put indium under new export restrictions [2]. The same month it began requiring export permits for indium phosphide substrates, the wafers that lasers for optical links are grown on [3].

Where the US stands

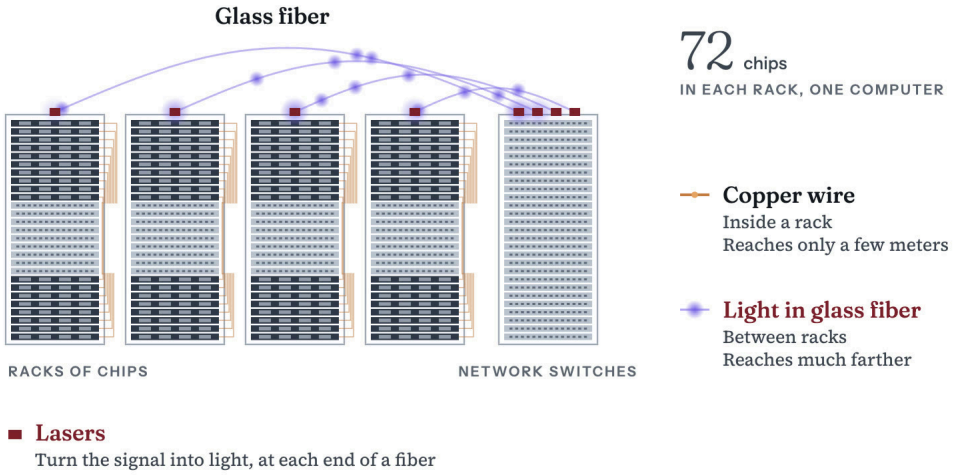
Coherent broke ground in Sherman, Texas, in June 2026 on an expansion of its factory there, which makes optical devices from indium phosphide [15]. In August 2026 Nvidia said that Wistron's new plant in Fort Worth, Texas, was producing its GB300 superchips and that Foxconn was building a factory in Houston [16].

01

How it works

Figure 13.1 · Explainer, 20 seconds

How AI chips are wired together



1. **One rack.** One AI chip is not enough to train a model. So 72 of them are wired together in one rack, a cabinet the size of a wardrobe, and run as one computer.
2. **Copper inside.** Inside the rack, copper wires link each tray of chips to the switch trays, which pass data between any two chips. Copper carries a signal only a few meters.
3. **Light between.** Training a large model takes many racks, most of them too far apart for copper. The links between them carry the signal as flashes of light in glass fiber.
4. **Lasers.** At each end of a fiber, a laser turns the electrical signal into flashes of light, and a detector turns them back. That lets thousands of chips in many racks work on one model.

Note: Simplified. The trays follow the layout of an Nvidia NVL72 rack. Racks and fibers are not drawn to scale.

An accelerator (the AI chip) ships as a module, the chip package on a small board with the circuits that feed it power. Nvidia calls its version SXM, and the open-standards equivalent is the Open Compute Project's accelerator module. Eight modules bolt onto a baseboard that carries their power and the links between them. On Nvidia's baseboard, switch chips pass data between any two modules at full speed.

2 Mineral Commodity Summaries 2026: Indium. U.S. Geological Survey, 5 February 2026. pubs.usgs.gov

3 AXT, quarterly report on Form 10-Q for the quarter ended 30 June 2026. U.S. Securities and Exchange Commission, 13 August 2026. sec.gov

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15 Coherent Announces a CHIPS Letter of Intent for $50 Million to Expand... Coherent, 16 June 2026. coherent.com

16 NVIDIA and Partners Build in America, for America. Nvidia, 5 August 2026. blogs.nvidia.com

Rack-scale systems drop the baseboard and put the chips on trays stacked in the rack. An Nvidia GB300 NVL72 holds 18 compute trays plus nine switch trays, 72 GPUs in all, and draws up to 142 kW[17]. The newer Vera Rubin NVL72 uses the same rack footprint[18]. The research firm SemiAnalysis estimated in August 2025 that a rack of the earlier GB200 NVL72 model cost a typical hyperscaler $3.1 million, or about $3.9 million with networking, storage and other items[6].

Inside the rack the trays reach each other over copper[17]. Between neighboring racks, the signal can travel up to 7 meters over active electrical cables, a type of copper cable sold by firms such as Credo[19]. Longer links run on glass fiber, with an optical transceiver at each end. A laser in the sending transceiver turns the electrical signal into pulses of light, and a detector in the receiving one turns the pulses back into current.

The accelerators in an AI cluster exchange data over two networks that are not interchangeable. Scale-up connects anywhere from a few dozen accelerators to thousands closely enough that they can read each other's memory. It runs at terabytes per second. Scale-out joins those groups to one another at hundreds of gigabits per second. Scale-up speeds are quoted in bytes and scale-out speeds in bits, and a byte is eight bits. Each GPU in a GB300 NVL72 can send 900 gigabytes per second each way over the scale-up network, or 7,200 gigabits per second[20]. Over the scale-out network it can send 800 gigabits per second[20]. So each GPU has nine times as much bandwidth for scale-up as for scale-out, SemiAnalysis calculated in January 2026[20].

Scale-up bandwidth limits how efficiently a large model trains. A model too big for one accelerator is split across many. The parts that exchange data constantly stay inside the scale-up group, because they need its high bandwidth[21]. Whole blocks of layers, the stages a model's calculation passes through in turn, can sit on different groups across the slower scale-out network[21].

02

Variants and trade-offs

Scale-up: NVLink, UALink and Google's torus

The designs differ in how many accelerators can share one pool of memory.

- **NVLink and NVSwitch.** Nvidia's NVLink bandwidth per GPU, counting both directions, went from 900 GB/s in NVLink 4 to 1,800 GB/s in NVLink 5[22]. Across a whole Blackwell NVL72 rack, NVLink moves 130 TB/s[22]. Nvidia's Rubin launch release of January 2026 gives NVLink 6 as 3,600 GB/s per GPU and 260 TB/s for a Vera

6 H100 vs GB200 NVL72 Training Benchmarks - Power, TCO, and Reliability Analysis... SemiAnalysis, 20 August 2025. newsletter.semianalysis.com

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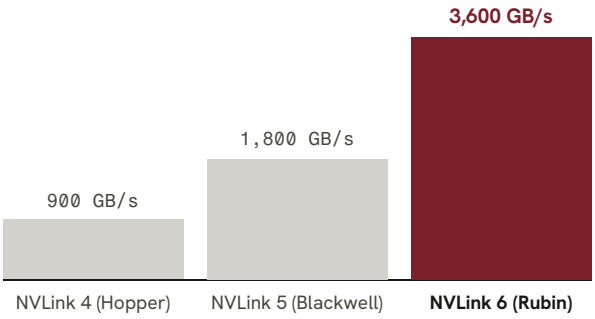
Rubin NVL72 rack[23]. As of October 2026, Nvidia's NVLink product page still lists preliminary figures of 3,000 GB/s and 216 TB/s[22].

- **UALink.** The open specification, from the UALink Consortium, runs 200 Gb/s per lane, and four lanes, which it calls a station, carry 800 Gb/s[12]. It can address up to 1,024 endpoints, such as accelerators, in one group[12]. AMD's Helios rack, launched in July 2026, joins 72 GPUs and runs UALink over Ethernet for its scale-up network[13].
- **Google's 3D torus.** Google wires 64 of its Ironwood TPU accelerators into each rack, called a cube[24]. Inside the cube each chip links directly to six others, in a three-dimensional grid that Google calls a 3D torus[24]. Optical circuit switches, which steer light from one fiber to another, join cubes into superpods of 9,216 chips and can route around a failed cube or link[24].

Figure 13.2

NVLink per GPU rose from 900 to 3,600 GB/s, Hopper to Rubin

NVLink bandwidth per GPU by generation



Source: Nvidia NVLink product page for NVLink 4 and 5, and Nvidia's Rubin launch release of January 2026 for NVLink 6 [23]

Scale-out: InfiniBand versus Ethernet

Scale-out networks typically use InfiniBand or Ethernet, SemiAnalysis wrote in January 2026[20]. InfiniBand is a specialized network long used in supercomputers. But Ethernet switch capacity has grown quickly. Broadcom's Tomahawk 4 switched 25.6 Tbps on one chip in December 2019[25]. Tomahawk 5 reached 51.2 Tbps in August 2022, and Broadcom said it had doubled the bandwidth about every two years since 2014[26]. Tomahawk 6 doubled it again to 102.4 Tbps in June 2025[27]. In June 2025 the Ultra Ethernet Consortium published its 1.0 specification, a set of Ethernet-based rules for communication in AI and supercomputing[28]. It covers how data is carried, how traffic jams are avoided and how one machine reads or writes another's memory directly[28].

12 UALink™ 200G 1.0 Specification Overview. UALink Consortium, 15 May 2025. ualinkconsortium.org

13 AAI 2026: AMD Launches AMD Helios Rackscale Solution for Frontier AI. AMD Newsroom, 23 July 2026. newsroom.amd.com

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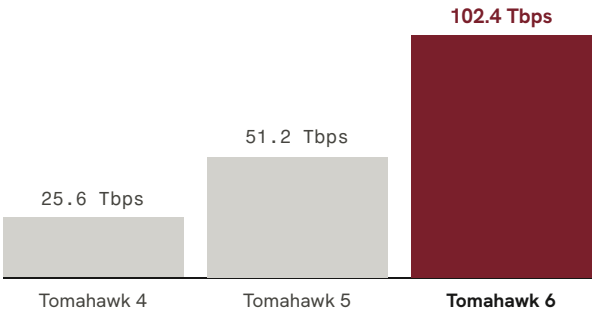
27 Broadcom Ships Tomahawk 6: World's First 102.4 Tbps Switch. Broadcom, 3 June 2025. investors.broadcom.com

28 Ultra Ethernet Consortium (UEC) Launches Specification 1.0 Transforming... Ultra Ethernet Consortium, 11 June 2025. ultraethernet.org

Figure 13.3

Tomahawk switch chip capacity rose from 25.6 Tbps to 102.4 Tbps

Broadcom Tomahawk switching capacity on one chip



Source: Broadcom product releases for Tomahawk 4, 5 and 6 [27]

Nvidia sells switches and network cards for both InfiniBand and Ethernet. Its data center revenue, which counts its GPUs as well as its networking gear, reached $89.0 billion in the quarter to July 2026, up 117 percent from a year earlier[29]. In the quarter before, it credited its data center growth to Blackwell 300 products and to demand for its "InfiniBand, Spectrum-X Ethernet, and NVLink" solutions[7].

Optics: pluggable, then co-packaged

Broadcom and Nvidia are moving the optics out of pluggable transceivers and into the switch chip's own package, a design called co-packaged optics. A cluster needs several transceivers, the modules that turn electricity into light and back, for each accelerator. Their power cost is one reason for the change. Broadcom says its Tomahawk 6 Davisson cuts the power used by optical links by 70 percent compared with traditional pluggable modules[30]. It keeps its lasers in field-replaceable modules[30]. Nvidia's Spectrum-X Photonics switches, announced in March 2025, have up to 512 ports of 800 gigabits per second each, and Nvidia says they use four times fewer lasers than traditional designs[31]. Nvidia builds them on its Spectrum-6 switch design[23]. In August 2026 Nvidia said its Spectrum-6 switch systems, with pluggable or co-packaged optics, were arriving at AI data centers around the world[32].

03

Who makes it

Chinese makers of optical modules have built plants outside China. InnoLight has run a plant in Thailand since 2019, alongside its sites in Suzhou and Taiwan[10]. The Thai

7 Nvidia, quarterly report on Form 10-Q for the quarter ended 26 April 2026. U.S. Securities and Exchange Commission, 20 May 2026. sec.gov

10 InnoLight Technology, company overview page. InnoLight Technology, accessed 9 September 2026. innolight.com

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32 NVIDIA Announces Financial Results for Second Quarter Fiscal 2027. Nvidia, 26 August 2026. nvidianews.nvidia.com

subsidiary of Eoptolink, another Chinese module maker, lists sites in Chonburi and Rayong[33].

A US rule from July 2026 may reach optical transceivers made by firms on a US security list. The list is the Federal Communications Commission's Covered List of equipment and services judged to pose an unacceptable risk to US national security or to the security and safety of US persons[34]. Under its July 2026 order, the FCC no longer approves new devices for the US market if they contain logic-bearing hardware components made by firms on the list[35]. The order counts as logic-bearing any part that uses digital logic, from a single chip to a whole module, and leaves out parts with no digital function, such as wiring and housings[36]. Commenters in the record named optical transceivers as one such component[37]. As of 16 October 2026, the Covered List did not include InnoLight[38].

China produces most of the world's indium, the raw material for indium phosphide lasers, and restricts its export. Indium phosphide is used in the high-speed laser diodes and photodetectors for optical communications[2]. China accounted for 70 percent of world indium production in 2025[2]. In February 2025 China's Ministry of Commerce placed indium under new export restrictions[2].

AI data centers were straining supplies of the lasers inside optical modules and pushing up their prices, Nikkei Asia reported in May 2026[5]. Nvidia had agreed in March 2026 to invest $2 billion in one laser maker, Lumentum, alongside a multibillion-dollar purchase commitment and rights to future capacity for advanced laser components[1]. Coherent, another laser maker, broke ground in Sherman, Texas, in June 2026 on an expansion of its factory there, which makes optical devices from indium phosphide[15].

Two American firms, Broadcom and Marvell, sell both custom AI chips and Ethernet switch chips. In its annual report for fiscal 2025, Broadcom said its AI chips include Ethernet switch chips as well as custom accelerators, which it builds to each customer's specifications[39]. Its AI semiconductor revenue reached $16.7 billion in the third quarter of fiscal 2026, up 221 percent from a year earlier[8]. Marvell also builds custom AI chips to each customer's specifications and sells Ethernet switch chips, its annual report for fiscal 2026 says[40]. Marvell's five-year agreement with Amazon Web

1 NVIDIA Announces Strategic Partnership With Lumentum to Develop... Nvidia, 2 March 2026. nvidianews.nvidia.com

2 Mineral Commodity Summaries 2026: Indium. U.S. Geological Survey, 5 February 2026. pubs.usgs.gov

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40 Marvell Technology, annual report on Form 10-K for the fiscal year ended 31... U.S. Securities and Exchange Commission, 11 March 2026. sec.gov

Services, announced in December 2024, covers custom AI chips alongside the optical and networking parts that go with them[41].

Taiwanese firms do much of the physical assembly. Foxconn booked NT$8.1 trillion of revenue in 2025, up 18 percent, and cloud and networking was its largest product category in the fourth quarter[9]. In the same results release, its chairman called Foxconn the world's largest AI server provider[9].

Some assembly is moving to the United States. In April 2025 Nvidia said it was building supercomputer manufacturing plants in Texas, with Foxconn in Houston and Wistron in Dallas[42]. It expected mass production to ramp up in the following 12 to 15 months[42]. In August 2026 Nvidia said Wistron's new plant in Fort Worth, Texas, near Dallas, was producing its GB300 superchips, which pair Nvidia's Grace processor with its Blackwell Ultra GPUs[16]. It said Foxconn was building a factory in Houston to make Nvidia's AI systems[16].

04

The chokepoint

Many firms can build the racks. In October 2025 Nvidia said more than 50 makers of systems and components were gearing up for the Vera Rubin NVL72, which follows MGX, Nvidia's standard design for trays and racks[18]. But in its filing for the quarter to July 2026, Nvidia said the complexity of producing its data center systems had caused production delays[29].

Four inputs to the rack are harder to source elsewhere.

- **Indium phosphide laser capacity.** Nvidia has secured rights to future capacity from laser makers. Alongside its Lumentum deal, in March 2026 Nvidia agreed to invest $2 billion in Coherent[11]. The deal also includes a multibillion-dollar purchase commitment and rights to future access and capacity for Coherent's advanced laser and optical networking products[11].
- **Indium phosphide substrates.** China added indium phosphide substrates, the wafers the lasers are grown on, to its export control list on 4 February 2025[3]. AXT, a US-listed maker of these wafers, does all its manufacturing in China[3]. In its filing for the quarter to June 2026, AXT called the need for Chinese export permits the most significant challenge it faces[3]. It said it could not estimate when it would resume shipping the wafers to the United States[3]. The same filing names Sumitomo and JX, both Japanese firms, as AXT's competitors in indium phosphide[3].

3 AXT, quarterly report on Form 10-Q for the quarter ended 30 June 2026. U.S. Securities and Exchange Commission, 13 August 2026. sec.gov

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42 NVIDIA to Manufacture American-Made AI Supercomputers in US for First Time. Nvidia, 14 April 2025. blogs.nvidia.com

- **NVSwitch.** Nvidia's annual report for fiscal 2026 lists NVLink interconnects and switches among its own networking products[4]. UALink is an open standard designed for accelerator communications within one scale-up group[12]. Since May 2025 Nvidia's NVLink Fusion program has let other firms' custom chips use NVLink, including CPUs that Fujitsu and Qualcomm plan to connect to Nvidia GPUs[14].
- **High-layer-count circuit boards.** TTM Technologies, an American circuit board maker, says its circuit board competitors are mostly based in China and Taiwan[43]. Nitto Boseki, a Japanese glass fiber maker, makes NE-glass, a glass cloth that absorbs less signal, for the boards in data center switches and routers[44]. See Substrates and PCBs.

Optical module assembly can move, as InnoLight's high-volume plant in Thailand, opened in 2019, shows[10].

Four questions to answer from memory

1. What is an AI rack? `see How it works`
2. Why do the longer links between AI chips carry signals as light? `see How it works`
3. Which part of an AI rack was scarce in 2026? `see The chokepoint`
4. How much of the world's indium did China produce in 2025? `see The chokepoint`

Four things to remember

— An AI rack wires 72 chips together to run as one computer.

— Almost every link longer than 7 meters carries its signal as light, because copper fades over longer runs.

— Nvidia's NVLink switch chips and the lasers are the hardest parts to replace. In October 2025 more than 50 makers of systems and components were gearing up to build Nvidia's next rack.

— China produced about 70 percent of the world's indium in 2025 and has required export permits for indium phosphide substrates since February 2025.

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The chain

01 Chip Design, EDA and IP
02 Silicon and Wafers
03 Chemicals, Gases and Photoresist
04 Lithography
05 Photomasks and Pellicles
06 Deposition and Etch
07 Metrology and Inspection
08 Foundries and Fabs
09 Memory and HBM
10 Advanced Packaging
11 Substrates and PCBs
12 Test and Assembly
13 Systems and Networking
14 Data Centers and Power

15 The Economics of AI Chips
16 Who Controls the Chain

Stage 14 of 14 · High concentration · Moderately hard to substitute

Data Centers and Power

SemiAnalysis forecast in October 2026 that 50 GW of data center capacity would be added in 2027, about as much as the world had built through 2024. The large transformers that connect such sites to the grid commonly took three years to arrive in 2024.

Specimen 14 · Data Centers and Power

A liquid-cooled rack of 72 GPUs, Nvidia's GB200 NVL72, drawn to scale. It is 2.24 m tall and 600 mm wide, taller than a door. It weighs about 1.36 metric tons, more than a small car, and draws about 120 kW. The newer GB300 NVL72 draws up to 142 kW. The gray steel power shelves and the copper busbar they feed are picked out.

In plain terms

A data center feeds electricity to the machines inside it and carries away the heat they give off. One rack of AI chips, a cabinet taller than a door, can draw 142 kilowatts[1]. Every watt of it comes back out as heat, and liquid piped through the rack carries the heat away[1]. A large training site already holds more than a hundred thousand chips[2]. The newest sites are planned in gigawatts[3]. One gigawatt is the whole output of a large power station, and Epoch AI projects that OpenAI's seven Stargate sites in the United States will add up to more than nine of them by 2029[3]. Getting that much electricity to the site is harder than putting up the building. Transformers and heavy switches connect a site to the grid, and turbines make the power. All of them come from a few suppliers and take years to arrive.

In short

Measured by capacity, US factories supplied 29.2 percent of the large power transformers bought in the US in 2023, and imports the rest[4]. GE Vernova's gas equipment backlog and reserved factory slots stood at 116 GW in mid-2026, against planned output of 20 GW a year from the third quarter[5].

ERCOT, the grid operator for most of Texas, counted 9,456 MW of big power users approved to energize in its overview for September 2026[6]. Judging by

1 System Hardware & Components. Nvidia, accessed 9 September 2026. docs.nvidia.com
2 Colossus 1, AI data centers directory. Epoch AI, accessed 9 September 2026. epoch.ai
3 OpenAI Stargate: where the US sites stand. Epoch AI, 17 April 2026. epoch.ai
4 Large Power Transformers from South Korea, Investigation No. 731-TA-1189... U.S. International Trade Commission, August 2024, p. 90. usitc.gov
5 GE Vernova, current report on Form 8-K dated 22 July 2026. U.S. Securities and Exchange Commission, 22 July 2026. sec.gov
6 Electric Reliability Council of Texas, ERCOT Monthly Operational Overview... Electric Reliability Council of Texas, 16 October 2026, p. 9. ercot.com

peak monthly use in September, it believed 4,132 MW of that was operating[6]. On 3 August 2026 Governor Greg Abbott ordered an audit of every data center seeking a connection before any data center project moves forward[7]. In September 2026 ERCOT said it had paused approvals to energize data centers of 75 MW or more until the audit and a review of their impact on local communities are complete[8].

A wafer passes through a leading-edge fab in months, but in 2024 a large power transformer commonly took three years to arrive[9]. Before the pandemic, the same order took under a year[10].

Chokepoint card Data Centers and Power

Concentration

High

Four to six firms could supply this stage today, and the top two take most of the revenue.

Substitutability

Moderate

Nvidia named nine data center power system providers in October 2025 for its 800-volt designs for gigawatt-scale data centers[11].

Price or market size

$580 billion

of data center investment expected worldwide in 2025, in the International Energy Agency's November 2025 outlook[12]

Who leads	US	**GE Vernova**	Gas equipment, transformers and switchgear, including through Prolec GE[13]
	CH	**Hitachi Energy**	Power transformers and switchgear[13]
	DE	**Siemens Energy**	Grid technologies, transformers, turbines[13]
	JP	**Mitsubishi Power**	Power generation equipment, including gas turbines[13]
	US	**Vertiv**	Data center power systems[11]
	FR	**Schneider Electric**	Cooling, through its Motivair division[14]
Where it is made	US	**United States**	The largest data center operators, and GE Vernova turbines and grid equipment
	CH	**Switzerland**	Hitachi Energy transformers and switchgear, and ABB switchgear
	DE	**Germany**	Siemens Energy grid technologies and turbines
	JP	**Japan**	Mitsubishi Power equipment, including gas turbines
Why substitution is possible	Buyers of grid equipment can choose among suppliers, and GE Vernova's 2025 annual report names Hitachi Energy, Siemens Energy, Schneider Electric and ABB among its key competitors[13]. But in the United States in 2024 the longest large power transformer lead times reached 60 months[9]. An International Energy Agency survey published in 2025 found waits of up to four years worldwide[15].		

6 Electric Reliability Council of Texas, ERCOT Monthly Operational Overview... Electric Reliability Council of Texas, 16 October 2026, p. 9. ercot.com

7 Governor Abbott Directs Comprehensive Data Center Audit. Office of the Texas Governor, 3 August 2026. gov.texas.gov

8 Electric Reliability Council of Texas, Item 14: Batch Zero Update, Board of... Electric Reliability Council of Texas, 11 September 2026. ercot.com

9 U.S. Department of Energy Large Power Transformer Resilience Report to... U.S. Department of Energy, 10 July 2024, p. 8. energy.gov

10 U.S. Department of Energy Large Power Transformer Resilience Report to... U.S. Department of Energy, 10 July 2024, p. 18. energy.gov

11 NVIDIA, Partners Drive Next-Gen Efficient Gigawatt AI Factories in Buildup for... Nvidia, 13 October 2025. blogs.nvidia.com

12 Executive summary – World Energy Outlook 2025. International Energy Agency, 12 November 2025. iea.org

13 GE Vernova, annual report on Form 10-K for the fiscal year ended 31 December... U.S. Securities and Exchange Commission, 29 January 2026. sec.gov

14 Hotter Than a Hot Tub: The 45°C Breakthrough to Cool AI's Biggest Machines. Nvidia, 22 June 2026. blogs.nvidia.com

15 Executive summary – Building the Future Transmission Grid. International Energy Agency, 2025. iea.org

Where China stands

China used 25 percent of the world's data center electricity in 2024, second to the United States, by the International Energy Agency's estimate [16]. Data centers made up around 3 percent of the growth in China's electricity demand from 2022 to 2024 [17].

Where the US stands

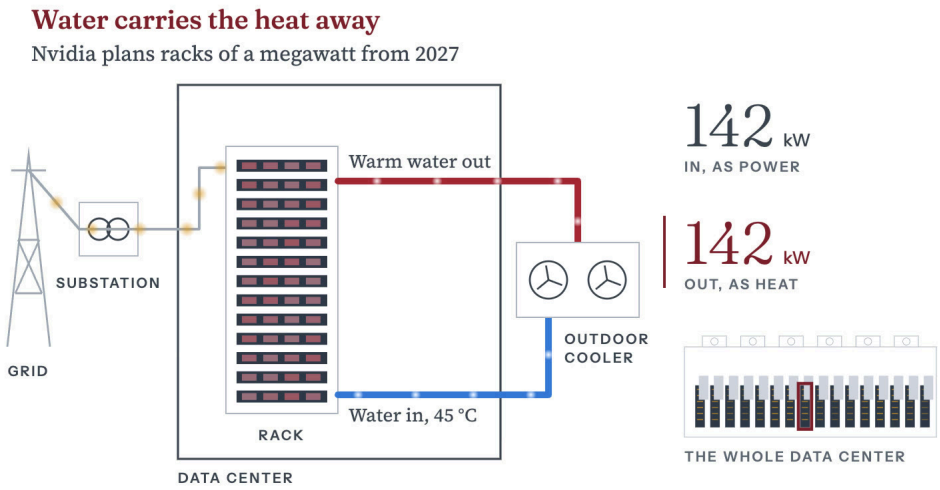
US factories supplied 29.2 percent of the large power transformers bought in the United States in 2023, measured by capacity [4]. Constellation said in August 2026 that it expects to restart the Crane nuclear plant in 2027 [18]. Microsoft has agreed to buy the plant's power [19].

01

How it works

Figure 14.1 · Explainer, 21 seconds

Power in, heat out



1. **The site.** Data centers are where AI chips go to work. A large training site holds a hundred thousand of them, in rows of racks.
2. **Power in.** Power comes in from the grid, stepped down by a substation, and runs through the data center to every rack. One rack of AI chips can draw 142 kilowatts.
3. **Heat.** Every watt of it comes back out as heat, more than air can carry away.
4. **Water out.** So water runs through plates on the chips, carries the heat to coolers outdoors and comes back. Nvidia plans for racks of a megawatt each from 2027, so the power coming in and the heat going out keep growing.

Note: Simplified diagram, with illustrative sizes. An Nvidia GB300 NVL72 rack draws at most 142 kW. Nvidia gives 45 °C as the supply-water temperature for its Vera Rubin racks. The megawatt racks from 2027 are Nvidia's plan.

The rack is the basic unit of AI compute. Nvidia's GB300 NVL72 rack draws up to 142 kW through eight 33 kW power shelves, the rack's own power supplies [1]. Large sites hold

1 System Hardware & Components. Nvidia, accessed 9 September 2026. docs.nvidia.com

4 Large Power Transformers from South Korea, Investigation No. 731-TA-1189... U.S. International Trade Commission, August 2024, p. 90. usitc.gov

16 Executive summary – Energy and AI. International Energy Agency, 10 April 2025. iea.org

17 Executive summary – Electricity 2025. International Energy Agency, 14 February 2025. iea.org

18 Constellation Reports Second Quarter 2026 Results. Constellation Energy, 6 August 2026. constellationenergy.com

19 Constellation Ahead of Schedule for Launch of Crane Clean Energy Center. Constellation Energy, 19 February 2025. constellationenergy.com

many such racks. Colossus 1, an AI data center in Memphis, Tennessee, held an estimated 230,000 Nvidia GPUs by July 2025[2].

Power reaches the rack through a chain of heavy equipment, and each piece has its own lead time, the wait between order and delivery. Between the grid and the rack, a substation steps the high voltage down with transformers. Switchgear, the building's heavy circuit breakers, can cut any part of the site off in a fault. Battery-backed supplies keep the racks running for the seconds it takes backup generators to start. Distribution boards split the feed among the rows of racks.

At the back of each rack, a busbar carries the current down to the server trays. It is a solid copper bar, because an ordinary cable would overheat at that current.

Heat leaves in water. Cold plates, metal blocks with coolant running through them, sit on the GPU packages and pass the heat to the building's water loop, which releases it outdoors[14]. Nvidia's Rubin racks run coolant at up to 45 °C, warm enough that in many climates the loop can release its heat without turning on chillers, the refrigeration machines that cool water[14].

Nvidia is also moving the power that feeds its racks to 800 volts of direct current. Raising the voltage cuts the current needed for the same power, and the heat in copper comes from current. Nvidia said in October 2025 that at 800 volts the same wire can carry 157 percent more power than at 415 volts of alternating current[20]. Inside racks such as the GB300 NVL72, power runs at 54 volts, and at that voltage a one-megawatt rack would need up to 200 kg of copper busbar[21].

02

Variants and trade-offs

Cooling

- **Air.** Heat exchangers in the rack door work when density, the power each rack draws, is moderate, but Nvidia's GB300 NVL72 rack is liquid cooled[1].
- **Direct-to-chip liquid.** Nvidia's Rubin platform is 100 percent liquid cooled, and Nvidia says every cloud provider and data center operator building for it is moving to liquid[14].
- **Immersion.** The servers sit in a bath of fluid, which makes them hard to service.

Nvidia says the cooling supply chain is keeping pace with the move to liquid[14]. Motivair, Schneider Electric's cooling division, has worked with Nvidia for nearly a decade, following Nvidia's product roadmap[14].

Rack power

Nvidia's 800-volt DC architecture is designed for racks from 100 kW to more than a megawatt, with megawatt racks from 2027[21]. Nvidia is making the change because the

1 System Hardware & Components. Nvidia, accessed 9 September 2026. docs.nvidia.com
2 Colossus 1, AI data centers directory. Epoch AI, accessed 9 September 2026. epoch.ai
14 Hotter Than a Hot Tub: The 45°C Breakthrough to Cool AI's Biggest Machines. Nvidia, 22 June 2026. blogs.nvidia.com
20 Building the 800 VDC Ecosystem for Efficient, Scalable AI Factories. Nvidia, 13 October 2025. developer.nvidia.com
21 NVIDIA 800 VDC Architecture Will Power the Next Generation of AI Factories. Nvidia, 20 May 2025. developer.nvidia.com

54-volt distribution in existing racks begins to hit physical limits once racks exceed 200 kW[21].

Nvidia names nine data center power system providers for its 800-volt designs for gigawatt-scale data centers, among them ABB, Eaton, Hitachi Energy, Mitsubishi Electric, Schneider Electric, Siemens and Vertiv[11].

A megawatt rack changes the busbar, the floor loading, the coolant flow, the leak detection and the energy stored in the rack. Nvidia's Vera Rubin NVL72 rack design adds energy storage to keep power steady[11].

Where the electrons come from

- **Grid interconnection.** The slow path in the United States, where the queues to connect are long[3].
- **Behind-the-meter gas**, generation on the site itself. The fast path, which at least three of OpenAI's seven US Stargate sites will use to avoid grid queues[3].
- **Nuclear.** Existing plants already sell power to data centers[22]. Talen Energy supplies Amazon's data center campus next to the Susquehanna nuclear plant in Pennsylvania under a June 2025 contract expected to reach 1,920 MW no later than 2032[22]. Restarts and new reactors are still to come. Constellation signed a power purchase agreement with Microsoft to restart Three Mile Island Unit 1 as the 835 MW Crane Clean Energy Center[19]. The plant shut in 2019[23]. A $1 billion federal loan that closed in November 2025 partly finances its restart, which is subject to licensing approvals from the US Nuclear Regulatory Commission[23].

03

Who makes it

The largest data center operators pay for the buildout. Capital spending by Alphabet, Amazon, Meta, Microsoft and Oracle came to $448.2 billion in 2025, the sum of quarterly figures compiled by Epoch AI, a research group[24]. If the trend continues, it will reach $770 billion in 2026[24]. The International Energy Agency expected worldwide investment in data centers to reach $580 billion in 2025[12].

3 OpenAI Stargate: where the US sites stand. Epoch AI, 17 April 2026. epoch.ai

11 NVIDIA, Partners Drive Next-Gen Efficient Gigawatt AI Factories in Buildup for... Nvidia, 13 October 2025. blogs.nvidia.com

12 Executive summary – World Energy Outlook 2025. International Energy Agency, 12 November 2025. iea.org

19 Constellation Ahead of Schedule for Launch of Crane Clean Energy Center. Constellation Energy, 19 February 2025. constellationenergy.com

21 NVIDIA 800 VDC Architecture Will Power the Next Generation of AI Factories. Nvidia, 20 May 2025. developer.nvidia.com

22 Talen Energy, current report on Form 8-K dated 11 June 2025. U.S. Securities and Exchange Commission, 11 June 2025. sec.gov

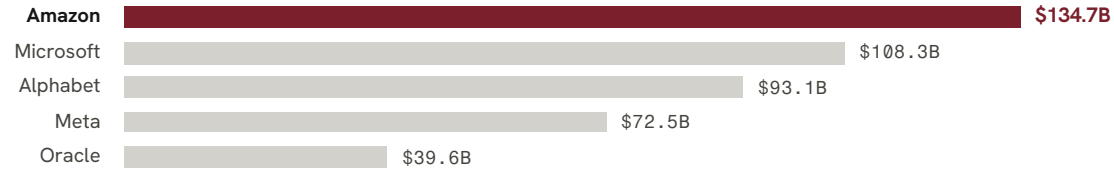
23 Energy Department Closes Loan to Restart Nuclear Power Plant in Pennsylvania. U.S. Department of Energy, 17 November 2025. energy.gov

24 Hyperscaler capex has quadrupled since GPT-4's release. Epoch AI, 26 February 2026. epoch.ai

Figure 14.2

Amazon spent the most of five large operators, $134.7 billion in 2025

Capital spending by five large US data center operators, 2025



Source: Epoch AI, hyperscaler capex trend, February 2026, quarterly figures from company filings summed for 2025 [24]

Epoch AI also projects that OpenAI's seven US Stargate sites will exceed 9 GW by 2029[3]. It expects the site in Abilene, Texas, to grow from an estimated 0.3 GW in April 2026 to 1.2 GW in the fourth quarter of 2026[3].

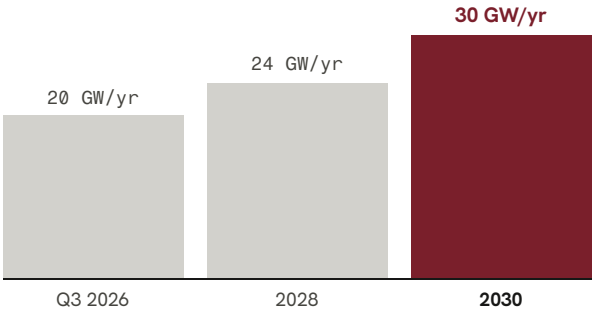
A few large firms make the turbines, transformers and switchgear. GE Vernova builds large gas turbines, and its 2025 annual report names Siemens Energy and Mitsubishi Power among the key competitors of its power generation business[13]. Its electrification business makes power transformers and switchgear, and the report names Hitachi Energy, Siemens Energy, Schneider Electric and ABB among that business's key competitors[13].

Equipment makers limit how fast the buildout can go. GE Vernova's gas equipment backlog and reserved factory slots grew from 100 GW to 116 GW in the second quarter of 2026[5]. In the first half of 2026, data center orders for its electrification business, which makes grid equipment, passed $5 billion, more than double that business's data center orders for all of 2025[5]. GE Vernova bought the remaining half of Prolec GE, a transformer maker, in the first half of 2026[5].

Figure 14.3

GE Vernova plans 30 GW of annual gas turbine output

GE Vernova's planned annual gas turbine output



Source: GE Vernova second-quarter 2026 results, July 2026 [5]

3 OpenAI Stargate: where the US sites stand. Epoch AI, 17 April 2026. epoch.ai

5 GE Vernova, current report on Form 8-K dated 22 July 2026. U.S. Securities and Exchange Commission, 22 July 2026. sec.gov

13 GE Vernova, annual report on Form 10-K for the fiscal year ended 31 December... U.S. Securities and Exchange Commission, 29 January 2026. sec.gov

24 Hyperscaler capex has quadrupled since GPT-4's release. Epoch AI, 26 February 2026. epoch.ai

04

Capacity and demand

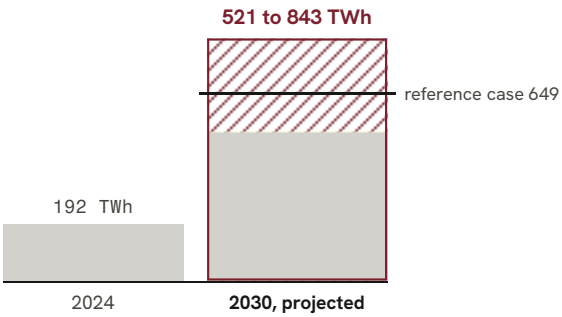
US data center electricity use is projected to rise steeply by 2030. US data centers used 192 TWh in 2024, 4.7 percent of national electricity, according to a June 2026 estimate by Lawrence Berkeley National Laboratory that leaves out cryptocurrency mining[25]. The lab's reference case, its main scenario, projects 464 TWh in 2028 and 649 TWh in 2030, which would be 11.8 percent of forecast US electricity use[25]. Its scenarios for 2030 range from 521 to 843 TWh, or 9.5 percent to 15.3 percent[26].

The US Energy Information Administration expects electricity demand to keep rising, but in September 2026 it cut its 2027 forecast for the region that includes Texas. In its October 2026 Short-Term Energy Outlook, it expects US electricity sales of 4,152.0 TWh in 2026 and 4,216.6 TWh in 2027[27]. It expects electricity demand from the commercial sector, which includes data centers, to grow 2.8 percent in 2027 and demand from industry to grow 2.7 percent[28]. Its September outlook had lowered the 2027 electricity sales forecast for the region that includes Texas by 4.7 percent from August[29]. That outlook noted that Texas had paused connecting new data centers to the grid while the state audits proposed projects[30]. The pause, set by the grid operator for most of Texas, covers data centers and crypto mining sites of 75 MW or more[8].

Figure 14.4

US data centers' projected 2030 use ranges from 521 to 843 TWh

US data center electricity use, 2024 and 2030 projection range



Source: Lawrence Berkeley National Laboratory, June 2026 [26]

By the International Energy Agency's estimate, China accounted for 25 percent of the world's data center electricity use in 2024, second to the United States at 45 percent[16]. Data centers made up around 3 percent of the growth in China's electricity demand from 2022 to 2024, the agency found[17].

8 Electric Reliability Council of Texas, Item 14: Batch Zero Update, Board of... Electric Reliability Council of Texas, 11 September 2026. ercot.com

16 Executive summary – Energy and AI. International Energy Agency, 10 April 2025. iea.org

17 Executive summary – Electricity 2025. International Energy Agency, 14 February 2025. iea.org

25 United States Data Center Energy Usage Report: 2025 Update. Lawrence Berkeley National Laboratory, 18 June 2026, p. 10. energy.gov

26 United States Data Center Energy Usage Report: 2025 Update. Lawrence Berkeley National Laboratory, 18 June 2026, p. 11. energy.gov

27 Short-Term Energy Outlook, October 2026. U.S. Energy Information Administration, 6 October 2026, p. 46. eia.gov

28 Short-Term Energy Outlook, October 2026. U.S. Energy Information Administration, 6 October 2026, p. 12. eia.gov

29 Short-Term Energy Outlook, September 2026. U.S. Energy Information Administration, 8 September 2026, p. 4. eia.gov

30 Short-Term Energy Outlook, September 2026. U.S. Energy Information Administration, 8 September 2026, p. 10. eia.gov

SemiAnalysis, a research firm, forecast in October 2026 that 50 GW of data center capacity would be added in 2027, which it said was about as much as the world's data center market had built in total through 2024[31]. Just 40 campuses in an advanced stage of construction held more than 24 GW, most of it to be delivered in 2027, it found[31].

Better chips do not by themselves cut total energy use. Google says its Ironwood TPU, an AI chip, delivers twice the performance per watt of Trillium, the TPU before it[32]. A token is the chunk of text a model reads or writes. Energy per token can fall while total energy use still rises, if the number of tokens rises faster.

05

The chokepoint

On the Texas grid, big power users have asked for far more electricity than is connected and running. ERCOT, the grid operator for most of Texas, was tracking about 474 GW of large loads (big power users) seeking a connection in June 2026, roughly 90 percent of it data centers[33]. In its operational overview for September 2026, ERCOT counted 9,456 MW of large loads approved to energize, that is, cleared to switch on[6]. Judging by peak monthly use in September, it believed only 4,132 MW of it was operating[6].

Before a big user can connect, engineers must study whether the nearby grid can carry its load. In a preliminary review in July 2026, ERCOT found that about 205 GW of the requests already had such a study, which made them eligible for Batch Zero, the first batch of requests it studies together[34]. In September 2026 ERCOT conditionally classed 66.4 GW of Batch Zero projects as base load and 127.9 GW as studied load, pending final decisions in December[8]. Under ERCOT's Batch Zero rules, base load is demand from projects already running or far enough along to be counted without further study[35]. Studied load is demand that ERCOT will study before giving each project a share of the grid's capacity[35].

Texas has paused new connections for large data centers while it audits them. On 3 August 2026 Texas Governor Greg Abbott ordered the state's Public Utility Commission and ERCOT to audit every data center waiting to connect to the grid before any data center project moves forward[7]. ERCOT then paused approvals to energize data centers and crypto mining sites of 75 MW or more until the audit and a review of their impact on local communities are complete[8]. It will report on the audit and the community review to the Public Utility Commission by 10 December 2026[8].

Projects in that queue also wait on equipment.

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6 Electric Reliability Council of Texas, ERCOT Monthly Operational Overview... Electric Reliability Council of Texas, 16 October 2026, p. 9. ercot.com

7 Governor Abbott Directs Comprehensive Data Center Audit. Office of the Texas Governor, 3 August 2026. gov.texas.gov

8 Electric Reliability Council of Texas, Item 14: Batch Zero Update, Board of... Electric Reliability Council of Texas, 11 September 2026. ercot.com

31 Datacenters 2027: The Year of Acceleration, +50GW in a Single Year. SemiAnalysis, 19 October 2026. newsletter.semianalysis.com

32 Inside the Ironwood TPU codesigned AI stack. Google Cloud, 6 November 2025. cloud.google.com

33 Electric Reliability Council of Texas, ERCOT update to the Senate Committee on... Electric Reliability Council of Texas, 29 July 2026, p. 6. ercot.com

34 Electric Reliability Council of Texas, ERCOT update to the Senate Committee on... Electric Reliability Council of Texas, 29 July 2026, p. 10. ercot.com

35 ercot.com

- **Large power transformers.** In the United States in 2024, lead times of 36 months were commonly quoted and the longest reached 60 months, against under a year before the pandemic[9]. By a US Commerce Department estimate, 82 percent of the units put into US service in 2019 were imported[36]. In an industry survey published in 2025, the International Energy Agency found that large power transformers took up to four years to secure worldwide[15]. It also found that their average lead times had almost doubled since 2021[15]. A transformer's size is given as a power rating in megavolt-amperes (MVA)[9]. Measured by capacity, US factories supplied 29.2 percent of the transformers rated 60 MVA and above bought in the US in 2023, and imports supplied the rest[4].
- **Gas turbines.** GE Vernova's gas equipment backlog and reserved factory slots stood at 116 GW in mid-2026[5]. In July 2026 it expected to build gas turbines at a yearly rate of 20 GW in the third quarter of 2026, 24 GW in 2028 and 30 GW in 2030[5].
- **Firm generation**, the kind that runs whatever the weather. The Nuclear Regulatory Commission has approved a fuel license amendment for the Crane plant[18]. Constellation called the approval a major milestone toward restarting the plant in 2027[18].

Memory chips are also in short supply. Micron's quarterly report for the period ended 28 May 2026 says AI-driven growth in demand for memory and storage is outpacing industry supply[37]. The memory shortage and the wait for power equipment both limit how much of the 2026 capital spending turns into working compute.

Four questions to answer from memory

1. Why are AI racks cooled with liquid? `see Cooling`
2. How big are the newest AI data centers? `see How it works`
3. What slows down building an AI data center besides chips? `see The chokepoint`
4. Which two countries used the most data center electricity in 2024? `see Capacity and demand`

Four things to remember

— One AI rack can draw 142 kilowatts, too much heat for air, so racks are cooled with liquid.

— The newest AI sites are planned in gigawatts, and each gigawatt is the output of a large power station.

— Power equipment is one of the slow parts of building a data center. In 2024 lead times of 36 months were commonly quoted for a large transformer.

— The United States used 45 percent of the world's data center electricity in 2024 and China 25 percent, by the International Energy Agency's estimate.

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5 GE Vernova, current report on Form 8-K dated 22 July 2026. U.S. Securities and Exchange Commission, 22 July 2026. `sec.gov`

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The chain

01 Chip Design, EDA and IP
02 Silicon and Wafers
03 Chemicals, Gases and Photoresist
04 Lithography
05 Photomasks and Pellicles
06 Deposition and Etch
07 Metrology and Inspection
08 Foundries and Fabs
09 Memory and HBM
10 Advanced Packaging
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The Economics of AI Chips

Epoch AI estimated in December 2025 that Nvidia's B200 AI chip cost about $6,400 to build, mostly for memory and packaging. Its processor silicon cost about $900.

Specimen 15 · The Economics of AI Chips

Epoch AI's December 2025 estimate of what it cost to build one B200, drawn as a column with each layer as tall as its share of the bill. From the bottom: $2,900 of memory in garnet, $1,100 of packaging, $1,000 of packaging yield loss drawn hollow, $900 of logic dies, $480 of power delivery, board and test. The total is about $6,400, against a selling price of $30,000 to $40,000 per chip reported in March 2024.

In plain terms

Most of what it costs to build Nvidia's B200, an AI chip from its Blackwell line, goes to memory and packaging that a few firms supply. A bill of materials is the list of parts inside a product and what each one costs. Epoch AI estimated in December 2025 that the B200 cost about $6,400 to build[1]. The two logic dies, the pieces of silicon that do the calculating, cost about $900 of that[1]. The stacked memory beside them accounted for about half, and the packaging that wires the dies and memory together cost more than the dies[1]. Three firms, SK hynix, Samsung and Micron, dominate the world market for that memory[2]. One Taiwanese firm, TSMC, does that packaging[1].

In short

Epoch AI says Nvidia's margin on Blackwell may be lower than the gap between the B200's build cost and its selling price suggests[1]. It cites reports from March 2024 that put the B200's selling price at $30,000 to $40,000 per chip[1]. Epoch AI notes that most Blackwell revenue comes from servers and rack-scale systems, which may carry lower margins[1].

In 2026 Micron's gross margin, the share of sales left after production costs, passed Nvidia's. Micron's rose from 39.8 percent for the year to August 2025 to 80.7 percent for the year to September 2026[3]. It reached 86.8 percent in the last quarter of that year[3]. Nvidia's was 75.0 percent in

1 NVIDIA's B200 costs around $6,400 to produce. Epoch AI, 10 December 2025. epoch.ai

2 Understanding the Biden Administration's Updated Export Controls. Center for Strategic and International Studies, 11 December 2024. csis.org

3 Micron Technology, Inc. Reports Record Fiscal Fourth-Quarter and Full-Year 2026... Micron Technology, 30 September 2026. investors.micron.com

the quarter to July 2026[4]. Epoch AI's estimate priced memory at $14 to $17 per gigabyte, before the 2026 rise in memory makers' sales and margins[5].

A B200, one of Nvidia's Blackwell AI chips, sells for far more than it costs to build. In December 2025 Epoch AI estimated that building one cost about $6,400[1]. Epoch AI cites reports from March 2024 that put its selling price at $30,000 to $40,000 per chip[1]. Most B200s are sold inside Nvidia's GB200 rack systems[6]. Epoch AI's chip sales model, with data through October 2025, puts what Nvidia earns per GPU for the computing hardware in those racks at $33,000 to $42,000[6]. That figure leaves out the networking gear in the racks[6].

01

How it works

Three numbers set the cost of a die, the finished rectangle of circuitry cut out of a wafer.

- **Wafer price.** TSMC does not publish it. TSMC makes Nvidia's Blackwell chips on a custom-built process called 4NP[7]. Epoch AI estimates $17,000 for a 12-inch (300 mm) wafer on that process[1].
- **Dies per wafer.** A wafer is a circle and dies are rectangles, so the edge of the wafer goes to waste. The bigger the die, the fewer fit. Each of the two logic dies in a B200 is about 800 mm^2[1]. Nvidia calls them reticle-limited, meaning they are near the largest size a scanner can print in one shot[7].
- **Yield.** Yield is the share of dies that work. Defects land at random, so a bigger die is more likely to catch one. Epoch AI models Blackwell die yield at 40 to 70 percent, centered on 60[1].

Together, those three numbers put the two dies inside a B200 at about $900[1]. Epoch AI's estimate for the whole B200 counts only variable costs, which rise with each chip made[1]. It leaves out fixed costs such as research and development[1].

02

Variants and trade-offs

Memory is the largest line

The largest line on the bill is the 192 GB of high-bandwidth memory beside those dies, a generation called HBM3E[1]. At $14 to $17 per gigabyte it comes to about $2,900, which Epoch AI counts as half of the build cost[1]. Epoch AI says HBM prices are negotiated deal by deal and can vary significantly with volume and contract length[1]. The $14 to $17 range dates from December 2025, before the 2026 rise in memory makers' sales and margins described below[5]. SK hynix said DRAM and NAND flash prices rose

1 NVIDIA's B200 costs around $6,400 to produce. Epoch AI, 10 December 2025. epoch.ai
4 NVIDIA Announces Financial Results for Second Quarter Fiscal 2027. Nvidia, 26 August 2026. nvidianews.nvidia.com
5 SK hynix Announces 2Q26 Financial Results. SK hynix, 29 July 2026. news.skhynix.com
6 AI chip sales documentation. Epoch AI, 8 January 2026. epoch.ai
7 NVIDIA Blackwell Architecture. Nvidia, 18 December 2025. nvidia.com

significantly in the second quarter of 2026 from the quarter before, led by high-performance products for AI servers[5].

Few firms make this memory, and one of them says demand has outrun its supply. SK hynix, Samsung and Micron dominate the world market for it[2]. SK hynix said in July 2026 that demand for its products had exceeded its available supply in recent quarters[8].

Packaging, substrate and test

CoWoS-L, the TSMC process that mounts dies and memory on one carrier and wires them together, adds about $1,100, more than the logic dies it carries[1]. Epoch AI models packaging yield at 65 to 95 percent[1]. A package that fails testing loses its logic dies and memory stacks[1]. Spreading that loss over the good units adds about $1,000 to each B200[1]. Power delivery hardware, the circuit board, and final assembly and testing add $480[1]. See Advanced packaging for how tight packaging capacity has been.

The B200 bill of materials

Line	Estimate	Range
HBM3E, 192 GB	$2,900	$2,800–3,100
CoWoS-L packaging	$1,100	$1,000–1,200
Packaging yield loss	$1,000	$430–1,700
Logic die, two on TSMC 4NP	$900	$720–1,200
Power delivery, board, assembly and test	$480	$370–600
Manufacturing cost	~$6,400	$5,700–7,300
Reported selling price per chip, March 2024	$30,000–40,000	
Nvidia compute revenue per GPU in GB200 racks, data to October 2025	$33,000–42,000	

Epoch AI models each line as a range, deriving the packaging line from TSMC's advanced packaging revenue and Nvidia's share of CoWoS capacity[1].

Epoch AI says Nvidia's actual margin on Blackwell may be lower than the gap between the B200's build cost and its reported chip price suggests[1]. The estimate covers the accelerator module only and leaves out servers and rack-scale systems[1]. Epoch AI notes that most Blackwell revenue comes from those systems, which may carry lower margins[1].

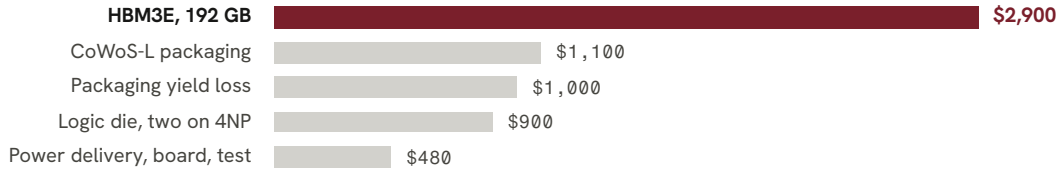
Epoch AI's line-by-line estimate covers the B200, which is no longer Nvidia's newest chip. In its results for the quarter to July 2026, Nvidia said its Vera Rubin platform was ramping into full production[4]. Each Rubin GPU carries up to 288 GB of HBM4, a newer generation of the stacked memory[9].

1 NVIDIA's B200 costs around $6,400 to produce. Epoch AI, 10 December 2025. epoch.ai
2 Understanding the Biden Administration's Updated Export Controls. Center for Strategic and International Studies, 11 December 2024. csis.org
4 NVIDIA Announces Financial Results for Second Quarter Fiscal 2027. Nvidia, 26 August 2026. nvidianews.nvidia.com
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8 SK hynix, prospectus on Form 424B4 dated 10 July 2026. U.S. Securities and Exchange Commission, 10 July 2026. sec.gov
9 Inside NVIDIA Rubin GPU Architecture: Powering the Era of Agentic AI. Nvidia, 21 July 2026. developer.nvidia.com

Figure 15.1

Memory, $2,900, is the largest line in Epoch AI's 2025 B200 estimate

Where the ~$6,400 of B200 manufacturing cost goes, Epoch AI estimate, December 2025



Source: Epoch AI B200 cost model, from a $17,000 4NP wafer, 40-70% die yield, $14-17 per GB of memory, and a packaging line derived from TSMC's advanced packaging revenue [1]

From package to rack to cluster

Each new generation of Nvidia rack costs its owner more per hour than the last[10]. At rack scale, cost is measured per GPU-hour, one GPU running for one hour. For an operator that owns the machine, SemiAnalysis put a Vera Rubin NVL72 at $3.57 per GPU-hour in July 2026, against $1.84 for a GB200 and $2.36 for a GB300[10].

But SemiAnalysis found Rubin cheaper than GB200 and GB300 per token, a short chunk of text[10]. Rubin stayed cheaper at every level of interactivity SemiAnalysis compared, where interactivity is the number of tokens per second each user receives[10]. That finding rests on early results from the cloud provider CoreWeave running the DeepSeek R1 model, and SemiAnalysis said it had not verified them[10].

One level up, a cluster needs power and a network as well as servers. SemiAnalysis estimated in June 2024 that a cluster of 100,000 of Nvidia's earlier H100 GPUs cost over $4 billion in servers[11]. The cluster needs more than 150 MW of data center capacity and uses 1.59 TWh a year, which costs $123.9 million at $0.078 per kWh[11]. In the network design Nvidia recommended, 98,304 optical transceivers link the cluster's GPUs to the first layer of switches, called leaf switches[11].

03

Who captures the value

Firm	Stage	Quarter ended	Gross margin
Micron	HBM and DRAM	September 2026	86.8%[3]
Nvidia	Accelerator design	July 2026	75.0%[4]
TSMC	Wafers and CoWoS	September 2026	66.4%[12]
KLA (non-GAAP)	Inspection	June 2026	62.4%[13]

1 NVIDIA's B200 costs around $6,400 to produce. Epoch AI, 10 December 2025. epoch.ai

3 Micron Technology, Inc. Reports Record Fiscal Fourth-Quarter and Full-Year 2026... Micron Technology, 30 September 2026. investors.micron.com

4 NVIDIA Announces Financial Results for Second Quarter Fiscal 2027. Nvidia, 26 August 2026. nvidianews.nvidia.com

10 Vera Rubin NVL72 vs GB200 NVL72? Inference TCO & Architecture Analysis. SemiAnalysis, 23 July 2026. inferencex.semianalysis.com

11 100,000 H100 Clusters: Power, Network Topology, Ethernet vs InfiniBand... SemiAnalysis, 17 June 2024. newsletter.semianalysis.com

12 TSMC Reports Third Quarter EPS of NT$29.43. U.S. Securities and Exchange Commission, 15 October 2026. sec.gov

13 Letter to Shareholders Q4 Fiscal 2026. KLA Corporation, 28 July 2026, p. 3. d1io3yog0oux5.cloudfront.net

ASML	Lithography	September 2026	57.5%[14]
Lam Research	Etch	June 2026	51.7%[15]
Applied Materials	Deposition	July 2026	50.3%[16]
ASE Technology	Packaging and test segment	June 2026	27.3%[17]
Amkor	OSAT	June 2026	16.8%[18]

Gross margin is highest at the memory maker Micron and lowest in assembly and test. The firms in design, equipment and leading-edge wafers earn gross margins of 50.3 to 75.0 percent, and those in assembly and test 16.8 to 27.3 percent. Micron's gross margin rose from 39.8 percent for the year to August 2025 to 80.7 percent for the year to September 2026[3]. It reached 86.8 percent in the last quarter of that year[3].

Each figure except KLA's comes from the firm's own earnings release. KLA's shareholder letter gives only a non-GAAP gross margin for the quarter, 62.4 percent[13]. KLA's non-GAAP figures adjust its results under US accounting rules, known as GAAP, to leave out items such as acquisition-related charges and restructuring, severance and other charges[19]. For ASE the figure is its packaging and test segment[17]. The company as a whole had a gross margin of 21.0 percent, because about a third of its revenue came from electronics manufacturing services, which earned 8.9 percent[17].

Two other firms' results give no gross margin. SK hynix's operating margin was 76 percent in the quarter to June 2026, and its revenue was up 257 percent from a year earlier[5]. The substrate maker Ibiden's operating profit was 21.8 percent of net sales in the quarter to June 2026[20].

World chip sales are forecast to jump in 2026, with memory growing far faster than logic. The Semiconductor Industry Association, using monthly data compiled by World Semiconductor Trade Statistics, counted $791.7 billion of global chip sales in 2025[21]. In its June 2026 forecast, World Semiconductor Trade Statistics put 2025 slightly higher, at $795.6 billion[22]. It forecast $1.51 trillion for 2026, with memory growing about 250 percent and logic 37 percent[22].

In 2025 capital spending at five large tech firms far exceeded the world's sales of chipmaking equipment. The industry group SEMI counted $135.1 billion of semiconductor equipment billings in 2025[23]. Capital spending at Alphabet, Amazon, Meta, Microsoft and Oracle came to $448.2 billion in 2025, the sum of Epoch AI's

3 Micron Technology, Inc. Reports Record Fiscal Fourth-Quarter and Full-Year 2026... Micron Technology, 30 September 2026. investors.micron.com

5 SK hynix Announces 2Q26 Financial Results. SK hynix, 29 July 2026. news.skhynix.com

13 Letter to Shareholders Q4 Fiscal 2026. KLA Corporation, 28 July 2026, p. 3. d1io3yog0oux5.cloudfront.net

14 ASML reports €11.7 billion total net sales and €5.0 billion net income in Q3... U.S. Securities and Exchange Commission, 14 October 2026. sec.gov

15 Lam Research Corporation Reports Financial Results for the Quarter Ended June... Lam Research, 29 July 2026. newsroom.lamresearch.com

16 Applied Materials Announces Third Quarter 2026 Results. Applied Materials, 13 August 2026. ir.appliedmaterials.com

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18 Amkor Technology, current report on Form 8-K dated 27 July 2026. U.S. Securities and Exchange Commission, 27 July 2026. sec.gov

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21 Global Annual Semiconductor Sales Increase 25.6% to $791.7 Billion in 2025. Semiconductor Industry Association, 6 February 2026. semiconductors.org

22 Global Semiconductor Market Surges Beyond USD 1.5 Trillion in 2026 Driven by... World Semiconductor Trade Statistics, 2 June 2026. wsts.org

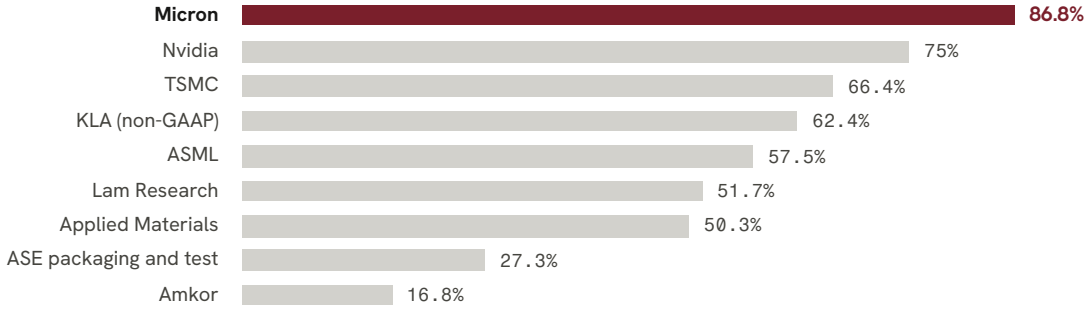
23 SEMI Reports Global Semiconductor Equipment Billings Reached $135 Billion in... SEMI, 7 April 2026. semi.org

quarterly figures[24]. Epoch AI says that spending would reach $770 billion in 2026 if the trend continues[24]. These totals overlap, so they cannot be added together.

Figure 15.2

Micron's gross margin, 86.8 percent, was highest of nine firms in 2026

Gross margin, quarters ended June to September 2026

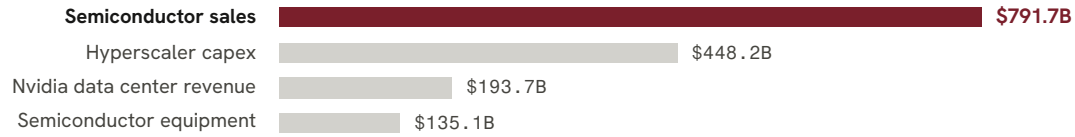


Source: Each firm's own quarterly earnings release, for quarters ending between June and September 2026 (TSMC's linked). ASE's bar is its packaging and test segment. KLA's bar is the non-GAAP figure from its shareholder letter [12]

Figure 15.3

World chip sales reached $791.7 billion in 2025, the largest total shown

Chip sales, capital spending at five tech firms, Nvidia data center revenue and equipment sales, 2025



Source: Semiconductor Industry Association for chip sales. Epoch AI's quarterly capital spending at Alphabet, Amazon, Meta, Microsoft and Oracle, summed for 2025. Nvidia's data center revenue for its fiscal year to January 2026. SEMI for equipment billings [21]

04

Where the money goes

Each stage falls into one of three classes, by whether its cost falls as volume grows. A structurally expensive stage stays costly because the work is hard, capacity is short or one firm sets the price. A stage compressible with scale gets cheaper as volume grows. An already commoditized stage has several competing suppliers. The classes are a judgment drawn from the sources in this chapter and the stage chapters.

Stage	Class	Why
EUV lithography	Structurally expensive	One supplier, ASML. See Lithography.
Masks and pellicles	Structurally expensive	Every design needs its own mask set, and every EUV layer a flawless mask. See Photomasks and pellicles.

12 TSMC Reports Third Quarter EPS of NT$29.43. U.S. Securities and Exchange Commission, 15 October 2026. sec.gov

21 Global Annual Semiconductor Sales Increase 25.6% to $791.7 Billion in 2025. Semiconductor Industry Association, 6 February 2026. semiconductors.org

24 Hyperscaler capex has quadrupled since GPT-4's release. Epoch AI, 26 February 2026. epoch.ai

Metrology and inspection	Structurally expensive	KLA held 56.8 percent of the market for these tools in 2025, by CSET's analysis of TechInsights data[25]. See Metrology and inspection.
HBM	Structurally expensive	SK hynix says HBM needs more wafer input than ordinary DRAM[8]. See Memory and HBM.
Advanced packaging, CoWoS-L	Structurally expensive	Modeled packaging yield is 65 to 95 percent, and a failed package loses its dies and memory.
ABF substrates	Structurally expensive	Investment can add substrate lines, but Ajinomoto said in June 2026 that the insulating film it makes for these substrates has held over 95 percent of its market since launch[26]. See Substrates and PCBs.
Power and cooling plant	Structurally expensive	In the United States, grid connections face long queues, so at least three of OpenAI's seven Stargate sites will use on-site natural gas plants[27]. GE Vernova's backlog of gas power equipment, together with agreements reserving future production slots, grew from 100 GW to 116 GW in the second quarter of 2026[28]. See Data centers and power.
Optical transceivers	Structurally expensive	In May 2026 Nikkei reported that building AI data centers had caused shortages and price rises for lasers, optical fibers and other optical parts[29]. Nvidia agreed in March 2026 to invest $2 billion in the laser maker Lumentum, with rights to future capacity[30]. It agreed a similar $2 billion deal with Coherent[31]. See Systems and networking.
Leading-edge logic wafers	Compressible with scale	A very large die yields well only once a process matures. See Foundries and fabs.
Design and EDA	Compressible with scale	A design is paid for once, and its cost is spread across every chip sold. See Chip design, EDA and IP.
Deposition and etch tools	Compressible with scale	Most deposition and etch steps have three or four established tool makers. See Deposition and etch.
Test and assembly (OSAT)	Already commoditized	Gross margins of 16.8 to 27.3 percent at Amkor and ASE's packaging and test segment in the quarter to June 2026.
Wafers and bulk gases	Already commoditized	Several firms make approved 300 mm wafers. By Siltronic's estimate, the five largest wafer makers served around 75 percent of the market in the first three quarters of 2025[32]. See Silicon and wafers.
Rack integration (ODM)	Already commoditized	In October 2025 Nvidia said more than 50 makers of systems and components were gearing up for the Vera Rubin NVL72[33]. See Systems and networking.

Three questions to answer from memory

1. What does an Nvidia B200 cost to build? `see The B200 bill of materials`
2. What is the most expensive part of an Nvidia B200? `see Memory is the largest line`
3. Why does packaging add so much to the cost of a B200?
   `see Packaging, substrate and test`

Three things to remember

— Epoch AI estimated in December 2025 that a B200 cost about $6,400 to build, against a selling price of $30,000 to $40,000 per chip reported in March 2024.

8 SK hynix, prospectus on Form 424B4 dated 10 July 2026. U.S. Securities and Exchange Commission, 10 July 2026. `sec.gov`

25 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO), accessed 9 September 2026. `chipexplorer.eto.tech`

26 Ajinomoto, Growth Strategy for the Electronic Materials Business, 30 June 2026. Ajinomoto, 30 June 2026, p. 3. `ajinomoto.co.jp`

27 OpenAI Stargate: where the US sites stand. Epoch AI, 17 April 2026. `epoch.ai`

28 GE Vernova, current report on Form 8-K dated 22 July 2026. U.S. Securities and Exchange Commission, 22 July 2026. `sec.gov`

29 AI demand strains supplies of lasers, fiber and other optical tech. Nikkei Asia, 27 May 2026. `asia.nikkei.com`

30 NVIDIA Announces Strategic Partnership With Lumentum to Develop... Nvidia, 2 March 2026. `nvidianews.nvidia.com`

31 NVIDIA and Coherent Announce Strategic Partnership to Develop Optics Technology... Nvidia, 2 March 2026. `nvidianews.nvidia.com`

32 Investor Presentation: Foundation of Digital Life. Siltronic, April 2026, p. 6. `siltronic.com`

33 NVIDIA, Partners Drive Next-Gen Efficient Gigawatt AI Factories in Buildup for... Nvidia, 13 October 2025. `blogs.nvidia.com`

— The stacked memory is the most expensive part, about half of the cost.

— Packaging costs more than the logic chips, and each package that fails wastes the chips and memory inside it.

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33. A NVIDIA, Partners Drive Next-Gen Efficient Gigawatt AI Factories in Buildup for Vera Rubin. Nvidia · 13 October 2025
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The chain

01 Chip Design, EDA and IP
02 Silicon and Wafers
03 Chemicals, Gases and Photoresist
04 Lithography
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15 The Economics of AI Chips
16 Who Controls the Chain

Who Controls the Chain

Since 2018 the United States has used export controls to limit China's access to advanced chips and chipmaking tools, and China has been learning to make both.

Specimen 16 · Who Controls the Chain

The world, with nineteen chokepoint towns pinned in garnet. Five towns building alternatives to those chokepoints are pinned in ink: Beijing, Wuhan, Shenzhen, Hefei and Dresden. The taller the garnet pin, the tighter the chokepoint at that town's stage of the chain. The pins are not to scale. If the globe were the size of the Earth, the tallest pin would stand about 1,000 km high.

In plain terms

Export controls work only when a country's own firms sell something the buyer cannot get elsewhere, so few governments can use them to much effect. An export control is a rule that requires the government's permission for a sale. The government names an item and a country or a buyer. From then on its own companies must hold a license before they ship that item to that country or buyer. The American government extends some of these rules to goods made abroad with American technology. The machines, software and chemicals used to make an advanced chip come from a short list of firms in the United States, the Netherlands and Japan, so those three governments can control who gets them.

In short

Japan and the Netherlands have aligned their own export controls with the American rules on chipmaking equipment[1]. China has answered by building its own tools and by restricting sales of US chips in China. Chinese toolmakers supplied 35 percent of China's own chipmaking equipment market in 2025, up from 10 to 15 percent before 2022[2]. Nvidia says the Chinese government restricted the sales of its H200 chip that the United States licensed in February 2026[3].

The most advanced chips come mainly from Taiwan. It held 69 percent of the world's capacity for logic chips below 10 nanometers in 2022, by the

1 Where the Chips Fall: U.S. Export Controls Under the Biden Administration from... Center for Strategic and International Studies, 12 December 2024. csis.org

2 China's Localization Drive in Semiconductors Gains Impetus from Allied Chip... Center for Strategic and International Studies, 24 March 2026. csis.org

3 Nvidia, quarterly report on Form 10-Q for the quarter ended 26 July 2026. U.S. Securities and Exchange Commission, 26 August 2026. sec.gov

count of the Semiconductor Industry Association and Boston Consulting Group[4].

01

The shape of the contest

China cannot yet make the most advanced lithography machines, which print a chip's finest circuits[5]. China's leading lithography maker, SMEE, produces only older, less precise machines, among them some deep-ultraviolet types[5]. China has also yet to show it can make high-bandwidth memory, the stacked memory beside an AI chip, in volume.

In September 2022 US National Security Advisor Jake Sullivan said the United States had previously aimed to stay "only a couple of generations ahead" in some key technologies[6]. He said it must now keep "as large of a lead as possible" in foundational technologies such as advanced logic and memory chips[6]. The export rules of October 2022 were the first to apply his doctrine of a small yard with a high fence, meaning strict controls on a narrow set of technologies[1]. The Center for Strategic and International Studies (CSIS), a research institute in Washington, judged that those rules went beyond the old aim of staying two technology nodes ahead of China and tried to push China's capability back below the level it had already reached[7].

US license rules can target a single buyer or a whole country, and in 2022 the rules on advanced AI chips and the tools for advanced fabs began to cover all of China. In October 2018, for example, the Bureau of Industry and Security (BIS), the US Commerce Department's export control office, put the Chinese chip company Fujian Jinhua Integrated Circuit on its Entity List[8]. From then on any item under US export rules needed a license to reach the company[8]. The 2022 rules went further and required a license for any advanced AI chip, and any tool for an advanced fab, bound for China[9]. BIS reviewed most applications under a presumption of denial, meaning it would normally refuse them[9]. It reviewed applications for equipment bound for the China fabs of firms headquartered in the United States or allied and partner countries case by case[9].

02

Chokepoints and who holds them

Segment	Leader	Share	Basis
EUV lithography	Netherlands	100.0%	revenue, 2025

1 Where the Chips Fall: U.S. Export Controls Under the Biden Administration from... Center for Strategic and International Studies, 12 December 2024. csis.org

4 Emerging Resilience in the Semiconductor Supply Chain. Boston Consulting Group and Semiconductor Industry..., May 2024, p. 14. semiconductors.org

5 Breakthroughs or Boasts? Assessing Recent Chinese Lithography Advancements. Center for Strategic and International Studies, 24 September 2025. csis.org

6 Remarks by National Security Advisor Jake Sullivan at the Special Competitive... The White House, 16 September 2022. bidenwhitehouse.archives.gov

7 Choking off China's Access to the Future of AI. Center for Strategic and International Studies, 11 October 2022. csis.org

8 Addition of an Entity to the Entity List. Federal Register, 30 October 2018. federalregister.gov

9 Implementation of Additional Export Controls: Certain Advanced Computing and... Federal Register, 13 October 2022. federalregister.gov

Immersion DUV lithography	Netherlands	99.3%	revenue, 2025
Metrology and inspection	United States	71.7%	revenue, 2025
Deposition	United States	60.3%	revenue, 2025
Etch and clean	United States	52.6%	revenue, 2025
Photoresist	Japan	78%	revenue, 2023
Silicon wafers	Japan	53%	revenue, 2023
Leading-edge logic, below 10 nm	Taiwan	69%	capacity, 2022
Foundry, all nodes	Taiwan	81.9%	market share, 2025
HBM	South Korea	82.5%	revenue, 2025: SK hynix 63.2, Samsung 19.3, Micron 17.4
Logic chip design	United States	65%	sales, 2022
Mature logic, 28 nm and above	China	33%	capacity, 2022

No country leads every segment. The United States leads four of the table's segments. The Netherlands, Japan and Taiwan lead two each, and South Korea and China one each.

- **Equipment.** TechInsights 2025 revenue by parent-company headquarters, from CSET's ETO Supply Chain Explorer[10].
- **Materials.** Fuji Keizai's survey by parent-company headquarters, as Japan's Ministry of Economy, Trade and Industry published it in December 2025[11].
- **Memory.** Figures from the market research firm IDC, as SK hynix's US listing prospectus relays them[12].
- **Logic capacity.** The Semiconductor Industry Association and Boston Consulting Group's 2022 measurement of logic capacity, below 10 nm for Taiwan and at 28 nm and above for China[4].

Advanced packaging is missing from the table because no public source splits its capacity by country. The Taiwanese chipmaker TSMC does not publish one either.

03

US export controls, 2018 to today

US chip controls grew tighter from 2020 to 2024. In August 2020 BIS widened its Foreign Direct Product Rule to require a license for any foreign-made item produced with US technology and bound for Huawei[13]. In 2022 BIS blocked China's access to the most advanced AI chips and to the tools to make them[1]. It tightened those rules in 2023 and 2024[1]. But in August 2025 the US government began licensing some sales of Nvidia's H20 chip to China[14]. In February 2026 it began licensing small amounts of the H200[14].

1 Where the Chips Fall: U.S. Export Controls Under the Biden Administration from... Center for Strategic and International Studies, 12 December 2024. csis.org

4 Emerging Resilience in the Semiconductor Supply Chain. Boston Consulting Group and Semiconductor Industry..., May 2024, p. 14. semiconductors.org

10 Documentation: Semiconductor Supply Chain Dataset. Emerging Technology Observatory (ETO), accessed 9 September 2026. eto.tech

11 METI, Semiconductor and digital industry strategy: future direction... Ministry of Economy, Trade and Industry (Japan), 23 December 2025, p. 8. meti.go.jp

12 SK hynix, prospectus on Form 424B4 dated 10 July 2026. U.S. Securities and Exchange Commission, 10 July 2026. sec.gov

13 Addition of Huawei Non-U.S. Affiliates to the Entity List, the Removal of... Federal Register, 20 August 2020. federalregister.gov

14 Nvidia, annual report on Form 10-K for the fiscal year ended 25 January 2026. U.S. Securities and Exchange Commission, 25 February 2026. sec.gov

The United States has also approved large AI chip sales in the Gulf. In November 2025 Commerce authorized G42 of the United Arab Emirates and HUMAIN of Saudi Arabia to buy the equivalent of up to 35,000 Nvidia GB300 chips each under security and reporting conditions[15]. On 10 July 2026 BIS moved the United Arab Emirates into Country Group A:5, the export-control group that holds close US allies[16]. It named the Emirati government, G42 and eight US-headquartered AI companies as recipients that need no license for controlled AI chips and systems[16].

The Export Controls page has the full timeline, with the legal citation for each rule.

Figure 16.1

US chip export controls tightened through January 2025, then partly eased

Eight US actions that shaped chip export controls

- 16 May 2019
  Huawei and 68 affiliates added to the Entity List
- 7 October 2022
  China-wide controls on AI chips and advanced-fab tools
- 17 October 2023
  Performance-density test added, covering Nvidia's A800 and H800
- 2 December 2024
  24 tool types, HBM controls, 140 entity listings
- 13 January 2025
  AI Diffusion rule announced, three country tiers
- 13 May 2025
  BIS stops enforcing the AI Diffusion rule
- 11 August 2025
  US officials expect 15 percent or more of licensed H20 revenue
- 15 January 2026
  H200 and MI325X move to case-by-case review

Source: Federal Register rules, BIS press releases and Nvidia filings, cited for each entry on the Export Controls timeline

04

Ways around the controls

Firms can use controlled chips without any chip crossing a border, because renting out time on a chip does not count as exporting it[17]. A Singapore data center selling time on an H200 already installed there, for example, moves nothing across a border[17]. The AI Diffusion rule of January 2025 required a license for advanced AI chips bound anywhere in the world[18]. For the data centers it approved, it said that API access to a model, and rented capacity for inference, "are not prohibited"[18]. On 13 May 2025 the

15 Statement on UAE and Saudi Chip Exports. U.S. Department of Commerce, November 2025. commerce.gov

16 Enhanced Favorable Treatment for the United Arab Emirates Under the Export... Federal Register, 14 July 2026. federalregister.gov

17 15 CFR 734.13 -- Export. Electronic Code of Federal Regulations, 3 June 2016. ecfr.gov

18 Framework for Artificial Intelligence Diffusion. Federal Register, 15 January 2025. federalregister.gov

Commerce Department said it would rescind the rule and told BIS officials not to enforce it[19].

The American rules reach that trade through the firms holding the chips. A license under the 15 January 2026 rule for H200-class chips bound for China or Macau requires the applicant to obtain a description of the recipient's know-your-customer procedures[20]. These are the recipient's checks on who its customers are. The applicant must also obtain the recipient's list of customers renting computing power in China and seven other places, among them Russia and Iran[20]. The recipient must also promise that no algorithm trained on the chips is served to blacklisted or military end users[20].

Smugglers route chips past the controls through Southeast Asia. Since 14 July 2025 Malaysia has required a permit to export, transship or transit any high-performance US-origin AI chip[21]. On 25 March 2026 the Justice Department announced charges against three people accused of using Thai companies' names to order servers with export-controlled chips for China[22]. The orders included one for $170 million that was never completed[22].

In July 2025 the US House of Representatives' select committee on China cited estimates that a median of 140,000 chips were smuggled to China in 2024[23]. A bill in Congress, the Chip Security Act, would require location verification on controlled chips within 180 days of enactment[24]. The House Foreign Affairs Committee approved the House version 42-0 on 26 March 2026[25].

05

Allies: the Netherlands, Japan, Korea and Europe

US tool controls alone would only shift the sales to Dutch and Japanese suppliers, so US controls depend on those two governments. Both have aligned their own controls on chipmaking equipment with the American rules[1].

- **The Netherlands.** The Dutch government began requiring export licenses for advanced deep-ultraviolet tools on 1 September 2023 and widened the controls in September 2024[26]. On 1 April 2025 it extended them to measurement and inspection tools[27].

1 Where the Chips Fall: U.S. Export Controls Under the Biden Administration from... Center for Strategic and International Studies, 12 December 2024. csis.org

19 Department of Commerce Announces Rescission of Biden-Era Artificial... U.S. Bureau of Industry and Security, 13 May 2025. bis.gov

20 Revision to License Review Policy for Advanced Computing Commodities. U.S. Government Publishing Office, 15 January 2026. govinfo.gov

21 Export, transshipment and transit of high-performance AI chips of US origin now... Ministry of Investment, Trade and Industry, 14 July 2025. miti.gov.my

22 Chinese National and Two U.S. Citizens Charged with Conspiring to Smuggle... U.S. Department of Justice, 25 March 2026. justice.gov

23 Protecting U.S. Tech: China Committee and Bipartisan, Bicameral Leaders Unite... U.S. House Select Committee on the Chinese Communist Party, 7 July 2025. chinaselectcommittee.house.gov

24 S.1705 – Chip Security Act, 119th Congress (2025-2026), bill text. U.S. Congress, 8 May 2025. congress.gov

25 UNANIMOUS: Bipartisan Huizenga Legislation to Curb AI Chip Smuggling Passes... U.S. House of Representatives, 26 March 2026. huizenga.house.gov

26 The Netherlands expands export control measure for advanced semiconductor... Government of the Netherlands, 6 September 2024. government.nl

27 Klever: export controls on advanced semiconductor manufacturing equipment to be... Government of the Netherlands, 15 January 2025. government.nl

- **Japan.** Japan's controls on advanced manufacturing equipment took effect on 23 July 2023, apply to all destinations and name no country[28].
- **US rules for South Korean fabs in China.** BIS revoked the Validated End-User status that let Samsung's and SK hynix's China subsidiaries and Intel Semiconductor (Dalian) receive US equipment without individual licenses, effective 31 December 2025[29]. BIS said it would license those fabs to keep running, but not to expand capacity or upgrade technology[30]. SK hynix, which bought the Dalian fab from Intel, received an annual BIS license in December 2025 to ship approved US equipment to its China fabs during 2026[31].

A BIS rule effective 29 September 2025 extended Entity List restrictions to any company at least 50 percent owned by one or more listed firms[32]. The rule reached Nexperia, a Nijmegen chipmaker that supplies Europe's car industry, and the Dutch government said it worked to secure an exemption for the company[33]. On 30 September the Dutch economy minister invoked the Goods Availability Act against Nexperia[34]. He cited governance failures that risked moving technology out of Europe[34]. The order aimed to keep Nexperia's production capabilities available to Europe in an emergency[33].

On 4 October the Chinese authorities imposed an export control on Nexperia's operations in China, so deliveries from those operations to customers needed an export license[35]. The Dutch government said uncertainty over licensing and long processing times caused chip shortages among Nexperia's customers and stopped production at some companies[35].

Both the US rule and the Dutch order were suspended in November. On 10 November BIS suspended its rule until 9 November 2026[36]. The minister suspended his order on 19 November[33]. He cited constructive meetings with the Chinese authorities and the measures they had taken to ensure the supply of chips to Europe and the rest of the world[33]. Europe holds a chokepoint in leading-edge lithography, but none in the mature-node chips its car industry uses.

A bill in the US Congress would push allies to match US limits on chipmaking tools sold to China. The MATCH Act, a bill introduced on 8 April 2026, would ban sales of immersion deep-ultraviolet scanners to China[37]. It would also restrict exports to, and servicing of, every chipmaking plant run by five Chinese companies, CXMT, Hua Hong, Huawei, SMIC and YMTC[37]. It would give allies 150 days to match those limits before the United States extends its own controls to foreign-made items that use US software, technology or components[37].

28 CSIS Translation: Updated Japanese Export Controls on High-Performance... Center for Strategic and International Studies, 18 July 2023. csis.org

29 Revocation of Validated End-User Authorizations in the People's Republic of... Federal Register, 2 September 2025. federalregister.gov

30 Department of Commerce Closes Export Controls Loophole for Foreign-Owned... U.S. Bureau of Industry and Security, 29 August 2025. bis.gov

31 SK hynix, prospectus on Form 424B4 dated 10 July 2026. U.S. Securities and Exchange Commission, 10 July 2026. sec.gov

32 Expansion of End-User Controls To Cover Affiliates of Certain Listed Entities. Federal Register, 30 September 2025. federalregister.gov

33 Update on invoking Goods Availability Act - 19 november 2025. Government of the Netherlands, 19 November 2025. government.nl

34 Minister of Economic Affairs invokes Goods Availability Act. Government of the Netherlands, 12 October 2025. government.nl

35 Developments concerning Nexperia, letter to the House of Representatives. Government of the Netherlands, 19 November 2025. government.nl

36 One Year Suspension of Expansion of End-User Controls for Affiliates of Certain... Federal Register, 12 November 2025. federalregister.gov

37 Risch, Ricketts, Kim Introduce MATCH Act; Level the Global Playing Field for... U.S. Senate Committee on Foreign Relations, 8 April 2026. foreign.senate.gov

06

China's response

China has answered by replacing foreign chips at home, restricting some American suppliers, limiting its exports of minerals and building its own chipmaking tools. In 2024 China's Ministry of Industry and Information Technology reportedly told Chinese telecom operators to strip foreign chips out of their networks by 2027[2].

China has also taken measures against American suppliers. On 21 May 2023 the Cyberspace Administration of China (CAC) said that products sold in China by Micron, the American memory maker, had failed a cybersecurity review[38]. It said operators of critical information infrastructure in China should stop buying them[38]. In September 2025 the CAC reportedly told firms including ByteDance and Alibaba to cancel Nvidia orders[39]. Nvidia says the Chinese government restricted the H200 sales that the United States licensed in February 2026[3]. It says China had earlier restricted licensed H20 sales too[3].

China's Ministry of Commerce (MOFCOM) has also restricted exports of minerals.

- **3 December 2024.** MOFCOM banned exports of gallium, germanium, antimony and superhard materials to the United States, one day after the American chip rules[40].
- **4 April 2025.** It required export licenses for seven medium and heavy rare earths[41].
- **9 October 2025.** It announced that from 1 December 2025 foreign firms would need its license to export listed goods made abroad with Chinese rare earths worth 0.1 percent or more of their value[42]. MOFCOM suspended that rule on 7 November 2025 until 10 November 2026[43].
- **9 November 2025.** MOFCOM suspended its ban on exports of gallium, germanium, antimony and superhard materials to the United States until 27 November 2026[44]. It acted after Donald Trump and Xi Jinping met at the Asia-Pacific summit in October 2025[45].

China's suspension of its October 2025 controls did not cover the April 2025 licensing rules for seven rare earths[43].

2 China's Localization Drive in Semiconductors Gains Impetus from Allied Chip... Center for Strategic and International Studies, 24 March 2026. csis.org

3 Nvidia, quarterly report on Form 10-Q for the quarter ended 26 July 2026. U.S. Securities and Exchange Commission, 26 August 2026. sec.gov

38 Cyberspace Administration of China, notice on the cybersecurity review of... Cyberspace Administration of China, 21 May 2023. cac.gov.cn

39 Beijing blocks Nvidia sales in vote of confidence for Chinese chipmakers, the... Center for Security and Emerging Technology, 18 September 2025. cset.georgetown.edu

40 Ministry of Commerce Notice 2024 No. 46: Notice Concerning Strengthening... Center for Security and Emerging Technology, 3 December 2024. cset.georgetown.edu

41 Announcement No.18 of 2025 of The Ministry of Commerce and The General... Ministry of Commerce of the People's Republic of China, 4 April 2025. english.mofcom.gov.cn

42 Ministry of Commerce Notice 2025 No. 61: Announcement of the Decision to... Center for Security and Emerging Technology, 9 October 2025. cset.georgetown.edu

43 MOFCOM and GAC Announcement 2025 No. 70, suspending the October 2025... Ministry of Commerce of the People's Republic of China, 7 November 2025. mofcom.gov.cn

44 商务部公告2025年第72号 公布调整实施商务部公告2024年第46号的决定. Ministry of Commerce of the People's Republic of China, 9 November 2025. mofcom.gov.cn

45 The U.S.-China Trade Truce Has Not Solved the Gallium Problem. Center for Strategic and International Studies, 11 May 2026. csis.org

Chinese toolmakers are winning a growing share of their home market. They held 10 to 15 percent of it before 2022, 25 percent in 2024 and 35 percent in 2025[2]. In March 2026 CSIS put their share of the Chinese market for etch and deposition tools at 40 percent[2]. But in September 2025 CSIS judged that making its own extreme-ultraviolet machines remained firmly out of China's reach despite massive state investment[5].

Chinese chipmakers remain behind at the leading edge, where TSMC makes 2 nm chips[46]. SMIC's most advanced process, N+3, reaches the transistor density of TSMC's 6 nm-class process without extreme-ultraviolet scanners[47]. It prints its tightest layers in several passes on deep-ultraviolet tools[47].

In September 2025 the research firm SemiAnalysis conservatively put SMIC's capacity at 7 nm and below at 45,000 wafers a month by the end of 2025[48]. It expected 60,000 in 2026 and 80,000 in 2027[48]. Those figures count wafers, and how many working chips they give depends on yield, the share of chips on a wafer that work.

SemiAnalysis judged that memory would limit China's output of AI chips before logic wafers did[48]. It expected Huawei to ship about 805,000 of its Ascend AI chips in 2025, with dies made by both TSMC and SMIC[48]. SemiAnalysis says Huawei got Ascend dies made at TSMC, in violation of export controls, while SMIC ramped up its own production[48]. By its count, China had procured roughly 13 million memory stacks, enough for about 1.6 million Ascend 910C packages[48]. It expected CXMT, China's own memory maker, to make only about 2 million stacks in 2026, enough for 250,000 to 300,000 more packages[48].

Huawei said in September 2025 that its Ascend 950 chips, due in 2026, would carry high-bandwidth memory of its own design[49]. It gave no figure for how much of that memory it would make[49].

2 China's Localization Drive in Semiconductors Gains Impetus from Allied Chip... Center for Strategic and International Studies, 24 March 2026. csis.org

5 Breakthroughs or Boasts? Assessing Recent Chinese Lithography Advancements. Center for Strategic and International Studies, 24 September 2025. csis.org

46 Why Taiwan Fears 'America First' Risks Eroding Its 'Silicon Shield'. Stimson Center, 10 October 2025. stimson.org

47 Is SMIC N+3's Metal Pitch Smaller than Intel 18A's? SemiAnalysis, 14 June 2026. newsletter.semianalysis.com

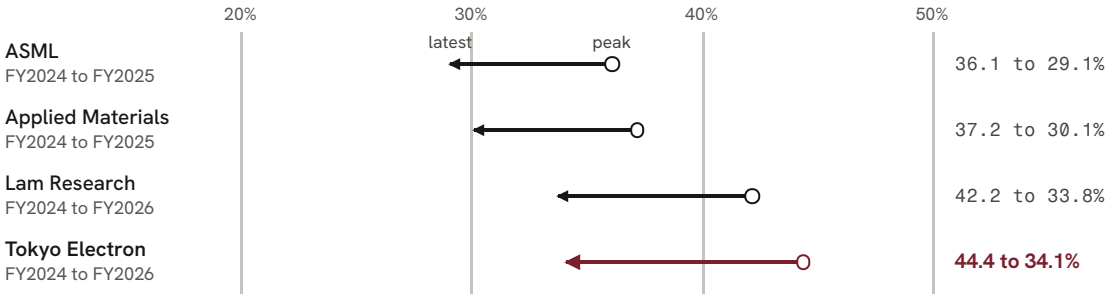
48 Huawei Ascend Production Ramp: Die Banks, TSMC Continued Production, HBM is The... SemiAnalysis, 8 September 2025. newsletter.semianalysis.com

49 Groundbreaking SuperPoD Interconnect: Leading a New Paradigm for AI... Huawei, 18 September 2025. huawei.com

Figure 16.2

China's revenue share fell at all four largest toolmakers after fiscal 2024

China's share of revenue at the four largest chip-tool makers, peak fiscal year and latest



Source: ASML 2025 Form 20-F (total net sales), Applied Materials and Lam Research 10-K filings and Tokyo Electron FY2024 and FY2026 consolidated financial statements, each China share computed from the filing's sales by region [50]

07

Industrial policy: CHIPS, Rapidus, Big Fund and the rest

The US government helps pay for new chip plants with grants and a tax credit. The CHIPS and Science Act authorized about $52.7 billion for manufacturing incentives, research and workforce support[2]. By July 2025 Commerce had awarded $30.9 billion of direct funding to 19 companies for 40 projects[51]. The grants covered a median 14.2 percent of each project's capital cost[51]. Commerce took a federal tax credit, the Advanced Manufacturing Investment Credit, into account when it set the size of those grants[51]. Since the One Big Beautiful Bill Act of 2025, the credit covers 35 percent of qualified investment in an advanced chip plant placed in service after 2025, up from 25 percent[51].

On 22 August 2025 the US government agreed to take a 9.9 percent stake in Intel by buying $8.9 billion of its stock[52]. Of that sum, $5.7 billion was to come from CHIPS Act grants that Intel had been awarded but not yet paid[52]. The other $3.2 billion was to come from Intel's award under Secure Enclave, a program to supply trusted and secure chips to the US Department of Defense[52].

The United States has also used tariffs to draw chip investment. In January 2026 the American Institute in Taiwan and the Taipei Economic and Cultural Representative Office in the United States signed a memorandum on investment[53]. Under it, Taiwanese chip and technology firms are to invest $250 billion in US semiconductor, energy and AI production[53]. In the same memorandum the United States agreed to cap some of its national-security tariffs on Taiwanese goods at 15 percent[53]. On 16 July 2026

2 China's Localization Drive in Semiconductors Gains Impetus from Allied Chip... Center for Strategic and International Studies, 24 March 2026. csis.org

50 ASML Holding, annual report on Form 20-F for the fiscal year ended 31 December... U.S. Securities and Exchange Commission, 25 February 2026. sec.gov

51 GAO-26-107882, SEMICONDUCTORS: Information on Projects Funded to Strengthen... U.S. Government Accountability Office, 11 December 2025. files.gao.gov

52 Intel and Trump Administration Reach Historic Agreement to Accelerate American... Intel, 22 August 2025. intc.com

53 Implementing Certain Tariff-Related Elements of a Trade and Security Agreement... Federal Register, 28 May 2026. federalregister.gov

TSMC announced another $100 billion for four more Arizona fabs, bringing its total Arizona investment to $265 billion[54].

The European Commission says the EU still depends on other countries for advanced chip manufacturing[55]. On 3 June 2026 the Commission proposed a Chips Act 2.0 to support advanced chip production in the EU[55].

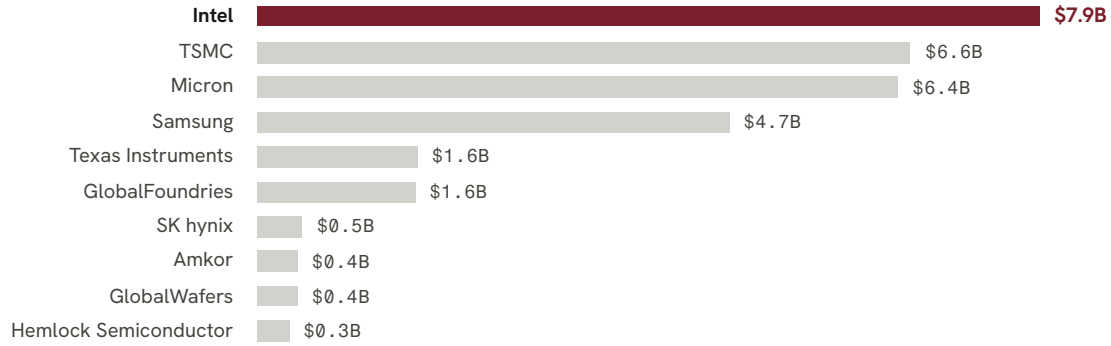
Japan backs a homegrown chipmaker, Rapidus. Rapidus began prototyping 2 nm gate-all-around transistors at its IIM-1 plant in July 2025[56]. In February 2026 it raised another 267.6 billion yen, about $1.7 billion, from Japan's Information-Technology Promotion Agency (IPA) and 32 companies, aiming for 2 nm mass production by 2027[57]. By June 2026 it had raised 424.95 billion yen in equity in all, after a further 150 billion yen from the IPA[58].

CSIS puts China's cumulative state funding for chips since 2014 at an estimated $150 billion and calls the CHIPS Act substantially smaller in scale[2]. The latest round of China's National Integrated Circuit Industry Investment Fund, the Big Fund, reached $47 billion[59]. That round, the fund's third, was set up in May 2024[60].

Figure 16.3

Intel led CHIPS Act direct funding with $7.865 billion by July 2025

CHIPS Act direct funding awarded, by firm, as of July 2025



Source: GAO-26-107882, Semiconductors: Information on Projects Funded to Strengthen U.S. Supply Chain [51]

08

Taiwan

The world relies on Taiwan for most of its advanced chips. Taiwan's foundries held 81.9 percent of the world's contract chipmaking market in 2025, according to the Taiwan

2 China's Localization Drive in Semiconductors Gains Impetus from Allied Chip... Center for Strategic and International Studies, 24 March 2026. csis.org

51 GAO-26-107882, SEMICONDUCTORS: Information on Projects Funded to Strengthen... U.S. Government Accountability Office, 11 December 2025. files.gao.gov

54 TSMC Announces Additional $100 Billion Investment In Arizona. City of Phoenix, 16 July 2026. phoenix.gov

55 Proposal for the Chips Act 2.0. European Commission, 3 June 2026. digital-strategy.ec.europa.eu

56 Rapidus Achieves Significant Milestone at its State-of-the-Art Foundry with... Rapidus, 18 July 2025. rapidus.inc

57 Rapidus Secures 267.6 Billion Yen in Funding from Japan Government and Private... Rapidus, 27 February 2026. rapidus.inc

58 Rapidus Completes 150 Billion Yen Funding Round from Japan Government. Rapidus, 5 June 2026. rapidus.inc

59 China's Evolving Industrial Policy for AI. RAND Corporation, 26 June 2025. rand.org

60 Six banks to invest in big way in IC fund. The State Council of the People's Republic of China, 29 May 2024. english.www.gov.cn

Semiconductor Industry Association's August 2026 report[61]. TSMC makes its 2 nm chips only in Taiwan[46]. Taiwan held 69 percent of the world's logic capacity below 10 nm in 2022[4].

TSMC's advanced packaging capacity could not keep up with demand. Asked in October 2025 how much CoWoS packaging capacity he would add, TSMC's chairman said only that the company was "working very hard to narrow the gap between the demand and supply"[62]. He called wafer and packaging capacity both "very tight"[62].

The silicon shield argument holds that this concentration deters an attack on Taiwan, because so many countries rely on its chips[46]. The Stimson Center calls that open to debate, though many Taiwanese believe it[46].

With an eye to that shield, Taiwan's rules keep TSMC's newest process at home first[46]. As a result, TSMC's Arizona fabs have run a generation behind its fabs in Taiwan[46]. The Stimson Center expected 2 nm production in Arizona in 2028 at the earliest[46]. TSMC said in October 2025 that it was "preparing to upgrade our technologies faster to N2 and more advanced process technologies in Arizona"[63]. The four Arizona fabs it announced in July 2026 are to run at 2 nm or more advanced[54].

TSMC has also announced two advanced packaging fabs for Arizona[64]. But TSMC's work in Taiwan rests on a local supply chain of more than 300 firms, which the Stimson Center judged extremely difficult to copy in the United States[46]. The US push for chip plants at home has raised fears in Taiwan that the shield could weaken[46].

09

Who gains and who pays

- **Taiwan and Korea.** Fabs built abroad add capacity without taking any away from Taiwan and Korea. TSMC makes its 2 nm chips only in Taiwan[46]. Korean firms took 82.5 percent of the world's high-bandwidth memory revenue in 2025, by the market research firm IDC's count[12].
- **The Netherlands and Japan.** Their toolmakers build the controlled machines, so the two governments gain a say in who gets them. China accounts for a falling share of the firms' sales. At ASML, China's share of net system sales fell from 41 percent in 2024 to 33 percent in 2025[65]. That measure leaves out ASML's service and field option sales, which come from servicing and upgrading machines already installed[66]. Against total net sales, the measure in the chart, it fell from 36.1 percent to 29.1

4 Emerging Resilience in the Semiconductor Supply Chain. Boston Consulting Group and Semiconductor Industry..., May 2024, p. 14. semiconductors.org

12 SK hynix, prospectus on Form 424B4 dated 10 July 2026. U.S. Securities and Exchange Commission, 10 July 2026. sec.gov

46 Why Taiwan Fears 'America First' Risks Eroding Its 'Silicon Shield'. Stimson Center, 10 October 2025. stimson.org

54 TSMC Announces Additional $100 Billion Investment In Arizona. City of Phoenix, 16 July 2026. phoenix.gov

61 Overview on Taiwan Semiconductor Industry (2026 Edition). Taiwan Semiconductor Industry Association, 7 August 2026, p. 9. tsia.org.tw

62 Q3 2025 Taiwan Semiconductor Manufacturing Co Ltd Earnings Call (Chinese... Refinitiv StreetEvents, 16 October 2025, p. 7. investor.tsmc.com

63 Q3 2025 Taiwan Semiconductor Manufacturing Co Ltd Earnings Call (Chinese... Refinitiv StreetEvents, 16 October 2025, p. 5. investor.tsmc.com

64 Q3 2025 Taiwan Semiconductor Manufacturing Co Ltd Earnings Call (Chinese... Refinitiv StreetEvents, 16 October 2025, p. 13. investor.tsmc.com

65 ASML 2025 fourth-quarter and full-year results (investor presentation). ASML, 28 January 2026, p. 9. ourbrand.asml.com

66 ASML Holding, 2025 annual report, filed on Form 6-K. U.S. Securities and Exchange Commission, 25 February 2026. sec.gov

percent[50]. At Tokyo Electron, China's share of sales fell from 44.4 percent in the year to March 2024 to 34.1 percent in the year to March 2026[67]. The company said China's share of its sales dropped to 26.8 percent in the quarter to March 2026 because spending on leading-edge nodes grew faster than spending on mature nodes[68].

- **The United States.** The Semiconductor Industry Association, using data from World Semiconductor Trade Statistics, Gartner and Omdia, put US firms' share of all chip sales in 2025 at 53.4 percent, the highest since 1984[69]. US firms also took 65 percent of the world's logic chip sales in 2022[70]. Commerce expects the CHIPS-funded projects to lift the US share of global leading-edge logic chip manufacturing from 0 percent in 2022 to 20 percent by 2030[51].
- **China.** Chinese suppliers are replacing foreign ones at home, most visibly in mature-node and commodity chips[2]. In September 2025 SemiAnalysis expected China to make fewer Ascend chips in 2026 than in 2025 unless memory was smuggled in, because its own memory maker could supply too little[48].

Five questions to answer from memory

1. What does a country need for its export controls to work? see Chokepoints and who holds them
2. How do US export controls reach goods made in other countries? see US export controls, 2018 to today
3. What share of China's own chipmaking equipment market did Chinese toolmakers supply in 2025? see China's response
4. What share of ASML's system sales went to China in 2025? see Who gains and who pays
5. Where are most leading-edge chips made? see Taiwan

Five things to remember

— An export control works only when a country's firms sell something the buyer cannot get elsewhere.

— US controls also cover some foreign goods made with American technology.

— Chinese toolmakers' share of China's own chipmaking equipment market rose from 10 to 15 percent before 2022 to 35 percent in 2025.

— China's share of ASML's and Tokyo Electron's sales has fallen since 2024.

— Taiwan held 69 percent of the world's logic capacity below 10 nm in 2022, and TSMC makes its 2 nm chips only there.

2 China's Localization Drive in Semiconductors Gains Impetus from Allied Chip... Center for Strategic and International Studies, 24 March 2026. csis.org

48 Huawei Ascend Production Ramp: Die Banks, TSMC Continued Production, HBM is The... SemiAnalysis, 8 September 2025. newsletter.semianalysis.com

50 ASML Holding, annual report on Form 20-F for the fiscal year ended 31 December... U.S. Securities and Exchange Commission, 25 February 2026. sec.gov

51 GAO-26-107882, SEMICONDUCTORS: Information on Projects Funded to Strengthen... U.S. Government Accountability Office, 11 December 2025. files.gao.gov

67 FY2026 (April 2025-March 2026) Financial Announcement, transcript. Tokyo Electron, 30 April 2026, p. 27. tel.com

68 FY2026 (April 2025-March 2026) Financial Announcement, transcript. Tokyo Electron, 30 April 2026, p. 5. tel.com

69 2026 State of the U.S. Semiconductor Industry. Semiconductor Industry Association, 2 July 2026, p. 7. semiconductors.org

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A company filings, governments, standards bodies, universities and research institutions; B SemiAnalysis, Reuters, the Financial Times and Nikkei Asia.

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